

.gitkeep

@@\_WebAppEntry.gs

/\*\*

\* Host dependency wiring and DB client bootstrap.

\*/

```
function resolveHostConfig(config) {
  var base = ConfigBootstrap.load();
  if (!config) return base;
```

```
  var merged = {};
  Object.keys(base).forEach(function(key) {
    if (Object.prototype.hasOwnProperty.call(config, key)) {
      merged[key] = config[key];
    } else {
      merged[key] = base[key];
    }
  });
}
```

```
  return merged;
}
```

```
function createHostDbClient(config) {
  var cfg = resolveHostConfig(config);
  var db = SheetDb.createClient({ spreadsheetId: cfg.spreadsheetId });
  var schema = function(columns, fields) { return { columns: columns, fields: fields || {} }; };
```

```
  db.schema('learners', schema(['id', 'slackUserId', 'email', 'name', 'status', 'createdAt', 'updatedAt', 'deletedAt']));
  db.schema('enrollment', schema(['id', 'learnerId', 'courseId', 'track', 'status', 'createdAt', 'updatedAt', 'deletedAt']));
  db.schema('lessons', schema(
    [
      'id', 'courseId', 'moduleId', 'sequenceNumber', 'track',
      'title', 'topic', 'objective', 'difficulty',
      'hook', 'coreContent', 'insight', 'takeaway',
      'mission', 'missionType', 'missionDuration', 'verification', 'submitBlock',
      'contentRef', 'slackPayload', 'active', 'createdAt', 'updatedAt', 'deletedAt'
    ]
  ));
  db.schema('learner_progress', schema(['id', 'learnerId', 'lessonId', 'state', 'dueAt', 'completedAt', 'createdAt', 'updatedAt']));
  db.schema('submission_log', schema(['id', 'learnerId', 'lessonId', 'submitKey', 'payload', 'createdAt', 'updatedAt']));
  db.schema('delivery_queue', schema(['id', 'learnerId', 'lessonId', 'status', 'runAt', 'priority', 'attempts', 'availability']));
  db.schema('retry_queue', schema(['id', 'jobType', 'payload', 'attempts', 'nextRunAt', 'status', 'lastError', 'correlationId']));
  db.schema('audit_log', schema(['id', 'actor', 'action', 'resourceType', 'resourceId', 'status', 'message', 'metadata']));
  db.schema('courses', schema(['id', 'courseTitle', 'brandScope', 'moduleOrder', 'durationMonths', 'status', 'createdAt']));
  db.schema('modules', schema(['id', 'courseId', 'moduleTitle', 'monthNumber', 'lessonCount', 'status', 'createdAt']));
  db.schema('onboarding_requests', schema(['id', 'requestorUserId', 'targetEmail', 'targetName', 'targetBrand', 'targetResource']));
  db.schema('onboarding_checklists', schema(['id', 'learnerId', 'taskTitle', 'taskOwner', 'dueDate', 'status', 'createdAt']));
  db.schema('onboarding_task_log', schema(['id', 'checklistItemId', 'learnerId', 'eventType', 'eventBy', 'note', 'createdAt']));
  db.schema('lesson_qa_records', schema(['id', 'lessonId', 'qaStatus', 'qaScore', 'qaReviewer', 'qaDate', 'qaNotes', 'createdAt']));
  db.schema('app_config', schema(['id', 'configKey', 'configValue', 'description', 'updatedAt', 'createdAt', 'deletedAt']));

  return db;
}
```

```
function createHostRouter() {
  return SlackRouter;
}
```

```
function createHostDependencies(config) {
  var cfg = resolveHostConfig(config);
  var db = createHostDbClient(cfg);
  var dbAdapter = new SheetsDBAdapter(db);
  var sheetsAudit = createSheetsAuditLogger(db);
  db._audit = sheetsAudit;

  var blocks = new SlackBlockKitBuilder();
  var stateMachine = new LearnerProgressStateMachine();
  var configRepo = new ConfigRepository(db);
  var workflow = new WorkflowEngine(function(action, metadata) { db.audit(action, 'workflow', metadata || {}); });
  var repositories = {
    learnerRepo: new LearnerRepository(db),
    progressRepo: new ProgressRepository(db),
    lessonRepo: new LessonRepository(db),
    configRepo: configRepo
  };

  var retryResolver = new RetryResolver(db, cfg, configRepo);
  var slackApi = new SlackApiClient(cfg, retryResolver);
}
```

```

var deps = {
  config: cfg,
  db: db,
  dbAdapter: dbAdapter,
  parser: SlackPayloadParser,
  security: SlackSecurity,
  router: createHostRouter(),
  slackApiClient: slackApi,
  blockKitBuilder: blocks,
  stateMachine: stateMachine,
  repositories: repositories,
  enrollmentService: new LmsEnrollmentService(db, slackApi, blocks, stateMachine, cfg, repositories, workflow, SecurityService),
  lessonService: new LmsLessonService(db, slackApi, blocks, stateMachine, cfg, repositories, workflow),
  progressService: new LmsProgressService(db, slackApi, blocks, cfg, repositories, workflow),
  completionService: new LmsCompletionService(db, slackApi, blocks, stateMachine, cfg, repositories, workflow, SecurityService),
  reminderService: new LmsReminderService(db, slackApi, blocks, cfg),
  reportService: new ReportService(db, cfg, repositories),
  onboardingService: new OnboardingService(db, slackApi, blocks, cfg),
  backupService: new BackupService(db, cfg),
  healthMonitor: new HealthMonitor(db, cfg),
  retryResolver: retryResolver,
  audit: function(action, metadata) {
    db.table('audit_log').insert({
      actor: 'system',
      action: action,
      resourceType: 'slack_host',
      resourceId: '',
      status: 'info',
      message: action,
      metadata: JSON.stringify(metadata || {})
    });
    db.audit(action, 'audit_log', metadata || {});
  }
};

deps.slackService = new SlackService({
  lessonService: deps.lessonService,
  completionService: deps.completionService,
  progressService: deps.progressService,
  enrollmentService: deps.enrollmentService,
  reportService: deps.reportService,
  onboardingService: deps.onboardingService
}, deps.blockKitBuilder, deps.config, SecurityService, configRepo);

return deps;
}

function migrateLearnerProgressStates() {
  var deps = createHostDependencies();
  return deps.stateMachine.migrateLegacyStatesInDb(deps.db);
}

01_SlackRouter.gs
/**
 * Central request router for normalized Slack envelopes.
 */
var SlackRouter = {
  _routeRegistry: {
    url_verification: 'handleUrlVerification',
    slash_command: 'handleSlashCommand',
    interactivity: 'handleInteractivity',
    event_callback: 'handleEventCallback',
    workflow_webhook: 'handleWorkflowWebhook'
  },
  route: function(parsed, deps) {
    if (!parsed || !parsed.ok) {
      return { ok: false, code: 'PARSE_FAILED', response: ErrorService.create('PARSE_FAILED', 'invalid_request', false) }
    }

    if (parsed.routeType === 'url_verification') {
      return { ok: true, code: 'URL_VERIFICATION', response: { challenge: parsed.body.challenge } };
    }

    var handlerName = this._routeRegistry[parsed.routeType];
    if (!handlerName || !deps.slackService || typeof deps.slackService[handlerName] !== 'function') {
      return {
        ok: false,
        code: 'UNSUPPORTED_ROUTE',

```

```

        response: ErrorService.create('UNSUPPORTED_ROUTE', 'unsupported_route: ' + parsed.routeType, false)
    };
}

return { ok: true, code: parsed.routeType.toUpperCase(), response: deps.slackService[handlerName](parsed) };
}
};

02_SlackPayloadParser.gs
/**
 * Normalizes Slack requests to a single envelope.
 */
var SlackPayloadParser = {
    parse: function(e) {
        var rawBody = (e && e.postData && e.postData.contents) ? String(e.postData.contents) : '';
        var params = (e && e.parameter) ? e.parameter : {};
        var headers = (e && e.headers) ? e.headers : {};
        var body = this._parseJson(rawBody);
        var interaction = params.payload ? this._parseJson(params.payload) : null;
        var parseOk = true;

        if (!body && rawBody && String((e.postData && e.postData.type) || '').indexOf('application/json') !== -1) {
            parseOk = false;
        }

        body = body || {};

        var routeType = 'unknown';
        if (body.type === 'url_verification') routeType = 'url_verification';
        else if (params.command) routeType = 'slash_command';
        else if (interaction) routeType = 'interactivity';
        else if (body.type === 'event_callback') routeType = 'event_callback';
        else if (body.workflow || body.workflow_step || params.workflow) routeType = 'workflow_webhook';

        var source = interaction || body || params;
        return {
            ok: parseOk,
            routeType: routeType,
            rawBody: rawBody,
            body: body,
            params: params,
            command: String(params.command || ''),
            payloadType: String((interaction && interaction.type) || body.type || ''),
            userId: this._pick(source, ['user.id', 'user_id', 'event.user']),
            channelId: this._pick(source, ['channel.id', 'channel_id', 'event.channel']),
            teamId: this._pick(source, ['team.id', 'team_id']),
            triggerId: this._pick(source, ['trigger_id']),
            responseUrl: this._pick(source, ['response_url']),
            slackSignature: String(headers['x-slack-signature'] || headers['X-Slack-Signature'] || params.x_slack_signature || ''),
            slackTimestamp: String(headers['x-slack-request-timestamp'] || headers['X-Slack-Request-Timestamp'] || params.x_slack_request_timestamp || ''),
            interaction: interaction,
            parseError: parseOk ? '' : 'Invalid JSON body'
        };
    },

    _parseJson: function(text) {
        try { return JSON.parse(text); } catch (err) { return null; }
    },

    _pick: function(obj, paths) {
        var i;
        for (i = 0; i < paths.length; i++) {
            var value = this._get(obj, paths[i]);
            if (value) return String(value);
        }
        return '';
    },

    _get: function(obj, path) {
        return String(path || '').split('.').reduce(function(acc, key) {
            return (acc && acc[key] !== null) ? acc[key] : '';
        }, obj || {});
    }
};

03_SlackService.gs
/**
 * Slack dispatch layer for slash/event/interactivity registries.
 */
class SlackService {

```



```

handleWorkflowWebhook(parsed) {
  var data = (parsed.body && parsed.body.data) || {};
  var slackUserId = this._security.sanitizeInput(data.user_id || parsed.params.user_id || '');
  var email = this._security.sanitizeInput(data.email || parsed.params.email || '');
  var name = this._security.sanitizeInput(data.name || parsed.params.name || '');
  var courseId = this._security.sanitizeInput(data.course_id || parsed.params.course_id || this._config.defaultCourseId);

  if (!slackUserId) {
    return ErrorService.create('INVALID_WORKFLOW_PAYLOAD', 'Missing user_id in workflow payload', false);
  }

  return this._enrollment.enrollLearner({
    slackUserId: slackUserId,
    email: email,
    name: name,
    courseId: courseId
  });
}

_handleHelp() {
  return { response_type: 'ephemeral', text: 'Commands: /learn, /submit <lesson_id> complete, /progress, /help, /enroll' };
}

_handleEnroll(parsed) {
  return this._enrollment.enrollLearner({
    slackUserId: parsed.userId,
    courseId: this._security.sanitizeInput(parsed.params.text || '') || this._config.defaultCourseId
  });
}

_handleOnboard(parsed) {
  var adminAccess = this._requireAdmin(parsed, '/onboard');
  if (adminAccess) return adminAccess;
  var email = this._security.sanitizeInput((parsed.params && parsed.params.text) || '');
  if (!email) {
    return { response_type: 'ephemeral', text: 'Usage: /onboard [email]' };
  }

  if (this._configRepo && !this._configRepo.getFlag('enable_onboarding', true)) {
    return { response_type: 'ephemeral', text: ':warning: Onboarding feature flag is disabled.' };
  }

  return this._onboarding.startOnboarding({
    requestorUserId: parsed.userId,
    targetEmail: email,
    source: 'slash_command'
  });
}

_handleReport(parsed) {
  var adminAccess = this._requireAdmin(parsed, '/report');
  if (adminAccess) return adminAccess;
  if (this._configRepo && !this._configRepo.getFlag('enable_reporting', true)) {
    return { response_type: 'ephemeral', text: ':warning: Reporting feature flag is disabled.' };
  }
  var dashboard = this._report.buildAdminDashboard(parsed && parsed.userId);
  return { response_type: 'ephemeral', text: 'Admin report', blocks: this._blocks.buildAdminSummary(dashboard) };
}

_handleGaps(parsed) {
  return this._report.handleGaps(parsed);
}

_handleAudit(parsed) {
  var adminAccess = this._requireAdmin(parsed, '/audit');
  if (adminAccess) return adminAccess;
  return this._onboarding.handleAuditQuery(parsed);
}

_handleMix(parsed) {
  return this._lesson.handleMix(parsed);
}

_handleReinforce(parsed) {
  return this._progress.handleReinforce(parsed);
}

_handleOffboard(parsed) {
  var adminAccess = this._requireAdmin(parsed, '/offboard');

```

```

    if (adminAccess) return adminAccess;
    return this._onboarding.handleOffboard(parsed);
}

_requireAdmin(parsed, commandName) {
    var adminIds = this._config.adminUserIds || [];
    if (adminIds.indexOf(parsed.userId) !== -1) return null;
    return { response_type: 'ephemeral', text: ':lock: ' + commandName + ' is restricted to admins.' };
}

_handleDm(event) {
    var text = String(event.text || '').toLowerCase();
    if (text.indexOf('progress') !== -1) {
        return { ok: true, text: 'Use '/progress' for your learner snapshot.' };
    }
    if (text.indexOf('help') !== -1) {
        return { ok: true, text: 'Need help? Try '/help' for all commands.' };
    }
    return { ok: true, text: 'Welcome! Use '/learn' to get your current lesson, or '/help' for all commands.' };
}

_handleModalSubmission(payload) {
    var values = (payload.view && payload.view.state && payload.view.state.values) || {};
    var lessonValue = this._extractModalValue(values, 'lesson', 'lesson_id');
    var notesValue = this._extractModalValue(values, 'notes', 'notes_value');
    var learnerId = payload.user && payload.user.id;

    var persisted = this._completion.recordSubmission({
        slackUserId: learnerId,
        lessonId: lessonValue,
        payload: JSON.stringify({ notes: notesValue, source: 'modal_submission' })
    });

    if (!persisted.ok) {
        return {
            response_action: 'errors',
            errors: { lesson: persisted.message || 'Unable to submit lesson.' }
        };
    }

    return { response_action: 'clear' };
}

_extractModalValue(values, blockId, actionId) {
    if (!values[blockId] || !values[blockId][actionId]) return '';
    return this._security.sanitizeInput(values[blockId][actionId].value || '');
}
}

04_SlackSecurity.gs
/**
 * Backward-compatible Slack security facade.
 */
var SlackSecurity = {
    verify: function(parsed, config) {
        return SecurityService.verifySlackRequest(parsed, config);
    }
};

05_AppSheetWebhook.gs
/**
 * AppSheet webhook contract handler.
 * All inbound AppSheet actions arrive here after token verification in the router.
 */
var AppSheetWebhook = {
    verifyToken: function(parsed, config) {
        var expected = String(config.appsheetWebhookToken || '');
        if (!expected) {
            return { ok: false, code: 'APPSHEET_TOKEN_NOT_CONFIGURED', message: 'AppSheet webhook token not set in script prop' };
        }
        var supplied = String(
            (parsed.body && parsed.body.appsheet_token) ||
            (parsed.params && parsed.params.appsheet_token) || ''
        );
        if (!supplied || supplied !== expected) {
            return { ok: false, code: 'APPSHEET_TOKEN_INVALID', message: 'AppSheet webhook token mismatch.' };
        }
        return { ok: true, code: 'OK', message: 'AppSheet token verified.' };
    },

```

```

route: function(action, payload, deps) {
  if (action === 'enroll_learner') {
    return deps.enrollmentService.enrollLearner({
      slackUserId: payload.slackUserId || '',
      email: payload.email || '',
      name: payload.name || '',
      courseId: payload.courseId || deps.config.defaultCourseId
    });
  }
  if (action === 'send_reminder') {
    return deps.reminderService.sendReminderForLearner(payload.learnerId || '');
  }
  if (action === 'archive_learner') {
    return deps.onboardingService.offboardLearner({
      learnerId: payload.learnerId || '',
      slackUserId: payload.slackUserId || '',
      email: payload.email || '',
      requestorUserId: payload.requestorUserId || 'appsheet',
      source: 'appsheet_archive_learner'
    });
  }
  return { ok: false, code: 'UNKNOWN_ACTION', message: 'Unrecognised AppSheet action: ' + action };
}
};

```

#### 07\_AuditLogger\_Sheets.gs

```

/**
 * Creates an AuditLogger instance wired to the audit_log Sheets table.
 * Use this in createHostDependencies() instead of relying on the library default.
 * @param {DbClient} db
 * @return {AuditLogger}
 */
function createSheetsAuditLogger(db) {
  return new AuditLogger({
    sink: function(entry) {
      try {
        db.table('audit_log').insert({
          actor: entry.actor || 'system',
          action: entry.action,
          resourceType: entry.table || '',
          resourceId: (entry.metadata && entry.metadata.id) ? String(entry.metadata.id) : '',
          status: 'info',
          message: entry.action,
          metadata: JSON.stringify(entry.metadata || {})
        });
      } catch (sinkErr) {
        Logger.log('AuditLogger Sheets sink failed: ' + String(sinkErr));
      }
    }
  });
}
};

```

#### 08\_DbSchema.gs

```

var DbSchema = {
  LEARNERS: { TABLE: 'learners', PK: 'id' },
  ENROLLMENT: { TABLE: 'enrollment', PK: 'id' },
  LESSONS: { TABLE: 'lessons', PK: 'id' },
  LEARNER_PROGRESS: { TABLE: 'learner_progress', PK: 'id' },
  SUBMISSION_LOG: { TABLE: 'submission_log', PK: 'id' },
  DELIVERY_QUEUE: { TABLE: 'delivery_queue', PK: 'id' },
  RETRY_QUEUE: { TABLE: 'retry_queue', PK: 'id' },
  AUDIT_LOG: { TABLE: 'audit_log', PK: 'id' },
  COURSES: { TABLE: 'courses', PK: 'id' },
  MODULES: { TABLE: 'modules', PK: 'id' },
  ONBOARDING_REQUESTS: { TABLE: 'onboarding_requests', PK: 'id' },
  ONBOARDING_CHECKLISTS: { TABLE: 'onboarding_checklists', PK: 'id' },
  ONBOARDING_TASK_LOG: { TABLE: 'onboarding_task_log', PK: 'id' },
  LESSON_QA_RECORDS: { TABLE: 'lesson_qa_records', PK: 'id' },
  APP_CONFIG: { TABLE: 'app_config', PK: 'id' }
};

```

#### 09\_LmsEnrollmentService.gs

```

class LmsEnrollmentService {
  constructor(db, slackApiClient, blocks, stateMachine, config, repositories, workflowEngine, securityService) {
    this._db = db;
    this._slack = slackApiClient;
    this._blocks = blocks;
    this._state = stateMachine;
    this._config = config || {};
    this._repos = repositories || {};
  }
}

```

```

    this._workflow = workflowEngine;
    this._security = securityService || SecurityService;
}

enrollLearner(input) {
    var self = this;
    return this._workflow.run('enrollment', input || {}, {
        validate: function(ctx) {
            var slackUserId = self._security.sanitizeInput(ctx.trigger.slackUserId);
            if (!slackUserId) throw { code: 'INVALID_INPUT', message: 'slackUserId is required.' };
            ctx.data.slackUserId = slackUserId;
        },
        process: function(ctx) {
            ctx.data.learner = self.ensureLearnerRecord({
                slackUserId: ctx.data.slackUserId,
                email: self._security.sanitizeInput(ctx.trigger.email || ''),
                name: self._security.sanitizeInput(ctx.trigger.name || '')
            });

            ctx.data.enrollment = self.ensureEnrollmentRecord({
                learnerId: ctx.data.learner.id,
                courseId: self._security.sanitizeInput(ctx.trigger.courseId || self._config.defaultCourseId),
                track: self._config.defaultTrack
            });

            ctx.data.queue = self.queueFirstLesson({
                learnerId: ctx.data.learner.id,
                courseId: ctx.data.enrollment.courseId
            });
        },
        persist: function(ctx) {
            self.sendWelcomeDm({
                slackUserId: ctx.data.learner.slackUserId,
                learnerId: ctx.data.learner.id,
                courseId: ctx.data.enrollment.courseId
            });
        },
        respond: function(ctx) {
            ctx.result = {
                ok: true,
                code: 'ENROLLED',
                learnerId: ctx.data.learner.id,
                enrollmentId: ctx.data.enrollment.id,
                queueId: ctx.data.queue.queueId || '',
                correlationId: ctx.correlationId
            };
        },
        audit: function(ctx) {
            self._db.audit('enroll_learner', 'enrollment', { learnerId: ctx.data.learner.id, enrollmentId: ctx.data.enrollment.id });
        }
    });
}

findLearnerBySlackUserId(slackUserId) {
    return this._repos.learnerRepo.findBySlackUserId(slackUserId);
}

ensureLearnerRecord(input) {
    var existing = this.findLearnerBySlackUserId(input.slackUserId);
    if (existing) return existing;
    return this._repos.learnerRepo.insert({
        slackUserId: input.slackUserId,
        email: input.email || '',
        name: input.name || '',
        status: 'active'
    });
}

ensureEnrollmentRecord(input) {
    var existing = this._db.table('enrollment').findAll().filter(function(row) {
        return row.learnerId === input.learnerId && row.courseId === input.courseId;
    })[0];
    if (existing) return existing;
    return this._db.table('enrollment').insert({
        learnerId: input.learnerId,
        courseId: input.courseId,
        track: input.track || 'ONBOARDING',
        status: 'active'
    });
}

```





```

    this._notifyManager(input.managerSlackId, input.targetName || input.targetEmail, requestRow.id);
  }

  this._db.audit('start_onboarding', 'onboarding_requests', { requestId: requestRow.id, targetEmail: input.targetEmail

  return {
    ok: true,
    code: 'ONBOARDING_STARTED',
    requestId: requestRow.id,
    checklistCount: itemRows.length
  };
}

_getDefaultChecklistItems(learnerId) {
  var now = new Date();
  var dueDate = new Date(now.getTime() + 7 * 24 * 60 * 60 * 1000).toISOString();
  return [
    { learnerId: learnerId, taskTitle: 'Complete IT setup and system access', taskOwner: 'IT', dueDate: dueDate, status: 'pending' },
    { learnerId: learnerId, taskTitle: 'Complete pre-onboarding modules (PRE-M01 to PRE-M05)', taskOwner: 'learner', dueDate: dueDate, status: 'pending' },
    { learnerId: learnerId, taskTitle: 'Meet with manager for orientation', taskOwner: 'manager', dueDate: dueDate, status: 'pending' },
    { learnerId: learnerId, taskTitle: 'Review RWR Group values and culture', taskOwner: 'learner', dueDate: dueDate, status: 'pending' },
    { learnerId: learnerId, taskTitle: 'Complete Slack and Google Workspace training', taskOwner: 'learner', dueDate: dueDate, status: 'pending' }
  ];
}

_notifyManager(managerSlackId, learnerName, requestId) {
  try {
    var dm = this._slack.openDm(managerSlackId);
    if (!dm.ok) return;
    this._slack.postMessage(dm.channelId, 'New hire onboarding started',
      [{ type: 'section', text: { type: 'mrkdwn', text: ':wave: Onboarding has been started for *' + learnerName + '*.' } }]);
  } catch (err) {
    Logger.log('Manager notify failed: ' + String(err));
  }
}

advanceOnboardingState(learnerId, completedTaskId, status) {
  try {
    this._db.table('onboarding_task_log').insert({
      checklistItemId: completedTaskId,
      learnerId: learnerId,
      eventType: status || 'completed',
      eventBy: learnerId,
      note: 'Checklist updated via Slack action'
    });
    this._db.table('onboarding_checklists').update(completedTaskId, { status: status || 'complete' });
    return { ok: true, code: 'TASK_ADVANCED' };
  } catch (err) {
    return { ok: false, code: 'ADVANCE_FAILED', message: String(err.message || err) };
  }
}

handleOffboard(ctx) {
  var email = SecurityService.sanitizeInput((ctx && ctx.params && ctx.params.text) || '');
  if (!email) {
    return { response_type: 'ephemeral', text: 'Usage: /offboard [email]' };
  }
  var result = this.offboardLearner({
    requestorUserId: (ctx && ctx.userId) || 'system',
    email: email,
    source: 'slash_command'
  });
  if (!result.ok) {
    return { response_type: 'ephemeral', text: ':warning: ' + (result.message || result.code || 'Unable to offboard learner') };
  }
  return { response_type: 'ephemeral', text: ':white_check_mark: Offboarded ' + email + '. Cancelled queue items: ' + result.cancelledQueueItems };
}

detectInactiveLearners(daysThreshold) {
  var thresholdDays = Number(daysThreshold || 3);
  var cutoffMs = Date.now() - (thresholdDays * 24 * 60 * 60 * 1000);
  var learners = this._db.table('learners').findAll();
  var progressRows = this._db.table('learner_progress').findAll();
  var submissions = this._db.table('submission_log').findAll();

  var latestByLearner = {};
  var pushDate = function(learnerId, isoDate) {
    var parsed = Date.parse(String(isoDate || ''));
    if (!learnerId || isNaN(parsed)) return;
    latestByLearner[learnerId] = parsed > latestByLearner[learnerId] ? parsed : latestByLearner[learnerId];
  };
  learners.forEach(learner => {
    pushDate(learner.id, learner.created_at);
    learner.progress.forEach(progress => {
      pushDate(learner.id, progress.updated_at);
    });
    learner.submissions.forEach(submission => {
      pushDate(learner.id, submission.created_at);
    });
  });
  var inactiveLearners = learners.filter(learner => !latestByLearner[learner.id]);
  return { ok: true, code: 'DETECT_INACTIVE_LEARNERS', inactiveLearners: inactiveLearners };
}

```

```

    if (!latestByLearner[learnerId] || latestByLearner[learnerId] < parsed) latestByLearner[learnerId] = parsed;
  };

  learners.forEach(function(row) { pushDate(row.id, row.updatedAt || row.createdAt); });
  progressRows.forEach(function(row) { pushDate(row.learnerId, row.updatedAt || row.completedAt || row.createdAt); });
  submissions.forEach(function(row) { pushDate(row.learnerId, row.createdAt || row.updatedAt); });

  var inactive = learners.filter(function(learner) {
    if (learner.status === 'offboarded' || learner.status === 'inactive') return false;
    var lastActivityMs = latestByLearner[learner.id] || Date.parse(String(learner.createdAt || ''));
    return !!lastActivityMs && lastActivityMs < cutoffMs;
  }).map(function(learner) {
    return {
      learnerId: learner.id,
      email: learner.email || '',
      slackUserId: learner.slackUserId || '',
      lastActivityAt: new Date(latestByLearner[learner.id]).toISOString()
    };
  });

  return {
    ok: true,
    thresholdDays: thresholdDays,
    cutoffAt: new Date(cutoffMs).toISOString(),
    count: inactive.length,
    learners: inactive
  };
}

queryAuditLog(input) {
  var filter = input || {};
  var fromMs = Date.parse(String(filter.from || ''));
  var toMs = Date.parse(String(filter.to || ''));
  var actor = String(filter.actor || '').trim();
  var resourceType = String(filter.resourceType || '').trim();
  var resourceId = String(filter.resourceId || '').trim();

  var rows = this._db.table('audit_log').findAll().filter(function(row) {
    var rowMs = Date.parse(String(row.createdAt || ''));
    if (!isNaN(fromMs) && (isNaN(rowMs) || rowMs < fromMs)) return false;
    if (!isNaN(toMs) && (isNaN(rowMs) || rowMs > toMs)) return false;
    if (actor && row.actor !== actor) return false;
    if (resourceType && row.resourceType !== resourceType) return false;
    if (resourceId && row.resourceId !== resourceId) return false;
    return true;
  }).sort(function(a, b) { return Date.parse(String(b.createdAt || '')) - Date.parse(String(a.createdAt || '')); });

  return {
    ok: true,
    filters: {
      from: isNaN(fromMs) ? '' : new Date(fromMs).toISOString(),
      to: isNaN(toMs) ? '' : new Date(toMs).toISOString(),
      actor: actor,
      resourceType: resourceType,
      resourceId: resourceId
    },
    count: rows.length,
    items: rows
  };
}

handleAuditQuery(ctx) {
  var text = SecurityService.sanitizeInput((ctx && ctx.params && ctx.params.text) || '');
  var parsed = this._parseAuditFilterText(text);
  var audit = this.queryAuditLog(parsed);
  var preview = audit.items.slice(0, 5).map(function(row) {
    return 'â€¢ ' + String(row.createdAt || '') + ' | ' + String(row.actor || '-') + ' | ' + String(row.action || '-');
  }).join('\n');
  return {
    response_type: 'ephemeral',
    text: '*Audit results:* ' + String(audit.count) + (preview ? '\n' + preview : '\nNo matching records.')
  };
}

offboardLearner(input) {
  var params = input || {};
  var learner = null;
  if (params.learnerId) learner = this._db.table('learners').findById(params.learnerId);
  if (!learner && params.slackUserId) {
    learner = this._db.table('learners').findAll().filter(function(row) { return row.slackUserId === params.slackUserI

```

```

    }
    if (!learner && params.email) {
        learner = this._db.table('learners').findAll().filter(function(row) {
            return String(row.email || '').toLowerCase() === String(params.email || '').toLowerCase();
        })[0] || null;
    }
    if (!learner) return { ok: false, code: 'LEARNER_NOT_FOUND', message: 'Learner not found for offboarding request.' };

    this._db.table('learners').update(learner.id, { status: 'offboarded' });

    var enrollmentRows = this._db.table('enrollment').findAll().filter(function(row) { return row.learnerId === learner.id });
    for (var i = 0; i < enrollmentRows.length; i++) {
        this._db.table('enrollment').update(enrollmentRows[i].id, { status: 'inactive' });
    }

    var nowMs = Date.now();
    var cancelled = 0;
    var deliveryRows = this._db.table('delivery_queue').findAll().filter(function(row) {
        var runAt = Date.parse(String(row.runAt || ''));
        var isFuture = !isNaN(runAt) ? runAt > nowMs : true;
        return row.learnerId === learner.id && row.status === 'queued' && isFuture;
    });
    for (var d = 0; d < deliveryRows.length; d++) {
        this._db.table('delivery_queue').update(deliveryRows[d].id, { status: 'cancelled' });
        cancelled++;
    }

    var retryRows = this._db.table('retry_queue').findAll().filter(function(row) {
        if (row.status !== 'queued' && row.status !== 'pending') return false;
        var nextRun = Date.parse(String(row.nextRunAt || ''));
        var isFuture = !isNaN(nextRun) ? nextRun > nowMs : true;
        if (!isFuture) return false;
        var payloadText = String(row.payload || '');
        return payloadText.indexOf(learner.id) !== -1;
    });
    for (var r = 0; r < retryRows.length; r++) {
        this._db.table('retry_queue').update(retryRows[r].id, { status: 'cancelled', lastError: 'Cancelled due to learner cancelled' });
    }

    this._db.audit('offboard_learner', 'learners', {
        requestorUserId: params.requestorUserId || 'system',
        learnerId: learner.id,
        email: learner.email || '',
        source: params.source || 'manual',
        cancelledQueueItems: cancelled
    });

    return {
        ok: true,
        code: 'LEARNER_OFFBOARDED',
        learnerId: learner.id,
        email: learner.email || '',
        cancelledQueueItems: cancelled
    };
}

_parseAuditFilterText(text) {
    var out = {};
    String(text || '').split(/\s+/).forEach(function(part) {
        var bits = part.split('=');
        if (bits.length < 2) return;
        var key = String(bits[0] || '').toLowerCase();
        var value = bits.slice(1).join('=');
        if (key === 'from' || key === 'to') out[key] = value;
        if (key === 'actor') out.actor = value;
        if (key === 'resource' || key === 'resourcetype') out.resourceType = value;
        if (key === 'resourceid') out.resourceId = value;
    });
    return out;
}

11_SlackApiClient.gs
class SlackApiClient {
    constructor(config, retryResolver) {
        this._config = config || {};
        this._retry = retryResolver || null;
        this._token = String(this._config.slackBotToken || this._config.slackBotTokenFallback || '');
    }
}

```

```

postMessage(channel, text, blocks) {
  if (this._isQuietHours()) {
    return { ok: false, code: 'QUIET_HOURS', message: 'Message suppressed during quiet hours.', retryable: false };
  }
  return this._call('chat.postMessage', { channel: channel, text: text, blocks: blocks || [] });
}

postEphemeral(channel, user, text, blocks) {
  return this._call('chat.postEphemeral', { channel: channel, user: user, text: text, blocks: blocks || [] });
}

openView(triggerId, view) { return this._call('views.open', { trigger_id: triggerId, view: view }); }
updateMessage(channel, ts, text, blocks) { return this._call('chat.update', { channel: channel, ts: ts, text: text, bl

openDm(userId) {
  var res = this._call('conversations.open', { users: userId });
  return {
    ok: !!res.ok,
    code: res.code,
    message: res.message,
    retryable: !!res.retryable,
    method: res.method || 'conversations.open',
    status: res.status,
    channelId: res.data && res.data.channel ? res.data.channel.id : ''
  };
}

_isQuietHours() {
  var start = Number(this._config.quietHoursStart || 21);
  var end = Number(this._config.quietHoursEnd || 7);
  var hour = new Date().getUTCHours() + 12;
  hour = hour % 24;
  if (start > end) return hour >= start || hour < end;
  return hour >= start && hour < end;
}

_call(method, payload) {
  if (!this._token) {
    return ErrorService.create('MISSING_BOT_TOKEN', 'SLACK_BOT_TOKEN missing', false);
  }

  var response;
  try {
    response = UrlFetchApp.fetch('https://slack.com/api/' + method, {
      method: 'post',
      contentType: 'application/json; charset=utf-8',
      headers: { Authorization: 'Bearer ' + this._token },
      payload: JSON.stringify(payload || {}),
      muteHttpExceptions: true
    });
  } catch (err) {
    var fetchErr = { ok: false, code: 'HTTP_FETCH_FAILED', message: String(err.message || err), retryable: true, metho
    this._queueRetry(method, payload, fetchErr);
    return fetchErr;
  }

  var status = response.getResponseCode();
  var body = {};
  try { body = JSON.parse(response.getContentText() || '{}'); } catch (e) { body = { ok: false, error: 'invalid_json'

  if (status < 200 || status >= 300 || !body.ok) {
    var errShape = {
      ok: false,
      code: body.error || 'SLACK_API_ERROR',
      message: 'Slack API call failed for ' + method,
      retryable: status >= 500 || body.error === 'ratelimited',
      method: method,
      status: status,
      data: body
    };
    this._queueRetry(method, payload, errShape);
    return errShape;
  }

  return { ok: true, code: 'OK', message: 'success', data: body };
}

_queueRetry(method, payload, errShape) {
  if (!this._retry || !errShape.retryable) return;

```



```

    return { ok: true, learnerId: learner.id, lesson: lesson, progress: progress };
}

deliverLessonToLearner(learnerId, lessonId) {
  var self = this;
  return this._workflow.run('lesson_delivery', { learnerId: learnerId, lessonId: lessonId }, {
    validate: function(ctx) {
      ctx.data.learner = self._repos.learnerRepo.findById(ctx.trigger.learnerId);
      if (!ctx.data.learner) throw { code: 'LEARNER_NOT_FOUND', message: 'Learner not found' };
      ctx.data.lesson = self._repos.lessonRepo.findById(ctx.trigger.lessonId) || { id: ctx.trigger.lessonId, title: 'L
    },
    process: function(ctx) {
      var payload = self._buildLessonMessagePayload(ctx.data.lesson, 'New lesson available');
      ctx.data.text = payload.text;
      ctx.data.blocks = payload.blocks;
      ctx.data.dm = self._slack.openDm(ctx.data.learner.slackUserId);
      if (!ctx.data.dm.ok) throw { code: ctx.data.dm.code, message: ctx.data.dm.message, retryable: !!ctx.data.dm.retr
      ctx.data.sent = self._slack.postMessage(ctx.data.dm.channelId, ctx.data.text, ctx.data.blocks);
      if (!ctx.data.sent.ok) throw { code: ctx.data.sent.code, message: ctx.data.sent.message, retryable: !!ctx.data.s
    },
    persist: function(ctx) {
      self._db.table('delivery_queue').insert({ learnerId: learnerId, lessonId: lessonId, status: 'delivered', runAt:
      var active = self._repos.progressRepo.findByLearnerAndLesson(learnerId, lessonId);
      if (active) {
        var transitioned = self._state.transition(active, self._state.states.IN_PROGRESS, { source: 'delivery' });
        if (transitioned.ok) self._repos.progressRepo.update(active.id, { state: transitioned.record.state });
      }
    },
    respond: function(ctx) {
      ctx.result = { ok: true, code: 'DELIVERED', learnerId: learnerId, lessonId: lessonId, correlationId: ctx.correla
    },
    audit: function(ctx) {
      self._db.audit('lesson_delivered', 'delivery_queue', { learnerId: learnerId, lessonId: lessonId, correlationId:
    }
  });
}

_parseSlackPayload(lesson) {
  var raw = lesson && (lesson.slackPayload || lesson.slack_payload);
  if (!raw) return null;

  if (typeof raw === 'object') {
    return raw;
  }

  try {
    var parsed = JSON.parse(raw);
    return parsed && typeof parsed === 'object' ? parsed : null;
  } catch (err) {
    return null;
  }
}

_buildLessonMessagePayload(lesson, defaultText) {
  var slackPayload = this._parseSlackPayload(lesson);
  if (slackPayload) {
    return {
      text: slackPayload.text || defaultText,
      blocks: slackPayload.blocks || []
    };
  }
  return {
    text: defaultText,
    blocks: this._blocks.buildLessonCard(lesson)
  };
}

deliverPendingLessons() {
  var pending = this._db.table('delivery_queue').findAll().filter(function(row) { return row.status === 'queued'; })
    .sort(function(a, b) {
      var pa = a.priority === 'high' ? 0 : 1;
      var pb = b.priority === 'high' ? 0 : 1;
      return pa - pb;
    })
    .slice(0, 25);
  var delivered = [];
  for (var i = 0; i < pending.length; i++) {
    if (pending[i].availableAt && new Date(pending[i].availableAt).getTime() > Date.now()) continue;
    var res = this.deliverLessonToLearner(pending[i].learnerId, pending[i].lessonId);
    delivered.push({ id: pending[i].id, ok: !!res.ok });
  }
}

```

```

        if (res.ok) {
            this._db.table('delivery_queue').update(pending[i].id, { status: 'delivered' });
        } else {
            this._db.table('delivery_queue').update(pending[i].id, { status: 'retry', attempts: String(Number(pending[i].attempts) + 1) });
        }
    }
    return { ok: true, total: pending.length, delivered: delivered };
}
}

12_SlackBlockKitBuilder.gs
class SlackBlockKitBuilder {
    buildWelcomeMessage(input) {
        return [{ type: 'section', text: { type: 'mrkdwn', text: '*Welcome to RWR LMS*\nCourse: ' + (input.courseId || '') } }];
    }

    buildLessonCard(input) {
        var lessonId = input.id || input.lessonId || '';
        var title = input.title || 'Lesson';
        var contentRef = input.contentRef || '';
        var objective = input.objective || '';
        var coreContent = input.coreContent || '';
        var submitCmd = '/submit ' + lessonId + ' complete';
        var lessonBody = contentRef
            ? ':books: *Complete the training here:* <' + contentRef + '|Open Lesson>'
            : ':books: *Lesson content:* \n' + (coreContent || objective || 'No lesson content provided yet. ');

        return [
            { type: 'header', text: { type: 'plain_text', text: title, emoji: true } },
            { type: 'divider' },
            {
                type: 'section',
                text: {
                    type: 'mrkdwn',
                    text: lessonBody + '\n\n:bulb: *Tip:* Take notes as you go â you\'ll need them for the verification question'
                }
            },
            { type: 'divider' },
            {
                type: 'section',
                text: {
                    type: 'mrkdwn',
                    text: 'When done, type: \'' + submitCmd + '\'' or react with :white_check_mark:'
                }
            },
            { type: 'context', elements: [{ type: 'mrkdwn', text: lessonId }] }
        ];
    }

    buildProgressSummary(input) {
        return [{ type: 'section', text: { type: 'mrkdwn', text: '*Progress*\nCompleted: ' + input.completedCount + '\nOverdue: ' + input.overdueCount } }];
    }

    buildReminder(input) {
        return [{ type: 'section', text: { type: 'mrkdwn', text: ':bell: Reminder for learner ' + (input.learnerId || '') } }];
    }

    buildAdminSummary(input) {
        return [{ type: 'section', text: { type: 'mrkdwn', text: '*Weekly Summary*\n' + JSON.stringify(input || {}) } }];
    }

    buildChannelAnnouncement(input) {
        return [{
            type: 'section',
            text: {
                type: 'mrkdwn',
                text: ':tada: Please welcome *' + (input.name || 'our new team member') + '* to the team!\nTheir first day is ' + (input.start || 'tomorrow')
            }
        }];
    }

    buildReflectionPrompt(input) {
        return [
            {
                type: 'section',
                text: { type: 'mrkdwn', text: ':thought_balloon: *Reflection Prompt*\n\nTake a moment to reflect on your progress' }
            },
            {
                type: 'context',
                elements: [{ type: 'mrkdwn', text: 'No submission required â this is for your own reflection.' }]
            }
        ];
    }
}

```



```

    }
  ];
}

buildProTip(input) {
  return [{
    type: 'section',
    text: { type: 'mrkdwn', text: ':bulb: *Pro Tip*\n\n' + (input.tip || 'Keep building your skills and ask questions') }
  ]};

buildShadowingPrompt(input) {
  return [
    { type: 'header', text: { type: 'plain_text', text: ':eyes: Shadowing Session: ' + (input.topic || 'Observation') },
    { type: 'divider' },
    { type: 'section', text: { type: 'mrkdwn', text: (input.instructions || 'Post your observations in the thread below') }
  ];
}

buildSubmitModal(input) {
  return {
    type: 'modal',
    title: { type: 'plain_text', text: 'Submit Lesson' },
    private_metadata: JSON.stringify(input || {}),
    submit: { type: 'plain_text', text: 'Submit' },
    close: { type: 'plain_text', text: 'Cancel' },
    blocks: [{ type: 'input', block_id: 'lesson', element: { type: 'plain_text_input', action_id: 'lesson_id' }, label: '' }
  ];
}

buildGenericErrorResponse(message) {
  return [{ type: 'section', text: { type: 'mrkdwn', text: ':warning: ' + (message || 'Unexpected error') } }];
}
}

13_LearnerProgressStateMachine.gs
class LearnerProgressStateMachine {
  constructor() {
    this.states = {
      NOT_STARTED: 'not_started',
      IN_PROGRESS: 'in_progress',
      SUBMITTED: 'submitted',
      COMPLETED: 'completed'
    };
    this._legacyStateMap = {
      queued: this.states.NOT_STARTED,
      delivered: this.states.NOT_STARTED,
      started: this.states.IN_PROGRESS,
      overdue: this.states.IN_PROGRESS,
      not_started: this.states.NOT_STARTED,
      in_progress: this.states.IN_PROGRESS,
      submitted: this.states.SUBMITTED,
      completed: this.states.COMPLETED
    };
    this._allowed = {};
    this._allowed[this.states.NOT_STARTED] = [this.states.IN_PROGRESS];
    this._allowed[this.states.IN_PROGRESS] = [this.states.SUBMITTED];
    this._allowed[this.states.SUBMITTED] = [this.states.COMPLETED];
    this._allowed[this.states.COMPLETED] = [];
  }

  normalizeState(state) {
    var raw = String(state || '').trim().toLowerCase();
    if (!raw) return this.states.NOT_STARTED;
    return this._legacyStateMap[raw] || this.states.NOT_STARTED;
  }

  isCanonicalState(state) {
    var normalized = this.normalizeState(state);
    return [
      this.states.NOT_STARTED,
      this.states.IN_PROGRESS,
      this.states.SUBMITTED,
      this.states.COMPLETED
    ].indexOf(normalized) !== -1;
  }

  canTransition(fromState, toState) {
    var from = this.normalizeState(fromState);
    var to = this.normalizeState(toState);
  }
}

```

```

    return (this._allowed[from] || []).indexOf(to) !== -1;
}

transition(record, toState, meta) {
    var source = record || {};
    var fromState = this.normalizeState(source.state);
    var normalizedTo = this.normalizeState(toState);
    if (!this.canTransition(fromState, normalizedTo)) {
        return {
            ok: false,
            code: 'INVALID_TRANSITION',
            message: fromState + ' -> ' + normalizedTo + ' not allowed.',
            record: source,
            meta: meta || {}
        };
    }

    var next = {};
    Object.keys(source).forEach(function(k) { next[k] = source[k]; });
    next.state = normalizedTo;
    next.updatedAt = new Date().toISOString();
    if (normalizedTo === this.states.COMPLETED && !next.completedAt) {
        next.completedAt = next.updatedAt;
    }

    return { ok: true, code: 'TRANSITION_OK', message: 'State updated', record: next, meta: meta || {} };
}

migrateLegacyStatesInDb(db) {
    var rows = db.table('learner_progress').findAll();
    var migration = this.migrateLegacyStates(rows);
    for (var i = 0; i < migration.updates.length; i++) {
        var patch = { state: migration.updates[i].toState, updatedAt: new Date().toISOString() };
        if (migration.updates[i].toState === this.states.COMPLETED) patch.completedAt = patch.updatedAt;
        db.table('learner_progress').update(migration.updates[i].id, patch);
    }
    return migration;
}

migrateLegacyStates(progressRows) {
    var rows = progressRows || [];
    var updates = [];

    for (var i = 0; i < rows.length; i++) {
        var row = rows[i] || {};
        var currentState = String(row.state || '').trim().toLowerCase();
        var normalizedState = this.normalizeState(currentState);
        if (currentState !== normalizedState) {
            updates.push({ id: row.id, fromState: currentState || '', toState: normalizedState });
        }
    }

    return { ok: true, total: rows.length, updated: updates.length, updates: updates };
}

13_LmsProgressService.gs
class LmsProgressService {
    constructor(db, slackApiClient, blocks, config, repositories, workflowEngine) {
        this._db = db;
        this._slack = slackApiClient;
        this._blocks = blocks;
        this._config = config || {};
        this._repos = repositories || {};
        this._workflow = workflowEngine;
    }

    handleProgress(ctx) {
        var snapshot = this.getProgressSnapshot(ctx.userId);
        if (!snapshot.ok) return { response_type: 'ephemeral', text: snapshot.message };
        return {
            response_type: 'ephemeral',
            text: 'Progress summary',
            blocks: this._blocks.buildProgressSummary(snapshot)
        };
    }

    handleReinforce(ctx) {
        var snapshot = this.getProgressSnapshot(ctx.userId);

```



```

    respond: function(ctx) {
      ctx.result = {
        ok: true,
        code: ctx.data.existing ? 'DUPLICATE_SUBMISSION' : 'SUBMISSION_RECORDED',
        learnerId: ctx.data.learner.id,
        submissionId: ctx.data.existing ? ctx.data.existing.id : ctx.data.submission.id,
        message: ctx.data.existing ? 'Submission already recorded.' : 'Submission recorded.',
        correlationId: ctx.correlationId
      };
    },
    audit: function(ctx) {
      self._db.audit('record_submission', 'submission_log', { learnerId: ctx.data.learner.id, lessonId: ctx.trigger.lessonId });
    }
  });
}

advanceLessonState(input) {
  var progress = this._repos.progressRepo.findById(input.learnerProgressId);
  if (!progress) return { ok: false, code: 'PROGRESS_NOT_FOUND', message: 'Progress not found.' };
  var transition = this._state.transition(progress, input.toState, { source: 'submission' });
  if (!transition.ok) return transition;
  this._repos.progressRepo.update(progress.id, { state: transition.record.state });
  return { ok: true, code: 'STATE_UPDATED', recordId: progress.id };
}

queueNextLesson(input) {
  var currentLesson = this._repos.lessonRepo.findById(input.currentLessonId);
  if (!currentLesson) {
    return { ok: false, code: 'CURRENT_LESSON_NOT_FOUND' };
  }

  var nextLesson = this._repos.lessonRepo.findNextLesson(currentLesson.courseId, currentLesson.sequenceNumber || 0);

  if (!nextLesson) {
    return { ok: true, code: 'COURSE_COMPLETE', message: 'No further lessons in this course.' };
  }

  var existingProgress = this._repos.progressRepo.findByLearnerAndLesson(input.learnerId, nextLesson.id);
  if (existingProgress && this._state.normalizeState(existingProgress.state) !== this._state.states.COMPLETED) {
    return { ok: true, code: 'ALREADY_QUEUED' };
  }

  this._repos.progressRepo.insert({
    learnerId: input.learnerId,
    lessonId: nextLesson.id,
    state: this._state.states.NOT_STARTED,
    dueAt: ''
  });

  this._db.table('delivery_queue').insert({
    learnerId: input.learnerId,
    lessonId: nextLesson.id,
    status: 'queued',
    priority: 'normal',
    runAt: new Date().toISOString(),
    attempts: '0',
    availableAt: new Date().toISOString(),
    conditionExpr: ''
  });

  return { ok: true, code: 'NEXT_LESSON_QUEUED', lessonId: nextLesson.id };
}
}

14_LmsReminderService.gs
class LmsReminderService {
  constructor(db, slackApiClient, blocks, config) {
    this._db = db;
    this._slack = slackApiClient;
    this._blocks = blocks;
    this._config = config || {};
  }

  sendOverdueReminders() {
    var needsReminder = this._db.table('learner_progress').findAll().filter(function(row) {
      return row.state === 'in_progress' || row.state === 'submitted';
    });
    var results = [];
    for (var i = 0; i < needsReminder.length; i++) {
      results.push(this.sendReminderForLearner(needsReminder[i].learnerId));
    }
  }
}

```

```

    }
    return { ok: true, total: needsReminder.length, results: results };
}

sendReminderForLearner(learnerId) {
    var learner = this._db.table('learners').findById(learnerId);
    if (!learner) return { ok: false, code: 'LEARNER_NOT_FOUND', learnerId: learnerId };
    var dm = this._slack.openDm(learner.slackUserId);
    if (!dm.ok) return dm;
    return this._slack.postMessage(dm.channelId, 'Lesson reminder', this._blocks.buildReminder({ learnerId: learnerId }));
}

escalateStalledLearner(learnerId) {
    return {
        ok: true,
        learnerId: learnerId,
        escalated: !!this._config.opsAlertChannel,
        message: this._config.opsAlertChannel ? 'Escalation target configured.' : 'No ops channel configured.'
    };
}
}

14_Scheduler.gs
function setupTriggers() {
    removeTriggers();
    ScriptApp.newTrigger('runDailyLessonDelivery').timeBased().everyDays(1).atHour(8).create();
    ScriptApp.newTrigger('runHourlyReminderCheck').timeBased().everyHours(1).create();
    ScriptApp.newTrigger('runWeeklyAdminReport').timeBased().onWeekDay(ScriptApp.WeekDay.MONDAY).atHour(9).create();
    ScriptApp.newTrigger('runHealthCheck').timeBased().everyMinutes(30).create();
    ScriptApp.newTrigger('runDailyBackup').timeBased().everyDays(1).atHour(2).create();
    return { ok: true, code: 'TRIGGERS_CREATED' };
}

function removeTriggers() {
    ScriptApp.getProjectTriggers().forEach(function(t) { ScriptApp.deleteTrigger(t); });
    return { ok: true, code: 'TRIGGERS_REMOVED' };
}

function runDailyLessonDelivery() {
    var lock = LockService.getScriptLock();
    lock.waitLock(30000);
    try {
        var deps = createHostDependencies();
        return deps.lessonService.deliverPendingLessons();
    } finally {
        lock.releaseLock();
    }
}

function runHourlyReminderCheck() {
    var lock = LockService.getScriptLock();
    lock.waitLock(30000);
    try {
        var deps = createHostDependencies();
        return deps.reminderService.sendOverdueReminders();
    } finally {
        lock.releaseLock();
    }
}

function runWeeklyAdminReport() {
    var lock = LockService.getScriptLock();
    lock.waitLock(30000);
    try {
        var deps = createHostDependencies();
        return deps.reportService.buildWeeklySummary();
    } finally {
        lock.releaseLock();
    }
}

function runHealthCheck() {
    var lock = LockService.getScriptLock();
    lock.waitLock(30000);
    try {
        return createHostDependencies().healthMonitor.getSnapshot();
    } finally {
        lock.releaseLock();
    }
}

```

```

function runDailyBackup() {
  var lock = LockService.getScriptLock();
  lock.waitLock(30000);
  try {
    return createHostDependencies().backupService.runDailyBackup();
  } finally {
    lock.releaseLock();
  }
}

15_SheetsDataSync.gs
function syncApprovedLessonsToRuntime() {
  // Lesson source mapping is intentionally constrained to the current schema contract.
  return { ok: true, upserted: 0, retired: deactivateRetiredLessons().retired };
}

function upsertLessonRuntimeRecord(row) {
  var db = createHostDbClient();
  var table = db.table('lessons');
  var source = row || {};
  var slackPayload = source.slackPayload || source.slack_payload || '';
  if (slackPayload && typeof slackPayload !== 'string') {
    slackPayload = JSON.stringify(slackPayload);
  }
  var lessonPayload = {
    courseId: source.courseId || '',
    moduleId: source.moduleId || '',
    sequenceNumber: String(source.sequenceNumber == null ? '0' : source.sequenceNumber),
    track: source.track || source.topic || '',
    title: source.title || '',
    topic: source.topic || source.track || '',
    objective: source.objective || '',
    difficulty: String(source.difficulty || 'independent').toLowerCase(),
    hook: source.hook || '',
    coreContent: source.coreContent || '',
    insight: source.insight || '',
    takeaway: source.takeaway || '',
    mission: source.mission || '',
    missionType: source.missionType || 'text',
    missionDuration: source.missionDuration || '3 min',
    verification: source.verification || '',
    submitBlock: source.submitBlock || '',
    contentRef: source.contentRef || '',
    slackPayload: slackPayload || '',
    active: String(source.active == null ? 'true' : source.active)
  };
  var existing = source.id ? table.findById(source.id) : null;

  if (existing) {
    var updated = table.update(existing.id, lessonPayload);
    return { ok: true, code: 'UPDATED', lessonId: updated.id };
  }

  lessonPayload.id = source.id || undefined;
  var inserted = table.insert(lessonPayload);
  return { ok: true, code: 'INSERTED', lessonId: inserted.id };
}

function deactivateRetiredLessons() {
  return { ok: true, retired: 0 };
}

16_ConfigBootstrap.gs
var ConfigBootstrap = {
  load: function() {
    var helper = new ScriptPropertiesHelper();
    return {
      slackBotToken: helper.getRequired('SLACK_BOT_TOKEN'),
      slackBotTokenFallback: helper.get('SLACK_BOT_TOKEN_FALLBACK', ''),
      slackSigningSecret: helper.getRequired('SLACK_SIGNING_SECRET'),
      spreadsheetId: helper.get('SHEET_DB_SPREADSHEET_ID', helper.getRequired('SPREADSHEET_ID')),
      adminUserIds: this._csv(helper.get('ADMIN_USER_IDS', '')),
      defaultCourseId: helper.get('DEFAULT_COURSE_ID', 'C001'),
      defaultTrack: helper.get('DEFAULT_TRACK', 'ONBOARDING'),
      opsAlertChannel: helper.get('OPS_ALERT_CHANNEL', ''),
      appsheetWebhookToken: helper.get('APPSHEET_WEBHOOK_TOKEN', ''),
      onCallUserId: helper.get('ON_CALL_USER_ID', ''),
      backupFolderId: helper.get('BACKUP_FOLDER_ID', ''),
      backupRetentionDays: Number(helper.get('BACKUP_RETENTION_DAYS', '30')),
    };
  }
};

```

```

        backupSchemaVersion: helper.get('BACKUP_SCHEMA_VERSION', '1'),
        backupTables: this._csv(helper.get('BACKUP_TABLES', '')),
        qaPassThreshold: Number(helper.get('QA_PASS_THRESHOLD', '70')),
        qaStrongPassThreshold: Number(helper.get('QA_STRONG_PASS_THRESHOLD', '85')),
        pipelineMaxRetries: Number(helper.get('PIPELINE_MAX_RETRIES', '3')),
        latencyWarningMs: Number(helper.get('LATENCY_WARNING_MS', '8000')),
        latencyCriticalMs: Number(helper.get('LATENCY_CRITICAL_MS', '15000')),
        maintenanceMode: helper.get('MAINTENANCE_MODE', 'false') === 'true',
        healthCheckEnabled: helper.get('HEALTH_CHECK_ENABLED', 'true') === 'true',
        quietHoursStart: Number(helper.get('QUIET_HOURS_START', '21')),
        quietHoursEnd: Number(helper.get('QUIET_HOURS_END', '7')),
        enableRetry: helper.get('ENABLE_RETRY', 'true') === 'true',
        enableReporting: helper.get('ENABLE_REPORTING', 'true') === 'true',
        enableOnboarding: helper.get('ENABLE_ONBOARDING', 'true') === 'true'
    };
},

validate: function() {
    try {
        this.load();
        return { ok: true, code: 'OK', message: 'Configuration valid.' };
    } catch (err) {
        return { ok: false, code: 'CONFIG_INVALID', message: String(err.message || err) };
    }
},

get: function(key, fallback) {
    var cfg = this.load();
    return cfg.hasOwnProperty(key) ? cfg[key] : fallback;
},

_csv: function(text) {
    return String(text || '').split(',').map(function(v) { return v.trim(); }).filter(function(v) { return !!v; });
}
};

17_HealthMonitor.gs
class HealthMonitor {
    constructor(db, config) {
        this._db = db;
        this._config = config || {};
    }

    recordJobStatus(jobName, status, metadata) {
        var row = this._db.table('audit_log').insert({
            actor: 'scheduler',
            action: jobName,
            resourceType: 'job',
            resourceId: '',
            status: status,
            message: jobName + ':' + status,
            metadata: JSON.stringify(metadata || {})
        });
        return { ok: true, auditId: row.id };
    }

    recordSlackApiFailure(errorShape) {
        return this.recordJobStatus('slack_api_failure', 'error', errorShape || {});
    }

    getQueueDepthReport() {
        return {
            ok: true,
            deliveryQueue: this._db.table('delivery_queue').findAll().length,
            retryQueue: this._db.table('retry_queue').findAll().length
        };
    }

    getOverdueMetrics() {
        var inProgress = this._db.table('learner_progress').findAll().filter(function(r) { return r.state === 'in_progress' });
        return { ok: true, overdueCount: inProgress.length };
    }

    getSnapshot() {
        return {
            ok: true,
            timestamp: new Date().toISOString(),
            config: ConfigBootstrap.validate(),
            sheetDb: SheetDb.healthCheck(),
            queues: this.getQueueDepthReport(),

```





```

        id: csvFile.getId(),
        name: csvFile.getName(),
        table: tableName,
        type: 'csv',
        rowCount: rows.length
    });
}

var runManifest = {
    timestamp: timestampIso,
    schemaVersion: schemaVersion,
    tableCounts: tableStats,
    artifacts: artifacts
};
runManifest.checksum = this._checksum(JSON.stringify(runManifest));

var manifestFile = folder.createFile(
    'backup_manifest_' + timestampToken + '.json',
    JSON.stringify(runManifest, null, 2),
    MimeType.PLAIN_TEXT
);
artifacts.push({
    id: manifestFile.getId(),
    name: manifestFile.getName(),
    table: 'manifest',
    type: 'json',
    rowCount: 0
});

var retentionResult = this._applyRetention(folder, retention);

this._audit('success', 'Daily backup completed', {
    code: 'BACKUP_SUCCESS',
    folderId: backupFolderId,
    retentionDays: retention,
    deletedArtifactCount: retentionResult.deletedCount,
    deletedArtifactIds: retentionResult.deletedIds,
    artifacts: artifacts,
    tableCounts: tableStats
});

return {
    ok: true,
    code: 'BACKUP_SUCCESS',
    folderId: backupFolderId,
    artifactCount: artifacts.length,
    tableCounts: tableStats,
    deletedArtifactCount: retentionResult.deletedCount
};
} catch (err2) {
    this._audit('error', 'Daily backup failed', {
        code: 'BACKUP_FAILED',
        folderId: backupFolderId,
        artifacts: artifacts,
        tableCounts: tableStats,
        error: String(err2 && err2.message ? err2.message : err2)
    });
    return {
        ok: false,
        code: 'BACKUP_FAILED',
        message: String(err2 && err2.message ? err2.message : err2)
    };
}
}

_resolveTables() {
    if (Array.isArray(this._config.backupTables) && this._config.backupTables.length > 0) {
        return this._config.backupTables;
    }
    return [
        'learners',
        'enrollment',
        'lessons',
        'learner_progress',
        'submission_log',
        'delivery_queue',
        'retry_queue',
        'audit_log',
        'courses',
        'modules',

```

```

        'onboarding_requests',
        'onboarding_checklists',
        'onboarding_task_log',
        'lesson_qa_records',
        'app_config'
    ];
}

_artifactName(tableName, timestampToken, ext) {
    return 'backup_' + tableName + '_' + timestampToken + '.' + ext;
}

_normalizeRetentionDays(raw) {
    var n = Number(raw);
    if (isNaN(n) || n < 1) {
        return 30;
    }
    return Math.floor(n);
}

_applyRetention(folder, retentionDays) {
    var deletedIds = [];
    var millis = retentionDays * 24 * 60 * 60 * 1000;
    var cutoff = new Date(Date.now() - millis);
    var files = folder.GetFiles();

    while (files.hasNext()) {
        var file = files.next();
        var name = file.getName();
        if (name.indexOf('backup_') !== 0) {
            continue;
        }
        if (file.getDateCreated().getTime() < cutoff.getTime()) {
            deletedIds.push(file.getId());
            file.setTrashed(true);
        }
    }

    return { deletedCount: deletedIds.length, deletedIds: deletedIds };
}

_checksum(text) {
    var input = String(text || '');
    var hash = 0;
    for (var i = 0; i < input.length; i++) {
        hash = ((hash << 5) - hash) + input.charCodeAt(i);
        hash |= 0;
    }
    return 'h' + Math.abs(hash);
}

_toCsv(rows) {
    if (!rows || rows.length === 0) {
        return '';
    }
    var headers = Object.keys(rows[0]);
    var out = [headers.join(',')];
    for (var i = 0; i < rows.length; i++) {
        var row = rows[i];
        var cols = headers.map(function(h) {
            var v = row[h];
            if (v === null || v === undefined) {
                return '';
            }
            var text = String(v);
            if (text.indexOf('"') !== -1) {
                text = text.replace(/"/g, '""');
            }
            if (text.indexOf(',') !== -1 || text.indexOf('\n') !== -1 || text.indexOf('"') !== -1) {
                text = '"' + text + '"';
            }
            return text;
        });
        out.push(cols.join(','));
    }
    return out.join('\n');
}

_audit(status, message, metadata) {
    this._db.table('audit_log').insert({

```

```

        actor: 'scheduler',
        action: 'daily_backup',
        resourceType: 'job',
        resourceId: '',
        status: status,
        message: message,
        metadata: JSON.stringify(metadata || {})
    });
}
}

18_RetryResolver.gs
class RetryResolver {
    constructor(db, config, configRepo) {
        this._db = db;
        this._config = config || {};
        this._configRepo = configRepo || null;
        this._maxAttempts = Number(this._config.pipelineMaxRetries || 5);
    }

    isRetryable(errorShape) {
        if (!errorShape) return false;
        if (errorShape.retryable) return true;
        return ['HTTP_FETCH_FAILED', 'ratelimited', 'internal_error', 'SLACK_API_ERROR'].indexOf(String(errorShape.code || '')) > -1;
    }

    scheduleRetry(job) {
        if (this._configRepo && !this._configRepo.getFlag('enable_retry', true)) {
            return { ok: false, code: 'RETRY_DISABLED' };
        }

        var payload = job || {};
        var attempts = Number(payload.attempts || 0);
        return this._db.table('retry_queue').insert({
            jobType: payload.jobType || 'unknown',
            payload: JSON.stringify(payload.payload || {}),
            attempts: String(attempts),
            nextRunAt: payload.nextRunAt || this._computeNextRunAt(attempts),
            status: 'queued',
            lastError: payload.lastError || '',
            correlationId: payload.correlationId || ''
        });
    }

    resolveDueRetries(limit) {
        var max = Number(limit || 20);
        var now = Date.now();
        var due = this._db.table('retry_queue').findAll().filter(function(r) {
            return r.status === 'queued' && (!r.nextRunAt || new Date(r.nextRunAt).getTime() <= now);
        }).slice(0, max);
        return { ok: true, dueCount: due.length, rows: due };
    }

    markAttempt(rowId, attempts, lastError) {
        var nextAttempts = Number(attempts || 0) + 1;
        var dead = nextAttempts >= this._maxAttempts;
        return this._db.table('retry_queue').update(rowId, {
            attempts: String(nextAttempts),
            status: dead ? 'dead_letter' : 'queued',
            lastError: lastError || '',
            nextRunAt: this._computeNextRunAt(nextAttempts)
        });
    }

    _computeNextRunAt(attempts) {
        var waitMs = Math.min(3600000, Math.pow(2, Number(attempts || 0)) * 60000);
        return new Date(Date.now() + waitMs).toISOString();
    }
}

19_HostTestHarness.gs
function hostTest_fakeSlashLearn() {
    return doPost(_fakeSignedSlash('/learn', '')).getContent();
}

function hostTest_fakeSlashSubmit() {
    return doPost(_fakeSignedSlash('/submit', 'PRE-M01 complete')).getContent();
}

function hostTest_fakeSlashProgress() {

```

```

    return doPost(_fakeSignedSlash('/progress', '')).getContent();
}

function hostTest_workflowWebhookEnroll() {
    var body = JSON.stringify({
        workflow: 'lms_onboarding',
        data: { user_id: 'U_TEST_WF', email: 'test@rwrgroup.com', name: 'Test Learner', course_id: 'C001' }
    });
    return doPost(_fakeSignedJson(body)).getContent();
}

function hostTest_duplicateSubmissionEdgeCase() {
    var deps = createHostDependencies();
    var first = deps.completionService.recordSubmission({ slackUserId: 'U123', lessonId: 'PRE-M01', payload: '{}' });
    var second = deps.completionService.recordSubmission({ slackUserId: 'U123', lessonId: 'PRE-M01', payload: '{}' });
    return JSON.stringify({ first: first, second: second });
}

function hostTest_missingLearnerEdgeCase() {
    var deps = createHostDependencies();
    return JSON.stringify(deps.completionService.recordSubmission({ slackUserId: 'U_MISSING', lessonId: 'PRE-M01', payload: '' }));
}

function hostTest_invalidPayloadEdgeCase() {
    return doPost({ postData: { type: 'application/json', contents: '{invalid-json}', parameter: {}, headers: {} }).getContent();
}

function hostTest_onboardAdmin() {
    return doPost(_fakeSignedSlash('/onboard', 'test@rwrgroup.com')).getContent();
}

function hostTest_fakeSlashGaps() {
    return doPost(_fakeSignedSlash('/gaps', '')).getContent();
}

function hostTest_fakeSlashAudit() {
    return doPost(_fakeSignedSlash('/audit', '')).getContent();
}

function hostTest_fakeSlashMix() {
    return doPost(_fakeSignedSlash('/mix', '')).getContent();
}

function hostTest_fakeSlashReinforce() {
    return doPost(_fakeSignedSlash('/reinforce', '')).getContent();
}

function hostTest_fakeSlashOffboard() {
    return doPost(_fakeSignedSlash('/offboard', 'departing.user@rwrgroup.com')).getContent();
}

function hostTest_fakeInteractive() {
    var payload = {
        type: 'block_actions',
        user: { id: 'U123' },
        channel: { id: 'C123' },
        team: { id: 'T123' },
        actions: [{ action_id: 'submit_lesson', value: 'PRE-M01' }]
    };
    var form = 'payload=' + encodeURIComponent(JSON.stringify(payload));
    return doPost(_fakeSignedForm(form, { payload: JSON.stringify(payload) })).getContent();
}

function hostTest_fakeAppMention() {
    return doPost(_fakeSignedEvent('app_mention')).getContent();
}

function hostTest_fakeMessageIm() {
    return doPost(_fakeSignedEvent('message', 'im')).getContent();
}

function _fakeSignedSlash(command, text) {
    var body = 'command=' + encodeURIComponent(command) + '&text=' + encodeURIComponent(text || '') + '&user_id=U123&channel_id=C123&team_id=T123';
    return _fakeSignedForm(body, { command: command, text: text || '', user_id: 'U123', channel_id: 'C123', team_id: 'T123' });
}

function _fakeSignedEvent(eventType, channelType) {
    var bodyObj = {
        type: 'event_callback',
        team_id: 'T123',
    };

```



```

    super(message, 'VALIDATION_ERROR');
  }
}

/**
 * Raised when underlying sheet operations fail.
 */
class StorageError extends SheetDbError {
  /**
   * @param {string} message
   */
  constructor(message) {
    super(message, 'STORAGE_ERROR');
  }
}

/**
 * Raised for optimistic transaction failures.
 */
class TransactionError extends SheetDbError {
  /**
   * @param {string} message
   */
  constructor(message) {
    super(message, 'TRANSACTION_ERROR');
  }
}

21_Util.gs
/**
 * Utility helpers for Sheet DB.
 */
class Util {
  /**
   * @return {string} RFC3339 timestamp in UTC.
   */
  static nowIso() {
    return new Date().toISOString();
  }

  /**
   * Creates a shallow clone of an object.
   * @param {Object} source
   * @return {Object}
   */
  static clone(source) {
    return Object.assign({}, source || {});
  }

  /**
   * Creates a stable row ID with a prefix.
   * @param {string=} prefix
   * @return {string}
   */
  static generateId(prefix) {
    var p = prefix || 'row';
    var random = Math.random().toString(36).slice(2, 10);
    return p + '_' + new Date().getTime() + '_' + random;
  }

  /**
   * Returns true when value is null or undefined.
   * @param {*} value
   * @return {boolean}
   */
  static isNil(value) {
    return value === null || value === undefined;
  }

  /**
   * Returns true when value is empty string, null, or undefined.
   * @param {*} value
   * @return {boolean}
   */
  static isBlank(value) {
    return value === '' || Util.isNil(value);
  }

  /**
   * Throws when a required condition is not met.

```

```

    * @param {boolean} condition
    * @param {string} message
    */
    static assert(condition, message) {
        if (!condition) {
            throw new ValidationError(message);
        }
    }
}

/**
 * Converts a sheet row to an object using the header order.
 * @param {string[]} headers
 * @param {Array<*>} row
 * @return {Object}
 */
static rowToObject(headers, row) {
    var out = {};
    for (var i = 0; i < headers.length; i++) {
        out[headers[i]] = row[i];
    }
    return out;
}

/**
 * Converts an object to row array by header order.
 * @param {string[]} headers
 * @param {Object} obj
 * @return {Array<*>}
 */
static objectToRow(headers, obj) {
    return headers.map(function(h) {
        return obj[h];
    });
}
}

22_ScriptProperties.gs
/**
 * Script Properties helper for secure and environment-specific configuration.
 */
class ScriptPropertiesHelper {
    /**
     * Creates a helper bound to Script Properties store.
     */
    constructor() {
        this._props = PropertiesService.getScriptProperties();
    }

    /**
     * Returns a property value, or a default when missing.
     * @param {string} key
     * @param {*=} defaultValue
     * @return {*}
     */
    get(key, defaultValue) {
        var value = this._props.getProperty(key);
        return value === null ? defaultValue : value;
    }

    /**
     * Returns a required property value or throws when missing/blank.
     * @param {string} key
     * @return {string}
     */
    getRequired(key) {
        var value = this.get(key, '');
        if (value === '') {
            throw new ConfigError('Missing required script property: ' + key);
        }
        return String(value);
    }

    /**
     * Returns a boolean property value.
     * Accepted true values: true, 1, yes, y, on.
     * Accepted false values: false, 0, no, n, off.
     * @param {string} key
     * @param {boolean=} defaultValue
     * @return {boolean}
     */
}

```

```

getBoolean(key, defaultValue) {
    var raw = this.get(key, null);
    if (raw === null || raw === '') {
        return defaultValue === undefined ? false : !!defaultValue;
    }
    var normalized = String(raw).trim().toLowerCase();
    if (['true', '1', 'yes', 'y', 'on'].indexOf(normalized) !== -1) {
        return true;
    }
    if (['false', '0', 'no', 'n', 'off'].indexOf(normalized) !== -1) {
        return false;
    }
    throw new ConfigError('Script property is not a valid boolean: ' + key);
}

/**
 * Returns a numeric property value.
 * @param {string} key
 * @param {number=} defaultValue
 * @return {number}
 */
getNumber(key, defaultValue) {
    var raw = this.get(key, null);
    if (raw === null || raw === '') {
        return defaultValue === undefined ? 0 : Number(defaultValue);
    }
    var n = Number(raw);
    if (isNaN(n)) {
        throw new ConfigError('Script property is not a valid number: ' + key);
    }
    return n;
}

/**
 * Returns all script properties as key-value object.
 * @return {Object<string, string>}
 */
getAll() {
    return this._props.getProperties();
}

/**
 * Returns all script properties with sensitive values redacted.
 * @return {Object<string, string>}
 */
getAllRedacted() {
    var all = this.getAll();
    var out = {};
    Object.keys(all).forEach(function(key) {
        out[key] = ScriptPropertiesHelper.shouldRedactKey(key) ? '[REDACTED]' : all[key];
    });
    return out;
}

/**
 * Sets a script property value.
 * @param {string} key
 * @param {*} value
 */
set(key, value) {
    this._props.setProperty(key, String(value));
}

/**
 * Sets multiple script properties.
 * @param {Object<string, *>} obj
 */
setMany(obj) {
    var stringified = {};
    Object.keys(obj || {}).forEach(function(key) {
        stringified[key] = String(obj[key]);
    });
    this._props.setProperties(stringified, false);
}

/**
 * Deletes a script property.
 * @param {string} key
 */
delete(key) {

```



```

    this._props.deleteProperty(key);
}

/**
 * Returns true when a script property exists and is non-empty.
 * @param {string} key
 * @return {boolean}
 */
has(key) {
    var value = this._props.getProperty(key);
    return value !== null && value !== '';
}

/**
 * Returns true when key should be redacted in debug output.
 * @param {string} key
 * @return {boolean}
 */
static shouldRedactKey(key) {
    var upper = String(key || '').toUpperCase();
    if (upper === 'APP_PASSCODE' || upper === 'SIGNING_SECRET' || upper === 'WEBHOOK_SECRET' || upper === 'API_KEY') {
        return true;
    }
    return upper.indexOf('TOKEN') !== -1 || upper.indexOf('SECRET') !== -1 || upper.indexOf('PASSWORD') !== -1;
}

/**
 * Reads SheetDb client-related config from Script Properties.
 *
 * Supported keys:
 * - SHEET_DB_SPREADSHEET_ID
 * - SHEET_DB_ID_COLUMN
 * - SHEET_DB_CREATED_AT_COLUMN
 * - SHEET_DB_UPDATED_AT_COLUMN
 * - SHEET_DB_DELETED_AT_COLUMN
 * - SHEET_DB_LOCK_TIMEOUT_MS
 *
 * Also reserved for host-layer secure use:
 * - APP_PASSCODE
 * - SIGNING_SECRET
 * - WEBHOOK_SECRET
 * - API_KEY
 *
 * @return {Object}
 */
function readSheetDbConfigFromScriptProperties() {
    var helper = new ScriptPropertiesHelper();
    var config = {};

    if (helper.has('SHEET_DB_SPREADSHEET_ID')) {
        config.spreadsheetId = helper.get('SHEET_DB_SPREADSHEET_ID', '');
    }
    if (helper.has('SHEET_DB_ID_COLUMN')) {
        config.idColumn = helper.get('SHEET_DB_ID_COLUMN', 'id');
    }
    if (helper.has('SHEET_DB_CREATED_AT_COLUMN')) {
        config.createdAtColumn = helper.get('SHEET_DB_CREATED_AT_COLUMN', 'createdAt');
    }
    if (helper.has('SHEET_DB_UPDATED_AT_COLUMN')) {
        config.updatedAtColumn = helper.get('SHEET_DB_UPDATED_AT_COLUMN', 'updatedAt');
    }
    if (helper.has('SHEET_DB_DELETED_AT_COLUMN')) {
        config.deletedAtColumn = helper.get('SHEET_DB_DELETED_AT_COLUMN', 'deletedAt');
    }
    if (helper.has('SHEET_DB_LOCK_TIMEOUT_MS')) {
        config.lockTimeoutMs = helper.getNumber('SHEET_DB_LOCK_TIMEOUT_MS', 30000);
    }

    return config;
}

/**
 * Merges explicit client options with script properties-based config.
 * Precedence: explicit options > script properties > hardcoded defaults.
 *
 * @param {Object=} options
 * @return {Object}
 */
function mergeClientOptionsWithScriptProperties(options) {

```



```

* Thin wrapper around SpreadsheetApp APIs.
*/
class SheetsGateway {
  /**
   * @param {Config} config
   */
  constructor(config) {
    this._config = config;
  }

  /**
   * @return {GoogleAppsScript.Spreadsheet.Spreadsheet}
   */
  getSpreadsheet() {
    if (this._config.spreadsheetId) {
      return SpreadsheetApp.openById(this._config.spreadsheetId);
    }
    return SpreadsheetApp.getActiveSpreadsheet();
  }

  /**
   * @param {string} tableName
   * @param {boolean=} createWhenMissing
   * @return {GoogleAppsScript.Spreadsheet.Sheet}
   */
  getSheet(tableName, createWhenMissing) {
    var ss = this.getSpreadsheet();
    var sheet = ss.getSheetByName(tableName);
    if (!sheet && createWhenMissing) {
      sheet = ss.insertSheet(tableName);
    }
    if (!sheet) {
      throw new StorageError('Sheet not found for table: ' + tableName);
    }
    return sheet;
  }

  /**
   * @param {string} tableName
   * @return {{headers: string[], rows: Array<Array<*>>}}
   */
  readTable(tableName) {
    var sheet = this.getSheet(tableName, false);
    var range = sheet.getDataRange();
    var values = range.getValues();
    if (values.length === 0) {
      return { headers: [], rows: [] };
    }
    return {
      headers: values[0],
      rows: values.slice(1)
    };
  }

  /**
   * Ensures table exists with given headers.
   * @param {string} tableName
   * @param {string[]} headers
   */
  ensureTable(tableName, headers) {
    var sheet = this.getSheet(tableName, true);
    var lastRow = sheet.getLastRow();
    if (lastRow === 0) {
      sheet.getRange(1, 1, 1, headers.length).setValues([headers]);
      return;
    }

    var existingHeaders = sheet.getRange(1, 1, 1, sheet.getLastColumn()).getValues()[0]
      .map(function(value) { return String(value); });
    var expectedHeaders = headers.map(function(value) { return String(value); });
    if (existingHeaders.join('|') !== expectedHeaders.join('|')) {
      throw new StorageError('Header mismatch for table: ' + tableName);
    }
  }

  /**
   * Writes all rows for a table, replacing existing body rows.
   * @param {string} tableName
   * @param {string[]} headers
   * @param {Array<Array<*>>} rows

```

```

    */
    writeAllRows(tableName, headers, rows) {
        var sheet = this.getSheet(tableName, true);
        this.ensureTable(tableName, headers);

        var maxRows = Math.max(sheet.getMaxRows(), rows.length + 1);
        if (sheet.getMaxRows() < maxRows) {
            sheet.insertRowsAfter(sheet.getMaxRows(), maxRows - sheet.getMaxRows());
        }

        if (sheet.getLastRow() > 1) {
            sheet.getRange(2, 1, sheet.getLastRow() - 1, headers.length).clearContent();
        }

        if (rows.length > 0) {
            sheet.getRange(2, 1, rows.length, headers.length).setValues(rows);
        }
    }
}

25_SchemaRegistry.gs
/**
 * Holds table schemas and applies field defaults/validation.
 */
class SchemaRegistry {
    /**
     * Creates a new in-memory schema registry.
     */
    constructor() {
        this._schemas = {};
    }

    /**
     * Registers a table schema.
     * @param {string} tableName
     * @param {Object} schema
     * @param {Array<string>} schema.columns Ordered columns for persistence.
     * @param {Object<string, Object>=} schema.fields Optional field constraints.
     * @return {SchemaRegistry}
     */
    register(tableName, schema) {
        if (!tableName) {
            throw new SchemaError('Table name is required.');
```

```

    var schema = this._schemas[tableName];
    if (!schema) {
        throw new SchemaError('No schema registered for table: ' + tableName);
    }
    return {
        columns: schema.columns.slice(),
        fields: Util.clone(schema.fields)
    };
}

/**
 * Validates and normalizes a record.
 * @param {string} tableName
 * @param {Object} record
 * @param {boolean=} isPartial True for patch-like updates.
 * @return {Object}
 */
normalizeRecord(tableName, record, isPartial) {
    var schema = this.get(tableName);
    var out = Util.clone(record || {});
    schema.columns.forEach(function(column) {
        var fieldRules = schema.fields[column] || {};
        var hasValue = !Util.isNil(out[column]);

        if (!hasValue && fieldRules.hasOwnProperty('default')) {
            var defaultValue = fieldRules.default;
            out[column] = typeof defaultValue === 'function' ? defaultValue() : defaultValue;
        }

        if (!isPartial && fieldRules.required && Util.isNil(out[column])) {
            throw new ValidationError('Field "' + column + '" is required.');
```

```

/**
 * @param {string} id
 * @return {Object|null}
 */
findById(id) {
  var rows = this.findAll();
  for (var i = 0; i < rows.length; i++) {
    if (rows[i][this._config.idColumn] === id) {
      return rows[i];
    }
  }
  return null;
}

/**
 * @param {Object} payload
 * @return {Object}
 */
insert(payload) {
  var schema = this._schemas.get(this._tableName);
  this._sheets.ensureTable(this._tableName, schema.columns);
  var record = this._schemas.normalizeRecord(this._tableName, payload, false);
  var now = Util.nowIso();
  record[this._config.idColumn] = record[this._config.idColumn] || Util.generateId(this._tableName);
  record[this._config.createdAtColumn] = record[this._config.createdAtColumn] || now;
  record[this._config.updatedAtColumn] = now;
  if (!record.hasOwnProperty(this._config.deletedAtColumn)) {
    record[this._config.deletedAtColumn] = '';
  }

  var tableData = this._sheets.readTable(this._tableName);
  var existingRows = tableData.rows.map(function(row) { return Util.rowToObject(tableData.headers, row); });
  var idColumn = this._config.idColumn;
  var duplicate = existingRows.some(function(row) {
    return row[idColumn] === record[idColumn];
  });
  if (duplicate) {
    throw new ValidationError('Record ID already exists: ' + record[idColumn]);
  }
  existingRows.push(record);
  var rowValues = existingRows.map(function(r) { return Util.objectToRow(schema.columns, r); });
  this._sheets.writeAllRows(this._tableName, schema.columns, rowValues);
  this._audit.log('insert', this._tableName, { id: record[this._config.idColumn] });
  return record;
}

/**
 * @param {string} id
 * @param {Object} patch
 * @return {Object}
 */
update(id, patch) {
  var schema = this._schemas.get(this._tableName);
  var normalizedPatch = this._schemas.normalizeRecord(this._tableName, patch, true);
  var tableData = this._sheets.readTable(this._tableName);
  var records = tableData.rows.map(function(row) { return Util.rowToObject(tableData.headers, row); });
  var found = null;

  for (var i = 0; i < records.length; i++) {
    if (records[i][this._config.idColumn] === id) {
      found = records[i];
      Object.keys(normalizedPatch).forEach(function(key) {
        if (key !== this._config.idColumn && key !== this._config.createdAtColumn) {
          found[key] = normalizedPatch[key];
        }
      }, this);
      found[this._config.updatedAtColumn] = Util.nowIso();
      break;
    }
  }

  if (!found) {
    throw new StorageError('Record not found for id: ' + id);
  }

  var rowValues = records.map(function(r) { return Util.objectToRow(schema.columns, r); });
  this._sheets.writeAllRows(this._tableName, schema.columns, rowValues);
  this._audit.log('update', this._tableName, { id: id, fields: Object.keys(normalizedPatch) });
  return found;
}

```



29\_DbClient.gs

```
/**
 * Main entrypoint for interacting with sheet-backed tables.
 */
class DbClient {
  /**
   * @param {Config=} config
   */
  constructor(config) {
    this._config = config || new Config();
    this._schemas = new SchemaRegistry();
    this._sheets = new SheetsGateway(this._config);
    this._audit = new AuditLogger();
    this._tx = new TransactionManager(this._config.lockTimeoutMs);
  }

  /**
   * Registers a table schema.
   * @param {string} tableName
   * @param {Object} schema
   * @return {DbClient}
   */
  registerTable(tableName, schema) {
    this._schemas.register(tableName, schema);
    this._sheets.ensureTable(tableName, schema.columns);
    return this;
  }

  /**
   * Returns a table repository.
   * @param {string} tableName
   * @return {TableRepository}
   */
  table(tableName) {
    this._schemas.get(tableName);
    return new TableRepository(
      tableName,
      this._config,
      this._schemas,
      this._sheets,
      this._audit
    );
  }

  /**
   * Gets or registers a table schema.
   *
   * - When 'schema' is provided, registers/overwrites the schema and ensures the table.
   * - When 'schema' is omitted, returns the current schema.
   *
   * @param {string} tableName
   * @param {Object=} schema
   * @return {Object|DbClient}
   */
  schema(tableName, schema) {
    if (schema) {
      this.registerTable(tableName, schema);
      return this;
    }
    return this._schemas.get(tableName);
  }

  /**
   * Writes an audit entry through the configured audit logger.
   * @param {string} action
   * @param {string} tableName
   * @param {Object=} metadata
   */
  audit(action, tableName, metadata) {
    this._audit.log(action, tableName, metadata || {});
  }

  /**
   * Runs a set of operations under lock.
   * @template T
   * @param {function(DbClient): T} callback
   * @return {T}
   */
  transaction(callback) {
    var self = this;
```



```

        return this._tx.run(function() {
            return callback(self);
        });
    }
}

30_SheetDb.gs
/**
 * @fileoverview Public, library-visible exports for SheetDb.
 */

/**
 * Library semantic version.
 *
 * NOTE: Keep this in sync with release tags when publishing as an Apps Script library.
 * @type {string}
 */
var SHEET_DB_VERSION = '0.1.0';

/**
 * Creates a new SheetDb client instance.
 *
 * @param {Object=} options Client configuration options.
 * @return {DbClient}
 */
function _sheetDbCreateClient(options) {
    return new DbClient(new Config(options || {}));
}

/**
 * Bootstraps a client with one or more table schemas.
 *
 * @param {Object=} options
 * @param {Array<{name: string, schema: Object}>=} options.tables Tables to register.
 * @param {Object=} options.clientOptions Configuration passed to 'createClient'.
 * @return {DbClient}
 */
function _sheetDbBootstrap(options) {
    var opts = options || {};
    var client = _sheetDbCreateClient(opts.clientOptions || {});
    var tables = opts.tables || [];

    tables.forEach(function(table) {
        if (table && table.name && table.schema) {
            client.registerTable(table.name, table.schema);
        }
    });

    return client;
}

/**
 * Basic health check that can be called by consuming projects.
 *
 * @return {{ok: boolean, version: string, timestamp: string, details: Object}}
 */
function _sheetDbHealthCheck() {
    try {
        var client = _sheetDbCreateClient();
        return {
            ok: !!client,
            version: SHEET_DB_VERSION,
            timestamp: Util.nowIso(),
            details: {
                clientReady: true,
                spreadsheetMode: client._config && client._config.spreadsheetId ? 'by_id' : 'active'
            }
        };
    } catch (err) {
        return {
            ok: false,
            version: SHEET_DB_VERSION,
            timestamp: new Date().toISOString(),
            details: {
                code: err.code || 'HEALTH_CHECK_FAILED',
                message: err.message || 'Unknown error.'
            }
        };
    }
}

```

```

/**
 * Creates a client using Script Properties with optional explicit overrides.
 * Precedence: explicit options > script properties > defaults.
 * @param {Object=} options
 * @return {DbClient}
 */
function _sheetDbCreateClientFromScriptProperties(options) {
  var merged = mergeClientOptionsWithScriptProperties(options || {});
  return _sheetDbCreateClient(merged);
}

/**
 * Public Apps Script library namespace.
 *
 * Consumers should call library exports through this object:
 * 'LibraryIdentifier.SheetDb.createClient(...)'.
 *
 * @type {{
 *   createClient: function(Object=): DbClient,
 *   createClientFromScriptProperties: function(Object=): DbClient,
 *   bootstrap: function(Object=): DbClient,
 *   healthCheck: function(): Object,
 *   version: string
 * }}
 */
var SheetDb = {
  createClient: function(options) {
    return _sheetDbCreateClient(options);
  },

  createClientFromScriptProperties: function(options) {
    return _sheetDbCreateClientFromScriptProperties(options);
  },

  bootstrap: function(options) {
    return _sheetDbBootstrap(options);
  },

  healthCheck: function() {
    return _sheetDbHealthCheck();
  },

  version: SHEET_DB_VERSION
};

/**
 * Backward-compatible factory export.
 * @deprecated Use 'SheetDb.createClient' instead.
 * @param {Object=} options
 * @return {DbClient}
 */
function createSheetDbClient(options) {
  return SheetDb.createClient(options);
}

33_SecurityService.gs
var SecurityService = {
  verifySlackRequest: function(parsed, config) {
    if (!parsed || !parsed.ok) {
      return ErrorService.create('PARSE_FAILED', parsed && parsed.parseError ? parsed.parseError : 'Invalid payload', false);
    }

    var secret = String((config && config.slackSigningSecret) || '');
    var signature = String(parsed.slackSignature || '');
    var ts = String(parsed.slackTimestamp || '');

    if (!secret) {
      return ErrorService.create('SAFE_MODE_NO_SIGNING_SECRET', 'Signing secret missing; rejecting request in safe mode.', false);
    }

    if (!signature || !ts || !parsed.rawBody) {
      return ErrorService.create('UNVERIFIED', 'Slack signature headers missing.', false);
    }

    var verify = this._verifySignature(secret, signature, ts, parsed.rawBody);
    if (!verify.ok) return verify;

    var replay = this._checkReplay(signature, ts);
    if (!replay.ok) return replay;
  }
};

```

```

    return { ok: true, success: true, code: 'OK' };
  },

  sanitizeInput: function(value) {
    return String(value || '').replace(/[<>]/g, '').replace(/[\u0000-\u001f]/g, '').trim();
  },

  _checkReplay: function(signature, ts) {
    var cache = CacheService.getScriptCache();
    var key = 'slack_replay_' + ts + '_' + signature.slice(0, 24);
    if (cache.get(key)) {
      return ErrorService.create('REPLAY_DETECTED', 'Replay request blocked.', false);
    }
    cache.put(key, '1', 300);
    return { ok: true, success: true, code: 'OK' };
  },

  _verifySignature: function(secret, signature, ts, rawBody) {
    var tsNum = Number(ts);
    if (!tsNum || Math.abs(Math.floor(Date.now() / 1000) - tsNum) > 300) {
      return ErrorService.create('REPLAY_REJECTED', 'Timestamp outside replay window.', false);
    }

    var base = 'v0:' + ts + ':' + rawBody;
    var bytes = Utilities.computeHmacSha256Signature(base, secret);
    var hex = bytes.map(function(b) {
      var n = (b < 0 ? b + 256 : b).toString(16);
      return n.length === 1 ? '0' + n : n;
    }).join('');
    var expected = 'v0=' + hex;

    if (!this._constantTimeEquals(expected, signature)) {
      return ErrorService.create('SIGNATURE_MISMATCH', 'Slack signature mismatch.', false);
    }

    return { ok: true, success: true, code: 'OK' };
  },

  _constantTimeEquals: function(a, b) {
    var aa = String(a || '');
    var bb = String(b || '');
    if (!aa || !bb || aa.length !== bb.length) return false;
    var mismatch = 0;
    for (var i = 0; i < aa.length; i++) mismatch |= (aa.charCodeAt(i) ^ bb.charCodeAt(i));
    return mismatch === 0;
  }
};

33_WorkflowEngine.gs
class WorkflowEngine {
  constructor(auditFn, logger) {
    this._audit = auditFn || function() {};
    this._logger = logger || Logger;
  }

  run(name, payload, handlers) {
    var correlationId = this._buildCorrelationId(name);
    var ctx = {
      workflow: name,
      correlationId: correlationId,
      trigger: payload || {},
      data: {},
      result: null
    };

    try {
      this._executeStep('trigger', handlers.trigger, ctx);
      this._executeStep('validate', handlers.validate, ctx);
      this._executeStep('process', handlers.process, ctx);
      this._executeStep('persist', handlers.persist, ctx);
      this._executeStep('respond', handlers.respond, ctx);
      this._executeStep('audit', handlers.audit, ctx);
      return ctx.result || { ok: true, correlationId: correlationId };
    } catch (err) {
      var normalized = ErrorService.fromException(err, 'WORKFLOW_FAILED', true, correlationId);
      this._safeAudit(name + '_failed', { correlationId: correlationId, error: normalized });
      return normalized;
    }
  }
}

```



```

    if (!slackUserId) return null;
    var rows = this.findAll();
    for (var i = 0; i < rows.length; i++) {
        if (rows[i].slackUserId === slackUserId) return rows[i];
    }
    return null;
}
}

class ProgressRepository extends BaseRepository {
    constructor(db) {
        super(db, 'learner_progress');
    }

    findByLearnerId(learnerId) {
        return this.findAll().filter(function(row) { return row.learnerId === learnerId; });
    }

    findActiveLesson(learnerId) {
        return this.findByLearnerId(learnerId).filter(function(row) { return row.state !== 'completed'; })[0] || null;
    }

    findByLearnerAndLesson(learnerId, lessonId) {
        return this.findAll().filter(function(row) {
            return row.learnerId === learnerId && row.lessonId === lessonId;
        })[0] || null;
    }
}

class LessonRepository extends BaseRepository {
    constructor(db, cacheService) {
        super(db, 'lessons');
        this._cacheService = cacheService || CacheService.getScriptCache();
    }

    findActiveByCourse(courseId) {
        var cacheKey = 'lessons_active_' + courseId;
        var cached = this._cacheService.get(cacheKey);
        if (cached) {
            try { return JSON.parse(cached); } catch (err) {}
        }

        var rows = this.findAll().filter(function(row) {
            return row.courseId === courseId && String(row.active) === 'true' && !String(row.deletedAt || '').trim();
        }).sort(function(a, b) {
            return Number(a.sequenceNumber || 0) - Number(b.sequenceNumber || 0);
        });

        this._cacheService.put(cacheKey, JSON.stringify(rows), 180);
        return rows;
    }

    findNextLesson(courseId, sequenceNumber) {
        var lessons = this.findActiveByCourse(courseId);
        for (var i = 0; i < lessons.length; i++) {
            if (Number(lessons[i].sequenceNumber || 0) > Number(sequenceNumber || 0)) return lessons[i];
        }
        return null;
    }
}

class ConfigRepository extends BaseRepository {
    constructor(db, cacheService) {
        super(db, 'app_config');
        this._cacheService = cacheService || CacheService.getScriptCache();
    }

    getValue(configKey, fallback) {
        var cacheKey = 'app_config_' + configKey;
        var cached = this._cacheService.get(cacheKey);
        if (cached !== null) return cached;
        var found = this.findAll().filter(function(row) { return row.configKey === configKey; })[0];
        var value = found ? String(found.configValue || '') : String(fallback || '');
        this._cacheService.put(cacheKey, value, 180);
        return value;
    }

    getFlag(configKey, fallbackBool) {
        var fallback = fallbackBool ? 'true' : 'false';
        return this.getValue(configKey, fallback) === 'true';
    }
}

```

```

    }
  }

36_DbAdapter.gs
class DBAdapter {
  table(name) { throw new Error('Not implemented'); }
  audit(action, resourceType, metadata) { throw new Error('Not implemented'); }
}

class SheetsDBAdapter extends DBAdapter {
  constructor(db) {
    super();
    this._db = db;
  }

  table(name) {
    return this._db.table(name);
  }

  audit(action, resourceType, metadata) {
    return this._db.audit(action, resourceType, metadata || {});
  }
}

```

README.md  
# Slack LMS Automation

## ## 1. Overview

This repository implements a Slack LMS automation host on Google Apps Script and Google Sheets, with shared utilities pr

## ## 2. Features

- Single-ingress Slack webhook host ('doPost(e)').
- Slash command routing for:
  - '/learn'
  - '/submit'
  - '/progress'
  - '/report'
  - '/onboard'
- Event callback routing for:
  - 'message.im'
  - 'app\_mention'
- Interactivity payload routing.
- Runtime data persistence in Google Sheets tables.
- Slack API wrapper with retry-friendly error shape.
- Installable scheduler jobs for delivery, reminders, and reporting.
- Health and retry scaffolds for operations.

## ## 3. Architecture Summary

High-level flow:

'Slack -> manifest.json endpoint -> code.gs#doPost -> SlackPayloadParser -> SlackSecurity -> SlackRouter -> SlackService'

Key principle: keep ingress thin and move behavior into modular services.

## ## 4. Repo Structure

- 'code.gs' â anchor web app entrypoint (thin host shell).
- '00\_WebAppEntry.gs' â dependency and DB bootstrap.
- '01'â '04' â router/parser/security/dispatch layers.
- Domain services â '09\_LmsEnrollmentService.gs', '12\_LmsLessonService.gs', '13\_LmsProgressService.gs', '13a\_LmsComple
- Integrations/state â '11\_SlackApiClient.gs', '12\_SlackBlockKitBuilder.gs', '13\_LearnerProgressStateMachine.gs'.
- Ops/support â '14\_Scheduler.gs', '15\_SheetsDataSync.gs', '16\_ConfigBootstrap.gs', '17\_HealthMonitor.gs', '18\_RetryRe
- 'manifest.json' â Slack app manifest.
- 'sheet\_db\_lib/' â shared library (DB/config/audit helpers).
- 'md\_mirror/', 'pdf\_mirror/' â mirrored reference docs.

## ## 5. Workflows

Supported runtime workflows:

- onboarding/enrollment scaffold
  - '/learn'
  - '/submit'
  - '/progress'
- interactivity handling scaffold
  - 'message.im' and 'app\_mention' events
- scheduled daily lesson delivery
- hourly reminders
- weekly reporting

See 'architecture.md' for the full workflow map.

## ## 6. Tech Stack

- Google Apps Script (V8 runtime)
- Google Sheets (runtime DB)
- Slack App + Events API + Interactivity
- Shared in-repo LMS library utilities ('SheetDb', '23\_Config.gs' DB config, script props helper, audit/transactions)

## ## 7. Configuration

Set Script Properties in Apps Script (loaded via '16\_ConfigBootstrap.gs'):

- Required:
  - 'SLACK\_BOT\_TOKEN'
  - 'SLACK\_SIGNING\_SECRET'
  - 'SPREADSHEET\_ID'
- Optional:
  - 'SLACK\_VERIFICATION\_TOKEN'
  - 'ADMIN\_USER\_IDS'
  - 'DEFAULT\_COURSE\_ID'
  - 'DEFAULT\_TRACK'
  - 'OPS\_ALERT\_CHANNEL'

## ## 8. Deployment

Use 'deployment.md' for the full deployment process:

1. Configure script properties.
2. Deploy Apps Script Web App.
3. Wire Slack manifest URLs to deployment URL.
4. Run 'setupTriggers()'.
5. Execute smoke tests.

## ## 9. Testing

Use host test harness functions in '19\_HostTestHarness.gs':

- 'hostTest\_fakeSlashLearn'
- 'hostTest\_fakeSlashSubmit'
- 'hostTest\_fakeSlashProgress'
- 'hostTest\_fakeInteractive'
- 'hostTest\_fakeAppMention'
- 'hostTest\_fakeMessageIm'
- 'hostTest\_reminderSmoke'
- 'hostTest\_dailyDeliverySmoke'

Also run syntax checks before deployment.

## ## 10. Extending

- Add new slash commands in '03\_SlackService.gs' registry.
- Extend event/interactivity branching in 'SlackService'.
- Implement final lesson sequencing and authored-content sync in '15\_SheetsDataSync.gs'.
- Add richer retry/dead-letter policies in '18\_RetryResolver.gs'.

## ## 11. Docs Index

- 'architecture.md' â‡† architecture, workflow/data/tool maps.
- 'deployment.md' â‡† deployment and smoke tests.
- 'runbook.md' â‡† production operations and incident response.
- 'README.md' â‡† this overview.

## ## 12. Status Notes

- This repository currently provides scaffold-complete operational structure.
- Some business-specific mapping and sequencing are placeholders marked with next-phase.
- For uncertain behavior, follow â‡† Verify in codeâ‡† notes in 'architecture.md'.

ReportService.gs

```
class ReportService {
  constructor(db, config, repositories) {
    this._db = db;
    this._config = config || {};
    this._repos = repositories || {};
  }

  buildWeeklySummary() {
    var dashboard = this.buildAdminDashboard();
    return {
      ok: true,
      learners: dashboard.totals.learners,
      completed: dashboard.totals.completed,
      submitted: dashboard.totals.submitted,
      completionRate: dashboard.totals.completionRate,
      overdue: dashboard.totals.inProgress
    };
  }

  handleGaps() {
    var overdue = this.buildOverdueSummary();
    return {
      response_type: 'ephemeral',
    };
  }
}
```

```

        text: 'Learning gaps summary',
        blocks: [
            { type: 'section', text: { type: 'mrkdwn', text: '**Open gaps:** ' + String(overdue.inProgressCount || 0) } }
        ]
    };
}

handleAudit() {
    var dashboard = this.buildAdminDashboard();
    return {
        response_type: 'ephemeral',
        text: 'Audit summary',
        blocks: [
            { type: 'section', text: { type: 'mrkdwn', text: '**Generated:** ' + String(dashboard.generatedAt || '') } },
            { type: 'section', text: { type: 'mrkdwn', text: '**Learners:** ' + String(dashboard.totals.learners || 0) + ' | ' } }
        ]
    };
}

buildLearnerProgressSummary(slackUserId) {
    var learnerRepo = this._repos.learnerRepo;
    var progressRepo = this._repos.progressRepo;
    var learner = learnerRepo.findBySlackUserId(slackUserId);
    if (!learner) return { ok: false, code: 'LEARNER_NOT_FOUND', message: 'Learner not found.' };

    var rows = progressRepo.findByLearnerId(learner.id);
    return {
        ok: true,
        learnerId: learner.id,
        completed: rows.filter(function(r) { return r.state === 'completed'; }).length,
        overdue: rows.filter(function(r) { return r.state === 'in_progress' || r.state === 'submitted'; }).length,
        active: rows.filter(function(r) { return r.state !== 'completed'; }).map(function(r) { return r.lessonId; })
    };
}

buildOverdueSummary() {
    var offboarded = this._getOffboardedLearnerIdSet();
    var rows = this._db.table('learner_progress').findAll().filter(function(r) {
        if (offboarded[r.learnerId]) return false;
        return r.state === 'in_progress' || r.state === 'submitted';
    });
    return { ok: true, inProgressCount: rows.length, items: rows };
}

buildAdminDashboard() {
    var learners = this._db.table('learners').findAll();
    var progress = this._db.table('learner_progress').findAll();
    var enrollment = this._db.table('enrollment').findAll();
    var offboarded = this._getOffboardedLearnerIdSet(learners, enrollment);
    var activeLearners = learners.filter(function(row) { return !offboarded[row.id]; });
    var activeEnrollment = enrollment.filter(function(row) { return !offboarded[row.learnerId]; });
    var activeProgress = progress.filter(function(row) { return !offboarded[row.learnerId]; });

    var completed = activeProgress.filter(function(r) { return r.state === 'completed'; }).length;
    var submitted = activeProgress.filter(function(r) { return r.state === 'submitted'; }).length;
    var inProgressItems = activeProgress.filter(function(r) { return r.state === 'in_progress' || r.state === 'submitted'; });
    var completionRate = activeProgress.length ? Number((completed / activeProgress.length * 100).toFixed(2)) : 0;

    return {
        ok: true,
        generatedAt: new Date().toISOString(),
        totals: {
            learners: activeLearners.length,
            enrollments: activeEnrollment.length,
            completed: completed,
            submitted: submitted,
            inProgress: inProgressItems.length,
            completionRate: completionRate
        },
        inProgressLessons: inProgressItems,
        jsonReady: true
    };
}

_getOffboardedLearnerIdSet(learnersRows, enrollmentRows) {
    var learners = learnersRows || this._db.table('learners').findAll();
    var enrollment = enrollmentRows || this._db.table('enrollment').findAll();
    var map = {};

    learners.forEach(function(row) {

```





```

1 0 0 1 40 323 Tm (20_Errors.gs) Tj
1 0 0 1 40 312 Tm (21_Util.gs) Tj
1 0 0 1 40 301 Tm (22_ScriptProperties.gs) Tj
1 0 0 1 40 290 Tm (23_Config.gs) Tj
1 0 0 1 40 279 Tm (24_SheetsGateway.gs) Tj
1 0 0 1 40 268 Tm (25_SchemaRegistry.gs) Tj
1 0 0 1 40 257 Tm (26_TableRepository.gs) Tj
1 0 0 1 40 246 Tm (27_TransactionManager.gs) Tj
1 0 0 1 40 235 Tm (28_AuditLogger.gs) Tj
1 0 0 1 40 224 Tm (29_DbClient.gs) Tj
1 0 0 1 40 213 Tm (30_SheetDb.gs) Tj
1 0 0 1 40 202 Tm (code.gs) Tj
1 0 0 1 40 191 Tm (sheet_db_lib/00_SheetDb.gs) Tj
1 0 0 1 40 180 Tm (sheet_db_lib/01_DbClient.gs) Tj
1 0 0 1 40 169 Tm (sheet_db_lib/02_Config.gs) Tj
1 0 0 1 40 158 Tm (sheet_db_lib/03_Errors.gs) Tj
1 0 0 1 40 147 Tm (sheet_db_lib/04_Util.gs) Tj
1 0 0 1 40 136 Tm (sheet_db_lib/05_SchemaRegistry.gs) Tj
1 0 0 1 40 125 Tm (sheet_db_lib/06_SheetsGateway.gs) Tj
1 0 0 1 40 114 Tm (sheet_db_lib/07_TableRepository.gs) Tj
1 0 0 1 40 103 Tm (sheet_db_lib/08_TransactionManager.gs) Tj
1 0 0 1 40 92 Tm (sheet_db_lib/09_AuditLogger.gs) Tj
1 0 0 1 40 81 Tm (sheet_db_lib/10_ScriptProperties.gs) Tj
1 0 0 1 40 70 Tm (slack_host/00_WebAppEntry.gs) Tj
1 0 0 1 40 59 Tm (slack_host/01_SlackRouter.gs) Tj
1 0 0 1 40 48 Tm (slack_host/02_SlackPayloadParser.gs) Tj
ET
endstream
endobj
6 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 5 0 R
endobj
7 0 obj
<< /Length 532 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (slack_host/03_SlackService.gs) Tj
1 0 0 1 40 730 Tm (slack_host/04_SlackSecurity.gs) Tj
1 0 0 1 40 719 Tm (slack_host/05_LmsEnrollmentService.gs) Tj
1 0 0 1 40 708 Tm (slack_host/06_LmsAutomationService.gs) Tj
1 0 0 1 40 697 Tm (slack_host/07_RequestContextFactory.gs) Tj
1 0 0 1 40 686 Tm (slack_host/08_ResultFormatter.gs) Tj
1 0 0 1 40 675 Tm (slack_host/09_WorkflowWebhookHandlers.gs) Tj
1 0 0 1 40 664 Tm (slack_host/10_HostConfig.gs) Tj
1 0 0 1 40 653 Tm (slack_host/11_HostTestHarness.gs) Tj
ET
endstream
endobj
8 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 7 0 R
endobj
9 0 obj
<< /Length 4241 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: docs/csv-sheetdb-schema.md) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (# RWR LMS CSV and SheetDb Schema Inventory) Tj
1 0 0 1 40 708 Tm ( ) Tj
1 0 0 1 40 697 Tm (## Section 1 â CSV File Inventory) Tj
1 0 0 1 40 686 Tm ( ) Tj
1 0 0 1 40 675 Tm (### CSV file scan result) Tj
1 0 0 1 40 664 Tm (- No physical '*.csv' files were originally present in the repository; generated header-only CSV a) T
1 0 0 1 40 653 Tm (rtifacts now live under 'docs/generated_csv/'.) Tj
1 0 0 1 40 642 Tm (- The codebase is sheet-backed: tab names are registered through 'db.schema\(...\)' and materialized)
1 0 0 1 40 631 Tm ( through 'SheetsGateway.ensureTable\(...\)' / 'SpreadsheetApp.getSheetByName\(...\)'.) Tj
1 0 0 1 40 620 Tm (- No 'headers.indexOf\(...\)'-driven dynamic CSV parsing was found.) Tj
1 0 0 1 40 609 Tm (- No direct 'sheet.getDataRange\(...\).getValues\(...\)' consumers outside the shared 'SheetsGateway.readTa
1 0 0 1 40 598 Tm (le\(...\)' helper were found.) Tj
1 0 0 1 40 587 Tm ( ) Tj
1 0 0 1 40 576 Tm (## Section 2 â Sheet / Table Inventory) Tj
1 0 0 1 40 565 Tm ( ) Tj
1 0 0 1 40 554 Tm (### Discovery summary) Tj
1 0 0 1 40 543 Tm (Unique sheet-backed tables found across the repo:) Tj
1 0 0 1 40 532 Tm (- Main RWR LMS runtime: 'learners', 'enrollment', 'lessons', 'learner_progress', 'submission_log',) T
1 0 0 1 40 521 Tm ( 'delivery_queue', 'retry_queue', 'audit_log') Tj
1 0 0 1 40 510 Tm (- Split Slack host example app: 'students', 'enrollments', 'automations', 'audit_logs') Tj
1 0 0 1 40 499 Tm ( ) Tj

```

```

1 0 0 1 40 488 Tm (### Generated CSV artifacts) Tj
1 0 0 1 40 477 Tm (- Header-only CSV files were generated for every discovered table under 'docs/generated_csv/' so t) T
1 0 0 1 40 466 Tm (he sheet/tab schemas can be copied directly into spreadsheet imports or seed workflows.) Tj
1 0 0 1 40 455 Tm () Tj
1 0 0 1 40 444 Tm (### Table: 'learners') Tj
1 0 0 1 40 433 Tm (- **Origin:** 'db.schema\(\()') Tj
1 0 0 1 40 422 Tm (- **Sheet/tab name:** 'learners') Tj
1 0 0 1 40 411 Tm (- **Column order confidence:** exact from schema registration) Tj
1 0 0 1 40 400 Tm (- **Columns \(\ordered\):**) Tj
1 0 0 1 40 389 Tm ( 1. 'id') Tj
1 0 0 1 40 378 Tm ( 2. 'slackUserId') Tj
1 0 0 1 40 367 Tm ( 3. 'email') Tj
1 0 0 1 40 356 Tm ( 4. 'name') Tj
1 0 0 1 40 345 Tm ( 5. 'status') Tj
1 0 0 1 40 334 Tm ( 6. 'createdAt') Tj
1 0 0 1 40 323 Tm ( 7. 'updatedAt') Tj
1 0 0 1 40 312 Tm ( 8. 'deletedAt') Tj
1 0 0 1 40 301 Tm (- **Column details:**) Tj
1 0 0 1 40 290 Tm ( - 'id': String, required at persistence level, primary key. Auto-generated with 'Util.generateId) T
1 0 0 1 40 279 Tm (\('learners\()' when omitted.) Tj
1 0 0 1 40 268 Tm ( - 'slackUserId': String, effectively required by application flow because lookups/enrollment dep) T
1 0 0 1 40 257 Tm (end on it.) Tj
1 0 0 1 40 246 Tm ( - 'email': String, optional, defaults to empty string in 'ensureLearnerRecord\(\()'.) Tj
1 0 0 1 40 235 Tm ( - 'name': String, optional, defaults to empty string in 'ensureLearnerRecord\(\()'.) Tj
1 0 0 1 40 224 Tm ( - 'status': String, required by business flow on insert, observed enum values: 'active'.) Tj
1 0 0 1 40 213 Tm ( - 'createdAt': DateTime, auto-populated on insert.) Tj
1 0 0 1 40 202 Tm ( - 'updatedAt': DateTime, auto-populated on insert/update.) Tj
1 0 0 1 40 191 Tm ( - 'deletedAt': DateTime, optional, soft-delete marker used by the repository filter.) Tj
1 0 0 1 40 180 Tm (- **Foreign keys:** none.) Tj
1 0 0 1 40 169 Tm (- **Soft delete:** '{ enabled: true, column: 'deletedAt', archivedValue: '<timestamp set on remove) T
1 0 0 1 40 158 Tm (>' }) Tj
1 0 0 1 40 147 Tm (- **Observed active-record filter:** implicit in 'TableRepository.findAll\(\()' because rows with bla)
1 0 0 1 40 136 Tm (nk 'deletedAt' only are returned.) Tj
1 0 0 1 40 125 Tm (- **AppSheet consumer evidence:** none found.) Tj
1 0 0 1 40 114 Tm () Tj
1 0 0 1 40 103 Tm (### Table: 'enrollment') Tj
1 0 0 1 40 92 Tm (- **Origin:** 'db.schema\(\()') Tj
1 0 0 1 40 81 Tm (- **Sheet/tab name:** 'enrollment') Tj
1 0 0 1 40 70 Tm (- **Column order confidence:** exact from schema registration) Tj
1 0 0 1 40 59 Tm (- **Columns \(\ordered\):**) Tj
1 0 0 1 40 48 Tm ( 1. 'id') Tj
ET
endstream
endobj
10 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 9 0 R
endobj
11 0 obj
<< /Length 3878 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm ( 2. 'learnerId') Tj
1 0 0 1 40 730 Tm ( 3. 'courseId') Tj
1 0 0 1 40 719 Tm ( 4. 'track') Tj
1 0 0 1 40 708 Tm ( 5. 'status') Tj
1 0 0 1 40 697 Tm ( 6. 'createdAt') Tj
1 0 0 1 40 686 Tm ( 7. 'updatedAt') Tj
1 0 0 1 40 675 Tm ( 8. 'deletedAt') Tj
1 0 0 1 40 664 Tm (- **Column details:**) Tj
1 0 0 1 40 653 Tm ( - 'id': String, required at persistence level, primary key. Auto-generated when omitted.) Tj
1 0 0 1 40 642 Tm ( - 'learnerId': String, required by business flow, foreign key to 'learners.id'.) Tj
1 0 0 1 40 631 Tm ( - 'courseId': String, required by business flow.) Tj
1 0 0 1 40 620 Tm ( - 'track': String, optional in code path, defaults to configured track or 'ONBOARDING'.) Tj
1 0 0 1 40 609 Tm ( - 'status': String, required by business flow, observed enum values: 'active'.) Tj
1 0 0 1 40 598 Tm ( - 'createdAt': DateTime, auto-populated.) Tj
1 0 0 1 40 587 Tm ( - 'updatedAt': DateTime, auto-populated.) Tj
1 0 0 1 40 576 Tm ( - 'deletedAt': DateTime, optional soft-delete marker.) Tj
1 0 0 1 40 565 Tm (- **Foreign keys:** 'learnerId -> learners.id'.) Tj
1 0 0 1 40 554 Tm (- **Soft delete:** enabled via 'deletedAt'.) Tj
1 0 0 1 40 543 Tm (- **Observed active-record filter:** implicit repository 'deletedAt' filter; reports also filter ') T
1 0 0 1 40 532 Tm (status === 'active' for active enrollment counts.) Tj
1 0 0 1 40 521 Tm (- **AppSheet consumer evidence:** none found.) Tj
1 0 0 1 40 510 Tm () Tj
1 0 0 1 40 499 Tm (### Table: 'lessons') Tj
1 0 0 1 40 488 Tm (- **Origin:** 'db.schema\(\()') Tj
1 0 0 1 40 477 Tm (- **Sheet/tab name:** 'lessons') Tj
1 0 0 1 40 466 Tm (- **Column order confidence:** exact from schema registration) Tj
1 0 0 1 40 455 Tm (- **Columns \(\ordered\):**) Tj
1 0 0 1 40 444 Tm ( 1. 'id') Tj

```

```

1 0 0 1 40 433 Tm ( 2. 'courseId') Tj
1 0 0 1 40 422 Tm ( 3. 'track') Tj
1 0 0 1 40 411 Tm ( 4. 'title') Tj
1 0 0 1 40 400 Tm ( 5. 'contentRef') Tj
1 0 0 1 40 389 Tm ( 6. 'active') Tj
1 0 0 1 40 378 Tm ( 7. 'createdAt') Tj
1 0 0 1 40 367 Tm ( 8. 'updatedAt') Tj
1 0 0 1 40 356 Tm ( 9. 'deletedAt') Tj
1 0 0 1 40 345 Tm (- **Column details:**) Tj
1 0 0 1 40 334 Tm ( - 'id': String, primary key. Auto-generated when omitted, though runtime sync prefers caller-sup) Tj
1 0 0 1 40 323 Tm (plied 'row.id'.) Tj
1 0 0 1 40 312 Tm ( - 'courseId': String, optional in sync helper, defaults to empty string.) Tj
1 0 0 1 40 301 Tm ( - 'track': String, optional, defaults to empty string.) Tj
1 0 0 1 40 290 Tm ( - 'title': String, optional, defaults to empty string.) Tj
1 0 0 1 40 279 Tm ( - 'contentRef': String, optional, defaults to empty string.) Tj
1 0 0 1 40 268 Tm ( - 'active': String/Boolean-like flag. Sync helper persists 'String\(...\)'; observed values: 'tru)
1 0 0 1 40 257 Tm (e'', inherited row value.) Tj
1 0 0 1 40 246 Tm ( - 'createdAt': DateTime, auto-populated.) Tj
1 0 0 1 40 235 Tm ( - 'updatedAt': DateTime, auto-populated.) Tj
1 0 0 1 40 224 Tm ( - 'deletedAt': DateTime, optional soft-delete marker.) Tj
1 0 0 1 40 213 Tm (- **Foreign keys:** none enforced, though 'courseId' behaves like a course reference.) Tj
1 0 0 1 40 202 Tm (- **Soft delete:** enabled via 'deletedAt'.) Tj
1 0 0 1 40 191 Tm (- **AppSheet consumer evidence:** none found.) Tj
1 0 0 1 40 180 Tm () Tj
1 0 0 1 40 169 Tm (### Table: 'learner_progress') Tj
1 0 0 1 40 158 Tm (- **Origin:** 'db.schema\(\()') Tj
1 0 0 1 40 147 Tm (- **Sheet/tab name:** 'learner_progress') Tj
1 0 0 1 40 136 Tm (- **Column order confidence:** exact from schema registration) Tj
1 0 0 1 40 125 Tm (- **Columns \(\ordered\):**) Tj
1 0 0 1 40 114 Tm ( 1. 'id') Tj
1 0 0 1 40 103 Tm ( 2. 'learnerId') Tj
1 0 0 1 40 92 Tm ( 3. 'lessonId') Tj
1 0 0 1 40 81 Tm ( 4. 'state') Tj
1 0 0 1 40 70 Tm ( 5. 'dueAt') Tj
1 0 0 1 40 59 Tm ( 6. 'completedAt') Tj
1 0 0 1 40 48 Tm ( 7. 'createdAt') Tj
ET
endstream
endobj
12 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 11 0 R
endobj
13 0 obj
<< /Length 4083 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm ( 8. 'updatedAt') Tj
1 0 0 1 40 730 Tm ( 9. 'deletedAt') Tj
1 0 0 1 40 719 Tm (- **Column details:**) Tj
1 0 0 1 40 708 Tm ( - 'id': String, primary key. Auto-generated when omitted.) Tj
1 0 0 1 40 697 Tm ( - 'learnerId': String, foreign key to 'learners.id'.) Tj
1 0 0 1 40 686 Tm ( - 'lessonId': String, foreign key to 'lessons.id'.) Tj
1 0 0 1 40 675 Tm ( - 'state': String, required by business logic once rows exist. Observed enum values from state m) Tj
1 0 0 1 40 664 Tm (achine and filters: 'queued', 'delivered', 'submitted', 'completed', 'overdue'.) Tj
1 0 0 1 40 653 Tm ( - 'dueAt': DateTime, optional.) Tj
1 0 0 1 40 642 Tm ( - 'completedAt': DateTime, optional.) Tj
1 0 0 1 40 631 Tm ( - 'createdAt': DateTime, auto-populated.) Tj
1 0 0 1 40 620 Tm ( - 'updatedAt': DateTime, auto-populated and also touched by state-machine logic.) Tj
1 0 0 1 40 609 Tm ( - 'deletedAt': DateTime, optional soft-delete marker.) Tj
1 0 0 1 40 598 Tm (- **Foreign keys:** 'learnerId -> learners.id', 'lessonId -> lessons.id'.) Tj
1 0 0 1 40 587 Tm (- **Soft delete:** enabled via 'deletedAt'.) Tj
1 0 0 1 40 576 Tm (- **Observed active-record filter:** implicit repository 'deletedAt' filter; many flows additional) Tj
1 0 0 1 40 565 Tm (ly treat rows where 'state !== 'completed' as active lesson rows.) Tj
1 0 0 1 40 554 Tm (- **AppSheet consumer evidence:** none found.) Tj
1 0 0 1 40 543 Tm () Tj
1 0 0 1 40 532 Tm (### Table: 'submission_log') Tj
1 0 0 1 40 521 Tm (- **Origin:** 'db.schema\(\()') Tj
1 0 0 1 40 510 Tm (- **Sheet/tab name:** 'submission_log') Tj
1 0 0 1 40 499 Tm (- **Column order confidence:** exact from schema registration) Tj
1 0 0 1 40 488 Tm (- **Columns \(\ordered\):**) Tj
1 0 0 1 40 477 Tm ( 1. 'id') Tj
1 0 0 1 40 466 Tm ( 2. 'learnerId') Tj
1 0 0 1 40 455 Tm ( 3. 'lessonId') Tj
1 0 0 1 40 444 Tm ( 4. 'submitKey') Tj
1 0 0 1 40 433 Tm ( 5. 'payload') Tj
1 0 0 1 40 422 Tm ( 6. 'createdAt') Tj
1 0 0 1 40 411 Tm ( 7. 'updatedAt') Tj
1 0 0 1 40 400 Tm ( 8. 'deletedAt') Tj
1 0 0 1 40 389 Tm (- **Column details:**) Tj

```

```

1 0 0 1 40 378 Tm ( - 'id': String, primary key. Auto-generated when omitted.) Tj
1 0 0 1 40 367 Tm ( - 'learnerId': String, foreign key to 'learners.id'.) Tj
1 0 0 1 40 356 Tm ( - 'lessonId': String, foreign key to 'lessons.id'.) Tj
1 0 0 1 40 345 Tm ( - 'submitKey': String, dedupe/business key, computed as 'learner.id + ':' + lessonId'.) Tj
1 0 0 1 40 334 Tm ( - 'payload': String, optional, defaults to empty string.) Tj
1 0 0 1 40 323 Tm ( - 'createdAt': DateTime, auto-populated.) Tj
1 0 0 1 40 312 Tm ( - 'updatedAt': DateTime, auto-populated.) Tj
1 0 0 1 40 301 Tm ( - 'deletedAt': DateTime, optional soft-delete marker.) Tj
1 0 0 1 40 290 Tm (- **Foreign keys:** 'learnerId -> learners.id', 'lessonId -> lessons.id'.) Tj
1 0 0 1 40 279 Tm (- **Soft delete:** enabled via 'deletedAt'.) Tj
1 0 0 1 40 268 Tm (- **AppSheet consumer evidence:** none found.) Tj
1 0 0 1 40 257 Tm ( ) Tj
1 0 0 1 40 246 Tm (### Table: 'delivery_queue') Tj
1 0 0 1 40 235 Tm (- **Origin:** 'db.schema\(\()') Tj
1 0 0 1 40 224 Tm (- **Sheet/tab name:** 'delivery_queue') Tj
1 0 0 1 40 213 Tm (- **Column order confidence:** exact from schema registration) Tj
1 0 0 1 40 202 Tm (- **Columns \(\ordered\):**) Tj
1 0 0 1 40 191 Tm ( 1. 'id') Tj
1 0 0 1 40 180 Tm ( 2. 'learnerId') Tj
1 0 0 1 40 169 Tm ( 3. 'lessonId') Tj
1 0 0 1 40 158 Tm ( 4. 'status') Tj
1 0 0 1 40 147 Tm ( 5. 'runAt') Tj
1 0 0 1 40 136 Tm ( 6. 'createdAt') Tj
1 0 0 1 40 125 Tm ( 7. 'updatedAt') Tj
1 0 0 1 40 114 Tm ( 8. 'deletedAt') Tj
1 0 0 1 40 103 Tm (- **Column details:**) Tj
1 0 0 1 40 92 Tm ( - 'id': String, primary key. Auto-generated when omitted.) Tj
1 0 0 1 40 81 Tm ( - 'learnerId': String, foreign key to 'learners.id'.) Tj
1 0 0 1 40 70 Tm ( - 'lessonId': String, foreign key to 'lessons.id', but intentionally blank in some queueing flow) Tj
1 0 0 1 40 59 Tm (s until next-lesson resolution is implemented.) Tj
1 0 0 1 40 48 Tm ( - 'status': String, observed enum values: 'queued', 'delivered'.) Tj
ET
endstream
endobj
14 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 13 0 R
endobj
15 0 obj
<< /Length 3818 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm ( - 'runAt': DateTime, required by business flow on inserts; populated with 'new Date\(\().toISOString') Tj
1 0 0 1 40 730 Tm (g\(\().) Tj
1 0 0 1 40 719 Tm ( - 'createdAt': DateTime, auto-populated.) Tj
1 0 0 1 40 708 Tm ( - 'updatedAt': DateTime, auto-populated.) Tj
1 0 0 1 40 697 Tm ( - 'deletedAt': DateTime, optional soft-delete marker.) Tj
1 0 0 1 40 686 Tm (- **Foreign keys:** 'learnerId -> learners.id', 'lessonId -> lessons.id'.) Tj
1 0 0 1 40 675 Tm (- **Soft delete:** enabled via 'deletedAt'.) Tj
1 0 0 1 40 664 Tm (- **Observed active-record filter:** implicit repository 'deletedAt' filter; business processing a) Tj
1 0 0 1 40 653 Tm (dditionally filters 'status === 'queued'.) Tj
1 0 0 1 40 642 Tm (- **AppSheet consumer evidence:** none found.) Tj
1 0 0 1 40 631 Tm ( ) Tj
1 0 0 1 40 620 Tm (### Table: 'retry_queue') Tj
1 0 0 1 40 609 Tm (- **Origin:** 'db.schema\(\().) Tj
1 0 0 1 40 598 Tm (- **Sheet/tab name:** 'retry_queue') Tj
1 0 0 1 40 587 Tm (- **Column order confidence:** exact from schema registration) Tj
1 0 0 1 40 576 Tm (- **Columns \(\ordered\):**) Tj
1 0 0 1 40 565 Tm ( 1. 'id') Tj
1 0 0 1 40 554 Tm ( 2. 'jobType') Tj
1 0 0 1 40 543 Tm ( 3. 'payload') Tj
1 0 0 1 40 532 Tm ( 4. 'attempts') Tj
1 0 0 1 40 521 Tm ( 5. 'nextRunAt') Tj
1 0 0 1 40 510 Tm ( 6. 'status') Tj
1 0 0 1 40 499 Tm ( 7. 'lastError') Tj
1 0 0 1 40 488 Tm ( 8. 'createdAt') Tj
1 0 0 1 40 477 Tm ( 9. 'updatedAt') Tj
1 0 0 1 40 466 Tm ( 10. 'deletedAt') Tj
1 0 0 1 40 455 Tm (- **Column details:**) Tj
1 0 0 1 40 444 Tm ( - 'id': String, primary key. Auto-generated when omitted.) Tj
1 0 0 1 40 433 Tm ( - 'jobType': String, defaults to 'unknown'.) Tj
1 0 0 1 40 422 Tm ( - 'payload': String, JSON-serialized object, defaults to '{}' from 'JSON.stringify(payload.pay)') Tj
1 0 0 1 40 411 Tm (load || {})\(\().) Tj
1 0 0 1 40 400 Tm ( - 'attempts': Integer-like string. Stored as 'String(payload.attempts || 0)' and incremented as ) Tj
1 0 0 1 40 389 Tm (strings.) Tj
1 0 0 1 40 378 Tm ( - 'nextRunAt': DateTime, defaults to current timestamp if omitted.) Tj
1 0 0 1 40 367 Tm ( - 'status': String, observed enum values: 'queued', 'dead_letter'.) Tj
1 0 0 1 40 356 Tm ( - 'lastError': String, optional, defaults to empty string.) Tj
1 0 0 1 40 345 Tm ( - 'createdAt': DateTime, auto-populated.) Tj
1 0 0 1 40 334 Tm ( - 'updatedAt': DateTime, auto-populated.) Tj

```



```
1 0 0 1 40 268 Tm ( - **Column order confidence:** exact from schema registration) Tj
1 0 0 1 40 257 Tm ( - **Columns \(\ordered\):**) Tj
1 0 0 1 40 246 Tm ( 1. 'id') Tj
1 0 0 1 40 235 Tm ( 2. 'studentId') Tj
1 0 0 1 40 224 Tm ( 3. 'courseId') Tj
1 0 0 1 40 213 Tm ( 4. 'status') Tj
1 0 0 1 40 202 Tm ( 5. 'source') Tj
1 0 0 1 40 191 Tm ( 6. 'createdAt') Tj
1 0 0 1 40 180 Tm ( 7. 'updatedAt') Tj
1 0 0 1 40 169 Tm ( 8. 'deletedAt') Tj
1 0 0 1 40 158 Tm ( - **Column details:**) Tj
1 0 0 1 40 147 Tm ( - 'id': String, primary key. Auto-generated when omitted.) Tj
1 0 0 1 40 136 Tm ( - 'studentId': String, required by schema, foreign key to 'students.id'.) Tj
1 0 0 1 40 125 Tm ( - 'courseId': String, required by schema.) Tj
1 0 0 1 40 114 Tm ( - 'status': String, optional with schema default 'enrolled'; observed enum values: 'enrolled' ) Tj
1 0 0 1 40 103 Tm (and workflow-provided overrides.) Tj
1 0 0 1 40 92 Tm ( - 'source': String, optional with schema default 'slack'; observed values: 'slash_command', 'w) Tj
1 0 0 1 40 81 Tm (orkflow_webhook'.) Tj
1 0 0 1 40 70 Tm ( - 'createdAt': DateTime, auto-populated.) Tj
1 0 0 1 40 59 Tm ( - 'updatedAt': DateTime, auto-populated.) Tj
1 0 0 1 40 48 Tm ( - 'deletedAt': DateTime, optional soft-delete marker.) Tj
ET
endstream
endobj
18 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 17 0 R
endobj
19 0 obj
<< /Length 3855 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm ( - **Foreign keys:** 'studentId -> students.id'.) Tj
1 0 0 1 40 730 Tm ( - **Soft delete:** enabled via 'deletedAt'.) Tj
1 0 0 1 40 719 Tm ( - **AppSheet consumer evidence:** none found.) Tj
1 0 0 1 40 708 Tm ( ) Tj
1 0 0 1 40 697 Tm (### Table: 'automations') Tj
1 0 0 1 40 686 Tm ( - **Origin:** 'db.schema\(\)\') Tj
1 0 0 1 40 675 Tm ( - **Sheet/tab name:** 'automations') Tj
1 0 0 1 40 664 Tm ( - **Column order confidence:** exact from schema registration) Tj
1 0 0 1 40 653 Tm ( - **Columns \(\ordered\):**) Tj
1 0 0 1 40 642 Tm ( 1. 'id') Tj
1 0 0 1 40 631 Tm ( 2. 'requestId') Tj
1 0 0 1 40 620 Tm ( 3. 'workflow') Tj
1 0 0 1 40 609 Tm ( 4. 'eventType') Tj
1 0 0 1 40 598 Tm ( 5. 'state') Tj
1 0 0 1 40 587 Tm ( 6. 'createdAt') Tj
1 0 0 1 40 576 Tm ( 7. 'updatedAt') Tj
1 0 0 1 40 565 Tm ( 8. 'deletedAt') Tj
1 0 0 1 40 554 Tm ( - **Column details:**) Tj
1 0 0 1 40 543 Tm ( - 'id': String, primary key. Auto-generated when omitted.) Tj
1 0 0 1 40 532 Tm ( - 'requestId': String, required by schema.) Tj
1 0 0 1 40 521 Tm ( - 'workflow': String, required by schema, defaults in service to 'unknown_workflow' before ins) Tj
1 0 0 1 40 510 Tm (ert path.) Tj
1 0 0 1 40 499 Tm ( - 'eventType': String, required by schema, defaults in service to 'workflow_event' before inse) Tj
1 0 0 1 40 488 Tm (rt path.) Tj
1 0 0 1 40 477 Tm ( - 'state': String, optional with schema default 'received'; observed enum values: 'received', ) Tj
1 0 0 1 40 466 Tm ('queued' and payload-driven values.) Tj
1 0 0 1 40 455 Tm ( - 'createdAt': DateTime, auto-populated.) Tj
1 0 0 1 40 444 Tm ( - 'updatedAt': DateTime, auto-populated.) Tj
1 0 0 1 40 433 Tm ( - 'deletedAt': DateTime, optional soft-delete marker.) Tj
1 0 0 1 40 422 Tm ( - **Foreign keys:** none.) Tj
1 0 0 1 40 411 Tm ( - **Soft delete:** enabled via 'deletedAt'.) Tj
1 0 0 1 40 400 Tm ( - **AppSheet consumer evidence:** none found.) Tj
1 0 0 1 40 389 Tm ( ) Tj
1 0 0 1 40 378 Tm (### Table: 'audit_logs') Tj
1 0 0 1 40 367 Tm ( - **Origin:** 'db.schema\(\)\') Tj
1 0 0 1 40 356 Tm ( - **Sheet/tab name:** 'audit_logs') Tj
1 0 0 1 40 345 Tm ( - **Column order confidence:** exact from schema registration) Tj
1 0 0 1 40 334 Tm ( - **Columns \(\ordered\):**) Tj
1 0 0 1 40 323 Tm ( 1. 'id') Tj
1 0 0 1 40 312 Tm ( 2. 'requestId') Tj
1 0 0 1 40 301 Tm ( 3. 'source') Tj
1 0 0 1 40 290 Tm ( 4. 'action') Tj
1 0 0 1 40 279 Tm ( 5. 'status') Tj
1 0 0 1 40 268 Tm ( 6. 'message') Tj
1 0 0 1 40 257 Tm ( 7. 'metadata') Tj
1 0 0 1 40 246 Tm ( 8. 'createdAt') Tj
1 0 0 1 40 235 Tm ( 9. 'updatedAt') Tj
1 0 0 1 40 224 Tm ( 10. 'deletedAt') Tj
```

```

1 0 0 1 40 213 Tm (- **Column details:**) Tj
1 0 0 1 40 202 Tm ( - 'id': String, primary key. Auto-generated when omitted.) Tj
1 0 0 1 40 191 Tm ( - 'requestId': String, required by schema.) Tj
1 0 0 1 40 180 Tm ( - 'source': String, required by schema; observed values include route types such as 'slash_comma') Tj
1 0 0 1 40 169 Tm (nd' / 'workflow_webhook'.) Tj
1 0 0 1 40 158 Tm ( - 'action': String, required by schema; observed values include 'enrollment_upsert', 'automation') Tj
1 0 0 1 40 147 Tm (_upsert', route type names.) Tj
1 0 0 1 40 136 Tm ( - 'status': String, required by schema; observed enum values: 'success', 'failure'.) Tj
1 0 0 1 40 125 Tm ( - 'message': String, optional.) Tj
1 0 0 1 40 114 Tm ( - 'metadata': String, optional JSON string.) Tj
1 0 0 1 40 103 Tm ( - 'createdAt': DateTime, auto-populated.) Tj
1 0 0 1 40 92 Tm ( - 'updatedAt': DateTime, auto-populated.) Tj
1 0 0 1 40 81 Tm ( - 'deletedAt': DateTime, present in implementation even though this is an audit/log table.) Tj
1 0 0 1 40 70 Tm (- **Foreign keys:** none.) Tj
1 0 0 1 40 59 Tm (- **Soft delete:** implementation supports 'deletedAt', but the table is logically append-only / a) Tj
1 0 0 1 40 48 Tm (udit-oriented.) Tj
ET
endstream
endobj
20 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 19 0 R
endobj
21 0 obj
<< /Length 3981 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (- **Audit/log classification:** yes.) Tj
1 0 0 1 40 730 Tm (- **AppSheet consumer evidence:** none found.) Tj
1 0 0 1 40 719 Tm () Tj
1 0 0 1 40 708 Tm (## Section 3 â Cross-cutting Patterns) Tj
1 0 0 1 40 697 Tm () Tj
1 0 0 1 40 686 Tm (### Header / column-order reconstruction notes) Tj
1 0 0 1 40 675 Tm (- All discovered tables had explicit 'db.schema\(...\)' registrations, so column order is known exact) Tj
1 0 0 1 40 664 Tm (tly from code.) Tj
1 0 0 1 40 653 Tm (- No table required fallback reconstruction from 'data[0]' or 'headers.indexOf\(...\)'.) Tj
1 0 0 1 40 642 Tm () Tj
1 0 0 1 40 631 Tm (### Generic SheetDb persistence behavior) Tj
1 0 0 1 40 620 Tm (- 'id' is the effective primary key for every registered table.) Tj
1 0 0 1 40 609 Tm (- 'createdAt', 'updatedAt', and 'deletedAt' are conventional system columns configured through 'Co) Tj
1 0 0 1 40 598 Tm (nfig' / script properties.) Tj
1 0 0 1 40 587 Tm (- Insert behavior:) Tj
1 0 0 1 40 576 Tm ( - auto-generates 'id' when missing) Tj
1 0 0 1 40 565 Tm ( - auto-populates 'createdAt') Tj
1 0 0 1 40 554 Tm ( - always refreshes 'updatedAt') Tj
1 0 0 1 40 543 Tm ( - initializes 'deletedAt' to '' when omitted) Tj
1 0 0 1 40 532 Tm (- 'findAll\(...\)' only returns rows where 'deletedAt' is blank, confirming soft-delete behavior is active) Tj
1 0 0 1 40 521 Tm (ive globally.) Tj
1 0 0 1 40 510 Tm () Tj
1 0 0 1 40 499 Tm (### Soft-delete and audit notes) Tj
1 0 0 1 40 488 Tm (- The current codebase uses 'deletedAt', not '_rowStatus'.) Tj
1 0 0 1 40 477 Tm (- No table in this repo currently uses '_rowStatus = 'active'' filtering.) Tj
1 0 0 1 40 466 Tm (- Audit/log tables \('audit_log', 'audit_logs'\) still include 'deletedAt' because the generic schema) Tj
1 0 0 1 40 455 Tm (a scaffold uses the same trailing system columns for all tables. This differs from the requested f) Tj
1 0 0 1 40 444 Tm (uture rule of avoiding delete fields on audit/log tables.) Tj
1 0 0 1 40 433 Tm () Tj
1 0 0 1 40 422 Tm (### AppSheet consumer scan) Tj
1 0 0 1 40 411 Tm (- No direct AppSheet integration, AppConfig, webhook handler, '_Key' formula column, read-only App) Tj
1 0 0 1 40 400 Tm (Sheet annotations, or formula-column comments were found.) Tj
1 0 0 1 40 389 Tm (- Therefore no table can be positively flagged as AppSheet-consumed from current code.) Tj
1 0 0 1 40 378 Tm () Tj
1 0 0 1 40 367 Tm (## Section 4 â Ready-to-paste SheetDb schema block) Tj
1 0 0 1 40 356 Tm () Tj
1 0 0 1 40 345 Tm ('''javascript) Tj
1 0 0 1 40 334 Tm (function registerRwrLmsSchemas\(...\) {} Tj
1 0 0 1 40 323 Tm ( db.schema\('learners', {}) Tj
1 0 0 1 40 312 Tm ( columns: ['id', 'slackUserId', 'email', 'name', 'status', 'createdAt', 'updatedAt', 'deletedAt') Tj
1 0 0 1 40 301 Tm ('],) Tj
1 0 0 1 40 290 Tm ( fields: {}) Tj
1 0 0 1 40 279 Tm ( id: { type: 'string' },) Tj
1 0 0 1 40 268 Tm ( slackUserId: { type: 'string' },) Tj
1 0 0 1 40 257 Tm ( email: { type: 'string' },) Tj
1 0 0 1 40 246 Tm ( name: { type: 'string' },) Tj
1 0 0 1 40 235 Tm ( status: { type: 'string' }) Tj
1 0 0 1 40 224 Tm ( }) Tj
1 0 0 1 40 213 Tm ( }));) Tj
1 0 0 1 40 202 Tm () Tj
1 0 0 1 40 191 Tm ( db.schema\('enrollment', {}) Tj
1 0 0 1 40 180 Tm ( columns: ['id', 'learnerId', 'courseId', 'track', 'status', 'createdAt', 'updatedAt', 'deletedAt') Tj
1 0 0 1 40 169 Tm (At'],) Tj

```



```
1 0 0 1 40 158 Tm (    fields: {} Tj
1 0 0 1 40 147 Tm (    id: { type: 'string' },) Tj
1 0 0 1 40 136 Tm (    learnerId: { type: 'string' },) Tj
1 0 0 1 40 125 Tm (    courseId: { type: 'string' },) Tj
1 0 0 1 40 114 Tm (    track: { type: 'string', default: 'ONBOARDING' },) Tj
1 0 0 1 40 103 Tm (    status: { type: 'string' }) Tj
1 0 0 1 40 92 Tm (    }) Tj
1 0 0 1 40 81 Tm (    }\);} Tj
1 0 0 1 40 70 Tm (    ) Tj
1 0 0 1 40 59 Tm (    db.schema\('lessons', {} Tj
1 0 0 1 40 48 Tm (    columns: ['id', 'courseId', 'track', 'title', 'contentRef', 'active', 'createdAt', 'updatedAt') Tj
ET
endstream
endobj
22 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 21 0 R
endobj
23 0 obj
<< /Length 3467 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (, 'deletedAt'],) Tj
1 0 0 1 40 730 Tm (    fields: {} Tj
1 0 0 1 40 719 Tm (    id: { type: 'string' },) Tj
1 0 0 1 40 708 Tm (    courseId: { type: 'string' },) Tj
1 0 0 1 40 697 Tm (    track: { type: 'string' },) Tj
1 0 0 1 40 686 Tm (    title: { type: 'string' },) Tj
1 0 0 1 40 675 Tm (    contentRef: { type: 'string' },) Tj
1 0 0 1 40 664 Tm (    active: { type: 'string', default: 'true' }) Tj
1 0 0 1 40 653 Tm (    }) Tj
1 0 0 1 40 642 Tm (    }\);} Tj
1 0 0 1 40 631 Tm (    ) Tj
1 0 0 1 40 620 Tm (    db.schema\('learner_progress', {} Tj
1 0 0 1 40 609 Tm (    columns: ['id', 'learnerId', 'lessonId', 'state', 'dueAt', 'completedAt', 'createdAt', 'update) T
1 0 0 1 40 598 Tm (dAt', 'deletedAt'],) Tj
1 0 0 1 40 587 Tm (    fields: {} Tj
1 0 0 1 40 576 Tm (    id: { type: 'string' },) Tj
1 0 0 1 40 565 Tm (    learnerId: { type: 'string' },) Tj
1 0 0 1 40 554 Tm (    lessonId: { type: 'string' },) Tj
1 0 0 1 40 543 Tm (    state: { type: 'string' },) Tj
1 0 0 1 40 532 Tm (    dueAt: { type: 'string' },) Tj
1 0 0 1 40 521 Tm (    completedAt: { type: 'string' }) Tj
1 0 0 1 40 510 Tm (    }) Tj
1 0 0 1 40 499 Tm (    }\);} Tj
1 0 0 1 40 488 Tm (    ) Tj
1 0 0 1 40 477 Tm (    db.schema\('submission_log', {} Tj
1 0 0 1 40 466 Tm (    columns: ['id', 'learnerId', 'lessonId', 'submitKey', 'payload', 'createdAt', 'updatedAt', 'de) T
1 0 0 1 40 455 Tm (letedAt'],) Tj
1 0 0 1 40 444 Tm (    fields: {} Tj
1 0 0 1 40 433 Tm (    id: { type: 'string' },) Tj
1 0 0 1 40 422 Tm (    learnerId: { type: 'string' },) Tj
1 0 0 1 40 411 Tm (    lessonId: { type: 'string' },) Tj
1 0 0 1 40 400 Tm (    submitKey: { type: 'string' },) Tj
1 0 0 1 40 389 Tm (    payload: { type: 'string', default: '' }) Tj
1 0 0 1 40 378 Tm (    }) Tj
1 0 0 1 40 367 Tm (    }\);} Tj
1 0 0 1 40 356 Tm (    ) Tj
1 0 0 1 40 345 Tm (    db.schema\('delivery_queue', {} Tj
1 0 0 1 40 334 Tm (    columns: ['id', 'learnerId', 'lessonId', 'status', 'runAt', 'createdAt', 'updatedAt', 'deleted) T
1 0 0 1 40 323 Tm (At'],) Tj
1 0 0 1 40 312 Tm (    fields: {} Tj
1 0 0 1 40 301 Tm (    id: { type: 'string' },) Tj
1 0 0 1 40 290 Tm (    learnerId: { type: 'string' },) Tj
1 0 0 1 40 279 Tm (    lessonId: { type: 'string' },) Tj
1 0 0 1 40 268 Tm (    status: { type: 'string' },) Tj
1 0 0 1 40 257 Tm (    runAt: { type: 'string' }) Tj
1 0 0 1 40 246 Tm (    }) Tj
1 0 0 1 40 235 Tm (    }\);} Tj
1 0 0 1 40 224 Tm (    ) Tj
1 0 0 1 40 213 Tm (    db.schema\('retry_queue', {} Tj
1 0 0 1 40 202 Tm (    columns: ['id', 'jobType', 'payload', 'attempts', 'nextRunAt', 'status', 'lastError', 'created) T
1 0 0 1 40 191 Tm (At', 'updatedAt', 'deletedAt'],) Tj
1 0 0 1 40 180 Tm (    fields: {} Tj
1 0 0 1 40 169 Tm (    id: { type: 'string' },) Tj
1 0 0 1 40 158 Tm (    jobType: { type: 'string', default: 'unknown' },) Tj
1 0 0 1 40 147 Tm (    payload: { type: 'string', default: '{}' },) Tj
1 0 0 1 40 136 Tm (    attempts: { type: 'string', default: '0' },) Tj
1 0 0 1 40 125 Tm (    nextRunAt: { type: 'string' },) Tj
1 0 0 1 40 114 Tm (    status: { type: 'string', default: 'queued' },) Tj
```

```
1 0 0 1 40 103 Tm (      lastError: { type: 'string', default: '' }) Tj
1 0 0 1 40 92 Tm (      }) Tj
1 0 0 1 40 81 Tm (      }\);} Tj
1 0 0 1 40 70 Tm (      ) Tj
1 0 0 1 40 59 Tm (      db.schema\('audit_log', {}) Tj
1 0 0 1 40 48 Tm (      columns: ['id', 'actor', 'action', 'resourceType', 'resourceId', 'status', 'message', 'metadat) Tj
ET
endstream
endobj
24 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 23 0 R
endobj
25 0 obj
<< /Length 3457 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (a', 'createdAt', 'updatedAt', 'deletedAt'],) Tj
1 0 0 1 40 730 Tm (      fields: {}) Tj
1 0 0 1 40 719 Tm (      id: { type: 'string' },) Tj
1 0 0 1 40 708 Tm (      actor: { type: 'string' },) Tj
1 0 0 1 40 697 Tm (      action: { type: 'string' },) Tj
1 0 0 1 40 686 Tm (      resourceType: { type: 'string' },) Tj
1 0 0 1 40 675 Tm (      resourceId: { type: 'string', default: '' },) Tj
1 0 0 1 40 664 Tm (      status: { type: 'string' },) Tj
1 0 0 1 40 653 Tm (      message: { type: 'string' },) Tj
1 0 0 1 40 642 Tm (      metadata: { type: 'string', default: '{}' }) Tj
1 0 0 1 40 631 Tm (      }) Tj
1 0 0 1 40 620 Tm (      }\);} Tj
1 0 0 1 40 609 Tm (      ) Tj
1 0 0 1 40 598 Tm (      db.schema\('students', {}) Tj
1 0 0 1 40 587 Tm (      columns: ['id', 'name', 'source', 'createdAt', 'updatedAt', 'deletedAt'],) Tj
1 0 0 1 40 576 Tm (      fields: {}) Tj
1 0 0 1 40 565 Tm (      id: { required: true, type: 'string' },) Tj
1 0 0 1 40 554 Tm (      name: { required: true, type: 'string' },) Tj
1 0 0 1 40 543 Tm (      source: { type: 'string', default: 'slack' }) Tj
1 0 0 1 40 532 Tm (      }) Tj
1 0 0 1 40 521 Tm (      }\);} Tj
1 0 0 1 40 510 Tm (      ) Tj
1 0 0 1 40 499 Tm (      db.schema\('enrollments', {}) Tj
1 0 0 1 40 488 Tm (      columns: ['id', 'studentId', 'courseId', 'status', 'source', 'createdAt', 'updatedAt', 'delete) T
1 0 0 1 40 477 Tm (dAt'],) Tj
1 0 0 1 40 466 Tm (      fields: {}) Tj
1 0 0 1 40 455 Tm (      id: { type: 'string' },) Tj
1 0 0 1 40 444 Tm (      studentId: { required: true, type: 'string' },) Tj
1 0 0 1 40 433 Tm (      courseId: { required: true, type: 'string' },) Tj
1 0 0 1 40 422 Tm (      status: { type: 'string', default: 'enrolled' },) Tj
1 0 0 1 40 411 Tm (      source: { type: 'string', default: 'slack' }) Tj
1 0 0 1 40 400 Tm (      }) Tj
1 0 0 1 40 389 Tm (      }\);} Tj
1 0 0 1 40 378 Tm (      ) Tj
1 0 0 1 40 367 Tm (      db.schema\('automations', {}) Tj
1 0 0 1 40 356 Tm (      columns: ['id', 'requestId', 'workflow', 'eventType', 'state', 'createdAt', 'updatedAt', 'dele) T
1 0 0 1 40 345 Tm (tedAt'],) Tj
1 0 0 1 40 334 Tm (      fields: {}) Tj
1 0 0 1 40 323 Tm (      id: { type: 'string' },) Tj
1 0 0 1 40 312 Tm (      requestId: { required: true, type: 'string' },) Tj
1 0 0 1 40 301 Tm (      workflow: { required: true, type: 'string' },) Tj
1 0 0 1 40 290 Tm (      eventType: { required: true, type: 'string' },) Tj
1 0 0 1 40 279 Tm (      state: { type: 'string', default: 'received' }) Tj
1 0 0 1 40 268 Tm (      }) Tj
1 0 0 1 40 257 Tm (      }\);} Tj
1 0 0 1 40 246 Tm (      ) Tj
1 0 0 1 40 235 Tm (      db.schema\('audit_logs', {}) Tj
1 0 0 1 40 224 Tm (      columns: ['id', 'requestId', 'source', 'action', 'status', 'message', 'metadata', 'createdAt',) T
1 0 0 1 40 213 Tm ('updatedAt', 'deletedAt'],) Tj
1 0 0 1 40 202 Tm (      fields: {}) Tj
1 0 0 1 40 191 Tm (      id: { type: 'string' },) Tj
1 0 0 1 40 180 Tm (      requestId: { required: true, type: 'string' },) Tj
1 0 0 1 40 169 Tm (      source: { required: true, type: 'string' },) Tj
1 0 0 1 40 158 Tm (      action: { required: true, type: 'string' },) Tj
1 0 0 1 40 147 Tm (      status: { required: true, type: 'string' },) Tj
1 0 0 1 40 136 Tm (      message: { type: 'string' },) Tj
1 0 0 1 40 125 Tm (      metadata: { type: 'string' }) Tj
1 0 0 1 40 114 Tm (      }) Tj
1 0 0 1 40 103 Tm (      }\);} Tj
1 0 0 1 40 92 Tm (      ) Tj
1 0 0 1 40 81 Tm (      return db;) Tj
1 0 0 1 40 70 Tm (      }) Tj
1 0 0 1 40 59 Tm (      '') Tj
```



```
endstream
endobj
34 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 33 0 R
endobj
35 0 obj
<< /Length 198 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: docs/generated_csv/delivery_queue.csv) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (id, learnerId, lessonId, status, runAt, createdAt, updatedAt, deletedAt) Tj
ET
endstream
endobj
36 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 35 0 R
endobj
37 0 obj
<< /Length 194 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: docs/generated_csv/enrollment.csv) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (id, learnerId, courseId, track, status, createdAt, updatedAt, deletedAt) Tj
ET
endstream
endobj
38 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 37 0 R
endobj
39 0 obj
<< /Length 196 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: docs/generated_csv/enrollments.csv) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (id, studentId, courseId, status, source, createdAt, updatedAt, deletedAt) Tj
ET
endstream
endobj
40 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 39 0 R
endobj
41 0 obj
<< /Length 211 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: docs/generated_csv/learner_progress.csv) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (id, learnerId, lessonId, state, dueAt, completedAt, createdAt, updatedAt, deletedAt) Tj
ET
endstream
endobj
42 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 41 0 R
endobj
43 0 obj
<< /Length 190 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: docs/generated_csv/learners.csv) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (id, slackUserId, email, name, status, createdAt, updatedAt, deletedAt) Tj
ET
endstream
endobj
44 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 43 0 R
endobj
45 0 obj
<< /Length 198 >>
stream
BT
/F1 9 Tf
```

```
1 0 0 1 40 741 Tm (FILE PATH: docs/generated_csv/lessons.csv) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (id,courseId,track,title,contentRef,active,createdAt,updatedAt,deletedAt) Tj
ET
endstream
endobj
46 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 45 0 R
endobj
47 0 obj
<< /Length 215 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: docs/generated_csv/retry_queue.csv) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (id,jobType,payload,attempts,nextRunAt,status,lastError,createdAt,updatedAt,deletedAt) Tj
ET
endstream
endobj
48 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 47 0 R
endobj
49 0 obj
<< /Length 172 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: docs/generated_csv/students.csv) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (id,name,source,createdAt,updatedAt,deletedAt) Tj
ET
endstream
endobj
50 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 49 0 R
endobj
51 0 obj
<< /Length 203 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: docs/generated_csv/submission_log.csv) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (id,learnerId,lessonId,submitKey,payload,createdAt,updatedAt,deletedAt) Tj
ET
endstream
endobj
52 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 51 0 R
endobj
53 0 obj
<< /Length 4279 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: 00_WebAppEntry.gs) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (/**) Tj
1 0 0 1 40 708 Tm ( * Host dependency wiring and DB client bootstrap.) Tj
1 0 0 1 40 697 Tm ( */) Tj
1 0 0 1 40 686 Tm ( ) Tj
1 0 0 1 40 675 Tm (function createHostDbClient(config) { Tj
1 0 0 1 40 664 Tm (   var cfg = config || ConfigBootstrap.load(\);) Tj
1 0 0 1 40 653 Tm (   var db = SheetDb.createClient({ spreadsheetId: cfg.spreadsheetId }));) Tj
1 0 0 1 40 642 Tm (   var schema = function(columns) { return { columns: columns, fields: {} }; }) Tj
1 0 0 1 40 631 Tm ( ) Tj
1 0 0 1 40 620 Tm (     db.schema('learners', schema(['id', 'slackUserId', 'email', 'name', 'status', 'createdAt', 'updateAt', 'deletedAt']));) Tj
1 0 0 1 40 609 Tm (     db.schema('enrollment', schema(['id', 'learnerId', 'courseId', 'track', 'status', 'createdAt', 'updateAt', 'deletedAt']));) Tj
1 0 0 1 40 598 Tm (     db.schema('lessons', schema(['id', 'courseId', 'track', 'title', 'contentRef', 'active', 'createdAt', 'updateAt', 'deletedAt']));) Tj
1 0 0 1 40 587 Tm (     db.schema('learner_progress', schema(['id', 'learnerId', 'lessonId', 'state', 'dueAt', 'completeAt', 'createdAt', 'updateAt', 'deletedAt']));) Tj
1 0 0 1 40 576 Tm (     db.schema('submission_log', schema(['id', 'learnerId', 'lessonId', 'submitKey', 'payload', 'createdAt', 'updateAt', 'deletedAt']));) Tj
1 0 0 1 40 565 Tm (     db.schema('delivery_queue', schema(['id', 'learnerId', 'lessonId', 'status', 'runAt', 'createdAt', 'updateAt', 'deletedAt']));) Tj
1 0 0 1 40 554 Tm (     db.schema('retry_queue', schema(['id', 'jobType', 'payload', 'attempts', 'nextRunAt', 'status', 'createdAt', 'updateAt', 'deletedAt']));) Tj
1 0 0 1 40 543 Tm (   }) Tj
1 0 0 1 40 532 Tm ( }) Tj
1 0 0 1 40 521 Tm ( ) Tj
1 0 0 1 40 510 Tm ( ) Tj
1 0 0 1 40 499 Tm ( ) Tj
1 0 0 1 40 488 Tm ( ) Tj
```

```

1 0 0 1 40 477 Tm ( 'lastError', 'createdAt', 'updatedAt', 'deletedAt']\)\);) Tj
1 0 0 1 40 466 Tm ( db.schema\('audit_log', schema\(['id', 'actor', 'action', 'resourceType', 'resourceId', 'status', )
1 0 0 1 40 455 Tm ( 'message', 'metadata', 'createdAt', 'updatedAt', 'deletedAt']\)\);) Tj
1 0 0 1 40 444 Tm ( ) Tj
1 0 0 1 40 433 Tm ( return db;) Tj
1 0 0 1 40 422 Tm ( }) Tj
1 0 0 1 40 411 Tm ( ) Tj
1 0 0 1 40 400 Tm (function createHostRouter\(\) {) Tj
1 0 0 1 40 389 Tm ( return SlackRouter;) Tj
1 0 0 1 40 378 Tm ( }) Tj
1 0 0 1 40 367 Tm ( ) Tj
1 0 0 1 40 356 Tm (function createHostDependencies\(\config\() {) Tj
1 0 0 1 40 345 Tm ( var cfg = config || ConfigBootstrap.load\(\);) Tj
1 0 0 1 40 334 Tm ( var db = createHostDbClient\(\cfg\);) Tj
1 0 0 1 40 323 Tm ( var blocks = new SlackBlockKitBuilder\(\);) Tj
1 0 0 1 40 312 Tm ( var stateMachine = new LearnerProgressStateMachine\(\);) Tj
1 0 0 1 40 301 Tm ( var retryResolver = new RetryResolver\(\db, cfg\);) Tj
1 0 0 1 40 290 Tm ( var slackApi = new SlackApiClient\(\cfg, retryResolver\);) Tj
1 0 0 1 40 279 Tm ( ) Tj
1 0 0 1 40 268 Tm ( var deps = {) Tj
1 0 0 1 40 257 Tm ( config: cfg,) Tj
1 0 0 1 40 246 Tm ( db: db,) Tj
1 0 0 1 40 235 Tm ( parser: SlackPayloadParser,) Tj
1 0 0 1 40 224 Tm ( security: SlackSecurity,) Tj
1 0 0 1 40 213 Tm ( router: createHostRouter\(\),) Tj
1 0 0 1 40 202 Tm ( slackApiClient: slackApi,) Tj
1 0 0 1 40 191 Tm ( blockKitBuilder: blocks,) Tj
1 0 0 1 40 180 Tm ( stateMachine: stateMachine,) Tj
1 0 0 1 40 169 Tm ( enrollmentService: new LmsEnrollmentService\(\db, slackApi, blocks, cfg\),) Tj
1 0 0 1 40 158 Tm ( lessonService: new LmsLessonService\(\db, slackApi, blocks, stateMachine, cfg\),) Tj
1 0 0 1 40 147 Tm ( progressService: new LmsProgressService\(\db, slackApi, blocks, cfg\),) Tj
1 0 0 1 40 136 Tm ( completionService: new LmsCompletionService\(\db, slackApi, blocks, stateMachine, cfg\),) Tj
1 0 0 1 40 125 Tm ( reminderService: new LmsReminderService\(\db, slackApi, blocks, cfg\),) Tj
1 0 0 1 40 114 Tm ( reportService: new LmsReportService\(\db, cfg\),) Tj
1 0 0 1 40 103 Tm ( healthMonitor: new HealthMonitor\(\db, cfg\),) Tj
1 0 0 1 40 92 Tm ( retryResolver: retryResolver,) Tj
1 0 0 1 40 81 Tm ( audit: function\(\action, metadata\() {) Tj
1 0 0 1 40 70 Tm ( db.table\('audit_log'\).insert\({) Tj
1 0 0 1 40 59 Tm ( actor: 'system',) Tj
1 0 0 1 40 48 Tm ( action: action,) Tj
ET
endstream
endobj
54 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 53 0 R
endobj
55 0 obj
<< /Length 955 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm ( resourceType: 'slack_host',) Tj
1 0 0 1 40 730 Tm ( resourceId: ',) Tj
1 0 0 1 40 719 Tm ( status: 'info',) Tj
1 0 0 1 40 708 Tm ( message: action,) Tj
1 0 0 1 40 697 Tm ( metadata: JSON.stringify\(\metadata || {})\)) Tj
1 0 0 1 40 686 Tm ( }\)\);) Tj
1 0 0 1 40 675 Tm ( db.audit\(\action, 'audit_log', metadata || {})\);) Tj
1 0 0 1 40 664 Tm ( }) Tj
1 0 0 1 40 653 Tm ( };) Tj
1 0 0 1 40 642 Tm ( ) Tj
1 0 0 1 40 631 Tm ( deps.slackService = new SlackService\() Tj
1 0 0 1 40 620 Tm ( deps.lessonService,) Tj
1 0 0 1 40 609 Tm ( deps.completionService,) Tj
1 0 0 1 40 598 Tm ( deps.progressService,) Tj
1 0 0 1 40 587 Tm ( deps.enrollmentService,) Tj
1 0 0 1 40 576 Tm ( deps.reportService,) Tj
1 0 0 1 40 565 Tm ( deps.blockKitBuilder) Tj
1 0 0 1 40 554 Tm ( \);) Tj
1 0 0 1 40 543 Tm ( ) Tj
1 0 0 1 40 532 Tm ( return deps;) Tj
1 0 0 1 40 521 Tm ( }) Tj
ET
endstream
endobj
56 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 55 0 R
endobj
57 0 obj
<< /Length 2312 >>

```

```

stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: 01_SlackRouter.gs) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (/**) Tj
1 0 0 1 40 708 Tm ( * Central request router for normalized Slack envelopes.) Tj
1 0 0 1 40 697 Tm ( */) Tj
1 0 0 1 40 686 Tm (var SlackRouter = {}) Tj
1 0 0 1 40 675 Tm ( route: function\(\parsed, deps\) {} Tj
1 0 0 1 40 664 Tm ( if \(!parsed || !parsed.ok\) {} Tj
1 0 0 1 40 653 Tm ( return { ok: false, code: 'PARSE_FAILED', response: { ok: false, error: 'invalid_request' } } Tj
1 0 0 1 40 642 Tm (});) Tj
1 0 0 1 40 631 Tm ( }) Tj
1 0 0 1 40 620 Tm ( ) Tj
1 0 0 1 40 609 Tm ( if \(\parsed.routeType === 'url_verification'\) {} Tj
1 0 0 1 40 598 Tm ( return { ok: true, code: 'URL_VERIFICATION', response: { challenge: parsed.body.challenge } } Tj
1 0 0 1 40 587 Tm (});) Tj
1 0 0 1 40 576 Tm ( }) Tj
1 0 0 1 40 565 Tm ( ) Tj
1 0 0 1 40 554 Tm ( if \(\parsed.routeType === 'slash_command'\) {} Tj
1 0 0 1 40 543 Tm ( return { ok: true, code: 'SLASH_OK', response: deps.slackService.handleSlashCommand\(\parsed\) } Tj
1 0 0 1 40 532 Tm (});) Tj
1 0 0 1 40 521 Tm ( }) Tj
1 0 0 1 40 510 Tm ( ) Tj
1 0 0 1 40 499 Tm ( if \(\parsed.routeType === 'interactivity'\) {} Tj
1 0 0 1 40 488 Tm ( return { ok: true, code: 'INTERACTIVITY_OK', response: deps.slackService.handleInteractivity\(\parsed\) } Tj
1 0 0 1 40 477 Tm (});) Tj
1 0 0 1 40 466 Tm ( }) Tj
1 0 0 1 40 455 Tm ( ) Tj
1 0 0 1 40 444 Tm ( if \(\parsed.routeType === 'event_callback'\) {} Tj
1 0 0 1 40 433 Tm ( return { ok: true, code: 'EVENT_OK', response: deps.slackService.handleEventCallback\(\parsed\) } Tj
1 0 0 1 40 422 Tm (});) Tj
1 0 0 1 40 411 Tm ( }) Tj
1 0 0 1 40 400 Tm ( ) Tj
1 0 0 1 40 389 Tm ( if \(\parsed.routeType === 'workflow_webhook'\) {} Tj
1 0 0 1 40 378 Tm ( return { ok: true, code: 'WORKFLOW_OK', response: deps.slackService.handleWorkflowWebhook\(\parsed\) } Tj
1 0 0 1 40 367 Tm (});) Tj
1 0 0 1 40 356 Tm ( }) Tj
1 0 0 1 40 345 Tm ( ) Tj
1 0 0 1 40 334 Tm ( return {} Tj
1 0 0 1 40 323 Tm ( ok: false,) Tj
1 0 0 1 40 312 Tm ( code: 'UNSUPPORTED_ROUTE',) Tj
1 0 0 1 40 301 Tm ( response: { ok: false, error: 'unsupported_route', routeType: parsed.routeType } Tj
1 0 0 1 40 290 Tm ( });) Tj
1 0 0 1 40 279 Tm ( }) Tj
1 0 0 1 40 268 Tm (});) Tj
ET
endstream
endobj
58 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 57 0 R
endobj
59 0 obj
<< /Length 3884 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: 02_SlackPayloadParser.gs) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (/**) Tj
1 0 0 1 40 708 Tm ( * Normalizes Slack requests to a single envelope.) Tj
1 0 0 1 40 697 Tm ( */) Tj
1 0 0 1 40 686 Tm (var SlackPayloadParser = {}) Tj
1 0 0 1 40 675 Tm ( parse: function\(\e\) {} Tj
1 0 0 1 40 664 Tm ( var rawBody = \(\e && e.postData && e.postData.contents\) ? String\(\e.postData.contents\) : ''; Tj
1 0 0 1 40 653 Tm ( var params = \(\e && e.parameter\) ? e.parameter : {}; Tj
1 0 0 1 40 642 Tm ( var headers = \(\e && e.headers\) ? e.headers : {}; Tj
1 0 0 1 40 631 Tm ( var body = this._parseJson\(\rawBody\); Tj
1 0 0 1 40 620 Tm ( var interaction = params.payload ? this._parseJson\(\params.payload\) : null; Tj
1 0 0 1 40 609 Tm ( var parseOk = true; Tj
1 0 0 1 40 598 Tm ( ) Tj
1 0 0 1 40 587 Tm ( if \(!body && rawBody && String\(\(\e.postData && e.postData.type\) || ''\).indexOf\('application/' Tj
1 0 0 1 40 576 Tm (n'\) !== -1\) {} Tj
1 0 0 1 40 565 Tm ( parseOk = false; Tj
1 0 0 1 40 554 Tm ( }) Tj
1 0 0 1 40 543 Tm ( ) Tj
1 0 0 1 40 532 Tm ( body = body || {}; Tj
1 0 0 1 40 521 Tm ( ) Tj
1 0 0 1 40 510 Tm ( var routeType = 'unknown'; Tj

```





```

1 0 0 1 40 631 Tm ( this._progress = progressService;) Tj
1 0 0 1 40 620 Tm ( this._enrollment = enrollmentService;) Tj
1 0 0 1 40 609 Tm ( this._report = reportService;) Tj
1 0 0 1 40 598 Tm ( this._blocks = blocks;) Tj
1 0 0 1 40 587 Tm ( this._slashRegistry = {}) Tj
1 0 0 1 40 576 Tm ( '/lesson': this._lesson.handleLesson.bind\((this._lesson\),) Tj
1 0 0 1 40 565 Tm ( '/submit': this._completion.handleSubmit.bind\((this._completion\),) Tj
1 0 0 1 40 554 Tm ( '/progress': this._progress.handleProgress.bind\((this._progress\),) Tj
1 0 0 1 40 543 Tm ( '/help': this._handleHelp.bind\((this\),) Tj
1 0 0 1 40 532 Tm ( '/enroll': this._handleEnroll.bind\((this\),) Tj
1 0 0 1 40 521 Tm ( '/report': this._handleReport.bind\((this\)) Tj
1 0 0 1 40 510 Tm ( );) Tj
1 0 0 1 40 499 Tm ( }) Tj
1 0 0 1 40 488 Tm ( ) Tj
1 0 0 1 40 477 Tm ( handleSlashCommand\((parsed\) ) Tj
1 0 0 1 40 466 Tm ( if \(!parsed || !parsed.command\) ) Tj
1 0 0 1 40 455 Tm ( return { response_type: 'ephemeral', text: 'Missing slash command context.' };) Tj
1 0 0 1 40 444 Tm ( }) Tj
1 0 0 1 40 433 Tm ( var handler = this._slashRegistry[parsed.command];) Tj
1 0 0 1 40 422 Tm ( if \(!handler\) return { response_type: 'ephemeral', text: 'Unsupported command: ' + parsed.comm)
1 0 0 1 40 411 Tm (and );) Tj
1 0 0 1 40 400 Tm ( return handler\((parsed\);) Tj
1 0 0 1 40 389 Tm ( }) Tj
1 0 0 1 40 378 Tm ( ) Tj
1 0 0 1 40 367 Tm ( handleInteractivity\((parsed\) ) Tj
1 0 0 1 40 356 Tm ( var payload = parsed.interaction || {};) Tj
1 0 0 1 40 345 Tm ( var actionId = \((payload.actions && payload.actions[0] && payload.actions[0].action_id\) || '');)
1 0 0 1 40 334 Tm ( var callbackId = payload.callback_id || ''); Tj
1 0 0 1 40 323 Tm ( var type = payload.type || ''); Tj
1 0 0 1 40 312 Tm ( ) Tj
1 0 0 1 40 301 Tm ( if \((type === 'view_submission'\) ) Tj
1 0 0 1 40 290 Tm ( return { response_action: 'clear' };) Tj
1 0 0 1 40 279 Tm ( }) Tj
1 0 0 1 40 268 Tm ( if \((actionId || callbackId\) ) Tj
1 0 0 1 40 257 Tm ( return { response_type: 'ephemeral', text: 'Interactive action received.' };) Tj
1 0 0 1 40 246 Tm ( }) Tj
1 0 0 1 40 235 Tm ( return { response_type: 'ephemeral', text: 'Unsupported interactive payload.' };) Tj
1 0 0 1 40 224 Tm ( }) Tj
1 0 0 1 40 213 Tm ( ) Tj
1 0 0 1 40 202 Tm ( handleEventCallback\((parsed\) ) Tj
1 0 0 1 40 191 Tm ( var event = \((parsed.body && parsed.body.event\) ? parsed.body.event : {};) Tj
1 0 0 1 40 180 Tm ( if \((event.type === 'app_mention'\) ) Tj
1 0 0 1 40 169 Tm ( return { ok: true, text: 'Thanks for the mention. Try /lesson, /submit, or /progress.' };) Tj
1 0 0 1 40 158 Tm ( }) Tj
1 0 0 1 40 147 Tm ( if \((event.type === 'message' && event.channel_type === 'im'\) ) Tj
1 0 0 1 40 136 Tm ( return { ok: true, text: 'DM received. Use /lesson to get started.' };) Tj
1 0 0 1 40 125 Tm ( }) Tj
1 0 0 1 40 114 Tm ( return { ok: true, text: 'Event ignored', eventType: event.type || '' };) Tj
1 0 0 1 40 103 Tm ( }) Tj
1 0 0 1 40 92 Tm ( ) Tj
1 0 0 1 40 81 Tm ( handleWorkflowWebhook\((parsed\) ) Tj
1 0 0 1 40 70 Tm ( return { ok: true, text: 'Workflow webhook received', payloadType: parsed.payloadType };) Tj
1 0 0 1 40 59 Tm ( }) Tj
1 0 0 1 40 48 Tm ( ) Tj
ET
endstream
endobj
64 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 63 0 R
endobj
65 0 obj
<< /Length 919 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm ( _handleHelp\(\) ) Tj
1 0 0 1 40 730 Tm ( return { response_type: 'ephemeral', text: 'Commands: /lesson, /submit <lessonId>, /progress' ) Tj
1 0 0 1 40 719 Tm ( );) Tj
1 0 0 1 40 708 Tm ( }) Tj
1 0 0 1 40 697 Tm ( ) Tj
1 0 0 1 40 686 Tm ( _handleEnroll\((parsed\) ) Tj
1 0 0 1 40 675 Tm ( return this._enrollment.enrollLearner\({} Tj
1 0 0 1 40 664 Tm ( slackUserId: parsed.userId,) Tj
1 0 0 1 40 653 Tm ( courseId: parsed.params.text || '') Tj
1 0 0 1 40 642 Tm ( );) Tj
1 0 0 1 40 631 Tm ( }) Tj
1 0 0 1 40 620 Tm ( ) Tj
1 0 0 1 40 609 Tm ( _handleReport\(\) ) Tj
1 0 0 1 40 598 Tm ( var summary = this._report.buildWeeklySummary\(\);) Tj
1 0 0 1 40 587 Tm ( return { response_type: 'ephemeral', text: 'Weekly summary', blocks: this._blocks.buildAdminSu) Tj

```

```

1 0 0 1 40 576 Tm (mmary\(summary\) );) Tj
1 0 0 1 40 565 Tm ( ) Tj
1 0 0 1 40 554 Tm (}) Tj
ET
endstream
endobj
66 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 65 0 R
endobj
67 0 obj
<< /Length 3931 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: 04_SlackSecurity.gs) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (/**) Tj
1 0 0 1 40 708 Tm ( * Slack security verifier \(\(\(signing secret + replay + fallback mode\)\)\)\) Tj
1 0 0 1 40 697 Tm ( */) Tj
1 0 0 1 40 686 Tm (var SlackSecurity = {}) Tj
1 0 0 1 40 675 Tm ( verify: function\(\(parsed, config\)\) {} Tj
1 0 0 1 40 664 Tm ( if \(!parsed || !parsed.ok\) {} Tj
1 0 0 1 40 653 Tm ( return { ok: false, code: 'PARSE_FAILED', message: parsed && parsed.parseError ? parsed.pars) T
1 0 0 1 40 642 Tm (eError : 'Invalid payload' );) Tj
1 0 0 1 40 631 Tm ( }) Tj
1 0 0 1 40 620 Tm ( ) Tj
1 0 0 1 40 609 Tm ( var secret = String\(\(config && config.slackSigningSecret\) || ''\);) Tj
1 0 0 1 40 598 Tm ( var signature = String\(\(parsed.slackSignature || ''\);) Tj
1 0 0 1 40 587 Tm ( var ts = String\(\(parsed.slackTimestamp || ''\);) Tj
1 0 0 1 40 576 Tm ( ) Tj
1 0 0 1 40 565 Tm ( if \(\(secret && signature && ts && parsed.rawBody\) {} Tj
1 0 0 1 40 554 Tm ( return this._verifySignature\(\(secret, signature, ts, parsed.rawBody\) );) Tj
1 0 0 1 40 543 Tm ( }) Tj
1 0 0 1 40 532 Tm ( ) Tj
1 0 0 1 40 521 Tm ( var token = String\(\(config && config.slackVerificationToken\) || ''\);) Tj
1 0 0 1 40 510 Tm ( if \(\(token\) {} Tj
1 0 0 1 40 499 Tm ( var presented = String\(\(parsed.params && parsed.params.token\) || \(\(parsed.body && parsed.bod
1 0 0 1 40 488 Tm ( .token\) || ''\);) Tj
1 0 0 1 40 477 Tm ( if \(\(this._constantTimeEquals\(\(token, presented\)\) {} Tj
1 0 0 1 40 466 Tm ( return { ok: true, code: 'OK_FALLBACK_TOKEN', message: 'Fallback token verification passed) T
1 0 0 1 40 455 Tm ( .' );) Tj
1 0 0 1 40 444 Tm ( }) Tj
1 0 0 1 40 433 Tm ( return { ok: false, code: 'FALLBACK_TOKEN_MISMATCH', message: 'Fallback token verification f) T
1 0 0 1 40 422 Tm ( ailed.' );) Tj
1 0 0 1 40 411 Tm ( }) Tj
1 0 0 1 40 400 Tm ( ) Tj
1 0 0 1 40 389 Tm ( return { ok: false, code: 'UNVERIFIED', message: 'Slack signature headers unavailable and fall) T
1 0 0 1 40 378 Tm ( back not configured.' );) Tj
1 0 0 1 40 367 Tm ( },) Tj
1 0 0 1 40 356 Tm ( ) Tj
1 0 0 1 40 345 Tm ( _verifySignature: function\(\(secret, signature, ts, rawBody\) {} Tj
1 0 0 1 40 334 Tm ( var tsNum = Number\(\(ts\) );) Tj
1 0 0 1 40 323 Tm ( if \(\(!tsNum || Math.abs\(\(Math.floor\(\(Date.now\(\\) / 1000\) - tsNum\) > 300\) {} Tj
1 0 0 1 40 312 Tm ( return { ok: false, code: 'REPLAY_REJECTED', message: 'Timestamp outside replay window.' );) Tj
1 0 0 1 40 301 Tm ( }) Tj
1 0 0 1 40 290 Tm ( ) Tj
1 0 0 1 40 279 Tm ( var base = 'v0:' + ts + ':' + rawBody;) Tj
1 0 0 1 40 268 Tm ( var bytes = Utilities.computeHmacSha256Signature\(\(base, secret\) );) Tj
1 0 0 1 40 257 Tm ( var hex = bytes.map\(\(function\(\(b\) {} Tj
1 0 0 1 40 246 Tm ( var n = \(\(b < 0 ? b + 256 : b\) .toString\(\(16\) );) Tj
1 0 0 1 40 235 Tm ( return n.length === 1 ? '0' + n : n;) Tj
1 0 0 1 40 224 Tm ( )\)\.join\(\(''\);) Tj
1 0 0 1 40 213 Tm ( var expected = 'v0=' + hex;) Tj
1 0 0 1 40 202 Tm ( ) Tj
1 0 0 1 40 191 Tm ( if \(\(!this._constantTimeEquals\(\(expected, signature\)\) {} Tj
1 0 0 1 40 180 Tm ( return { ok: false, code: 'SIGNATURE_MISMATCH', message: 'Slack signature mismatch.' );) Tj
1 0 0 1 40 169 Tm ( }) Tj
1 0 0 1 40 158 Tm ( ) Tj
1 0 0 1 40 147 Tm ( return { ok: true, code: 'OK', message: 'Signature verified.' );) Tj
1 0 0 1 40 136 Tm ( },) Tj
1 0 0 1 40 125 Tm ( ) Tj
1 0 0 1 40 114 Tm ( _constantTimeEquals: function\(\(a, b\) {} Tj
1 0 0 1 40 103 Tm ( var aa = String\(\(a || ''\) );) Tj
1 0 0 1 40 92 Tm ( var bb = String\(\(b || ''\) );) Tj
1 0 0 1 40 81 Tm ( if \(\(!aa || !bb || aa.length !== bb.length\) return false;) Tj
1 0 0 1 40 70 Tm ( var mismatch = 0;) Tj
1 0 0 1 40 59 Tm ( for \(\(var i = 0; i < aa.length; i++\) mismatch |= \(\(aa.charCodeAt\(\(i\) ^ bb.charCodeAt\(\(i\) )\);) Tj
1 0 0 1 40 48 Tm ( return mismatch === 0;) Tj
ET
endstream

```

```

endobj
68 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 67 0 R
endobj
69 0 obj
<< /Length 67 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm ( ) Tj
1 0 0 1 40 730 Tm ({};) Tj
ET
endstream
endobj
70 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 69 0 R
endobj
71 0 obj
<< /Length 3729 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: 05_LmsEnrollmentService.gs) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (class LmsEnrollmentService {) Tj
1 0 0 1 40 708 Tm ( constructor\ (db, slackApiClient, blocks, config\ ) {) Tj
1 0 0 1 40 697 Tm ( this._db = db;) Tj
1 0 0 1 40 686 Tm ( this._slack = slackApiClient;) Tj
1 0 0 1 40 675 Tm ( this._blocks = blocks;) Tj
1 0 0 1 40 664 Tm ( this._config = config || {};) Tj
1 0 0 1 40 653 Tm ( }) Tj
1 0 0 1 40 642 Tm ( ) Tj
1 0 0 1 40 631 Tm ( enrollLearner\ (input\ ) {) Tj
1 0 0 1 40 620 Tm ( var learner = this.ensureLearnerRecord\ (input || {} \);) Tj
1 0 0 1 40 609 Tm ( var enrollment = this.ensureEnrollmentRecord\ ({) Tj
1 0 0 1 40 598 Tm ( learnerId: learner.id,) Tj
1 0 0 1 40 587 Tm ( courseId: \ (input && input.courseId\ ) || this._config.defaultCourseId,) Tj
1 0 0 1 40 576 Tm ( track: this._config.defaultTrack) Tj
1 0 0 1 40 565 Tm ( } \);) Tj
1 0 0 1 40 554 Tm ( var queued = this.queueFirstLesson\ ({ learnerId: learner.id, courseId: enrollment.courseId } \);) Tj
1 0 0 1 40 543 Tm ( this.sendWelcomeDm\ ({ slackUserId: learner.slackUserId, learnerId: learner.id, courseId: enroll)
1 0 0 1 40 532 Tm (ment.courseId } \);) Tj
1 0 0 1 40 521 Tm ( this._db.audit\ ('enroll_learner', 'enrollment', { learnerId: learner.id, enrollmentId: enrollme)
1 0 0 1 40 510 Tm ( nt.id } \);) Tj
1 0 0 1 40 499 Tm ( return { ok: true, code: 'ENROLLED', learnerId: learner.id, enrollmentId: enrollment.id, queue) T
1 0 0 1 40 488 Tm ( Id: queued.id } ;) Tj
1 0 0 1 40 477 Tm ( }) Tj
1 0 0 1 40 466 Tm ( ) Tj
1 0 0 1 40 455 Tm ( findLearnerBySlackUserId\ (slackUserId\ ) {) Tj
1 0 0 1 40 444 Tm ( if \ (!slackUserId\ ) return null;) Tj
1 0 0 1 40 433 Tm ( return this._db.table\ ('learners' \).findAll\ ( \ ).filter\ (function\ (row\ ) { return row.slackUserId
1 0 0 1 40 422 Tm ( slackUserId; } \)[0] || null;) Tj
1 0 0 1 40 411 Tm ( }) Tj
1 0 0 1 40 400 Tm ( ) Tj
1 0 0 1 40 389 Tm ( ensureLearnerRecord\ (input\ ) {) Tj
1 0 0 1 40 378 Tm ( var existing = this.findLearnerBySlackUserId\ (input.slackUserId \);) Tj
1 0 0 1 40 367 Tm ( if \ (existing\ ) return existing;) Tj
1 0 0 1 40 356 Tm ( return this._db.table\ ('learners' \).insert\ ({) Tj
1 0 0 1 40 345 Tm ( slackUserId: input.slackUserId,) Tj
1 0 0 1 40 334 Tm ( email: input.email || '',) Tj
1 0 0 1 40 323 Tm ( name: input.name || '',) Tj
1 0 0 1 40 312 Tm ( status: 'active') Tj
1 0 0 1 40 301 Tm ( } \);) Tj
1 0 0 1 40 290 Tm ( }) Tj
1 0 0 1 40 279 Tm ( ) Tj
1 0 0 1 40 268 Tm ( ensureEnrollmentRecord\ (input\ ) {) Tj
1 0 0 1 40 257 Tm ( var existing = this._db.table\ ('enrollment' \).findAll\ ( \ ).filter\ (function\ (row\ ) {) Tj
1 0 0 1 40 246 Tm ( return row.learnerId === input.learnerId && row.courseId === input.courseId;) Tj
1 0 0 1 40 235 Tm ( } \)[0];) Tj
1 0 0 1 40 224 Tm ( if \ (existing\ ) return existing;) Tj
1 0 0 1 40 213 Tm ( return this._db.table\ ('enrollment' \).insert\ ({) Tj
1 0 0 1 40 202 Tm ( learnerId: input.learnerId,) Tj
1 0 0 1 40 191 Tm ( courseId: input.courseId,) Tj
1 0 0 1 40 180 Tm ( track: input.track || 'ONBOARDING',) Tj
1 0 0 1 40 169 Tm ( status: 'active') Tj
1 0 0 1 40 158 Tm ( } \);) Tj
1 0 0 1 40 147 Tm ( }) Tj
1 0 0 1 40 136 Tm ( ) Tj
1 0 0 1 40 125 Tm ( sendWelcomeDm\ (input\ ) {) Tj
1 0 0 1 40 114 Tm ( var dm = this._slack.openDm\ (input.slackUserId \);) Tj

```

```

1 0 0 1 40 103 Tm (    if \(!dm.ok\) return dm;) Tj
1 0 0 1 40 92 Tm (    return this._slack.postMessage(dm.channelId, 'Welcome to RWR LMS', this._blocks.buildWelcomeMe) T
1 0 0 1 40 81 Tm (ssage\(\input\) \);) Tj
1 0 0 1 40 70 Tm (    }) Tj
1 0 0 1 40 59 Tm (    ) Tj
1 0 0 1 40 48 Tm (    queueFirstLesson\(\input\) \}) Tj
ET
endstream
endobj
72 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 71 0 R
endobj
73 0 obj
<< /Length 487 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (    // TODO: resolve first lesson id from runtime lesson sequencing rules.) Tj
1 0 0 1 40 730 Tm (    return this._db.table\('delivery_queue'\).insert\({} Tj
1 0 0 1 40 719 Tm (        learnerId: input.learnerId,) Tj
1 0 0 1 40 708 Tm (        lessonId: ',') Tj
1 0 0 1 40 697 Tm (        status: 'queued',) Tj
1 0 0 1 40 686 Tm (        runAt: new Date\(\).toISOString\(\)) Tj
1 0 0 1 40 675 Tm (    }) \);) Tj
1 0 0 1 40 664 Tm (    }) Tj
1 0 0 1 40 653 Tm (}) Tj
ET
endstream
endobj
74 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 73 0 R
endobj
75 0 obj
<< /Length 4342 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: 06_LmsLessonService.gs) Tj
1 0 0 1 40 730 Tm (    ) Tj
1 0 0 1 40 719 Tm (class LmsLessonService {) Tj
1 0 0 1 40 708 Tm (    constructor\(\db, slackApiClient, blocks, stateMachine, config\) \}) Tj
1 0 0 1 40 697 Tm (        this._db = db;) Tj
1 0 0 1 40 686 Tm (        this._slack = slackApiClient;) Tj
1 0 0 1 40 675 Tm (        this._blocks = blocks;) Tj
1 0 0 1 40 664 Tm (        this._state = stateMachine;) Tj
1 0 0 1 40 653 Tm (        this._config = config || {};) Tj
1 0 0 1 40 642 Tm (    }) Tj
1 0 0 1 40 631 Tm (    ) Tj
1 0 0 1 40 620 Tm (    handleLesson\(\ctx\) \}) Tj
1 0 0 1 40 609 Tm (        var current = this.getCurrentLessonForLearner\(\ctx.userId\) \);) Tj
1 0 0 1 40 598 Tm (        if \(!current.ok\) return { response_type: 'ephemeral', text: current.message };) Tj
1 0 0 1 40 587 Tm (        return {} Tj
1 0 0 1 40 576 Tm (            response_type: 'ephemeral',) Tj
1 0 0 1 40 565 Tm (            text: 'Your current lesson',) Tj
1 0 0 1 40 554 Tm (            blocks: this._blocks.buildLessonCard\(\current.lesson\) \}) Tj
1 0 0 1 40 543 Tm (        }) Tj
1 0 0 1 40 532 Tm (    }) Tj
1 0 0 1 40 521 Tm (    ) Tj
1 0 0 1 40 510 Tm (    getCurrentLessonForLearner\(\slackUserId\) \}) Tj
1 0 0 1 40 499 Tm (        var learner = this._db.table\('learners'\).findAll\(\).filter\((function\(\r\) { return r.slackUser
1 0 0 1 40 488 Tm (== slackUserId; })[0];) Tj
1 0 0 1 40 477 Tm (        if \(!learner\) return { ok: false, code: 'LEARNER_NOT_FOUND', message: 'Learner not found.' };) Tj
1 0 0 1 40 466 Tm (    }) Tj
1 0 0 1 40 455 Tm (        var progress = this._db.table\('learner_progress'\).findAll\(\).filter\((function\(\r\) {) Tj
1 0 0 1 40 444 Tm (            return r.learnerId === learner.id && r.state !== 'completed';) Tj
1 0 0 1 40 433 Tm (        })[0];) Tj
1 0 0 1 40 422 Tm (        if \(!progress\) return { ok: false, code: 'NO_ACTIVE_LESSON', message: 'No lesson assigned yet.) Tj
1 0 0 1 40 411 Tm (        ' };) Tj
1 0 0 1 40 400 Tm (    }) Tj
1 0 0 1 40 389 Tm (        var lesson = this._db.table\('lessons'\).findById\(\progress.lessonId\) || { id: progress.lessonId
1 0 0 1 40 378 Tm (        title: 'Lesson', track: ' ' };) Tj
1 0 0 1 40 367 Tm (        return { ok: true, learnerId: learner.id, lesson: lesson, progress: progress };) Tj
1 0 0 1 40 356 Tm (    }) Tj
1 0 0 1 40 345 Tm (    ) Tj
1 0 0 1 40 334 Tm (    deliverLessonToLearner\(\learnerId, lessonId\) \}) Tj
1 0 0 1 40 323 Tm (        var learner = this._db.table\('learners'\).findById\(\learnerId\) \);) Tj
1 0 0 1 40 312 Tm (        if \(!learner\) return { ok: false, code: 'LEARNER_NOT_FOUND', message: 'Learner not found' };) Tj
1 0 0 1 40 301 Tm (    }) Tj
1 0 0 1 40 290 Tm (        var dm = this._slack.openDm\(\learner.slackUserId\) \);) Tj
1 0 0 1 40 279 Tm (        if \(!dm.ok\) return dm;) Tj

```



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1 0 0 1 40 367 Tm (      activeCourse: this._config.defaultCourseId || '',) Tj
1 0 0 1 40 356 Tm (      currentLesson: active ? active.lessonId : '',) Tj
1 0 0 1 40 345 Tm (      completedCount: completed,) Tj
1 0 0 1 40 334 Tm (      overdueCount: overdue,) Tj
1 0 0 1 40 323 Tm (      nextAction: active ? 'Complete lesson ' + active.lessonId : 'Request next lesson',) Tj
1 0 0 1 40 312 Tm (      totalCount: progress.length) Tj
1 0 0 1 40 301 Tm (      );) Tj
1 0 0 1 40 290 Tm (    }) Tj
1 0 0 1 40 279 Tm (}) Tj
ET
endstream
endobj
80 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 79 0 R
endobj
81 0 obj
<< /Length 4288 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: 08_LmsCompletionService.gs) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (class LmsCompletionService {) Tj
1 0 0 1 40 708 Tm (  constructor\(\db, slackApiClient, blocks, stateMachine, config\) {) Tj
1 0 0 1 40 697 Tm (    this._db = db;) Tj
1 0 0 1 40 686 Tm (    this._slack = slackApiClient;) Tj
1 0 0 1 40 675 Tm (    this._blocks = blocks;) Tj
1 0 0 1 40 664 Tm (    this._state = stateMachine;) Tj
1 0 0 1 40 653 Tm (    this._config = config || {};) Tj
1 0 0 1 40 642 Tm (  }) Tj
1 0 0 1 40 631 Tm ( ) Tj
1 0 0 1 40 620 Tm (  handleSubmit\(\ctx\) {) Tj
1 0 0 1 40 609 Tm (    var lessonId = String\(\(\ctx.params && ctx.params.text\) || ''\).trim\(\);) Tj
1 0 0 1 40 598 Tm (    if \(!lessonId\) return { response_type: 'ephemeral', text: 'Usage: /submit <lesson_id>' };) Tj
1 0 0 1 40 587 Tm ( ) Tj
1 0 0 1 40 576 Tm (    var recorded = this.recordSubmission\({ slackUserId: ctx.userId, lessonId: lessonId, payload: c)
1 0 0 1 40 565 Tm (tx.rawBody }));) Tj
1 0 0 1 40 554 Tm (    if \(!recorded.ok\) return { response_type: 'ephemeral', text: recorded.message };) Tj
1 0 0 1 40 543 Tm ( ) Tj
1 0 0 1 40 532 Tm (    this.advanceLessonState\({ learnerProgressId: recorded.learnerProgressId, toState: 'submitted' )
1 0 0 1 40 521 Tm (});) Tj
1 0 0 1 40 510 Tm (    this.queueNextLesson\({ learnerId: recorded.learnerId, currentLessonId: lessonId }));) Tj
1 0 0 1 40 499 Tm (    return { response_type: 'ephemeral', text: 'Submission received for ' + lessonId };) Tj
1 0 0 1 40 488 Tm (  }) Tj
1 0 0 1 40 477 Tm ( ) Tj
1 0 0 1 40 466 Tm (  recordSubmission\(\input\) {) Tj
1 0 0 1 40 455 Tm (    var learner = this._db.table\('learners'\).findAll\(\).filter\(\function\(\r\) { return r.slackUser
1 0 0 1 40 444 Tm (== input.slackUserId; }\)[0];) Tj
1 0 0 1 40 433 Tm (    if \(!learner\) return { ok: false, code: 'LEARNER_NOT_FOUND', message: 'Learner not found.' };)
1 0 0 1 40 422 Tm ( ) Tj
1 0 0 1 40 411 Tm (    var submitKey = learner.id + ':' + input.lessonId;) Tj
1 0 0 1 40 400 Tm (    var existing = this._db.table\('submission_log'\).findAll\(\).filter\(\function\(\r\) { return r.su
1 0 0 1 40 389 Tm (Key === submitKey; }\)[0];) Tj
1 0 0 1 40 378 Tm (    if \(\existing\) {) Tj
1 0 0 1 40 367 Tm (      return { ok: true, code: 'DUPLICATE_SUBMISSION', learnerId: learner.id, learnerProgressId: l) T
1 0 0 1 40 356 Tm (earner.id, submissionId: existing.id, message: 'Submission already recorded.' };) Tj
1 0 0 1 40 345 Tm (    }) Tj
1 0 0 1 40 334 Tm ( ) Tj
1 0 0 1 40 323 Tm (    var row = this._db.table\('submission_log'\).insert\({) Tj
1 0 0 1 40 312 Tm (      learnerId: learner.id,) Tj
1 0 0 1 40 301 Tm (      lessonId: input.lessonId,) Tj
1 0 0 1 40 290 Tm (      submitKey: submitKey,) Tj
1 0 0 1 40 279 Tm (      payload: input.payload || '') Tj
1 0 0 1 40 268 Tm (    });) Tj
1 0 0 1 40 257 Tm ( ) Tj
1 0 0 1 40 246 Tm (    this._db.audit\('record_submission', 'submission_log', { learnerId: learner.id, lessonId: input)
1 0 0 1 40 235 Tm ( .lessonId, submissionId: row.id }));) Tj
1 0 0 1 40 224 Tm (    return { ok: true, code: 'SUBMISSION_RECORDED', learnerId: learner.id, learnerProgressId: lear) T
1 0 0 1 40 213 Tm (ner.id, submissionId: row.id };) Tj
1 0 0 1 40 202 Tm (  }) Tj
1 0 0 1 40 191 Tm ( ) Tj
1 0 0 1 40 180 Tm (  advanceLessonState\(\input\) {) Tj
1 0 0 1 40 169 Tm (    var progress = this._db.table\('learner_progress'\).findById\(\input.learnerProgressId\);) Tj
1 0 0 1 40 158 Tm (    if \(!progress\) return { ok: false, code: 'PROGRESS_NOT_FOUND', message: 'Progress not found.' )
1 0 0 1 40 147 Tm ( ) Tj
1 0 0 1 40 136 Tm (    var transition = this._state.transition\(\progress, input.toState, { source: 'submission' }));) Tj
1 0 0 1 40 125 Tm (    if \(!transition.ok\) return transition;) Tj
1 0 0 1 40 114 Tm (    this._db.table\('learner_progress'\).update\(\progress.id, { state: transition.record.state }));)
1 0 0 1 40 103 Tm (    return { ok: true, code: 'STATE_UPDATED', recordId: progress.id };) Tj
1 0 0 1 40 92 Tm (  }) Tj

```

```

1 0 0 1 40 81 Tm ( ) Tj
1 0 0 1 40 70 Tm ( queueNextLesson\(\input\) {) Tj
1 0 0 1 40 59 Tm ( // TODO: determine next lesson ordering from course track.) Tj
1 0 0 1 40 48 Tm ( var row = this._db.table\('delivery_queue'\).insert\({) Tj
ET
endstream
endobj
82 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 81 0 R
endobj
83 0 obj
<< /Length 396 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm ( learnerId: input.learnerId,) Tj
1 0 0 1 40 730 Tm ( lessonId: ',) Tj
1 0 0 1 40 719 Tm ( status: 'queued',) Tj
1 0 0 1 40 708 Tm ( runAt: new Date\(\).toISOString\(\)) Tj
1 0 0 1 40 697 Tm ( }\\);) Tj
1 0 0 1 40 686 Tm ( return { ok: true, code: 'NEXT_QUEUED', queueId: row.id };) Tj
1 0 0 1 40 675 Tm ( }) Tj
1 0 0 1 40 664 Tm (}) Tj
ET
endstream
endobj
84 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 83 0 R
endobj
85 0 obj
<< /Length 2218 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: 09_LmsReminderService.gs) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (class LmsReminderService {) Tj
1 0 0 1 40 708 Tm ( constructor\(\db, slackApiClient, blocks, config\) {) Tj
1 0 0 1 40 697 Tm ( this._db = db;) Tj
1 0 0 1 40 686 Tm ( this._slack = slackApiClient;) Tj
1 0 0 1 40 675 Tm ( this._blocks = blocks;) Tj
1 0 0 1 40 664 Tm ( this._config = config || {};) Tj
1 0 0 1 40 653 Tm ( }) Tj
1 0 0 1 40 642 Tm ( ) Tj
1 0 0 1 40 631 Tm ( sendOverdueReminders\(\) {) Tj
1 0 0 1 40 620 Tm ( var overdue = this._db.table\('learner_progress'\).findAll\(\).filter\((function\((row) { return r
1 0 0 1 40 609 Tm (tate === 'overdue'; }\\);) Tj
1 0 0 1 40 598 Tm ( var results = [];) Tj
1 0 0 1 40 587 Tm ( for \((var i = 0; i < overdue.length; i++) {) Tj
1 0 0 1 40 576 Tm ( results.push\((this.sendReminderForLearner\((overdue[i].learnerId\\));) Tj
1 0 0 1 40 565 Tm ( }) Tj
1 0 0 1 40 554 Tm ( return { ok: true, total: overdue.length, results: results };) Tj
1 0 0 1 40 543 Tm ( }) Tj
1 0 0 1 40 532 Tm ( ) Tj
1 0 0 1 40 521 Tm ( sendReminderForLearner\((learnerId\) {) Tj
1 0 0 1 40 510 Tm ( var learner = this._db.table\('learners'\).findById\((learnerId\\);) Tj
1 0 0 1 40 499 Tm ( if \(!learner\) return { ok: false, code: 'LEARNER_NOT_FOUND', learnerId: learnerId };) Tj
1 0 0 1 40 488 Tm ( var dm = this._slack.openDm\((learner.slackUserId\\);) Tj
1 0 0 1 40 477 Tm ( if \(!dm.ok\) return dm;) Tj
1 0 0 1 40 466 Tm ( return this._slack.postMessage\((dm.channelId, 'Lesson reminder', this._blocks.buildReminder\(({ l
1 0 0 1 40 455 Tm (earnerId: learnerId }\\));) Tj
1 0 0 1 40 444 Tm ( }) Tj
1 0 0 1 40 433 Tm ( ) Tj
1 0 0 1 40 422 Tm ( escalateStalledLearner\((learnerId\) {) Tj
1 0 0 1 40 411 Tm ( return {}) Tj
1 0 0 1 40 400 Tm ( ok: true,) Tj
1 0 0 1 40 389 Tm ( learnerId: learnerId,) Tj
1 0 0 1 40 378 Tm ( escalated: !!this._config.opsAlertChannel,) Tj
1 0 0 1 40 367 Tm ( message: this._config.opsAlertChannel ? 'Escalation target configured.' : 'No ops channel co) T
1 0 0 1 40 356 Tm (nfigured.') Tj
1 0 0 1 40 345 Tm ( }\\);) Tj
1 0 0 1 40 334 Tm ( }) Tj
1 0 0 1 40 323 Tm (}) Tj
ET
endstream
endobj
86 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 85 0 R
endobj
87 0 obj

```

```

<< /Length 1932 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: 10_LmsReportService.gs) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (class LmsReportService {}) Tj
1 0 0 1 40 708 Tm ( constructor\ (db, config\ ) {} ) Tj
1 0 0 1 40 697 Tm ( this._db = db;) Tj
1 0 0 1 40 686 Tm ( this._config = config || {};) Tj
1 0 0 1 40 675 Tm ( ) Tj
1 0 0 1 40 664 Tm ( ) Tj
1 0 0 1 40 653 Tm ( buildWeeklySummary\ ( ) {} ) Tj
1 0 0 1 40 642 Tm ( var learners = this._db.table\('learners'\).findAll\ ( ).length;) Tj
1 0 0 1 40 631 Tm ( var progress = this._db.table\('learner_progress'\).findAll\ ( );) Tj
1 0 0 1 40 620 Tm ( return {} ) Tj
1 0 0 1 40 609 Tm ( ok: true,) Tj
1 0 0 1 40 598 Tm ( learners: learners,) Tj
1 0 0 1 40 587 Tm ( completed: progress.filter\ (function\ (r\ ) { return r.state === 'completed'; } ).length,) Tj
1 0 0 1 40 576 Tm ( submitted: progress.filter\ (function\ (r\ ) { return r.state === 'submitted'; } ).length) Tj
1 0 0 1 40 565 Tm ( );) Tj
1 0 0 1 40 554 Tm ( ) Tj
1 0 0 1 40 543 Tm ( ) Tj
1 0 0 1 40 532 Tm ( buildCohortSummary\ (courseId\ ) {} ) Tj
1 0 0 1 40 521 Tm ( var rows = this._db.table\('enrollment'\).findAll\ ( ).filter\ (function\ (r\ ) { return r.courseId =
1 0 0 1 40 510 Tm (courseId; } );) Tj
1 0 0 1 40 499 Tm ( return { ok: true, courseId: courseId, enrollmentCount: rows.length, activeCount: rows.filter\ (
1 0 0 1 40 488 Tm (function\ (r\ ) { return r.status === 'active'; } ).length };) Tj
1 0 0 1 40 477 Tm ( ) Tj
1 0 0 1 40 466 Tm ( ) Tj
1 0 0 1 40 455 Tm ( buildOverdueSummary\ ( ) {} ) Tj
1 0 0 1 40 444 Tm ( var rows = this._db.table\('learner_progress'\).findAll\ ( ).filter\ (function\ (r\ ) { return r.stat
1 0 0 1 40 433 Tm ( = 'overdue'; } );) Tj
1 0 0 1 40 422 Tm ( return { ok: true, overdueCount: rows.length, learnerIds: rows.map\ (function\ (r\ ) { return r.lear
1 0 0 1 40 411 Tm (nerId; } );) Tj
1 0 0 1 40 400 Tm ( ) Tj
1 0 0 1 40 389 Tm ( ) Tj
ET
endstream
endobj
88 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [ 0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 87 0 R
endobj
89 0 obj
<< /Length 3886 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: 11_SlackApiClient.gs) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (class SlackApiClient {}) Tj
1 0 0 1 40 708 Tm ( constructor\ (config, retryResolver\ ) {} ) Tj
1 0 0 1 40 697 Tm ( this._config = config || {};) Tj
1 0 0 1 40 686 Tm ( this._retry = retryResolver || null;) Tj
1 0 0 1 40 675 Tm ( this._token = String\ (this._config.slackBotToken || ''\ );) Tj
1 0 0 1 40 664 Tm ( ) Tj
1 0 0 1 40 653 Tm ( ) Tj
1 0 0 1 40 642 Tm ( postMessage\ (channel, text, blocks\ ) { return this._call\ ('chat.postMessage', { channel: channel, t
1 0 0 1 40 631 Tm (ext: text, blocks: blocks || [] } );) Tj
1 0 0 1 40 620 Tm ( postEphemeral\ (channel, user, text, blocks\ ) { return this._call\ ('chat.postEphemeral', { channel:
1 0 0 1 40 609 Tm (channel, user: user, text: text, blocks: blocks || [] } );) Tj
1 0 0 1 40 598 Tm ( openView\ (triggerId, view\ ) { return this._call\ ('views.open', { trigger_id: triggerId, view: view
1 0 0 1 40 587 Tm ( ); } );) Tj
1 0 0 1 40 576 Tm ( updateMessage\ (channel, ts, text, blocks\ ) { return this._call\ ('chat.update', { channel: channel,
1 0 0 1 40 565 Tm (ts: ts, text: text, blocks: blocks || [] } );) Tj
1 0 0 1 40 554 Tm ( openDm\ (userId\ ) {} ) Tj
1 0 0 1 40 543 Tm ( var res = this._call\ ('conversations.open', { users: userId } );) Tj
1 0 0 1 40 532 Tm ( return { ok: !!res.ok, code: res.code, message: res.message, channelId: res.data && res.data.c) T
1 0 0 1 40 521 Tm (channel ? res.data.channel.id : '' } );) Tj
1 0 0 1 40 510 Tm ( ) Tj
1 0 0 1 40 499 Tm ( ) Tj
1 0 0 1 40 488 Tm ( _call\ (method, payload\ ) {} ) Tj
1 0 0 1 40 477 Tm ( if \ (!this._token\ ) {} ) Tj
1 0 0 1 40 466 Tm ( return { ok: false, code: 'MISSING_BOT_TOKEN', message: 'SLACK_BOT_TOKEN missing', retryable) T
1 0 0 1 40 455 Tm (: false } );) Tj
1 0 0 1 40 444 Tm ( ) Tj
1 0 0 1 40 433 Tm ( ) Tj
1 0 0 1 40 422 Tm ( var response;) Tj
1 0 0 1 40 411 Tm ( try {} ) Tj
1 0 0 1 40 400 Tm ( response = UrlFetchApp.fetch\ ('https://slack.com/api/' + method, {} ) Tj

```



```

1 0 0 1 40 389 Tm (      method: 'post',) Tj
1 0 0 1 40 378 Tm (      contentType: 'application/json; charset=utf-8',) Tj
1 0 0 1 40 367 Tm (      headers: { Authorization: 'Bearer ' + this._token },) Tj
1 0 0 1 40 356 Tm (      payload: JSON.stringify\(payload || {}\) ,) Tj
1 0 0 1 40 345 Tm (      muteHttpExceptions: true) Tj
1 0 0 1 40 334 Tm (      }\);} Tj
1 0 0 1 40 323 Tm (      } catch \(err\) {) Tj
1 0 0 1 40 312 Tm (      var fetchErr = { ok: false, code: 'HTTP_FETCH_FAILED', message: String\(err.message || err\) , )
1 0 0 1 40 301 Tm (retryable: true, method: method ;);} Tj
1 0 0 1 40 290 Tm (      this._queueRetry\(method, payload, fetchErr\);} Tj
1 0 0 1 40 279 Tm (      return fetchErr;) Tj
1 0 0 1 40 268 Tm (      }) Tj
1 0 0 1 40 257 Tm ( ) Tj
1 0 0 1 40 246 Tm (      var status = response.getResponseCode\(\);} Tj
1 0 0 1 40 235 Tm (      var body = {};) Tj
1 0 0 1 40 224 Tm (      try { body = JSON.parse\(response.getContentText\(\) || '{}\) \); } catch \(e\) { body = { ok: fals
1 0 0 1 40 213 Tm (error: 'invalid_json' }; }) Tj
1 0 0 1 40 202 Tm ( ) Tj
1 0 0 1 40 191 Tm (      if \(status < 200 || status >= 300 || !body.ok\) {) Tj
1 0 0 1 40 180 Tm (      var errShape = {} ) Tj
1 0 0 1 40 169 Tm (      ok: false,) Tj
1 0 0 1 40 158 Tm (      code: body.error || 'SLACK_API_ERROR',) Tj
1 0 0 1 40 147 Tm (      message: 'Slack API call failed for ' + method,) Tj
1 0 0 1 40 136 Tm (      retryable: status >= 500 || body.error === 'ratelimited',) Tj
1 0 0 1 40 125 Tm (      method: method,) Tj
1 0 0 1 40 114 Tm (      status: status,) Tj
1 0 0 1 40 103 Tm (      data: body) Tj
1 0 0 1 40 92 Tm (      });} Tj
1 0 0 1 40 81 Tm (      this._queueRetry\(method, payload, errShape\);} Tj
1 0 0 1 40 70 Tm (      return errShape;) Tj
1 0 0 1 40 59 Tm (      }) Tj
1 0 0 1 40 48 Tm ( ) Tj
ET
endstream
endobj
90 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 89 0 R
endobj
91 0 obj
<< /Length 605 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (      return { ok: true, code: 'OK', message: 'success', data: body };) Tj
1 0 0 1 40 730 Tm (      }) Tj
1 0 0 1 40 719 Tm ( ) Tj
1 0 0 1 40 708 Tm (      _queueRetry\(method, payload, errShape\) {) Tj
1 0 0 1 40 697 Tm (      if \(!this._retry || !errShape.retryable\) return;) Tj
1 0 0 1 40 686 Tm (      this._retry.scheduleRetry\({) Tj
1 0 0 1 40 675 Tm (      jobType: 'slack_api:' + method,) Tj
1 0 0 1 40 664 Tm (      payload: payload,) Tj
1 0 0 1 40 653 Tm (      lastError: errShape.code) Tj
1 0 0 1 40 642 Tm (      }\);} Tj
1 0 0 1 40 631 Tm (      }) Tj
1 0 0 1 40 620 Tm (}) Tj
ET
endstream
endobj
92 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 91 0 R
endobj
93 0 obj
<< /Length 2616 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: 12_SlackBlockKitBuilder.gs) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (class SlackBlockKitBuilder {) Tj
1 0 0 1 40 708 Tm (  buildWelcomeMessage\(input\) {) Tj
1 0 0 1 40 697 Tm (      return [{ type: 'section', text: { type: 'mrkdwn', text: '**Welcome to RWR LMS*\nCourse: ' + \(i
1 0 0 1 40 686 Tm (nput.courseId || ''\) } }];) Tj
1 0 0 1 40 675 Tm (      }) Tj
1 0 0 1 40 664 Tm ( ) Tj
1 0 0 1 40 653 Tm (  buildLessonCard\(input\) {) Tj
1 0 0 1 40 642 Tm (      return [{ type: 'section', text: { type: 'mrkdwn', text: '**Lesson:* ' + \(input.title || input.)
1 0 0 1 40 631 Tm (lessonId || ''\) } }];) Tj
1 0 0 1 40 620 Tm (      }) Tj
1 0 0 1 40 609 Tm ( ) Tj
1 0 0 1 40 598 Tm (  buildProgressSummary\(input\) {) Tj

```



```
1 0 0 1 40 323 Tm (    };) Tj
1 0 0 1 40 312 Tm (    }) Tj
1 0 0 1 40 301 Tm ( ) Tj
1 0 0 1 40 290 Tm (    var next = {};) Tj
1 0 0 1 40 279 Tm (    Object.keys\source\).forEach(function(k) { next[k] = source[k]; }\\;) Tj
1 0 0 1 40 268 Tm (    next.state = toState;) Tj
1 0 0 1 40 257 Tm (    next.updatedAt = new Date\\).toISOString\\;) Tj
1 0 0 1 40 246 Tm ( ) Tj
1 0 0 1 40 235 Tm (    return { ok: true, code: 'TRANSITION_OK', message: 'State updated', record: next, meta: meta | } Tj
1 0 0 1 40 224 Tm (| {} };) Tj
1 0 0 1 40 213 Tm (    }) Tj
1 0 0 1 40 202 Tm (}) Tj
ET
endstream
endobj
96 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 95 0 R
endobj
97 0 obj
<< /Length 2821 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: 14_Scheduler.gs) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (function setupTriggers\\) {} Tj
1 0 0 1 40 708 Tm (    removeTriggers\\;) Tj
1 0 0 1 40 697 Tm (    ScriptApp.newTrigger('runDailyLessonDelivery').timeBased\\).everyDays\\(1\\).atHour\\(8\\).create\\(\\)
1 0 0 1 40 686 Tm (    ScriptApp.newTrigger('runHourlyReminderCheck').timeBased\\(\\).everyHours\\(1\\).create\\(\\);) Tj
1 0 0 1 40 675 Tm (    ScriptApp.newTrigger('runWeeklyAdminReport').timeBased\\(\\).onWeekDay\\(ScriptApp.WeekDay.MONDAY\\).
1 0 0 1 40 664 Tm (our\\(9\\).create\\(\\);) Tj
1 0 0 1 40 653 Tm (    return { ok: true, code: 'TRIGGERS_CREATED' };) Tj
1 0 0 1 40 642 Tm (}) Tj
1 0 0 1 40 631 Tm ( ) Tj
1 0 0 1 40 620 Tm (function removeTriggers\\) {} Tj
1 0 0 1 40 609 Tm (    ScriptApp.getProjectTriggers\\).forEach(function(t) { ScriptApp.deleteTrigger\\(t); }\\;) Tj
1 0 0 1 40 598 Tm (    return { ok: true, code: 'TRIGGERS_REMOVED' };) Tj
1 0 0 1 40 587 Tm (}) Tj
1 0 0 1 40 576 Tm ( ) Tj
1 0 0 1 40 565 Tm (function runDailyLessonDelivery\\) {} Tj
1 0 0 1 40 554 Tm (    var lock = LockService.getScriptLock\\;) Tj
1 0 0 1 40 543 Tm (    lock.waitForLock\\(30000\\);) Tj
1 0 0 1 40 532 Tm (    try {} Tj
1 0 0 1 40 521 Tm (        var deps = createHostDependencies\\(ConfigBootstrap.load\\(\\)\\);) Tj
1 0 0 1 40 510 Tm (        return deps.lessonService.deliverPendingLessons\\(\\);) Tj
1 0 0 1 40 499 Tm (    } finally {} Tj
1 0 0 1 40 488 Tm (    lock.releaseLock\\(\\);) Tj
1 0 0 1 40 477 Tm (    }) Tj
1 0 0 1 40 466 Tm (}) Tj
1 0 0 1 40 455 Tm ( ) Tj
1 0 0 1 40 444 Tm (function runHourlyReminderCheck\\) {} Tj
1 0 0 1 40 433 Tm (    var lock = LockService.getScriptLock\\;) Tj
1 0 0 1 40 422 Tm (    lock.waitForLock\\(30000\\);) Tj
1 0 0 1 40 411 Tm (    try {} Tj
1 0 0 1 40 400 Tm (        var deps = createHostDependencies\\(ConfigBootstrap.load\\(\\)\\);) Tj
1 0 0 1 40 389 Tm (        return deps.reminderService.sendOverdueReminders\\(\\);) Tj
1 0 0 1 40 378 Tm (    } finally {} Tj
1 0 0 1 40 367 Tm (    lock.releaseLock\\(\\);) Tj
1 0 0 1 40 356 Tm (    }) Tj
1 0 0 1 40 345 Tm (}) Tj
1 0 0 1 40 334 Tm ( ) Tj
1 0 0 1 40 323 Tm (function runWeeklyAdminReport\\) {} Tj
1 0 0 1 40 312 Tm (    var lock = LockService.getScriptLock\\;) Tj
1 0 0 1 40 301 Tm (    lock.waitForLock\\(30000\\);) Tj
1 0 0 1 40 290 Tm (    try {} Tj
1 0 0 1 40 279 Tm (        var deps = createHostDependencies\\(ConfigBootstrap.load\\(\\)\\);) Tj
1 0 0 1 40 268 Tm (        return deps.reportService.buildWeeklySummary\\(\\);) Tj
1 0 0 1 40 257 Tm (    } finally {} Tj
1 0 0 1 40 246 Tm (    lock.releaseLock\\(\\);) Tj
1 0 0 1 40 235 Tm (    }) Tj
1 0 0 1 40 224 Tm (}) Tj
1 0 0 1 40 213 Tm ( ) Tj
1 0 0 1 40 202 Tm (function runHealthCheck\\) {} Tj
1 0 0 1 40 191 Tm (    return createHostDependencies\\(ConfigBootstrap.load\\(\\)\\).healthMonitor.getSnapshot\\(\\);) Tj
1 0 0 1 40 180 Tm (}) Tj
ET
endstream
endobj
98 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 97 0 R
```

```

endobj
99 0 obj
<< /Length 1718 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: 15_SheetsDataSync.gs) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (function syncApprovedLessonsToRuntime\(\) {} ) Tj
1 0 0 1 40 708 Tm ( // TODO: map authored lesson source rows once source schema is finalized.) Tj
1 0 0 1 40 697 Tm ( return { ok: true, upserted: 0, retired: deactivateRetiredLessons\(\).retired };) Tj
1 0 0 1 40 686 Tm (}) Tj
1 0 0 1 40 675 Tm ( ) Tj
1 0 0 1 40 664 Tm (function upsertLessonRuntimeRecord\(\row\) {} ) Tj
1 0 0 1 40 653 Tm ( var db = createHostDbClient\(\ConfigBootstrap.load\(\)\);) Tj
1 0 0 1 40 642 Tm ( var table = db.table\('lessons'\);) Tj
1 0 0 1 40 631 Tm ( var existing = row && row.id ? table.findById\(\row.id\) : null;) Tj
1 0 0 1 40 620 Tm ( ) Tj
1 0 0 1 40 609 Tm ( if \(\existing\) {} ) Tj
1 0 0 1 40 598 Tm ( var updated = table.update\(\existing.id, row\);) Tj
1 0 0 1 40 587 Tm ( return { ok: true, code: 'UPDATED', lessonId: updated.id };) Tj
1 0 0 1 40 576 Tm ( }) Tj
1 0 0 1 40 565 Tm ( ) Tj
1 0 0 1 40 554 Tm ( var inserted = table.insert\(\{) Tj
1 0 0 1 40 543 Tm ( id: row.id || undefined,) Tj
1 0 0 1 40 532 Tm ( courseId: row.courseId || '',) Tj
1 0 0 1 40 521 Tm ( track: row.track || '',) Tj
1 0 0 1 40 510 Tm ( title: row.title || '',) Tj
1 0 0 1 40 499 Tm ( contentRef: row.contentRef || '',) Tj
1 0 0 1 40 488 Tm ( active: String\(\row.active == null ? 'true' : row.active\) ) Tj
1 0 0 1 40 477 Tm ( } \);) Tj
1 0 0 1 40 466 Tm ( return { ok: true, code: 'INSERTED', lessonId: inserted.id };) Tj
1 0 0 1 40 455 Tm (}) Tj
1 0 0 1 40 444 Tm ( ) Tj
1 0 0 1 40 433 Tm (function deactivateRetiredLessons\(\) {} ) Tj
1 0 0 1 40 422 Tm ( return { ok: true, retired: 0 };) Tj
1 0 0 1 40 411 Tm (}) Tj
ET
endstream
endobj
100 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 99 0 R
endobj
101 0 obj
<< /Length 2121 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: 16_ConfigBootstrap.gs) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (var ConfigBootstrap = {} ) Tj
1 0 0 1 40 708 Tm ( load: function\(\) {} ) Tj
1 0 0 1 40 697 Tm ( var helper = new ScriptPropertiesHelper\(\);) Tj
1 0 0 1 40 686 Tm ( return {} ) Tj
1 0 0 1 40 675 Tm ( slackBotToken: helper.getRequired\('SLACK_BOT_TOKEN'\),) Tj
1 0 0 1 40 664 Tm ( slackSigningSecret: helper.getRequired\('SLACK_SIGNING_SECRET'\),) Tj
1 0 0 1 40 653 Tm ( spreadsheetId: helper.get\('SHEET_DB_SPREADSHEET_ID', helper.getRequired\('SPREADSHEET_ID'\)\),) Tj
1 0 0 1 40 642 Tm ( slackVerificationToken: helper.get\('SLACK_VERIFICATION_TOKEN', '\'),) Tj
1 0 0 1 40 631 Tm ( adminUserIds: this._csv\(\helper.get\('ADMIN_USER_IDS', '\'),\) ) Tj
1 0 0 1 40 620 Tm ( defaultCourseId: helper.get\('DEFAULT_COURSE_ID', 'C001'\),) Tj
1 0 0 1 40 609 Tm ( defaultTrack: helper.get\('DEFAULT_TRACK', 'ONBOARDING'\),) Tj
1 0 0 1 40 598 Tm ( opsAlertChannel: helper.get\('OPS_ALERT_CHANNEL', '\') ) Tj
1 0 0 1 40 587 Tm ( );) Tj
1 0 0 1 40 576 Tm ( },) Tj
1 0 0 1 40 565 Tm ( ) Tj
1 0 0 1 40 554 Tm ( validate: function\(\) {} ) Tj
1 0 0 1 40 543 Tm ( try {} ) Tj
1 0 0 1 40 532 Tm ( this.load\(\);) Tj
1 0 0 1 40 521 Tm ( return { ok: true, code: 'OK', message: 'Configuration valid.' };) Tj
1 0 0 1 40 510 Tm ( } catch \(\err\) {} ) Tj
1 0 0 1 40 499 Tm ( return { ok: false, code: 'CONFIG_INVALID', message: String\(\err.message || err\) };) Tj
1 0 0 1 40 488 Tm ( }) Tj
1 0 0 1 40 477 Tm ( },) Tj
1 0 0 1 40 466 Tm ( ) Tj
1 0 0 1 40 455 Tm ( get: function\(\key, fallback\) {} ) Tj
1 0 0 1 40 444 Tm ( var cfg = this.load\(\);) Tj
1 0 0 1 40 433 Tm ( return cfg.hasOwnProperty\(\key\) ? cfg[key] : fallback;) Tj
1 0 0 1 40 422 Tm ( },) Tj
1 0 0 1 40 411 Tm ( ) Tj
1 0 0 1 40 400 Tm ( _csv: function\(\text\) {} ) Tj

```

```

1 0 0 1 40 389 Tm (    return String\(\text || ''\).split\(',\').map\(function\(\v\) { return v.trim\(\); }\).filter\(func
1 0 0 1 40 378 Tm ({ return !!v; }\);) Tj
1 0 0 1 40 367 Tm (    }) Tj
1 0 0 1 40 356 Tm (};) Tj
ET
endstream
endobj
102 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 101 0
endobj
103 0 obj
<< /Length 2506 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: 17_HealthMonitor.gs) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (class HealthMonitor {) Tj
1 0 0 1 40 708 Tm (    constructor\(\db, config\) {) Tj
1 0 0 1 40 697 Tm (        this._db = db;) Tj
1 0 0 1 40 686 Tm (        this._config = config || {};) Tj
1 0 0 1 40 675 Tm (    }) Tj
1 0 0 1 40 664 Tm ( ) Tj
1 0 0 1 40 653 Tm (    recordJobStatus\(\jobName, status, metadata\) {) Tj
1 0 0 1 40 642 Tm (        var row = this._db.table\('audit_log'\).insert\({) Tj
1 0 0 1 40 631 Tm (            actor: 'scheduler',) Tj
1 0 0 1 40 620 Tm (            action: jobName,) Tj
1 0 0 1 40 609 Tm (            resourceType: 'job',) Tj
1 0 0 1 40 598 Tm (            resourceId: '',) Tj
1 0 0 1 40 587 Tm (            status: status,) Tj
1 0 0 1 40 576 Tm (            message: jobName + ':' + status,) Tj
1 0 0 1 40 565 Tm (            metadata: JSON.stringify\(\metadata || {}\) ) Tj
1 0 0 1 40 554 Tm (        }\);) Tj
1 0 0 1 40 543 Tm (        return { ok: true, auditId: row.id };) Tj
1 0 0 1 40 532 Tm (    }) Tj
1 0 0 1 40 521 Tm ( ) Tj
1 0 0 1 40 510 Tm (    recordSlackApiFailure\(\errorShape\) {) Tj
1 0 0 1 40 499 Tm (        return this.recordJobStatus\('slack_api_failure', 'error', errorShape || {}\) );) Tj
1 0 0 1 40 488 Tm (    }) Tj
1 0 0 1 40 477 Tm ( ) Tj
1 0 0 1 40 466 Tm (    getQueueDepthReport\(\) {) Tj
1 0 0 1 40 455 Tm (        return {) Tj
1 0 0 1 40 444 Tm (            ok: true,) Tj
1 0 0 1 40 433 Tm (            deliveryQueue: this._db.table\('delivery_queue'\).findAll\(\).length,) Tj
1 0 0 1 40 422 Tm (            retryQueue: this._db.table\('retry_queue'\).findAll\(\).length) Tj
1 0 0 1 40 411 Tm (        );) Tj
1 0 0 1 40 400 Tm (    }) Tj
1 0 0 1 40 389 Tm ( ) Tj
1 0 0 1 40 378 Tm (    getOverdueMetrics\(\) {) Tj
1 0 0 1 40 367 Tm (        var overdue = this._db.table\('learner_progress'\).findAll\(\).filter\(function\(\r\) { return r.s
1 0 0 1 40 356 Tm (        === 'overdue'; }\);) Tj
1 0 0 1 40 345 Tm (        return { ok: true, overdueCount: overdue.length };) Tj
1 0 0 1 40 334 Tm (    }) Tj
1 0 0 1 40 323 Tm ( ) Tj
1 0 0 1 40 312 Tm (    getSnapshot\(\) {) Tj
1 0 0 1 40 301 Tm (        return {) Tj
1 0 0 1 40 290 Tm (            ok: true,) Tj
1 0 0 1 40 279 Tm (            timestamp: new Date\(\).toISOString\(\),) Tj
1 0 0 1 40 268 Tm (            config: ConfigBootstrap.validate\(\),) Tj
1 0 0 1 40 257 Tm (            sheetDb: SheetDb.healthCheck\(\),) Tj
1 0 0 1 40 246 Tm (            queues: this.getQueueDepthReport\(\),) Tj
1 0 0 1 40 235 Tm (            overdue: this.getOverdueMetrics\(\)) Tj
1 0 0 1 40 224 Tm (        );) Tj
1 0 0 1 40 213 Tm (    }) Tj
1 0 0 1 40 202 Tm (}) Tj
ET
endstream
endobj
104 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 103 0
endobj
105 0 obj
<< /Length 2584 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: 18_RetryResolver.gs) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (class RetryResolver {) Tj
1 0 0 1 40 708 Tm (    constructor\(\db, config\) {) Tj

```

```

1 0 0 1 40 697 Tm ( this._db = db;) Tj
1 0 0 1 40 686 Tm ( this._config = config || {};) Tj
1 0 0 1 40 675 Tm ( this._maxAttempts = 5;) Tj
1 0 0 1 40 664 Tm ( }) Tj
1 0 0 1 40 653 Tm ( ) Tj
1 0 0 1 40 642 Tm ( isRetryable\(\errorShape\) ) Tj
1 0 0 1 40 631 Tm ( if \(!errorShape\) return false;) Tj
1 0 0 1 40 620 Tm ( if \(\errorShape.retryable\) return true;) Tj
1 0 0 1 40 609 Tm ( return ['HTTP_FETCH_FAILED', 'ratelimited', 'internal_error'].indexOf\(\String\(\errorShape.code |\)
1 0 0 1 40 598 Tm ( '\)\)\) !== -1;) Tj
1 0 0 1 40 587 Tm ( }) Tj
1 0 0 1 40 576 Tm ( ) Tj
1 0 0 1 40 565 Tm ( scheduleRetry\(\job\) ) Tj
1 0 0 1 40 554 Tm ( var payload = job || {};) Tj
1 0 0 1 40 543 Tm ( return this._db.table\('retry_queue'\).insert\(\{) Tj
1 0 0 1 40 532 Tm ( jobType: payload.jobType || 'unknown',) Tj
1 0 0 1 40 521 Tm ( payload: JSON.stringify\(\payload.payload || {}\) ,) Tj
1 0 0 1 40 510 Tm ( attempts: String\(\payload.attempts || 0\) ,) Tj
1 0 0 1 40 499 Tm ( nextRunAt: payload.nextRunAt || new Date\(\\) .toISOString\(\\) ,) Tj
1 0 0 1 40 488 Tm ( status: 'queued',) Tj
1 0 0 1 40 477 Tm ( lastError: payload.lastError || '') Tj
1 0 0 1 40 466 Tm ( }\)\);) Tj
1 0 0 1 40 455 Tm ( }) Tj
1 0 0 1 40 444 Tm ( ) Tj
1 0 0 1 40 433 Tm ( resolveDueRetries\(\limit\) ) Tj
1 0 0 1 40 422 Tm ( var max = Number\(\limit || 20\) ;) Tj
1 0 0 1 40 411 Tm ( var due = this._db.table\('retry_queue'\).findAll\(\\) .filter\(\function\(\r\) { return r.status ===
1 0 0 1 40 400 Tm ( eued'; }\)\).slice\(\0, max\) ;) Tj
1 0 0 1 40 389 Tm ( return { ok: true, dueCount: due.length, rows: due } ;) Tj
1 0 0 1 40 378 Tm ( }) Tj
1 0 0 1 40 367 Tm ( ) Tj
1 0 0 1 40 356 Tm ( markAttempt\(\rowId, attempts, lastError\) ) Tj
1 0 0 1 40 345 Tm ( var nextAttempts = Number\(\attempts || 0\) + 1;) Tj
1 0 0 1 40 334 Tm ( var dead = nextAttempts >= this._maxAttempts;) Tj
1 0 0 1 40 323 Tm ( return this._db.table\('retry_queue'\).update\(\rowId, {) Tj
1 0 0 1 40 312 Tm ( attempts: String\(\nextAttempts\) ,) Tj
1 0 0 1 40 301 Tm ( status: dead ? 'dead_letter' : 'queued',) Tj
1 0 0 1 40 290 Tm ( lastError: lastError || ' ,) Tj
1 0 0 1 40 279 Tm ( nextRunAt: new Date\(\Date.now\(\\) + Math.pow\(\2, nextAttempts\) * 60000\) .toISOString\(\\) ) Tj
1 0 0 1 40 268 Tm ( }\)\);) Tj
1 0 0 1 40 257 Tm ( }) Tj
1 0 0 1 40 246 Tm (}) Tj
ET
endstream
endobj
106 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 105 0
endobj
107 0 obj
<< /Length 3400 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: 19_HostTestHarness.gs) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (function hostTest_fakeSlashLesson\(\\) ) Tj
1 0 0 1 40 708 Tm ( return doPost\(_fakeSlash\('/lesson', '\)\)\).getContent\(\\) ;) Tj
1 0 0 1 40 697 Tm (}) Tj
1 0 0 1 40 686 Tm ( ) Tj
1 0 0 1 40 675 Tm (function hostTest_fakeSlashSubmit\(\\) ) Tj
1 0 0 1 40 664 Tm ( return doPost\(_fakeSlash\('/submit', 'L001'\)\)\).getContent\(\\) ;) Tj
1 0 0 1 40 653 Tm (}) Tj
1 0 0 1 40 642 Tm ( ) Tj
1 0 0 1 40 631 Tm (function hostTest_fakeSlashProgress\(\\) ) Tj
1 0 0 1 40 620 Tm ( return doPost\(_fakeSlash\('/progress', '\)\)\).getContent\(\\) ;) Tj
1 0 0 1 40 609 Tm (}) Tj
1 0 0 1 40 598 Tm ( ) Tj
1 0 0 1 40 587 Tm (function hostTest_fakeInteractive\(\\) ) Tj
1 0 0 1 40 576 Tm ( var payload = {} ) Tj
1 0 0 1 40 565 Tm ( type: 'block_actions',) Tj
1 0 0 1 40 554 Tm ( user: { id: 'U123' },) Tj
1 0 0 1 40 543 Tm ( channel: { id: 'C123' },) Tj
1 0 0 1 40 532 Tm ( team: { id: 'T123' },) Tj
1 0 0 1 40 521 Tm ( actions: [{ action_id: 'submit_lesson' }]) Tj
1 0 0 1 40 510 Tm ( );) Tj
1 0 0 1 40 499 Tm ( return doPost\(\{) Tj
1 0 0 1 40 488 Tm ( parameter: { payload: JSON.stringify\(\payload\) , token: 'fallback' },) Tj
1 0 0 1 40 477 Tm ( postData: { type: 'application/x-www-form-urlencoded', contents: 'payload=' + encodeURIComponent) Tj
1 0 0 1 40 466 Tm ( nt\(\JSON.stringify\(\payload\)\) }) Tj
1 0 0 1 40 455 Tm ( }\)\).getContent\(\\) ;) Tj

```



```

1 0 0 1 40 554 Tm (/**) Tj
1 0 0 1 40 543 Tm ( * Raised when configuration is invalid or missing.) Tj
1 0 0 1 40 532 Tm ( */) Tj
1 0 0 1 40 521 Tm (class ConfigError extends SheetDbError {}) Tj
1 0 0 1 40 510 Tm ( /**) Tj
1 0 0 1 40 499 Tm ( * @param {string} message) Tj
1 0 0 1 40 488 Tm ( */) Tj
1 0 0 1 40 477 Tm ( constructor\ (message\ ) {} Tj
1 0 0 1 40 466 Tm ( super\ (message, 'CONFIG_ERROR'\);) Tj
1 0 0 1 40 455 Tm ( }) Tj
1 0 0 1 40 444 Tm (}) Tj
1 0 0 1 40 433 Tm ( ) Tj
1 0 0 1 40 422 Tm (/**) Tj
1 0 0 1 40 411 Tm ( * Raised when schema definitions are invalid.) Tj
1 0 0 1 40 400 Tm ( */) Tj
1 0 0 1 40 389 Tm (class SchemaError extends SheetDbError {}) Tj
1 0 0 1 40 378 Tm ( /**) Tj
1 0 0 1 40 367 Tm ( * @param {string} message) Tj
1 0 0 1 40 356 Tm ( */) Tj
1 0 0 1 40 345 Tm ( constructor\ (message\ ) {} Tj
1 0 0 1 40 334 Tm ( super\ (message, 'SCHEMA_ERROR'\);) Tj
1 0 0 1 40 323 Tm ( }) Tj
1 0 0 1 40 312 Tm (}) Tj
1 0 0 1 40 301 Tm ( ) Tj
1 0 0 1 40 290 Tm (/**) Tj
1 0 0 1 40 279 Tm ( * Raised when records fail validation.) Tj
1 0 0 1 40 268 Tm ( */) Tj
1 0 0 1 40 257 Tm (class ValidationError extends SheetDbError {}) Tj
1 0 0 1 40 246 Tm ( /**) Tj
1 0 0 1 40 235 Tm ( * @param {string} message) Tj
1 0 0 1 40 224 Tm ( */) Tj
1 0 0 1 40 213 Tm ( constructor\ (message\ ) {} Tj
1 0 0 1 40 202 Tm ( super\ (message, 'VALIDATION_ERROR'\);) Tj
1 0 0 1 40 191 Tm ( }) Tj
1 0 0 1 40 180 Tm (}) Tj
1 0 0 1 40 169 Tm ( ) Tj
1 0 0 1 40 158 Tm (/**) Tj
1 0 0 1 40 147 Tm ( * Raised when underlying sheet operations fail.) Tj
1 0 0 1 40 136 Tm ( */) Tj
1 0 0 1 40 125 Tm (class StorageError extends SheetDbError {}) Tj
1 0 0 1 40 114 Tm ( /**) Tj
1 0 0 1 40 103 Tm ( * @param {string} message) Tj
1 0 0 1 40 92 Tm ( */) Tj
1 0 0 1 40 81 Tm ( constructor\ (message\ ) {} Tj
1 0 0 1 40 70 Tm ( super\ (message, 'STORAGE_ERROR'\);) Tj
1 0 0 1 40 59 Tm ( }) Tj
1 0 0 1 40 48 Tm (}) Tj
ET
endstream
endobj
112 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 111 0
endobj
113 0 obj
<< /Length 509 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm ( ) Tj
1 0 0 1 40 730 Tm (/**) Tj
1 0 0 1 40 719 Tm ( * Raised for optimistic transaction failures.) Tj
1 0 0 1 40 708 Tm ( */) Tj
1 0 0 1 40 697 Tm (class TransactionError extends SheetDbError {}) Tj
1 0 0 1 40 686 Tm ( /**) Tj
1 0 0 1 40 675 Tm ( * @param {string} message) Tj
1 0 0 1 40 664 Tm ( */) Tj
1 0 0 1 40 653 Tm ( constructor\ (message\ ) {} Tj
1 0 0 1 40 642 Tm ( super\ (message, 'TRANSACTION_ERROR'\);) Tj
1 0 0 1 40 631 Tm ( }) Tj
1 0 0 1 40 620 Tm (}) Tj
ET
endstream
endobj
114 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 113 0
endobj
115 0 obj
<< /Length 2872 >>
stream
BT

```



```

/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: 21_Util.gs) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (/**) Tj
1 0 0 1 40 708 Tm ( * Utility helpers for Sheet DB.) Tj
1 0 0 1 40 697 Tm ( */) Tj
1 0 0 1 40 686 Tm (class Util {) Tj
1 0 0 1 40 675 Tm ( /**) Tj
1 0 0 1 40 664 Tm ( * @return {string} RFC3339 timestamp in UTC.) Tj
1 0 0 1 40 653 Tm ( */) Tj
1 0 0 1 40 642 Tm ( static nowIso\(\) {) Tj
1 0 0 1 40 631 Tm ( return new Date\(\).toISOString\(\);) Tj
1 0 0 1 40 620 Tm ( }) Tj
1 0 0 1 40 609 Tm ( ) Tj
1 0 0 1 40 598 Tm ( /**) Tj
1 0 0 1 40 587 Tm ( * Creates a shallow clone of an object.) Tj
1 0 0 1 40 576 Tm ( * @param {Object} source) Tj
1 0 0 1 40 565 Tm ( * @return {Object}) Tj
1 0 0 1 40 554 Tm ( */) Tj
1 0 0 1 40 543 Tm ( static clone\(\source\) {) Tj
1 0 0 1 40 532 Tm ( return Object.assign\({}, source || {});) Tj
1 0 0 1 40 521 Tm ( }) Tj
1 0 0 1 40 510 Tm ( ) Tj
1 0 0 1 40 499 Tm ( /**) Tj
1 0 0 1 40 488 Tm ( * Creates a stable row ID with a prefix.) Tj
1 0 0 1 40 477 Tm ( * @param {string=} prefix) Tj
1 0 0 1 40 466 Tm ( * @return {string}) Tj
1 0 0 1 40 455 Tm ( */) Tj
1 0 0 1 40 444 Tm ( static generateId\(\prefix\) {) Tj
1 0 0 1 40 433 Tm ( var p = prefix || 'row';) Tj
1 0 0 1 40 422 Tm ( var random = Math.random\(\).toString\(\36\).slice\(\2, 10\);) Tj
1 0 0 1 40 411 Tm ( return p + '_' + new Date\(\).getTime\(\) + '_' + random;) Tj
1 0 0 1 40 400 Tm ( }) Tj
1 0 0 1 40 389 Tm ( ) Tj
1 0 0 1 40 378 Tm ( /**) Tj
1 0 0 1 40 367 Tm ( * Returns true when value is null or undefined.) Tj
1 0 0 1 40 356 Tm ( * @param {*} value) Tj
1 0 0 1 40 345 Tm ( * @return {boolean}) Tj
1 0 0 1 40 334 Tm ( */) Tj
1 0 0 1 40 323 Tm ( static isNil\(\value\) {) Tj
1 0 0 1 40 312 Tm ( return value === null || value === undefined;) Tj
1 0 0 1 40 301 Tm ( }) Tj
1 0 0 1 40 290 Tm ( ) Tj
1 0 0 1 40 279 Tm ( /**) Tj
1 0 0 1 40 268 Tm ( * Returns true when value is empty string, null, or undefined.) Tj
1 0 0 1 40 257 Tm ( * @param {*} value) Tj
1 0 0 1 40 246 Tm ( * @return {boolean}) Tj
1 0 0 1 40 235 Tm ( */) Tj
1 0 0 1 40 224 Tm ( static isBlank\(\value\) {) Tj
1 0 0 1 40 213 Tm ( return value === '' || Util.isNil\(\value\);) Tj
1 0 0 1 40 202 Tm ( }) Tj
1 0 0 1 40 191 Tm ( ) Tj
1 0 0 1 40 180 Tm ( /**) Tj
1 0 0 1 40 169 Tm ( * Throws when a required condition is not met.) Tj
1 0 0 1 40 158 Tm ( * @param {boolean} condition) Tj
1 0 0 1 40 147 Tm ( * @param {string} message) Tj
1 0 0 1 40 136 Tm ( */) Tj
1 0 0 1 40 125 Tm ( static assert\(\condition, message\) {) Tj
1 0 0 1 40 114 Tm ( if \(!condition\) {) Tj
1 0 0 1 40 103 Tm ( throw new ValidationError\(\message\);) Tj
1 0 0 1 40 92 Tm ( }) Tj
1 0 0 1 40 81 Tm ( }) Tj
1 0 0 1 40 70 Tm ( ) Tj
1 0 0 1 40 59 Tm ( /**) Tj
1 0 0 1 40 48 Tm ( * Converts a sheet row to an object using the header order.) Tj
ET
endstream
endobj
116 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 115 0
endobj
117 0 obj
<< /Length 1078 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm ( * @param {string[]} headers) Tj
1 0 0 1 40 730 Tm ( * @param {Array<*>} row) Tj
1 0 0 1 40 719 Tm ( * @return {Object}) Tj
1 0 0 1 40 708 Tm ( */) Tj

```

```

1 0 0 1 40 697 Tm ( static rowToObject\(headers, row\ ) {} Tj
1 0 0 1 40 686 Tm ( var out = {};) Tj
1 0 0 1 40 675 Tm ( for \ (var i = 0; i < headers.length; i++) {} Tj
1 0 0 1 40 664 Tm ( out[headers[i]] = row[i];) Tj
1 0 0 1 40 653 Tm ( }) Tj
1 0 0 1 40 642 Tm ( return out;) Tj
1 0 0 1 40 631 Tm ( }) Tj
1 0 0 1 40 620 Tm ( ) Tj
1 0 0 1 40 609 Tm ( /**) Tj
1 0 0 1 40 598 Tm ( * Converts an object to row array by header order.) Tj
1 0 0 1 40 587 Tm ( * @param {string[]} headers) Tj
1 0 0 1 40 576 Tm ( * @param {Object} obj) Tj
1 0 0 1 40 565 Tm ( * @return {Array<*>} Tj
1 0 0 1 40 554 Tm ( */) Tj
1 0 0 1 40 543 Tm ( static objectToRow\(headers, obj\ ) {} Tj
1 0 0 1 40 532 Tm ( return headers.map\ (function\ (h\ ) {} Tj
1 0 0 1 40 521 Tm ( return obj[h];) Tj
1 0 0 1 40 510 Tm ( })\);) Tj
1 0 0 1 40 499 Tm ( }) Tj
1 0 0 1 40 488 Tm ( {} ) Tj
ET
endstream
endobj
118 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 117 0
endobj
119 0 obj
<< /Length 3258 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: 22_ScriptProperties.gs) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (/**) Tj
1 0 0 1 40 708 Tm ( * Script Properties helper for secure and environment-specific configuration.) Tj
1 0 0 1 40 697 Tm ( */) Tj
1 0 0 1 40 686 Tm (class ScriptPropertiesHelper {} Tj
1 0 0 1 40 675 Tm ( /**) Tj
1 0 0 1 40 664 Tm ( * Creates a helper bound to Script Properties store.) Tj
1 0 0 1 40 653 Tm ( */) Tj
1 0 0 1 40 642 Tm ( constructor\(\ ) {} Tj
1 0 0 1 40 631 Tm ( this._props = PropertiesService.getScriptProperties\(\);) Tj
1 0 0 1 40 620 Tm ( }) Tj
1 0 0 1 40 609 Tm ( ) Tj
1 0 0 1 40 598 Tm ( /**) Tj
1 0 0 1 40 587 Tm ( * Returns a property value, or a default when missing.) Tj
1 0 0 1 40 576 Tm ( * @param {string} key) Tj
1 0 0 1 40 565 Tm ( * @param {*=} defaultValue) Tj
1 0 0 1 40 554 Tm ( * @return {*} Tj
1 0 0 1 40 543 Tm ( */) Tj
1 0 0 1 40 532 Tm ( get\ (key, defaultValue\ ) {} Tj
1 0 0 1 40 521 Tm ( var value = this._props.getProperty\ (key\);) Tj
1 0 0 1 40 510 Tm ( return value === null ? defaultValue : value;) Tj
1 0 0 1 40 499 Tm ( }) Tj
1 0 0 1 40 488 Tm ( ) Tj
1 0 0 1 40 477 Tm ( /**) Tj
1 0 0 1 40 466 Tm ( * Returns a required property value or throws when missing/blank.) Tj
1 0 0 1 40 455 Tm ( * @param {string} key) Tj
1 0 0 1 40 444 Tm ( * @return {string} Tj
1 0 0 1 40 433 Tm ( */) Tj
1 0 0 1 40 422 Tm ( getRequired\ (key\ ) {} Tj
1 0 0 1 40 411 Tm ( var value = this.get\ (key, ''\);) Tj
1 0 0 1 40 400 Tm ( if \ (value === ''\ ) {} Tj
1 0 0 1 40 389 Tm ( throw new ConfigError\ ('Missing required script property: ' + key\);) Tj
1 0 0 1 40 378 Tm ( }) Tj
1 0 0 1 40 367 Tm ( return String\ (value\);) Tj
1 0 0 1 40 356 Tm ( }) Tj
1 0 0 1 40 345 Tm ( ) Tj
1 0 0 1 40 334 Tm ( /**) Tj
1 0 0 1 40 323 Tm ( * Returns a boolean property value.) Tj
1 0 0 1 40 312 Tm ( * Accepted true values: true, 1, yes, y, on.) Tj
1 0 0 1 40 301 Tm ( * Accepted false values: false, 0, no, n, off.) Tj
1 0 0 1 40 290 Tm ( * @param {string} key) Tj
1 0 0 1 40 279 Tm ( * @param {boolean=} defaultValue) Tj
1 0 0 1 40 268 Tm ( * @return {boolean} Tj
1 0 0 1 40 257 Tm ( */) Tj
1 0 0 1 40 246 Tm ( getBoolean\ (key, defaultValue\ ) {} Tj
1 0 0 1 40 235 Tm ( var raw = this.get\ (key, null\);) Tj
1 0 0 1 40 224 Tm ( if \ (raw === null || raw === ''\ ) {} Tj
1 0 0 1 40 213 Tm ( return defaultValue === undefined ? false : !!defaultValue;) Tj

```

```

1 0 0 1 40 202 Tm (    }) Tj
1 0 0 1 40 191 Tm (    var normalized = String\raw\).trim\(\).toLowerCase\(\);) Tj
1 0 0 1 40 180 Tm (    if \(['true', '1', 'yes', 'y', 'on'].indexOf(normalized\) !== -1\) {}) Tj
1 0 0 1 40 169 Tm (        return true;) Tj
1 0 0 1 40 158 Tm (    }) Tj
1 0 0 1 40 147 Tm (    if \(['false', '0', 'no', 'n', 'off'].indexOf(normalized\) !== -1\) {}) Tj
1 0 0 1 40 136 Tm (        return false;) Tj
1 0 0 1 40 125 Tm (    }) Tj
1 0 0 1 40 114 Tm (    throw new ConfigError\('Script property is not a valid boolean: ' + key\);) Tj
1 0 0 1 40 103 Tm (    }) Tj
1 0 0 1 40 92 Tm (    Tj
1 0 0 1 40 81 Tm (    /**) Tj
1 0 0 1 40 70 Tm (    * Returns a numeric property value.) Tj
1 0 0 1 40 59 Tm (    * @param {string} key) Tj
1 0 0 1 40 48 Tm (    * @param {number=} defaultValue) Tj
ET
endstream
endobj
120 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 119 0
endobj
121 0 obj
<< /Length 2984 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (    * @return {number}) Tj
1 0 0 1 40 730 Tm (    */) Tj
1 0 0 1 40 719 Tm (    getNumber\key, defaultValue\ {}) Tj
1 0 0 1 40 708 Tm (    var raw = this.get\key, null\);) Tj
1 0 0 1 40 697 Tm (    if \raw === null || raw === ''\ {}) Tj
1 0 0 1 40 686 Tm (        return defaultValue === undefined ? 0 : Number\defaultValue\);) Tj
1 0 0 1 40 675 Tm (    }) Tj
1 0 0 1 40 664 Tm (    var n = Number\raw\);) Tj
1 0 0 1 40 653 Tm (    if \isNaN\(\)\ {}) Tj
1 0 0 1 40 642 Tm (        throw new ConfigError\('Script property is not a valid number: ' + key\);) Tj
1 0 0 1 40 631 Tm (    }) Tj
1 0 0 1 40 620 Tm (    return n;) Tj
1 0 0 1 40 609 Tm (    }) Tj
1 0 0 1 40 598 Tm (    Tj
1 0 0 1 40 587 Tm (    /**) Tj
1 0 0 1 40 576 Tm (    * Returns all script properties as key-value object.) Tj
1 0 0 1 40 565 Tm (    * @return {Object<string, string>}) Tj
1 0 0 1 40 554 Tm (    */) Tj
1 0 0 1 40 543 Tm (    getAll\(\) {}) Tj
1 0 0 1 40 532 Tm (    return this._props.getProperties\(\);) Tj
1 0 0 1 40 521 Tm (    }) Tj
1 0 0 1 40 510 Tm (    Tj
1 0 0 1 40 499 Tm (    /**) Tj
1 0 0 1 40 488 Tm (    * Returns all script properties with sensitive values redacted.) Tj
1 0 0 1 40 477 Tm (    * @return {Object<string, string>}) Tj
1 0 0 1 40 466 Tm (    */) Tj
1 0 0 1 40 455 Tm (    getAllRedacted\(\) {}) Tj
1 0 0 1 40 444 Tm (    var all = this.getAll\(\);) Tj
1 0 0 1 40 433 Tm (    var out = {};) Tj
1 0 0 1 40 422 Tm (    Object.keys\all\).forEach\function\key\ {} Tj
1 0 0 1 40 411 Tm (        out[key] = ScriptPropertiesHelper.shouldRedactKey\key\ ? '[REDACTED]' : all[key];) Tj
1 0 0 1 40 400 Tm (    \);) Tj
1 0 0 1 40 389 Tm (    return out;) Tj
1 0 0 1 40 378 Tm (    }) Tj
1 0 0 1 40 367 Tm (    Tj
1 0 0 1 40 356 Tm (    /**) Tj
1 0 0 1 40 345 Tm (    * Sets a script property value.) Tj
1 0 0 1 40 334 Tm (    * @param {string} key) Tj
1 0 0 1 40 323 Tm (    * @param {*} value) Tj
1 0 0 1 40 312 Tm (    */) Tj
1 0 0 1 40 301 Tm (    set\key, value\ {}) Tj
1 0 0 1 40 290 Tm (        this._props.setProperty\key, String\value\);) Tj
1 0 0 1 40 279 Tm (    }) Tj
1 0 0 1 40 268 Tm (    Tj
1 0 0 1 40 257 Tm (    /**) Tj
1 0 0 1 40 246 Tm (    * Sets multiple script properties.) Tj
1 0 0 1 40 235 Tm (    * @param {Object<string, *>} obj) Tj
1 0 0 1 40 224 Tm (    */) Tj
1 0 0 1 40 213 Tm (    setMany\obj\ {}) Tj
1 0 0 1 40 202 Tm (        var stringified = {};) Tj
1 0 0 1 40 191 Tm (        Object.keys\obj || {}\.forEach\function\key\ {} Tj
1 0 0 1 40 180 Tm (            stringified[key] = String\obj[key\];) Tj
1 0 0 1 40 169 Tm (        \);) Tj
1 0 0 1 40 158 Tm (        this._props.setProperties\stringified, false\);) Tj

```



```

1 0 0 1 40 92 Tm (    config.updatedAtColumn = helper.get\('SHEET_DB_UPDATED_AT_COLUMN', 'updatedAt'\);) Tj
1 0 0 1 40 81 Tm (    }) Tj
1 0 0 1 40 70 Tm (    if \(helper.has\('SHEET_DB_DELETED_AT_COLUMN'\)\) \{) Tj
1 0 0 1 40 59 Tm (        config.deletedAtColumn = helper.get\('SHEET_DB_DELETED_AT_COLUMN', 'deletedAt'\);) Tj
1 0 0 1 40 48 Tm (    }) Tj
ET
endstream
endobj
124 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 123 0
endobj
125 0 obj
<< /Length 1892 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (    if \(helper.has\('SHEET_DB_LOCK_TIMEOUT_MS'\)\) \{) Tj
1 0 0 1 40 730 Tm (        config.lockTimeoutMs = helper.getNumber\('SHEET_DB_LOCK_TIMEOUT_MS', 30000\);) Tj
1 0 0 1 40 719 Tm (    }) Tj
1 0 0 1 40 708 Tm (    }) Tj
1 0 0 1 40 697 Tm (    return config;) Tj
1 0 0 1 40 686 Tm (}) Tj
1 0 0 1 40 675 Tm (    }) Tj
1 0 0 1 40 664 Tm (/**) Tj
1 0 0 1 40 653 Tm ( * Merges explicit client options with script properties-based config.) Tj
1 0 0 1 40 642 Tm ( * Precedence: explicit options > script properties > hardcoded defaults.) Tj
1 0 0 1 40 631 Tm ( *) Tj
1 0 0 1 40 620 Tm ( * @param {Object=} options) Tj
1 0 0 1 40 609 Tm ( * @return {Object}) Tj
1 0 0 1 40 598 Tm ( */) Tj
1 0 0 1 40 587 Tm (function mergeClientOptionsWithScriptProperties\(options\() Tj
1 0 0 1 40 576 Tm (    var fromProps = readSheetDbConfigFromScriptProperties\(\);) Tj
1 0 0 1 40 565 Tm (    var explicit = options || {};) Tj
1 0 0 1 40 554 Tm (    var merged = {};) Tj
1 0 0 1 40 543 Tm (    Object.keys\(\fromProps\).forEach\((function\(\key\() Tj
1 0 0 1 40 532 Tm (        merged[key] = fromProps[key];) Tj
1 0 0 1 40 521 Tm (    ));) Tj
1 0 0 1 40 510 Tm (    Object.keys\(\explicit\).forEach\((function\(\key\() Tj
1 0 0 1 40 499 Tm (        merged[key] = explicit[key];) Tj
1 0 0 1 40 488 Tm (    ));) Tj
1 0 0 1 40 477 Tm (    return merged;) Tj
1 0 0 1 40 466 Tm (}) Tj
1 0 0 1 40 455 Tm (    }) Tj
1 0 0 1 40 444 Tm (/**) Tj
1 0 0 1 40 433 Tm ( * Example setup helper for consumers \(\manual one-time setup\).) Tj
1 0 0 1 40 422 Tm ( *) Tj
1 0 0 1 40 411 Tm ( * Example:) Tj
1 0 0 1 40 400 Tm ( * PropertiesService.getScriptProperties\(\).setProperty\('SHEET_DB_SPREADSHEET_ID', '...\');) Tj
1 0 0 1 40 389 Tm ( */) Tj
1 0 0 1 40 378 Tm (function exampleSetSheetDbScriptProperties\(\() Tj
1 0 0 1 40 367 Tm (    // Intentionally no-op; serves as in-library usage documentation only.) Tj
1 0 0 1 40 356 Tm (}) Tj
ET
endstream
endobj
126 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 125 0
endobj
127 0 obj
<< /Length 3076 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: 23_Config.gs) Tj
1 0 0 1 40 730 Tm (    }) Tj
1 0 0 1 40 719 Tm (/**) Tj
1 0 0 1 40 708 Tm ( * Immutable runtime configuration for the Sheet DB client.) Tj
1 0 0 1 40 697 Tm ( */) Tj
1 0 0 1 40 686 Tm (class Config {}) Tj
1 0 0 1 40 675 Tm (/**) Tj
1 0 0 1 40 664 Tm (    * @param {Object=} options) Tj
1 0 0 1 40 653 Tm (    * @param {string=} options.spreadsheetId Spreadsheet ID; defaults to active spreadsheet.) Tj
1 0 0 1 40 642 Tm (    * @param {string=} options.idColumn Record primary key column name.) Tj
1 0 0 1 40 631 Tm (    * @param {string=} options.createdAtColumn Created timestamp column name.) Tj
1 0 0 1 40 620 Tm (    * @param {string=} options.updatedAtColumn Updated timestamp column name.) Tj
1 0 0 1 40 609 Tm (    * @param {string=} options.deletedAtColumn Soft-delete timestamp column name.) Tj
1 0 0 1 40 598 Tm (    * @param {number=} options.lockTimeoutMs Lock acquisition timeout for transactions.) Tj
1 0 0 1 40 587 Tm (    */) Tj
1 0 0 1 40 576 Tm (    constructor\(\options\() Tj
1 0 0 1 40 565 Tm (        var opts = options || {};) Tj

```

```

1 0 0 1 40 554 Tm ( this.spreadsheetId = opts.spreadsheetId || '';) Tj
1 0 0 1 40 543 Tm ( this.idColumn = opts.idColumn || 'id';) Tj
1 0 0 1 40 532 Tm ( this.createdAtColumn = opts.createdAtColumn || 'createdAt';) Tj
1 0 0 1 40 521 Tm ( this.updatedAtColumn = opts.updatedAtColumn || 'updatedAt';) Tj
1 0 0 1 40 510 Tm ( this.deletedAtColumn = opts.deletedAtColumn || 'deletedAt';) Tj
1 0 0 1 40 499 Tm ( this.lockTimeoutMs = Util.isNil\(\(opts.lockTimeoutMs\) ? 30000 : Number\(\(opts.lockTimeoutMs\) \);) Tj
1 0 0 1 40 488 Tm ( this.validate\(\);) Tj
1 0 0 1 40 477 Tm ( Object.freeze\(\(this\) \);) Tj
1 0 0 1 40 466 Tm ( }) Tj
1 0 0 1 40 455 Tm ( ) Tj
1 0 0 1 40 444 Tm ( /**) Tj
1 0 0 1 40 433 Tm ( * @return {void}) Tj
1 0 0 1 40 422 Tm ( */) Tj
1 0 0 1 40 411 Tm ( validate\(\) {}) Tj
1 0 0 1 40 400 Tm ( var columns = []) Tj
1 0 0 1 40 389 Tm ( this.idColumn,) Tj
1 0 0 1 40 378 Tm ( this.createdAtColumn,) Tj
1 0 0 1 40 367 Tm ( this.updatedAtColumn,) Tj
1 0 0 1 40 356 Tm ( this.deletedAtColumn) Tj
1 0 0 1 40 345 Tm ( );) Tj
1 0 0 1 40 334 Tm ( var unique = {};) Tj
1 0 0 1 40 323 Tm ( columns.forEach\(\(function\(\(column\)\) {}) Tj
1 0 0 1 40 312 Tm ( if \(!column || typeof column !== 'string'\) {}) Tj
1 0 0 1 40 301 Tm ( throw new ConfigError\('System column names must be non-empty strings.'\) \);) Tj
1 0 0 1 40 290 Tm ( }) Tj
1 0 0 1 40 279 Tm ( if \(\(unique[column]\\)\) {}) Tj
1 0 0 1 40 268 Tm ( throw new ConfigError\('System column names must be unique. Duplicate: ' + column\) \);) Tj
1 0 0 1 40 257 Tm ( }) Tj
1 0 0 1 40 246 Tm ( unique[column] = true;) Tj
1 0 0 1 40 235 Tm ( }) \);) Tj
1 0 0 1 40 224 Tm ( ) Tj
1 0 0 1 40 213 Tm ( if \(\(isNaN\(\(this.lockTimeoutMs\)\) || this.lockTimeoutMs <= 0\)\) {}) Tj
1 0 0 1 40 202 Tm ( throw new ConfigError\('lockTimeoutMs must be a positive number.'\) \);) Tj
1 0 0 1 40 191 Tm ( }) Tj
1 0 0 1 40 180 Tm ( }) Tj
1 0 0 1 40 169 Tm ( }) Tj
ET
endstream
endobj
128 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 127 0
endobj
129 0 obj
<< /Length 3051 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: 24_SheetsGateway.gs) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (/**) Tj
1 0 0 1 40 708 Tm ( * Thin wrapper around SpreadsheetApp APIs.) Tj
1 0 0 1 40 697 Tm ( */) Tj
1 0 0 1 40 686 Tm (class SheetsGateway {}) Tj
1 0 0 1 40 675 Tm ( /**) Tj
1 0 0 1 40 664 Tm ( * @param {Config} config) Tj
1 0 0 1 40 653 Tm ( */) Tj
1 0 0 1 40 642 Tm ( constructor\(\(config\)\) {}) Tj
1 0 0 1 40 631 Tm ( this._config = config;) Tj
1 0 0 1 40 620 Tm ( }) Tj
1 0 0 1 40 609 Tm ( ) Tj
1 0 0 1 40 598 Tm ( /**) Tj
1 0 0 1 40 587 Tm ( * @return {GoogleAppsScript.Spreadsheet.Spreadsheet}) Tj
1 0 0 1 40 576 Tm ( */) Tj
1 0 0 1 40 565 Tm ( getSpreadsheet\(\) {}) Tj
1 0 0 1 40 554 Tm ( if \(\(this._config.spreadsheetId\)\) {}) Tj
1 0 0 1 40 543 Tm ( return SpreadsheetApp.openById\(\(this._config.spreadsheetId\) \);) Tj
1 0 0 1 40 532 Tm ( }) Tj
1 0 0 1 40 521 Tm ( return SpreadsheetApp.getActiveSpreadsheet\(\);) Tj
1 0 0 1 40 510 Tm ( }) Tj
1 0 0 1 40 499 Tm ( ) Tj
1 0 0 1 40 488 Tm ( /**) Tj
1 0 0 1 40 477 Tm ( * @param {string} tableName) Tj
1 0 0 1 40 466 Tm ( * @param {boolean=} createWhenMissing) Tj
1 0 0 1 40 455 Tm ( * @return {GoogleAppsScript.Spreadsheet.Sheet}) Tj
1 0 0 1 40 444 Tm ( */) Tj
1 0 0 1 40 433 Tm ( getSheet\(\(tableName, createWhenMissing\)\) {}) Tj
1 0 0 1 40 422 Tm ( var ss = this.getSpreadsheet\(\);) Tj
1 0 0 1 40 411 Tm ( var sheet = ss.getSheetByName\(\(tableName\) \);) Tj
1 0 0 1 40 400 Tm ( if \(!sheet && createWhenMissing\) {}) Tj
1 0 0 1 40 389 Tm ( sheet = ss.insertSheet\(\(tableName\) \);) Tj

```

```

1 0 0 1 40 378 Tm (    }) Tj
1 0 0 1 40 367 Tm (    if \(!sheet\) {) Tj
1 0 0 1 40 356 Tm (        throw new StorageError\('Sheet not found for table: ' + tableName\);) Tj
1 0 0 1 40 345 Tm (    }) Tj
1 0 0 1 40 334 Tm (    return sheet;) Tj
1 0 0 1 40 323 Tm ( }) Tj
1 0 0 1 40 312 Tm ( ) Tj
1 0 0 1 40 301 Tm ( /**) Tj
1 0 0 1 40 290 Tm ( * @param {string} tableName) Tj
1 0 0 1 40 279 Tm ( * @return {{headers: string[], rows: Array<Array<*>>}}) Tj
1 0 0 1 40 268 Tm ( */) Tj
1 0 0 1 40 257 Tm ( readTable\(tableName\) {) Tj
1 0 0 1 40 246 Tm (     var sheet = this.getSheet\(tableName, false\);) Tj
1 0 0 1 40 235 Tm (     var range = sheet.getDataRange\(\);) Tj
1 0 0 1 40 224 Tm (     var values = range.getValues\(\);) Tj
1 0 0 1 40 213 Tm (     if \((values.length === 0\) {) Tj
1 0 0 1 40 202 Tm (         return { headers: [], rows: [] };) Tj
1 0 0 1 40 191 Tm (     }) Tj
1 0 0 1 40 180 Tm (     return {) Tj
1 0 0 1 40 169 Tm (         headers: values[0],) Tj
1 0 0 1 40 158 Tm (         rows: values.slice\((1\)) Tj
1 0 0 1 40 147 Tm (     );) Tj
1 0 0 1 40 136 Tm ( }) Tj
1 0 0 1 40 125 Tm ( ) Tj
1 0 0 1 40 114 Tm ( /**) Tj
1 0 0 1 40 103 Tm ( * Ensures table exists with given headers.) Tj
1 0 0 1 40 92 Tm ( * @param {string} tableName) Tj
1 0 0 1 40 81 Tm ( * @param {string[]} headers) Tj
1 0 0 1 40 70 Tm ( */) Tj
1 0 0 1 40 59 Tm ( ensureTable\(tableName, headers\) {) Tj
1 0 0 1 40 48 Tm (     var sheet = this.getSheet\(tableName, true\);) Tj
ET
endstream
endobj
130 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 129 0
endobj
131 0 obj
<< /Length 2220 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (     var lastRow = sheet.getLastRow\(\);) Tj
1 0 0 1 40 730 Tm (     if \((lastRow === 0\) {) Tj
1 0 0 1 40 719 Tm (         sheet.getRange\((1, 1, 1, headers.length\)).setValues\([headers]\);) Tj
1 0 0 1 40 708 Tm (         return;) Tj
1 0 0 1 40 697 Tm (     }) Tj
1 0 0 1 40 686 Tm ( ) Tj
1 0 0 1 40 675 Tm (     var existingHeaders = sheet.getRange\((1, 1, 1, sheet.getLastColumn\(\)\)).getValues\(\)[0]) Tj
1 0 0 1 40 664 Tm (         .map\((function\((value\) { return String\((value\) );})\);) Tj
1 0 0 1 40 653 Tm (         var expectedHeaders = headers.map\((function\((value\) { return String\((value\) );})\);) Tj
1 0 0 1 40 642 Tm (         if \((existingHeaders.join\('|\`'\) !== expectedHeaders.join\('|\`'\) ) {) Tj
1 0 0 1 40 631 Tm (             throw new StorageError\('Header mismatch for table: ' + tableName\);) Tj
1 0 0 1 40 620 Tm (         }) Tj
1 0 0 1 40 609 Tm (     }) Tj
1 0 0 1 40 598 Tm ( ) Tj
1 0 0 1 40 587 Tm ( /**) Tj
1 0 0 1 40 576 Tm ( * Writes all rows for a table, replacing existing body rows.) Tj
1 0 0 1 40 565 Tm ( * @param {string} tableName) Tj
1 0 0 1 40 554 Tm ( * @param {string[]} headers) Tj
1 0 0 1 40 543 Tm ( * @param {Array<Array<*>>} rows) Tj
1 0 0 1 40 532 Tm ( */) Tj
1 0 0 1 40 521 Tm ( writeAllRows\(tableName, headers, rows\) {) Tj
1 0 0 1 40 510 Tm (     var sheet = this.getSheet\(tableName, true\);) Tj
1 0 0 1 40 499 Tm (     this.ensureTable\(tableName, headers\);) Tj
1 0 0 1 40 488 Tm ( ) Tj
1 0 0 1 40 477 Tm (     var maxRows = Math.max\((sheet.getMaxRows\(\), rows.length + 1\);) Tj
1 0 0 1 40 466 Tm (     if \((sheet.getMaxRows\(\) < maxRows\) {) Tj
1 0 0 1 40 455 Tm (         sheet.insertRowsAfter\((sheet.getMaxRows\(\), maxRows - sheet.getMaxRows\(\)\);) Tj
1 0 0 1 40 444 Tm (     }) Tj
1 0 0 1 40 433 Tm ( ) Tj
1 0 0 1 40 422 Tm (     if \((sheet.getLastRow\(\) > 1\) {) Tj
1 0 0 1 40 411 Tm (         sheet.getRange\((2, 1, sheet.getLastRow\(\) - 1, headers.length\)).clearContent\(\);) Tj
1 0 0 1 40 400 Tm (     }) Tj
1 0 0 1 40 389 Tm ( ) Tj
1 0 0 1 40 378 Tm (     if \((rows.length > 0\) {) Tj
1 0 0 1 40 367 Tm (         sheet.getRange\((2, 1, rows.length, headers.length\)).setValues\((rows\) );) Tj
1 0 0 1 40 356 Tm (     }) Tj
1 0 0 1 40 345 Tm ( }) Tj
1 0 0 1 40 334 Tm ( }) Tj

```

```

ET
endstream
endobj
132 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 131 0
endobj
133 0 obj
<< /Length 3397 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: 25_SchemaRegistry.gs) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (/**) Tj
1 0 0 1 40 708 Tm ( * Holds table schemas and applies field defaults/validation.) Tj
1 0 0 1 40 697 Tm ( */) Tj
1 0 0 1 40 686 Tm (class SchemaRegistry {} Tj
1 0 0 1 40 675 Tm ( /**) Tj
1 0 0 1 40 664 Tm ( * Creates a new in-memory schema registry.) Tj
1 0 0 1 40 653 Tm ( */) Tj
1 0 0 1 40 642 Tm ( constructor\(\) {} Tj
1 0 0 1 40 631 Tm ( this._schemas = {};) Tj
1 0 0 1 40 620 Tm ( }) Tj
1 0 0 1 40 609 Tm ( ) Tj
1 0 0 1 40 598 Tm ( /**) Tj
1 0 0 1 40 587 Tm ( * Registers a table schema.) Tj
1 0 0 1 40 576 Tm ( * @param {string} tableName) Tj
1 0 0 1 40 565 Tm ( * @param {Object} schema) Tj
1 0 0 1 40 554 Tm ( * @param {Array<string>} schema.columns Ordered columns for persistence.) Tj
1 0 0 1 40 543 Tm ( * @param {Object<string, Object>} schema.fields Optional field constraints.) Tj
1 0 0 1 40 532 Tm ( * @return {SchemaRegistry}) Tj
1 0 0 1 40 521 Tm ( */) Tj
1 0 0 1 40 510 Tm ( register\(tableName, schema\) {} Tj
1 0 0 1 40 499 Tm ( if \(!tableName\) {} Tj
1 0 0 1 40 488 Tm ( throw new SchemaError\('Table name is required.\');

```





```

1 0 0 1 40 532 Tm (  }) Tj
1 0 0 1 40 521 Tm ( ) Tj
1 0 0 1 40 510 Tm ( /**) Tj
1 0 0 1 40 499 Tm ( * @return {Array<Object>}) Tj
1 0 0 1 40 488 Tm ( */) Tj
1 0 0 1 40 477 Tm ( findAll\(\) {} ) Tj
1 0 0 1 40 466 Tm ( var schema = this._schemas.get\((this._tableName\));) Tj
1 0 0 1 40 455 Tm ( this._sheets.ensureTable\((this._tableName, schema.columns\));) Tj
1 0 0 1 40 444 Tm ( var data = this._sheets.readTable\((this._tableName\));) Tj
1 0 0 1 40 433 Tm ( return data.rows) Tj
1 0 0 1 40 422 Tm ( .map\((function\((row\)) { return Util.rowToObject\((data.headers, row\)); })\)) Tj
1 0 0 1 40 411 Tm ( .filter\((function\((record\)) { return Util.isBlank\((record[this._config.deletedAtColumn]\)); }, t
1 0 0 1 40 400 Tm (s\));) Tj
1 0 0 1 40 389 Tm ( }) Tj
1 0 0 1 40 378 Tm ( ) Tj
1 0 0 1 40 367 Tm ( /**) Tj
1 0 0 1 40 356 Tm ( * @param {string} id) Tj
1 0 0 1 40 345 Tm ( * @return {Object|null}) Tj
1 0 0 1 40 334 Tm ( */) Tj
1 0 0 1 40 323 Tm ( findById\((id\)) {} ) Tj
1 0 0 1 40 312 Tm ( var rows = this.findAll\(\);) Tj
1 0 0 1 40 301 Tm ( for \((var i = 0; i < rows.length; i++\)) {} ) Tj
1 0 0 1 40 290 Tm ( if \((rows[i][this._config.idColumn] === id\)) {} ) Tj
1 0 0 1 40 279 Tm ( return rows[i];) Tj
1 0 0 1 40 268 Tm ( }) Tj
1 0 0 1 40 257 Tm ( }) Tj
1 0 0 1 40 246 Tm ( return null;) Tj
1 0 0 1 40 235 Tm ( }) Tj
1 0 0 1 40 224 Tm ( ) Tj
1 0 0 1 40 213 Tm ( /**) Tj
1 0 0 1 40 202 Tm ( * @param {Object} payload) Tj
1 0 0 1 40 191 Tm ( * @return {Object}) Tj
1 0 0 1 40 180 Tm ( */) Tj
1 0 0 1 40 169 Tm ( insert\((payload\)) {} ) Tj
1 0 0 1 40 158 Tm ( var schema = this._schemas.get\((this._tableName\));) Tj
1 0 0 1 40 147 Tm ( this._sheets.ensureTable\((this._tableName, schema.columns\));) Tj
1 0 0 1 40 136 Tm ( var record = this._schemas.normalizeRecord\((this._tableName, payload, false\));) Tj
1 0 0 1 40 125 Tm ( var now = Util.nowIso\(\);) Tj
1 0 0 1 40 114 Tm ( record[this._config.idColumn] = record[this._config.idColumn] || Util.generateId\((this._tableNa
1 0 0 1 40 103 Tm (me\));) Tj
1 0 0 1 40 92 Tm ( record[this._config.createdAtColumn] = record[this._config.createdAtColumn] || now;) Tj
1 0 0 1 40 81 Tm ( record[this._config.updatedAtColumn] = now;) Tj
1 0 0 1 40 70 Tm ( if \(!record.hasOwnProperty\((this._config.deletedAtColumn\)\)) {} ) Tj
1 0 0 1 40 59 Tm ( record[this._config.deletedAtColumn] = '');) Tj
1 0 0 1 40 48 Tm ( }) Tj
ET
endstream
endobj
138 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 137 0
endobj
139 0 obj
<< /Length 3795 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm ( ) Tj
1 0 0 1 40 730 Tm ( var tableData = this._sheets.readTable\((this._tableName\));) Tj
1 0 0 1 40 719 Tm ( var existingRows = tableData.rows.map\((function\((row\)) { return Util.rowToObject\((tableData.head
1 0 0 1 40 708 Tm (s, row\)); })\);) Tj
1 0 0 1 40 697 Tm ( var idColumn = this._config.idColumn;) Tj
1 0 0 1 40 686 Tm ( var duplicate = existingRows.some\((function\((row\)) {} ) Tj
1 0 0 1 40 675 Tm ( return row[idColumn] === record[idColumn];) Tj
1 0 0 1 40 664 Tm ( ));) Tj
1 0 0 1 40 653 Tm ( if \((duplicate\)) {} ) Tj
1 0 0 1 40 642 Tm ( throw new ValidationError\('Record ID already exists: ' + record[idColumn]\));) Tj
1 0 0 1 40 631 Tm ( }) Tj
1 0 0 1 40 620 Tm ( existingRows.push\((record\));) Tj
1 0 0 1 40 609 Tm ( var rowValues = existingRows.map\((function\((r\)) { return Util.objectToRow\((schema.columns, r\)); }) Tj
1 0 0 1 40 598 Tm ( this._sheets.writeAllRows\((this._tableName, schema.columns, rowValues\));) Tj
1 0 0 1 40 587 Tm ( this._audit.log\('insert', this._tableName, { id: record[this._config.idColumn] })\);) Tj
1 0 0 1 40 576 Tm ( return record;) Tj
1 0 0 1 40 565 Tm ( }) Tj
1 0 0 1 40 554 Tm ( ) Tj
1 0 0 1 40 543 Tm ( /**) Tj
1 0 0 1 40 532 Tm ( * @param {string} id) Tj
1 0 0 1 40 521 Tm ( * @param {Object} patch) Tj
1 0 0 1 40 510 Tm ( * @return {Object}) Tj
1 0 0 1 40 499 Tm ( */) Tj
1 0 0 1 40 488 Tm ( update\((id, patch\)) {} ) Tj

```

```

1 0 0 1 40 477 Tm (    var schema = this._schemas.get\(this._tableName\);) Tj
1 0 0 1 40 466 Tm (    var normalizedPatch = this._schemas.normalizeRecord\(this._tableName, patch, true\);) Tj
1 0 0 1 40 455 Tm (    var tableData = this._sheets.readTable\(this._tableName\);) Tj
1 0 0 1 40 444 Tm (    var records = tableData.rows.map\(function\ (row\ ) { return Util.rowToObject\ (tableData.headers, r
1 0 0 1 40 433 Tm (w\); }\);) Tj
1 0 0 1 40 422 Tm (    var found = null;) Tj
1 0 0 1 40 411 Tm ( ) Tj
1 0 0 1 40 400 Tm (    for \ (var i = 0; i < records.length; i++\ ) {) Tj
1 0 0 1 40 389 Tm (        if \ (records[i][this._config.idColumn] === id\ ) {) Tj
1 0 0 1 40 378 Tm (            found = records[i];) Tj
1 0 0 1 40 367 Tm (            Object.keys\ (normalizedPatch\ ).forEach\ (function\ (key\ ) {) Tj
1 0 0 1 40 356 Tm (                if \ (key !== this._config.idColumn && key !== this._config.createdAtColumn\ ) {) Tj
1 0 0 1 40 345 Tm (                    found[key] = normalizedPatch[key];) Tj
1 0 0 1 40 334 Tm (                }) Tj
1 0 0 1 40 323 Tm (            }, this\);) Tj
1 0 0 1 40 312 Tm (            found[this._config.updatedAtColumn] = Util.nowIso\ ();) Tj
1 0 0 1 40 301 Tm (            break;) Tj
1 0 0 1 40 290 Tm (        }) Tj
1 0 0 1 40 279 Tm (    }) Tj
1 0 0 1 40 268 Tm ( ) Tj
1 0 0 1 40 257 Tm (    if \ (!found\ ) {) Tj
1 0 0 1 40 246 Tm (        throw new StorageError\ ('Record not found for id: ' + id\);) Tj
1 0 0 1 40 235 Tm (    }) Tj
1 0 0 1 40 224 Tm ( ) Tj
1 0 0 1 40 213 Tm (    var rowValues = records.map\ (function\ (r\ ) { return Util.objectToRow\ (schema.columns, r\); }\);)
1 0 0 1 40 202 Tm (    this._sheets.writeAllRows\ (this._tableName, schema.columns, rowValues\);) Tj
1 0 0 1 40 191 Tm (    this._audit.log\ ('update', this._tableName, { id: id, fields: Object.keys\ (normalizedPatch\ ) }\);)
1 0 0 1 40 180 Tm (    return found;) Tj
1 0 0 1 40 169 Tm ( }) Tj
1 0 0 1 40 158 Tm ( ) Tj
1 0 0 1 40 147 Tm ( /**) Tj
1 0 0 1 40 136 Tm ( * Soft-deletes a record.) Tj
1 0 0 1 40 125 Tm ( * @param {string} id) Tj
1 0 0 1 40 114 Tm ( * @return {Object}) Tj
1 0 0 1 40 103 Tm ( */) Tj
1 0 0 1 40 92 Tm ( remove\ (id\ ) {) Tj
1 0 0 1 40 81 Tm (     var result = this.update\ (id, \ (function\ (col, now\ ) {) Tj
1 0 0 1 40 70 Tm (         var patch = {};) Tj
1 0 0 1 40 59 Tm (         patch[col] = now;) Tj
1 0 0 1 40 48 Tm (         return patch;) Tj
ET
endstream
endobj
140 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 139 0
endobj
141 0 obj
<< /Length 276 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (    }\)\ (this._config.deletedAtColumn, Util.nowIso\ (\)\)\);) Tj
1 0 0 1 40 730 Tm (    this._audit.log\ ('remove', this._tableName, { id: id }\);) Tj
1 0 0 1 40 719 Tm (    return result;) Tj
1 0 0 1 40 708 Tm ( }) Tj
1 0 0 1 40 697 Tm (}) Tj
ET
endstream
endobj
142 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 141 0
endobj
143 0 obj
<< /Length 1442 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: 27_TransactionManager.gs) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (/**) Tj
1 0 0 1 40 708 Tm ( * Runs operations under ScriptLock for coarse transaction semantics.) Tj
1 0 0 1 40 697 Tm ( */) Tj
1 0 0 1 40 686 Tm (class TransactionManager {) Tj
1 0 0 1 40 675 Tm ( /**) Tj
1 0 0 1 40 664 Tm ( * @param {number=} lockTimeoutMs) Tj
1 0 0 1 40 653 Tm ( */) Tj
1 0 0 1 40 642 Tm ( constructor\ (lockTimeoutMs\ ) {) Tj
1 0 0 1 40 631 Tm (     this._lockTimeoutMs = lockTimeoutMs || 30000;) Tj
1 0 0 1 40 620 Tm ( }) Tj
1 0 0 1 40 609 Tm ( ) Tj

```

```

1 0 0 1 40 598 Tm ( /**) Tj
1 0 0 1 40 587 Tm ( * Executes callback while holding a script lock.) Tj
1 0 0 1 40 576 Tm ( * @template T) Tj
1 0 0 1 40 565 Tm ( * @param {function\\): T} callback) Tj
1 0 0 1 40 554 Tm ( * @return {T}) Tj
1 0 0 1 40 543 Tm ( */) Tj
1 0 0 1 40 532 Tm ( run\\(callback\\) {}) Tj
1 0 0 1 40 521 Tm ( var lock = LockService.getScriptLock\\(\\);) Tj
1 0 0 1 40 510 Tm ( if \\(!lock.tryLock\\(this._lockTimeoutMs\\)\\) {}) Tj
1 0 0 1 40 499 Tm ( throw new TransactionError\\('Failed to acquire script lock within timeout.'\\);) Tj
1 0 0 1 40 488 Tm ( }) Tj
1 0 0 1 40 477 Tm ( try {} Tj
1 0 0 1 40 466 Tm ( return callback\\(\\);) Tj
1 0 0 1 40 455 Tm ( } finally {} Tj
1 0 0 1 40 444 Tm ( lock.releaseLock\\(\\);) Tj
1 0 0 1 40 433 Tm ( }) Tj
1 0 0 1 40 422 Tm ( }) Tj
1 0 0 1 40 411 Tm (}) Tj
ET
endstream
endobj
144 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 143 0
endobj
145 0 obj
<< /Length 1436 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: 28_AuditLogger.gs) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (/**) Tj
1 0 0 1 40 708 Tm ( * Minimal pluggable audit logger.) Tj
1 0 0 1 40 697 Tm ( */) Tj
1 0 0 1 40 686 Tm (class AuditLogger {}) Tj
1 0 0 1 40 675 Tm ( /**) Tj
1 0 0 1 40 664 Tm ( * @param {Object=} options) Tj
1 0 0 1 40 653 Tm ( * @param {Function=} options.sink Optional custom sink function.) Tj
1 0 0 1 40 642 Tm ( */) Tj
1 0 0 1 40 631 Tm ( constructor\\(options\\) {}) Tj
1 0 0 1 40 620 Tm ( var opts = options || {};) Tj
1 0 0 1 40 609 Tm ( this._sink = opts.sink || function\\(entry\\) {}) Tj
1 0 0 1 40 598 Tm ( Logger.log\\(JSON.stringify\\(entry\\)\\);) Tj
1 0 0 1 40 587 Tm ( );) Tj
1 0 0 1 40 576 Tm ( }) Tj
1 0 0 1 40 565 Tm ( ) Tj
1 0 0 1 40 554 Tm ( /**) Tj
1 0 0 1 40 543 Tm ( * Emits an audit entry.) Tj
1 0 0 1 40 532 Tm ( * @param {string} action) Tj
1 0 0 1 40 521 Tm ( * @param {string} tableName) Tj
1 0 0 1 40 510 Tm ( * @param {Object=} metadata) Tj
1 0 0 1 40 499 Tm ( */) Tj
1 0 0 1 40 488 Tm ( log\\(action, tableName, metadata\\) {}) Tj
1 0 0 1 40 477 Tm ( this._sink\\({} Tj
1 0 0 1 40 466 Tm ( at: Util.nowIso\\(\\),) Tj
1 0 0 1 40 455 Tm ( action: action,) Tj
1 0 0 1 40 444 Tm ( table: tableName,) Tj
1 0 0 1 40 433 Tm ( metadata: Util.clone\\(metadata || {}\\)) Tj
1 0 0 1 40 422 Tm ( );) Tj
1 0 0 1 40 411 Tm ( }) Tj
1 0 0 1 40 400 Tm (}) Tj
ET
endstream
endobj
146 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 145 0
endobj
147 0 obj
<< /Length 3000 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: 29_DbClient.gs) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (/**) Tj
1 0 0 1 40 708 Tm ( * Main entrypoint for interacting with sheet-backed tables.) Tj
1 0 0 1 40 697 Tm ( */) Tj
1 0 0 1 40 686 Tm (class DbClient {}) Tj
1 0 0 1 40 675 Tm ( /**) Tj
1 0 0 1 40 664 Tm ( * @param {Config=} config) Tj

```

```

1 0 0 1 40 653 Tm (    */) Tj
1 0 0 1 40 642 Tm (    constructor\(\config\) {}) Tj
1 0 0 1 40 631 Tm (        this._config = config || new Config\(\);) Tj
1 0 0 1 40 620 Tm (        this._schemas = new SchemaRegistry\(\);) Tj
1 0 0 1 40 609 Tm (        this._sheets = new SheetsGateway\(\this._config\);) Tj
1 0 0 1 40 598 Tm (        this._audit = new AuditLogger\(\);) Tj
1 0 0 1 40 587 Tm (        this._tx = new TransactionManager\(\this._config.lockTimeoutMs\);) Tj
1 0 0 1 40 576 Tm (    }) Tj
1 0 0 1 40 565 Tm (    Tj
1 0 0 1 40 554 Tm (    /**) Tj
1 0 0 1 40 543 Tm (        * Registers a table schema.) Tj
1 0 0 1 40 532 Tm (        * @param {string} tableName) Tj
1 0 0 1 40 521 Tm (        * @param {Object} schema) Tj
1 0 0 1 40 510 Tm (        * @return {DbClient}) Tj
1 0 0 1 40 499 Tm (        */) Tj
1 0 0 1 40 488 Tm (    registerTable\(\tableName, schema\) {}) Tj
1 0 0 1 40 477 Tm (        this._schemas.register\(\tableName, schema\);) Tj
1 0 0 1 40 466 Tm (        this._sheets.ensureTable\(\tableName, schema.columns\);) Tj
1 0 0 1 40 455 Tm (        return this;) Tj
1 0 0 1 40 444 Tm (    }) Tj
1 0 0 1 40 433 Tm (    Tj
1 0 0 1 40 422 Tm (    /**) Tj
1 0 0 1 40 411 Tm (        * Returns a table repository.) Tj
1 0 0 1 40 400 Tm (        * @param {string} tableName) Tj
1 0 0 1 40 389 Tm (        * @return {TableRepository}) Tj
1 0 0 1 40 378 Tm (        */) Tj
1 0 0 1 40 367 Tm (    table\(\tableName\) {}) Tj
1 0 0 1 40 356 Tm (        this._schemas.get\(\tableName\);) Tj
1 0 0 1 40 345 Tm (        return new TableRepository\() Tj
1 0 0 1 40 334 Tm (            tableName,) Tj
1 0 0 1 40 323 Tm (            this._config,) Tj
1 0 0 1 40 312 Tm (            this._schemas,) Tj
1 0 0 1 40 301 Tm (            this._sheets,) Tj
1 0 0 1 40 290 Tm (            this._audit) Tj
1 0 0 1 40 279 Tm (        \);) Tj
1 0 0 1 40 268 Tm (    }) Tj
1 0 0 1 40 257 Tm (    Tj
1 0 0 1 40 246 Tm (    /**) Tj
1 0 0 1 40 235 Tm (        * Gets or registers a table schema.) Tj
1 0 0 1 40 224 Tm (        *) Tj
1 0 0 1 40 213 Tm (        * - When 'schema' is provided, registers/overwrites the schema and ensures the table.) Tj
1 0 0 1 40 202 Tm (        * - When 'schema' is omitted, returns the current schema.) Tj
1 0 0 1 40 191 Tm (        *) Tj
1 0 0 1 40 180 Tm (        * @param {string} tableName) Tj
1 0 0 1 40 169 Tm (        * @param {Object=} schema) Tj
1 0 0 1 40 158 Tm (        * @return {Object|DbClient}) Tj
1 0 0 1 40 147 Tm (        */) Tj
1 0 0 1 40 136 Tm (    schema\(\tableName, schema\) {}) Tj
1 0 0 1 40 125 Tm (        if \(\schema\) {}) Tj
1 0 0 1 40 114 Tm (            this.registerTable\(\tableName, schema\);) Tj
1 0 0 1 40 103 Tm (            return this;) Tj
1 0 0 1 40 92 Tm (        }) Tj
1 0 0 1 40 81 Tm (        return this._schemas.get\(\tableName\);) Tj
1 0 0 1 40 70 Tm (    }) Tj
1 0 0 1 40 59 Tm (    Tj
1 0 0 1 40 48 Tm (    /**) Tj
ET
endstream
endobj
148 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 147 0
endobj
149 0 obj
<< /Length 1054 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (    * Writes an audit entry through the configured audit logger.) Tj
1 0 0 1 40 730 Tm (    * @param {string} action) Tj
1 0 0 1 40 719 Tm (    * @param {string} tableName) Tj
1 0 0 1 40 708 Tm (    * @param {Object=} metadata) Tj
1 0 0 1 40 697 Tm (    */) Tj
1 0 0 1 40 686 Tm (    audit\(\action, tableName, metadata\) {}) Tj
1 0 0 1 40 675 Tm (        this._audit.log\(\action, tableName, metadata || {} \);) Tj
1 0 0 1 40 664 Tm (    }) Tj
1 0 0 1 40 653 Tm (    Tj
1 0 0 1 40 642 Tm (    /**) Tj
1 0 0 1 40 631 Tm (        * Runs a set of operations under lock.) Tj
1 0 0 1 40 620 Tm (        * @template T) Tj
1 0 0 1 40 609 Tm (        * @param {function\(\DbClient\): T} callback) Tj

```



```

1 0 0 1 40 81 Tm (      spreadsheetMode: client._config && client._config.spreadsheetId ? 'by_id' : 'active') Tj
1 0 0 1 40 70 Tm (      }) Tj
1 0 0 1 40 59 Tm (      });) Tj
1 0 0 1 40 48 Tm (    } catch \(\err\) {}) Tj
ET
endstream
endobj
152 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 151 0
endobj
153 0 obj
<< /Length 3130 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (      return {}) Tj
1 0 0 1 40 730 Tm (      ok: false,) Tj
1 0 0 1 40 719 Tm (      version: SHEET_DB_VERSION,) Tj
1 0 0 1 40 708 Tm (      timestamp: new Date\(\).toISOString\(\),) Tj
1 0 0 1 40 697 Tm (      details: {}) Tj
1 0 0 1 40 686 Tm (      code: err.code || 'HEALTH_CHECK_FAILED',) Tj
1 0 0 1 40 675 Tm (      message: err.message || 'Unknown error.') Tj
1 0 0 1 40 664 Tm (      }) Tj
1 0 0 1 40 653 Tm (      });) Tj
1 0 0 1 40 642 Tm (    }) Tj
1 0 0 1 40 631 Tm (  }) Tj
1 0 0 1 40 620 Tm (  ) Tj
1 0 0 1 40 609 Tm ( /**) Tj
1 0 0 1 40 598 Tm ( * Creates a client using Script Properties with optional explicit overrides.) Tj
1 0 0 1 40 587 Tm ( * Precedence: explicit options > script properties > defaults.) Tj
1 0 0 1 40 576 Tm ( * @param {Object=} options) Tj
1 0 0 1 40 565 Tm ( * @return {DbClient}) Tj
1 0 0 1 40 554 Tm ( */) Tj
1 0 0 1 40 543 Tm (function _sheetDbCreateClientFromScriptProperties\(\options\) {}) Tj
1 0 0 1 40 532 Tm (  var merged = mergeClientOptionsWithScriptProperties\(\options || {}\);) Tj
1 0 0 1 40 521 Tm (  return _sheetDbCreateClient\(\merged\);) Tj
1 0 0 1 40 510 Tm (}) Tj
1 0 0 1 40 499 Tm ( ) Tj
1 0 0 1 40 488 Tm ( /**) Tj
1 0 0 1 40 477 Tm ( * Public Apps Script library namespace.) Tj
1 0 0 1 40 466 Tm ( *) Tj
1 0 0 1 40 455 Tm ( * Consumers should call library exports through this object:) Tj
1 0 0 1 40 444 Tm ( * 'LibraryIdentifier.SheetDb.createClient\(...\)'.) Tj
1 0 0 1 40 433 Tm ( *) Tj
1 0 0 1 40 422 Tm ( * @type {{}) Tj
1 0 0 1 40 411 Tm ( *   createClient: function\(\Object=\): DbClient,) Tj
1 0 0 1 40 400 Tm ( *   createClientFromScriptProperties: function\(\Object=\): DbClient,) Tj
1 0 0 1 40 389 Tm ( *   bootstrap: function\(\Object=\): DbClient,) Tj
1 0 0 1 40 378 Tm ( *   healthCheck: function\(\): Object,) Tj
1 0 0 1 40 367 Tm ( *   version: string) Tj
1 0 0 1 40 356 Tm ( * }) Tj
1 0 0 1 40 345 Tm ( */) Tj
1 0 0 1 40 334 Tm (var SheetDb = {}) Tj
1 0 0 1 40 323 Tm (  createClient: function\(\options\) {}) Tj
1 0 0 1 40 312 Tm (  return _sheetDbCreateClient\(\options\);) Tj
1 0 0 1 40 301 Tm (  },) Tj
1 0 0 1 40 290 Tm ( ) Tj
1 0 0 1 40 279 Tm (  createClientFromScriptProperties: function\(\options\) {}) Tj
1 0 0 1 40 268 Tm (  return _sheetDbCreateClientFromScriptProperties\(\options\);) Tj
1 0 0 1 40 257 Tm (  },) Tj
1 0 0 1 40 246 Tm ( ) Tj
1 0 0 1 40 235 Tm (  bootstrap: function\(\options\) {}) Tj
1 0 0 1 40 224 Tm (  return _sheetDbBootstrap\(\options\);) Tj
1 0 0 1 40 213 Tm (  },) Tj
1 0 0 1 40 202 Tm ( ) Tj
1 0 0 1 40 191 Tm (  healthCheck: function\(\) {}) Tj
1 0 0 1 40 180 Tm (  return _sheetDbHealthCheck\(\);) Tj
1 0 0 1 40 169 Tm (  },) Tj
1 0 0 1 40 158 Tm ( ) Tj
1 0 0 1 40 147 Tm (  version: SHEET_DB_VERSION) Tj
1 0 0 1 40 136 Tm (});) Tj
1 0 0 1 40 125 Tm ( ) Tj
1 0 0 1 40 114 Tm ( /**) Tj
1 0 0 1 40 103 Tm ( * Backward-compatible factory export.) Tj
1 0 0 1 40 92 Tm ( * @deprecated Use 'SheetDb.createClient' instead.) Tj
1 0 0 1 40 81 Tm ( * @param {Object=} options) Tj
1 0 0 1 40 70 Tm ( * @return {DbClient}) Tj
1 0 0 1 40 59 Tm ( */) Tj
1 0 0 1 40 48 Tm (function createSheetDbClient\(\options\) {}) Tj
ET

```

```

endstream
endobj
154 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 153 0
endobj
155 0 obj
<< /Length 104 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm ( return SheetDb.createClient\(options\);) Tj
1 0 0 1 40 730 Tm (}) Tj
ET
endstream
endobj
156 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 155 0
endobj
157 0 obj
<< /Length 3183 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: code.gs) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (/***** Tj
1 0 0 1 40 708 Tm ( * RWR LMS â Slack Web App Anchor \(\thin host shell\)) Tj
1 0 0 1 40 697 Tm ( *****/) Tj
1 0 0 1 40 686 Tm ( ) Tj
1 0 0 1 40 675 Tm (const CONFIG = Object.freeze({) Tj
1 0 0 1 40 664 Tm ( ACTOR_SYSTEM: 'system') Tj
1 0 0 1 40 653 Tm (}\);} Tj
1 0 0 1 40 642 Tm ( ) Tj
1 0 0 1 40 631 Tm (function doGet\(\) {) Tj
1 0 0 1 40 620 Tm ( return textResponse_\('alive'\);} Tj
1 0 0 1 40 609 Tm (}) Tj
1 0 0 1 40 598 Tm ( ) Tj
1 0 0 1 40 587 Tm (function doPost\(\e\) {) Tj
1 0 0 1 40 576 Tm ( try {) Tj
1 0 0 1 40 565 Tm ( var config = getHostConfig_\(\);} Tj
1 0 0 1 40 554 Tm ( var deps = getHostDependencies_\(config\);} Tj
1 0 0 1 40 543 Tm ( var parsed = SlackPayloadParser.parse\(\e\);} Tj
1 0 0 1 40 532 Tm ( var verified = SlackSecurity.verify\(\parsed, config\);} Tj
1 0 0 1 40 521 Tm ( ) Tj
1 0 0 1 40 510 Tm ( if \(!verified.ok\) {) Tj
1 0 0 1 40 499 Tm ( deps.audit\('request_denied', {) Tj
1 0 0 1 40 488 Tm ( code: verified.code,) Tj
1 0 0 1 40 477 Tm ( routeType: parsed.routeType) Tj
1 0 0 1 40 466 Tm ( }\);} Tj
1 0 0 1 40 455 Tm ( return jsonResponse_\({ ok: false, error: verified.code, message: verified.message }\);} Tj
1 0 0 1 40 444 Tm ( }) Tj
1 0 0 1 40 433 Tm ( ) Tj
1 0 0 1 40 422 Tm ( var routed = SlackRouter.route\(\parsed, deps\);} Tj
1 0 0 1 40 411 Tm ( deps.audit\('request_routed', {) Tj
1 0 0 1 40 400 Tm ( routeType: parsed.routeType,) Tj
1 0 0 1 40 389 Tm ( ok: !!routed.ok,) Tj
1 0 0 1 40 378 Tm ( code: routed.code || '') Tj
1 0 0 1 40 367 Tm ( }\);} Tj
1 0 0 1 40 356 Tm ( ) Tj
1 0 0 1 40 345 Tm ( return jsonResponse_\(routed.response || { ok: false, error: 'missing_response' }\);} Tj
1 0 0 1 40 334 Tm ( } catch \(\err\) {) Tj
1 0 0 1 40 323 Tm ( try {) Tj
1 0 0 1 40 312 Tm ( logAuditExact_\({) Tj
1 0 0 1 40 301 Tm ( actor: CONFIG.ACTOR_SYSTEM,) Tj
1 0 0 1 40 290 Tm ( action: 'doPost_error',) Tj
1 0 0 1 40 279 Tm ( status: 'error',) Tj
1 0 0 1 40 268 Tm ( message: safeError_\(err\)) Tj
1 0 0 1 40 257 Tm ( }\);} Tj
1 0 0 1 40 246 Tm ( } catch \(\logErr\) {) Tj
1 0 0 1 40 235 Tm ( ) Tj
1 0 0 1 40 224 Tm ( return jsonResponse_\({ ok: false, error: 'host_error', message: 'Something went wrong in the S)
1 0 0 1 40 213 Tm (lack handler.' }\);} Tj
1 0 0 1 40 202 Tm ( }) Tj
1 0 0 1 40 191 Tm (}) Tj
1 0 0 1 40 180 Tm ( ) Tj
1 0 0 1 40 169 Tm (function getHostConfig_\(\) {) Tj
1 0 0 1 40 158 Tm ( return ConfigBootstrap.load\(\);} Tj
1 0 0 1 40 147 Tm (}) Tj
1 0 0 1 40 136 Tm ( ) Tj
1 0 0 1 40 125 Tm (function getHostDependencies_\(config\) {) Tj

```



```

1 0 0 1 40 114 Tm ( return createHostDependencies\(\config\);) Tj
1 0 0 1 40 103 Tm (}) Tj
1 0 0 1 40 92 Tm ( ) Tj
1 0 0 1 40 81 Tm (function textResponse_\(text\) {) Tj
1 0 0 1 40 70 Tm ( return ContentService.createTextOutput\(\String\(\text || '\`)\)\.setMimeType\(\ContentService.MimeType.
1 0 0 1 40 59 Tm (EXT\);) Tj
1 0 0 1 40 48 Tm (}) Tj
ET
endstream
endobj
158 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 157 0
endobj
159 0 obj
<< /Length 1427 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm ( ) Tj
1 0 0 1 40 730 Tm (function jsonResponse_\(obj\) {) Tj
1 0 0 1 40 719 Tm ( return ContentService.createTextOutput\(\JSON.stringify\(\obj || {}\)\)\.setMimeType\(\ContentService.M
1 0 0 1 40 708 Tm (eType.JSON\);) Tj
1 0 0 1 40 697 Tm (}) Tj
1 0 0 1 40 686 Tm ( ) Tj
1 0 0 1 40 675 Tm (function logAuditExact_\(entry\) {) Tj
1 0 0 1 40 664 Tm ( logger.log\(\JSON.stringify\(\sanitizeObjectForLog_\(entry || {}\)\)\);) Tj
1 0 0 1 40 653 Tm (}) Tj
1 0 0 1 40 642 Tm ( ) Tj
1 0 0 1 40 631 Tm (function safeError_\(err\) {) Tj
1 0 0 1 40 620 Tm ( return \(\err && err.message\) ? String\(\err.message\) : String\(\err\);) Tj
1 0 0 1 40 609 Tm (}) Tj
1 0 0 1 40 598 Tm ( ) Tj
1 0 0 1 40 587 Tm (function sanitizeObjectForLog_\(obj\) {) Tj
1 0 0 1 40 576 Tm ( var clean = {};) Tj
1 0 0 1 40 565 Tm ( Object.keys\(\obj || {}\)\.forEach\(\function\(\key\) {) Tj
1 0 0 1 40 554 Tm ( var lower = String\(\key\)\.toLowerCase\(\);) Tj
1 0 0 1 40 543 Tm ( if \(\lower.indexOf\('token'\) !== -1 || lower.indexOf\('secret'\) !== -1 || lower.indexOf\('autho
1 0 0 1 40 532 Tm (ation'\) !== -1\) {) Tj
1 0 0 1 40 521 Tm ( clean[key] = '[REDACTED]';) Tj
1 0 0 1 40 510 Tm ( return;) Tj
1 0 0 1 40 499 Tm ( }) Tj
1 0 0 1 40 488 Tm ( clean[key] = String\(\obj[key] == null ? '' : obj[key]\);) Tj
1 0 0 1 40 477 Tm ( });) Tj
1 0 0 1 40 466 Tm ( return clean;) Tj
1 0 0 1 40 455 Tm (}) Tj
ET
endstream
endobj
160 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 159 0
endobj
161 0 obj
<< /Length 3145 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: sheet_db_lib/00_SheetDb.gs) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (/**) Tj
1 0 0 1 40 708 Tm ( * @fileoverview Public, library-visible exports for SheetDb.) Tj
1 0 0 1 40 697 Tm ( */) Tj
1 0 0 1 40 686 Tm ( ) Tj
1 0 0 1 40 675 Tm (/**) Tj
1 0 0 1 40 664 Tm ( * Library semantic version.) Tj
1 0 0 1 40 653 Tm ( *) Tj
1 0 0 1 40 642 Tm ( * NOTE: Keep this in sync with release tags when publishing as an Apps Script library.) Tj
1 0 0 1 40 631 Tm ( * @type {string}) Tj
1 0 0 1 40 620 Tm ( */) Tj
1 0 0 1 40 609 Tm (var SHEET_DB_VERSION = '0.1.0';) Tj
1 0 0 1 40 598 Tm ( ) Tj
1 0 0 1 40 587 Tm (/**) Tj
1 0 0 1 40 576 Tm ( * Creates a new SheetDb client instance.) Tj
1 0 0 1 40 565 Tm ( *) Tj
1 0 0 1 40 554 Tm ( * @param {Object=} options Client configuration options.) Tj
1 0 0 1 40 543 Tm ( * @return {DbClient}) Tj
1 0 0 1 40 532 Tm ( */) Tj
1 0 0 1 40 521 Tm (function _sheetDbCreateClient\(\options\) {) Tj
1 0 0 1 40 510 Tm ( return new DbClient\(\new Config\(\options || {}\)\);) Tj
1 0 0 1 40 499 Tm (}) Tj
1 0 0 1 40 488 Tm ( ) Tj

```

```

1 0 0 1 40 477 Tm (/**) Tj
1 0 0 1 40 466 Tm ( * Bootstraps a client with one or more table schemas.) Tj
1 0 0 1 40 455 Tm ( *) Tj
1 0 0 1 40 444 Tm ( * @param {Object=} options) Tj
1 0 0 1 40 433 Tm ( * @param {Array<{name: string, schema: Object}>=} options.tables Tables to register.) Tj
1 0 0 1 40 422 Tm ( * @param {Object=} options.clientOptions Configuration passed to 'createClient'.) Tj
1 0 0 1 40 411 Tm ( * @return {DbClient}) Tj
1 0 0 1 40 400 Tm ( */) Tj
1 0 0 1 40 389 Tm (function _sheetDbBootstrap\(\options\) {}) Tj
1 0 0 1 40 378 Tm ( var opts = options || {};) Tj
1 0 0 1 40 367 Tm ( var client = _sheetDbCreateClient\(\opts.clientOptions || {}\);) Tj
1 0 0 1 40 356 Tm ( var tables = opts.tables || [];) Tj
1 0 0 1 40 345 Tm ( ) Tj
1 0 0 1 40 334 Tm ( tables.forEach\((function\(\table\) {}) Tj
1 0 0 1 40 323 Tm ( if \(\table && table.name && table.schema\) {}) Tj
1 0 0 1 40 312 Tm ( client.registerTable\(\table.name, table.schema\);) Tj
1 0 0 1 40 301 Tm ( }) Tj
1 0 0 1 40 290 Tm ( }\);) Tj
1 0 0 1 40 279 Tm ( ) Tj
1 0 0 1 40 268 Tm ( return client;) Tj
1 0 0 1 40 257 Tm (}) Tj
1 0 0 1 40 246 Tm ( ) Tj
1 0 0 1 40 235 Tm (/**) Tj
1 0 0 1 40 224 Tm ( * Basic health check that can be called by consuming projects.) Tj
1 0 0 1 40 213 Tm ( *) Tj
1 0 0 1 40 202 Tm ( * @return {{ok: boolean, version: string, timestamp: string, details: Object}}) Tj
1 0 0 1 40 191 Tm ( */) Tj
1 0 0 1 40 180 Tm (function _sheetDbHealthCheck\(\) {}) Tj
1 0 0 1 40 169 Tm ( try {} Tj
1 0 0 1 40 158 Tm ( var client = _sheetDbCreateClient\(\);) Tj
1 0 0 1 40 147 Tm ( return {} Tj
1 0 0 1 40 136 Tm ( ok: !!client,) Tj
1 0 0 1 40 125 Tm ( version: SHEET_DB_VERSION,) Tj
1 0 0 1 40 114 Tm ( timestamp: Util.nowIso\(\),) Tj
1 0 0 1 40 103 Tm ( details: {} Tj
1 0 0 1 40 92 Tm ( clientReady: true,) Tj
1 0 0 1 40 81 Tm ( spreadsheetMode: client._config && client._config.spreadsheetId ? 'by_id' : 'active') Tj
1 0 0 1 40 70 Tm ( }) Tj
1 0 0 1 40 59 Tm ( };) Tj
1 0 0 1 40 48 Tm ( } catch \(\err\) {} Tj
ET
endstream
endobj
162 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 161 0
endobj
163 0 obj
<< /Length 3130 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm ( return {} Tj
1 0 0 1 40 730 Tm ( ok: false,) Tj
1 0 0 1 40 719 Tm ( version: SHEET_DB_VERSION,) Tj
1 0 0 1 40 708 Tm ( timestamp: new Date\(\).toISOString\(\),) Tj
1 0 0 1 40 697 Tm ( details: {} Tj
1 0 0 1 40 686 Tm ( code: err.code || 'HEALTH_CHECK_FAILED',) Tj
1 0 0 1 40 675 Tm ( message: err.message || 'Unknown error.') Tj
1 0 0 1 40 664 Tm ( }) Tj
1 0 0 1 40 653 Tm ( };) Tj
1 0 0 1 40 642 Tm ( }) Tj
1 0 0 1 40 631 Tm (}) Tj
1 0 0 1 40 620 Tm ( ) Tj
1 0 0 1 40 609 Tm (/**) Tj
1 0 0 1 40 598 Tm ( * Creates a client using Script Properties with optional explicit overrides.) Tj
1 0 0 1 40 587 Tm ( * Precedence: explicit options > script properties > defaults.) Tj
1 0 0 1 40 576 Tm ( * @param {Object=} options) Tj
1 0 0 1 40 565 Tm ( * @return {DbClient}) Tj
1 0 0 1 40 554 Tm ( */) Tj
1 0 0 1 40 543 Tm (function _sheetDbCreateClientFromScriptProperties\(\options\) {}) Tj
1 0 0 1 40 532 Tm ( var merged = mergeClientOptionsWithScriptProperties\(\options || {}\);) Tj
1 0 0 1 40 521 Tm ( return _sheetDbCreateClient\(\merged\);) Tj
1 0 0 1 40 510 Tm (}) Tj
1 0 0 1 40 499 Tm ( ) Tj
1 0 0 1 40 488 Tm (/**) Tj
1 0 0 1 40 477 Tm ( * Public Apps Script library namespace.) Tj
1 0 0 1 40 466 Tm ( *) Tj
1 0 0 1 40 455 Tm ( * Consumers should call library exports through this object:) Tj
1 0 0 1 40 444 Tm ( * 'LibraryIdentifier.SheetDb.createClient\(...\)'.) Tj
1 0 0 1 40 433 Tm ( *) Tj

```



```

1 0 0 1 40 510 Tm ( * @return {DbClient}) Tj
1 0 0 1 40 499 Tm ( */) Tj
1 0 0 1 40 488 Tm ( registerTable\(tableName, schema\) {}) Tj
1 0 0 1 40 477 Tm ( this._schemas.register\(tableName, schema\);) Tj
1 0 0 1 40 466 Tm ( this._sheets.ensureTable\(tableName, schema.columns\);) Tj
1 0 0 1 40 455 Tm ( return this;) Tj
1 0 0 1 40 444 Tm ( }) Tj
1 0 0 1 40 433 Tm ( ) Tj
1 0 0 1 40 422 Tm ( /**) Tj
1 0 0 1 40 411 Tm ( * Returns a table repository.) Tj
1 0 0 1 40 400 Tm ( * @param {string} tableName) Tj
1 0 0 1 40 389 Tm ( * @return {TableRepository}) Tj
1 0 0 1 40 378 Tm ( */) Tj
1 0 0 1 40 367 Tm ( table\(tableName\) {}) Tj
1 0 0 1 40 356 Tm ( this._schemas.get\(tableName\);) Tj
1 0 0 1 40 345 Tm ( return new TableRepository\() Tj
1 0 0 1 40 334 Tm ( tableName,) Tj
1 0 0 1 40 323 Tm ( this._config,) Tj
1 0 0 1 40 312 Tm ( this._schemas,) Tj
1 0 0 1 40 301 Tm ( this._sheets,) Tj
1 0 0 1 40 290 Tm ( this._audit) Tj
1 0 0 1 40 279 Tm ( \; ) Tj
1 0 0 1 40 268 Tm ( }) Tj
1 0 0 1 40 257 Tm ( ) Tj
1 0 0 1 40 246 Tm ( /**) Tj
1 0 0 1 40 235 Tm ( * Gets or registers a table schema.) Tj
1 0 0 1 40 224 Tm ( *) Tj
1 0 0 1 40 213 Tm ( * - When 'schema' is provided, registers/overwrites the schema and ensures the table.) Tj
1 0 0 1 40 202 Tm ( * - When 'schema' is omitted, returns the current schema.) Tj
1 0 0 1 40 191 Tm ( *) Tj
1 0 0 1 40 180 Tm ( * @param {string} tableName) Tj
1 0 0 1 40 169 Tm ( * @param {Object=} schema) Tj
1 0 0 1 40 158 Tm ( * @return {Object|DbClient}) Tj
1 0 0 1 40 147 Tm ( */) Tj
1 0 0 1 40 136 Tm ( schema\(tableName, schema\) {}) Tj
1 0 0 1 40 125 Tm ( if \(schema\) {}) Tj
1 0 0 1 40 114 Tm ( this.registerTable\(tableName, schema\);) Tj
1 0 0 1 40 103 Tm ( return this;) Tj
1 0 0 1 40 92 Tm ( }) Tj
1 0 0 1 40 81 Tm ( return this._schemas.get\(tableName\);) Tj
1 0 0 1 40 70 Tm ( }) Tj
1 0 0 1 40 59 Tm ( ) Tj
1 0 0 1 40 48 Tm ( /**) Tj
ET
endstream
endobj
168 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 167 0
endobj
169 0 obj
<< /Length 1054 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm ( * Writes an audit entry through the configured audit logger.) Tj
1 0 0 1 40 730 Tm ( * @param {string} action) Tj
1 0 0 1 40 719 Tm ( * @param {string} tableName) Tj
1 0 0 1 40 708 Tm ( * @param {Object=} metadata) Tj
1 0 0 1 40 697 Tm ( */) Tj
1 0 0 1 40 686 Tm ( audit\(action, tableName, metadata\) {}) Tj
1 0 0 1 40 675 Tm ( this._audit.log\(action, tableName, metadata || {} \; ) Tj
1 0 0 1 40 664 Tm ( }) Tj
1 0 0 1 40 653 Tm ( ) Tj
1 0 0 1 40 642 Tm ( /**) Tj
1 0 0 1 40 631 Tm ( * Runs a set of operations under lock.) Tj
1 0 0 1 40 620 Tm ( * @template T) Tj
1 0 0 1 40 609 Tm ( * @param {function(DbClient): T} callback) Tj
1 0 0 1 40 598 Tm ( * @return {T}) Tj
1 0 0 1 40 587 Tm ( */) Tj
1 0 0 1 40 576 Tm ( transaction\(callback\) {}) Tj
1 0 0 1 40 565 Tm ( var self = this;) Tj
1 0 0 1 40 554 Tm ( return this._tx.run\(function\(\) {}) Tj
1 0 0 1 40 543 Tm ( return callback\(self \; ) Tj
1 0 0 1 40 532 Tm ( } \; ) Tj
1 0 0 1 40 521 Tm ( }) Tj
1 0 0 1 40 510 Tm ( }) Tj
ET
endstream
endobj
170 0 obj

```

```

<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 169 0
endobj
171 0 obj
<< /Length 3089 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: sheet_db_lib/02_Config.gs) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (/**) Tj
1 0 0 1 40 708 Tm ( * Immutable runtime configuration for the Sheet DB client.) Tj
1 0 0 1 40 697 Tm ( */) Tj
1 0 0 1 40 686 Tm (class Config {} Tj
1 0 0 1 40 675 Tm ( /**) Tj
1 0 0 1 40 664 Tm ( * @param {Object=} options) Tj
1 0 0 1 40 653 Tm ( * @param {string=} options.spreadsheetId Spreadsheet ID; defaults to active spreadsheet.) Tj
1 0 0 1 40 642 Tm ( * @param {string=} options.idColumn Record primary key column name.) Tj
1 0 0 1 40 631 Tm ( * @param {string=} options.createdAtColumn Created timestamp column name.) Tj
1 0 0 1 40 620 Tm ( * @param {string=} options.updatedAtColumn Updated timestamp column name.) Tj
1 0 0 1 40 609 Tm ( * @param {string=} options.deletedAtColumn Soft-delete timestamp column name.) Tj
1 0 0 1 40 598 Tm ( * @param {number=} options.lockTimeoutMs Lock acquisition timeout for transactions.) Tj
1 0 0 1 40 587 Tm ( */) Tj
1 0 0 1 40 576 Tm ( constructor\(options\) {} Tj
1 0 0 1 40 565 Tm ( var opts = options || {};) Tj
1 0 0 1 40 554 Tm ( this.spreadsheetId = opts.spreadsheetId || '';) Tj
1 0 0 1 40 543 Tm ( this.idColumn = opts.idColumn || 'id';) Tj
1 0 0 1 40 532 Tm ( this.createdAtColumn = opts.createdAtColumn || 'createdAt';) Tj
1 0 0 1 40 521 Tm ( this.updatedAtColumn = opts.updatedAtColumn || 'updatedAt';) Tj
1 0 0 1 40 510 Tm ( this.deletedAtColumn = opts.deletedAtColumn || 'deletedAt';) Tj
1 0 0 1 40 499 Tm ( this.lockTimeoutMs = Util.isNil\ (opts.lockTimeoutMs\) ? 30000 : Number\ (opts.lockTimeoutMs\);) Tj
1 0 0 1 40 488 Tm ( this.validate\ ();) Tj
1 0 0 1 40 477 Tm ( Object.freeze\ (this\);) Tj
1 0 0 1 40 466 Tm ( }) Tj
1 0 0 1 40 455 Tm ( ) Tj
1 0 0 1 40 444 Tm ( /**) Tj
1 0 0 1 40 433 Tm ( * @return {void}) Tj
1 0 0 1 40 422 Tm ( */) Tj
1 0 0 1 40 411 Tm ( validate\ (\) {} Tj
1 0 0 1 40 400 Tm ( var columns = [] Tj
1 0 0 1 40 389 Tm ( this.idColumn,) Tj
1 0 0 1 40 378 Tm ( this.createdAtColumn,) Tj
1 0 0 1 40 367 Tm ( this.updatedAtColumn,) Tj
1 0 0 1 40 356 Tm ( this.deletedAtColumn) Tj
1 0 0 1 40 345 Tm ( ];) Tj
1 0 0 1 40 334 Tm ( var unique = {};) Tj
1 0 0 1 40 323 Tm ( columns.forEach\ (function\ (column\) {} Tj
1 0 0 1 40 312 Tm ( if \ (!column || typeof column !== 'string\) {} Tj
1 0 0 1 40 301 Tm ( throw new ConfigError\ ('System column names must be non-empty strings.\ ');) Tj
1 0 0 1 40 290 Tm ( }) Tj
1 0 0 1 40 279 Tm ( if \ (unique[column]\) {} Tj
1 0 0 1 40 268 Tm ( throw new ConfigError\ ('System column names must be unique. Duplicate: ' + column\);) Tj
1 0 0 1 40 257 Tm ( }) Tj
1 0 0 1 40 246 Tm ( unique[column] = true;) Tj
1 0 0 1 40 235 Tm ( }\ );) Tj
1 0 0 1 40 224 Tm ( ) Tj
1 0 0 1 40 213 Tm ( if \ (isNaN\ (this.lockTimeoutMs\) || this.lockTimeoutMs <= 0\) {} Tj
1 0 0 1 40 202 Tm ( throw new ConfigError\ ('lockTimeoutMs must be a positive number.\ ');) Tj
1 0 0 1 40 191 Tm ( }) Tj
1 0 0 1 40 180 Tm ( }) Tj
1 0 0 1 40 169 Tm ( {} Tj
ET
endstream
endobj
172 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 171 0
endobj
173 0 obj
<< /Length 2733 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: sheet_db_lib/03_Errors.gs) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (/**) Tj
1 0 0 1 40 708 Tm ( * Base error for the Sheet DB library.) Tj
1 0 0 1 40 697 Tm ( */) Tj
1 0 0 1 40 686 Tm (class SheetDbError extends Error {} Tj
1 0 0 1 40 675 Tm ( /**) Tj
1 0 0 1 40 664 Tm ( * @param {string} message Human-readable error message.) Tj
1 0 0 1 40 653 Tm ( * @param {string=} code Stable error code for programmatic handling.) Tj

```

```

1 0 0 1 40 642 Tm (    */) Tj
1 0 0 1 40 631 Tm (    constructor\(message, code\) {}) Tj
1 0 0 1 40 620 Tm (        super\(message\);) Tj
1 0 0 1 40 609 Tm (        this.name = this.constructor.name;) Tj
1 0 0 1 40 598 Tm (        this.code = code || 'SHEET_DB_ERROR';) Tj
1 0 0 1 40 587 Tm (    }) Tj
1 0 0 1 40 576 Tm (}) Tj
1 0 0 1 40 565 Tm ( ) Tj
1 0 0 1 40 554 Tm (/**) Tj
1 0 0 1 40 543 Tm ( * Raised when configuration is invalid or missing.) Tj
1 0 0 1 40 532 Tm ( */) Tj
1 0 0 1 40 521 Tm (class ConfigError extends SheetDbError {}) Tj
1 0 0 1 40 510 Tm (    /**) Tj
1 0 0 1 40 499 Tm (        * @param {string} message) Tj
1 0 0 1 40 488 Tm (        */) Tj
1 0 0 1 40 477 Tm (    constructor\(message\) {}) Tj
1 0 0 1 40 466 Tm (        super\(message, 'CONFIG_ERROR'\);) Tj
1 0 0 1 40 455 Tm (    }) Tj
1 0 0 1 40 444 Tm (}) Tj
1 0 0 1 40 433 Tm ( ) Tj
1 0 0 1 40 422 Tm (/**) Tj
1 0 0 1 40 411 Tm ( * Raised when schema definitions are invalid.) Tj
1 0 0 1 40 400 Tm ( */) Tj
1 0 0 1 40 389 Tm (class SchemaError extends SheetDbError {}) Tj
1 0 0 1 40 378 Tm (    /**) Tj
1 0 0 1 40 367 Tm (        * @param {string} message) Tj
1 0 0 1 40 356 Tm (        */) Tj
1 0 0 1 40 345 Tm (    constructor\(message\) {}) Tj
1 0 0 1 40 334 Tm (        super\(message, 'SCHEMA_ERROR'\);) Tj
1 0 0 1 40 323 Tm (    }) Tj
1 0 0 1 40 312 Tm (}) Tj
1 0 0 1 40 301 Tm ( ) Tj
1 0 0 1 40 290 Tm (/**) Tj
1 0 0 1 40 279 Tm ( * Raised when records fail validation.) Tj
1 0 0 1 40 268 Tm ( */) Tj
1 0 0 1 40 257 Tm (class ValidationError extends SheetDbError {}) Tj
1 0 0 1 40 246 Tm (    /**) Tj
1 0 0 1 40 235 Tm (        * @param {string} message) Tj
1 0 0 1 40 224 Tm (        */) Tj
1 0 0 1 40 213 Tm (    constructor\(message\) {}) Tj
1 0 0 1 40 202 Tm (        super\(message, 'VALIDATION_ERROR'\);) Tj
1 0 0 1 40 191 Tm (    }) Tj
1 0 0 1 40 180 Tm (}) Tj
1 0 0 1 40 169 Tm ( ) Tj
1 0 0 1 40 158 Tm (/**) Tj
1 0 0 1 40 147 Tm ( * Raised when underlying sheet operations fail.) Tj
1 0 0 1 40 136 Tm ( */) Tj
1 0 0 1 40 125 Tm (class StorageError extends SheetDbError {}) Tj
1 0 0 1 40 114 Tm (    /**) Tj
1 0 0 1 40 103 Tm (        * @param {string} message) Tj
1 0 0 1 40 92 Tm (        */) Tj
1 0 0 1 40 81 Tm (    constructor\(message\) {}) Tj
1 0 0 1 40 70 Tm (        super\(message, 'STORAGE_ERROR'\);) Tj
1 0 0 1 40 59 Tm (    }) Tj
1 0 0 1 40 48 Tm (}) Tj
ET
endstream
endobj
174 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 173 0
endobj
175 0 obj
<< /Length 509 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm ( ) Tj
1 0 0 1 40 730 Tm (/**) Tj
1 0 0 1 40 719 Tm ( * Raised for optimistic transaction failures.) Tj
1 0 0 1 40 708 Tm ( */) Tj
1 0 0 1 40 697 Tm (class TransactionError extends SheetDbError {}) Tj
1 0 0 1 40 686 Tm (    /**) Tj
1 0 0 1 40 675 Tm (        * @param {string} message) Tj
1 0 0 1 40 664 Tm (        */) Tj
1 0 0 1 40 653 Tm (    constructor\(message\) {}) Tj
1 0 0 1 40 642 Tm (        super\(message, 'TRANSACTION_ERROR'\);) Tj
1 0 0 1 40 631 Tm (    }) Tj
1 0 0 1 40 620 Tm (}) Tj
ET
endstream

```

```

endobj
176 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 175 0
endobj
177 0 obj
<< /Length 2885 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: sheet_db_lib/04_Util.gs) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (/**) Tj
1 0 0 1 40 708 Tm ( * Utility helpers for Sheet DB.) Tj
1 0 0 1 40 697 Tm ( */) Tj
1 0 0 1 40 686 Tm (class Util {) Tj
1 0 0 1 40 675 Tm ( /**) Tj
1 0 0 1 40 664 Tm ( * @return {string} RFC3339 timestamp in UTC.) Tj
1 0 0 1 40 653 Tm ( */) Tj
1 0 0 1 40 642 Tm ( static nowIso\(\) {) Tj
1 0 0 1 40 631 Tm ( return new Date\(\).toISOString\(\);) Tj
1 0 0 1 40 620 Tm ( }) Tj
1 0 0 1 40 609 Tm ( ) Tj
1 0 0 1 40 598 Tm ( /**) Tj
1 0 0 1 40 587 Tm ( * Creates a shallow clone of an object.) Tj
1 0 0 1 40 576 Tm ( * @param {Object} source) Tj
1 0 0 1 40 565 Tm ( * @return {Object}) Tj
1 0 0 1 40 554 Tm ( */) Tj
1 0 0 1 40 543 Tm ( static clone\(\source\) {) Tj
1 0 0 1 40 532 Tm ( return Object.assign\({}, source || {});) Tj
1 0 0 1 40 521 Tm ( }) Tj
1 0 0 1 40 510 Tm ( ) Tj
1 0 0 1 40 499 Tm ( /**) Tj
1 0 0 1 40 488 Tm ( * Creates a stable row ID with a prefix.) Tj
1 0 0 1 40 477 Tm ( * @param {string=} prefix) Tj
1 0 0 1 40 466 Tm ( * @return {string}) Tj
1 0 0 1 40 455 Tm ( */) Tj
1 0 0 1 40 444 Tm ( static generateId\(\prefix\) {) Tj
1 0 0 1 40 433 Tm ( var p = prefix || 'row';) Tj
1 0 0 1 40 422 Tm ( var random = Math.random\(\).toString\(\36\).slice\(\2, 10\);) Tj
1 0 0 1 40 411 Tm ( return p + '_' + new Date\(\).getTime\(\) + '_' + random;) Tj
1 0 0 1 40 400 Tm ( }) Tj
1 0 0 1 40 389 Tm ( ) Tj
1 0 0 1 40 378 Tm ( /**) Tj
1 0 0 1 40 367 Tm ( * Returns true when value is null or undefined.) Tj
1 0 0 1 40 356 Tm ( * @param {*} value) Tj
1 0 0 1 40 345 Tm ( * @return {boolean}) Tj
1 0 0 1 40 334 Tm ( */) Tj
1 0 0 1 40 323 Tm ( static isNil\(\value\) {) Tj
1 0 0 1 40 312 Tm ( return value === null || value === undefined;) Tj
1 0 0 1 40 301 Tm ( }) Tj
1 0 0 1 40 290 Tm ( ) Tj
1 0 0 1 40 279 Tm ( /**) Tj
1 0 0 1 40 268 Tm ( * Returns true when value is empty string, null, or undefined.) Tj
1 0 0 1 40 257 Tm ( * @param {*} value) Tj
1 0 0 1 40 246 Tm ( * @return {boolean}) Tj
1 0 0 1 40 235 Tm ( */) Tj
1 0 0 1 40 224 Tm ( static isBlank\(\value\) {) Tj
1 0 0 1 40 213 Tm ( return value === '' || Util.isNil\(\value\);) Tj
1 0 0 1 40 202 Tm ( }) Tj
1 0 0 1 40 191 Tm ( ) Tj
1 0 0 1 40 180 Tm ( /**) Tj
1 0 0 1 40 169 Tm ( * Throws when a required condition is not met.) Tj
1 0 0 1 40 158 Tm ( * @param {boolean} condition) Tj
1 0 0 1 40 147 Tm ( * @param {string} message) Tj
1 0 0 1 40 136 Tm ( */) Tj
1 0 0 1 40 125 Tm ( static assert\(\condition, message\) {) Tj
1 0 0 1 40 114 Tm ( if \(!condition\) {) Tj
1 0 0 1 40 103 Tm ( throw new ValidationError\(\message\);) Tj
1 0 0 1 40 92 Tm ( }) Tj
1 0 0 1 40 81 Tm ( }) Tj
1 0 0 1 40 70 Tm ( ) Tj
1 0 0 1 40 59 Tm ( /**) Tj
1 0 0 1 40 48 Tm ( * Converts a sheet row to an object using the header order.) Tj
ET
endstream
endobj
178 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 177 0
endobj
179 0 obj

```





```

1 0 0 1 40 290 Tm (      if \(!seen[fieldName]\) {} Tj
1 0 0 1 40 279 Tm (      throw new SchemaError\() Tj
1 0 0 1 40 268 Tm (      'Field rules defined for unknown column "' + fieldName + '" in table: ' + tableName) Tj
1 0 0 1 40 257 Tm (      \);) Tj
1 0 0 1 40 246 Tm (    }) Tj
1 0 0 1 40 235 Tm (  })\);) Tj
1 0 0 1 40 224 Tm () Tj
1 0 0 1 40 213 Tm (    this._schemas[tableName] = {} Tj
1 0 0 1 40 202 Tm (      columns: schema.columns.slice\(\),) Tj
1 0 0 1 40 191 Tm (      fields: fields) Tj
1 0 0 1 40 180 Tm (    );) Tj
1 0 0 1 40 169 Tm (    return this;) Tj
1 0 0 1 40 158 Tm (  }) Tj
1 0 0 1 40 147 Tm () Tj
1 0 0 1 40 136 Tm (  /**) Tj
1 0 0 1 40 125 Tm (    * Returns schema for a table.) Tj
1 0 0 1 40 114 Tm (    * @param {string} tableName) Tj
1 0 0 1 40 103 Tm (    * @return {Object}) Tj
1 0 0 1 40 92 Tm (    */) Tj
1 0 0 1 40 81 Tm (  get\(tableName\) {} Tj
1 0 0 1 40 70 Tm (    var schema = this._schemas[tableName];) Tj
1 0 0 1 40 59 Tm (    if \(!schema\) {} Tj
1 0 0 1 40 48 Tm (      throw new SchemaError\('No schema registered for table: ' + tableName\);) Tj
ET
endstream
endobj
182 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 181 0
endobj
183 0 obj
<< /Length 2445 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (    }) Tj
1 0 0 1 40 730 Tm (    return {} Tj
1 0 0 1 40 719 Tm (      columns: schema.columns.slice\(\),) Tj
1 0 0 1 40 708 Tm (      fields: Util.clone\(schema.fields\)) Tj
1 0 0 1 40 697 Tm (    );) Tj
1 0 0 1 40 686 Tm (  }) Tj
1 0 0 1 40 675 Tm () Tj
1 0 0 1 40 664 Tm (  /**) Tj
1 0 0 1 40 653 Tm (    * Validates and normalizes a record.) Tj
1 0 0 1 40 642 Tm (    * @param {string} tableName) Tj
1 0 0 1 40 631 Tm (    * @param {Object} record) Tj
1 0 0 1 40 620 Tm (    * @param {boolean=} isPartial True for patch-like updates.) Tj
1 0 0 1 40 609 Tm (    * @return {Object}) Tj
1 0 0 1 40 598 Tm (    */) Tj
1 0 0 1 40 587 Tm (  normalizeRecord\(tableName, record, isPartial\) {} Tj
1 0 0 1 40 576 Tm (    var schema = this.get\(tableName\);) Tj
1 0 0 1 40 565 Tm (    var out = Util.clone\record || {});) Tj
1 0 0 1 40 554 Tm (    schema.columns.forEach\(function\column\) {} Tj
1 0 0 1 40 543 Tm (      var fieldRules = schema.fields[column] || {};) Tj
1 0 0 1 40 532 Tm (      var hasValue = !Util.isNil\out[column]\);) Tj
1 0 0 1 40 521 Tm () Tj
1 0 0 1 40 510 Tm (    if \(!hasValue && fieldRules.hasOwnProperty\('default'\)\) {} Tj
1 0 0 1 40 499 Tm (      var defaultValue = fieldRules.default;) Tj
1 0 0 1 40 488 Tm (      out[column] = typeof defaultValue === 'function' ? defaultValue\(\) : defaultValue;) Tj
1 0 0 1 40 477 Tm (    }) Tj
1 0 0 1 40 466 Tm () Tj
1 0 0 1 40 455 Tm (    if \(!isPartial && fieldRules.required && Util.isNil\out[column]\)\) {} Tj
1 0 0 1 40 444 Tm (      throw new ValidationError\('Field "' + column + '" is required.');
```

```

endobj
184 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 183 0
endobj
185 0 obj
<< /Length 3064 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: sheet_db_lib/06_SheetsGateway.gs) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (/**) Tj
1 0 0 1 40 708 Tm ( * Thin wrapper around SpreadsheetApp APIs.) Tj
1 0 0 1 40 697 Tm ( */) Tj
1 0 0 1 40 686 Tm (class SheetsGateway {}) Tj
1 0 0 1 40 675 Tm ( /**) Tj
1 0 0 1 40 664 Tm ( * @param {Config} config) Tj
1 0 0 1 40 653 Tm ( */) Tj
1 0 0 1 40 642 Tm ( constructor\config\ ) Tj
1 0 0 1 40 631 Tm ( this._config = config;) Tj
1 0 0 1 40 620 Tm ( }) Tj
1 0 0 1 40 609 Tm ( ) Tj
1 0 0 1 40 598 Tm ( /**) Tj
1 0 0 1 40 587 Tm ( * @return {GoogleAppsScript.Spreadsheet.Spreadsheet}) Tj
1 0 0 1 40 576 Tm ( */) Tj
1 0 0 1 40 565 Tm ( getSpreadsheet\(\) {} ) Tj
1 0 0 1 40 554 Tm ( if \this._config.spreadsheetId\ ) Tj
1 0 0 1 40 543 Tm ( return SpreadsheetApp.openById\this._config.spreadsheetId\;) Tj
1 0 0 1 40 532 Tm ( }) Tj
1 0 0 1 40 521 Tm ( return SpreadsheetApp.getActiveSpreadsheet\(\;) Tj
1 0 0 1 40 510 Tm ( }) Tj
1 0 0 1 40 499 Tm ( ) Tj
1 0 0 1 40 488 Tm ( /**) Tj
1 0 0 1 40 477 Tm ( * @param {string} tableName) Tj
1 0 0 1 40 466 Tm ( * @param {boolean=} createWhenMissing) Tj
1 0 0 1 40 455 Tm ( * @return {GoogleAppsScript.Spreadsheet.Sheet}) Tj
1 0 0 1 40 444 Tm ( */) Tj
1 0 0 1 40 433 Tm ( getSheet\tableName, createWhenMissing\ ) Tj
1 0 0 1 40 422 Tm ( var ss = this.getSpreadsheet\(\;) Tj
1 0 0 1 40 411 Tm ( var sheet = ss.getSheetByName\tableName\;) Tj
1 0 0 1 40 400 Tm ( if \(!sheet && createWhenMissing\ ) Tj
1 0 0 1 40 389 Tm ( sheet = ss.insertSheet\tableName\;) Tj
1 0 0 1 40 378 Tm ( }) Tj
1 0 0 1 40 367 Tm ( if \(!sheet\ ) Tj
1 0 0 1 40 356 Tm ( throw new StorageError\('Sheet not found for table: ' + tableName\;) Tj
1 0 0 1 40 345 Tm ( }) Tj
1 0 0 1 40 334 Tm ( return sheet;) Tj
1 0 0 1 40 323 Tm ( }) Tj
1 0 0 1 40 312 Tm ( ) Tj
1 0 0 1 40 301 Tm ( /**) Tj
1 0 0 1 40 290 Tm ( * @param {string} tableName) Tj
1 0 0 1 40 279 Tm ( * @return {{headers: string[], rows: Array<Array<*>>}}) Tj
1 0 0 1 40 268 Tm ( */) Tj
1 0 0 1 40 257 Tm ( readTable\tableName\ ) Tj
1 0 0 1 40 246 Tm ( var sheet = this.getSheet\tableName, false\;) Tj
1 0 0 1 40 235 Tm ( var range = sheet.getDataRange\(\;) Tj
1 0 0 1 40 224 Tm ( var values = range.getValues\(\;) Tj
1 0 0 1 40 213 Tm ( if \values.length === 0\ ) Tj
1 0 0 1 40 202 Tm ( return { headers: [], rows: [] };) Tj
1 0 0 1 40 191 Tm ( }) Tj
1 0 0 1 40 180 Tm ( return {} ) Tj
1 0 0 1 40 169 Tm ( headers: values[0],) Tj
1 0 0 1 40 158 Tm ( rows: values.slice\1\)) Tj
1 0 0 1 40 147 Tm ( );) Tj
1 0 0 1 40 136 Tm ( }) Tj
1 0 0 1 40 125 Tm ( ) Tj
1 0 0 1 40 114 Tm ( /**) Tj
1 0 0 1 40 103 Tm ( * Ensures table exists with given headers.) Tj
1 0 0 1 40 92 Tm ( * @param {string} tableName) Tj
1 0 0 1 40 81 Tm ( * @param {string[]} headers) Tj
1 0 0 1 40 70 Tm ( */) Tj
1 0 0 1 40 59 Tm ( ensureTable\tableName, headers\ ) Tj
1 0 0 1 40 48 Tm ( var sheet = this.getSheet\tableName, true\;) Tj
ET
endstream
endobj
186 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 185 0
endobj
187 0 obj

```

```

<< /Length 2220 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (    var lastRow = sheet.getLastRow\(\);) Tj
1 0 0 1 40 730 Tm (    if \(\lastRow === 0\) \) Tj
1 0 0 1 40 719 Tm (        sheet.getRange\(\(1, 1, 1, headers.length\)\).setValues\([headers]\);) Tj
1 0 0 1 40 708 Tm (        return;) Tj
1 0 0 1 40 697 Tm (    }) Tj
1 0 0 1 40 686 Tm (    ) Tj
1 0 0 1 40 675 Tm (        var existingHeaders = sheet.getRange\(\(1, 1, 1, sheet.getLastColumn\(\)\)\).getValues\(\)[0]) Tj
1 0 0 1 40 664 Tm (        .map\((function\(\value\) { return String\(\value\); })\);) Tj
1 0 0 1 40 653 Tm (        var expectedHeaders = headers.map\((function\(\value\) { return String\(\value\); })\);) Tj
1 0 0 1 40 642 Tm (        if \(\existingHeaders.join\('|\`'\) !== expectedHeaders.join\('|\`'\)\) \) Tj
1 0 0 1 40 631 Tm (            throw new StorageError\('Header mismatch for table: ' + tableName\);) Tj
1 0 0 1 40 620 Tm (        }) Tj
1 0 0 1 40 609 Tm (    }) Tj
1 0 0 1 40 598 Tm (    ) Tj
1 0 0 1 40 587 Tm (    /**) Tj
1 0 0 1 40 576 Tm (        * Writes all rows for a table, replacing existing body rows.) Tj
1 0 0 1 40 565 Tm (        * @param {string} tableName) Tj
1 0 0 1 40 554 Tm (        * @param {string[]} headers) Tj
1 0 0 1 40 543 Tm (        * @param {Array<Array<*>>} rows) Tj
1 0 0 1 40 532 Tm (        */) Tj
1 0 0 1 40 521 Tm (        writeAllRows\(\(tableName, headers, rows\) \) Tj
1 0 0 1 40 510 Tm (        var sheet = this.getSheet\(\(tableName, true\);) Tj
1 0 0 1 40 499 Tm (        this.ensureTable\(\(tableName, headers\);) Tj
1 0 0 1 40 488 Tm (    ) Tj
1 0 0 1 40 477 Tm (        var maxRows = Math.max\(\(sheet.getMaxRows\(\)\), rows.length + 1\);) Tj
1 0 0 1 40 466 Tm (        if \(\(sheet.getMaxRows\(\) < maxRows\) \) Tj
1 0 0 1 40 455 Tm (            sheet.insertRowsAfter\(\(sheet.getMaxRows\(\)\), maxRows - sheet.getMaxRows\(\)\)\);) Tj
1 0 0 1 40 444 Tm (        }) Tj
1 0 0 1 40 433 Tm (    ) Tj
1 0 0 1 40 422 Tm (        if \(\(sheet.getLastRow\(\) > 1\) \) Tj
1 0 0 1 40 411 Tm (            sheet.getRange\(\(2, 1, sheet.getLastRow\(\) - 1, headers.length\)\).clearContent\(\);) Tj
1 0 0 1 40 400 Tm (        }) Tj
1 0 0 1 40 389 Tm (    ) Tj
1 0 0 1 40 378 Tm (        if \(\(rows.length > 0\) \) Tj
1 0 0 1 40 367 Tm (            sheet.getRange\(\(2, 1, rows.length, headers.length\)\).setValues\(\(rows\);) Tj
1 0 0 1 40 356 Tm (        }) Tj
1 0 0 1 40 345 Tm (    }) Tj
1 0 0 1 40 334 Tm (}) Tj
ET
endstream
endobj
188 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 187 0
endobj
189 0 obj
<< /Length 3451 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: sheet_db_lib/07_TableRepository.gs) Tj
1 0 0 1 40 730 Tm (    ) Tj
1 0 0 1 40 719 Tm (    /**) Tj
1 0 0 1 40 708 Tm (        * CRUD repository for a specific table.) Tj
1 0 0 1 40 697 Tm (        */) Tj
1 0 0 1 40 686 Tm (class TableRepository {) Tj
1 0 0 1 40 675 Tm (    /**) Tj
1 0 0 1 40 664 Tm (        * @param {string} tableName) Tj
1 0 0 1 40 653 Tm (        * @param {Config} config) Tj
1 0 0 1 40 642 Tm (        * @param {SchemaRegistry} schemaRegistry) Tj
1 0 0 1 40 631 Tm (        * @param {SheetsGateway} sheetsGateway) Tj
1 0 0 1 40 620 Tm (        * @param {AuditLogger} auditLogger) Tj
1 0 0 1 40 609 Tm (        */) Tj
1 0 0 1 40 598 Tm (        constructor\(\(tableName, config, schemaRegistry, sheetsGateway, auditLogger\) \) Tj
1 0 0 1 40 587 Tm (            this._tableName = tableName;) Tj
1 0 0 1 40 576 Tm (            this._config = config;) Tj
1 0 0 1 40 565 Tm (            this._schemas = schemaRegistry;) Tj
1 0 0 1 40 554 Tm (            this._sheets = sheetsGateway;) Tj
1 0 0 1 40 543 Tm (            this._audit = auditLogger;) Tj
1 0 0 1 40 532 Tm (        }) Tj
1 0 0 1 40 521 Tm (    ) Tj
1 0 0 1 40 510 Tm (    /**) Tj
1 0 0 1 40 499 Tm (        * @return {Array<Object>} Tj
1 0 0 1 40 488 Tm (        */) Tj
1 0 0 1 40 477 Tm (        findAll\(\) \) Tj
1 0 0 1 40 466 Tm (            var schema = this._schemas.get\(\(this._tableName\);) Tj
1 0 0 1 40 455 Tm (            this._sheets.ensureTable\(\(this._tableName, schema.columns\);) Tj

```

```

1 0 0 1 40 444 Tm (    var data = this._sheets.readTable\(this._tableName\);) Tj
1 0 0 1 40 433 Tm (    return data.rows) Tj
1 0 0 1 40 422 Tm (    .map\((function\((row\)) { return Util.rowToObject\((data.headers, row\)); })\)) Tj
1 0 0 1 40 411 Tm (    .filter\((function\((record\)) { return Util.isBlank\((record[this._config.deletedAtColumn]\); }, t
1 0 0 1 40 400 Tm (s\);) Tj
1 0 0 1 40 389 Tm (    }) Tj
1 0 0 1 40 378 Tm (    ) Tj
1 0 0 1 40 367 Tm (    /**) Tj
1 0 0 1 40 356 Tm (    * @param {string} id) Tj
1 0 0 1 40 345 Tm (    * @return {Object|null}) Tj
1 0 0 1 40 334 Tm (    */) Tj
1 0 0 1 40 323 Tm (    findById\((id\)) {) Tj
1 0 0 1 40 312 Tm (        var rows = this.findAll\(\);) Tj
1 0 0 1 40 301 Tm (        for \((var i = 0; i < rows.length; i++\) {) Tj
1 0 0 1 40 290 Tm (            if \((rows[i][this._config.idColumn] === id\) {) Tj
1 0 0 1 40 279 Tm (                return rows[i];) Tj
1 0 0 1 40 268 Tm (            }) Tj
1 0 0 1 40 257 Tm (        }) Tj
1 0 0 1 40 246 Tm (        return null;) Tj
1 0 0 1 40 235 Tm (    }) Tj
1 0 0 1 40 224 Tm (    ) Tj
1 0 0 1 40 213 Tm (    /**) Tj
1 0 0 1 40 202 Tm (    * @param {Object} payload) Tj
1 0 0 1 40 191 Tm (    * @return {Object}) Tj
1 0 0 1 40 180 Tm (    */) Tj
1 0 0 1 40 169 Tm (    insert\((payload\) {) Tj
1 0 0 1 40 158 Tm (        var schema = this._schemas.get\((this._tableName\);) Tj
1 0 0 1 40 147 Tm (        this._sheets.ensureTable\((this._tableName, schema.columns\);) Tj
1 0 0 1 40 136 Tm (        var record = this._schemas.normalizeRecord\((this._tableName, payload, false\);) Tj
1 0 0 1 40 125 Tm (        var now = Util.nowIso\(\);) Tj
1 0 0 1 40 114 Tm (        record[this._config.idColumn] = record[this._config.idColumn] || Util.generateId\((this._tableNa
1 0 0 1 40 103 Tm (me\);) Tj
1 0 0 1 40 92 Tm (        record[this._config.createdAtColumn] = record[this._config.createdAtColumn] || now;) Tj
1 0 0 1 40 81 Tm (        record[this._config.updatedAtColumn] = now;) Tj
1 0 0 1 40 70 Tm (        if \(!record.hasOwnProperty\((this._config.deletedAtColumn\)\)) {) Tj
1 0 0 1 40 59 Tm (            record[this._config.deletedAtColumn] = '';) Tj
1 0 0 1 40 48 Tm (        }) Tj
ET
endstream
endobj
190 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 189 0
endobj
191 0 obj
<< /Length 3795 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (    ) Tj
1 0 0 1 40 730 Tm (    var tableData = this._sheets.readTable\((this._tableName\);) Tj
1 0 0 1 40 719 Tm (    var existingRows = tableData.rows.map\((function\((row\)) { return Util.rowToObject\((tableData.headers,
1 0 0 1 40 708 Tm (s, row\); })\);) Tj
1 0 0 1 40 697 Tm (    var idColumn = this._config.idColumn;) Tj
1 0 0 1 40 686 Tm (    var duplicate = existingRows.some\((function\((row\)) {) Tj
1 0 0 1 40 675 Tm (        return row[idColumn] === record[idColumn];) Tj
1 0 0 1 40 664 Tm (    })\);) Tj
1 0 0 1 40 653 Tm (    if \((duplicate\) {) Tj
1 0 0 1 40 642 Tm (        throw new ValidationError\('Record ID already exists: ' + record[idColumn]\);) Tj
1 0 0 1 40 631 Tm (    }) Tj
1 0 0 1 40 620 Tm (    existingRows.push\((record\);) Tj
1 0 0 1 40 609 Tm (    var rowValues = existingRows.map\((function\((r\)) { return Util.objectToRow\((schema.columns, r\);) Tj
1 0 0 1 40 598 Tm (    this._sheets.writeAllRows\((this._tableName, schema.columns, rowValues\);) Tj
1 0 0 1 40 587 Tm (    this._audit.log\('insert', this._tableName, { id: record[this._config.idColumn] })\);) Tj
1 0 0 1 40 576 Tm (    return record;) Tj
1 0 0 1 40 565 Tm (    }) Tj
1 0 0 1 40 554 Tm (    ) Tj
1 0 0 1 40 543 Tm (    /**) Tj
1 0 0 1 40 532 Tm (    * @param {string} id) Tj
1 0 0 1 40 521 Tm (    * @param {Object} patch) Tj
1 0 0 1 40 510 Tm (    * @return {Object}) Tj
1 0 0 1 40 499 Tm (    */) Tj
1 0 0 1 40 488 Tm (    update\((id, patch\) {) Tj
1 0 0 1 40 477 Tm (        var schema = this._schemas.get\((this._tableName\);) Tj
1 0 0 1 40 466 Tm (        var normalizedPatch = this._schemas.normalizeRecord\((this._tableName, patch, true\);) Tj
1 0 0 1 40 455 Tm (        var tableData = this._sheets.readTable\((this._tableName\);) Tj
1 0 0 1 40 444 Tm (        var records = tableData.rows.map\((function\((row\)) { return Util.rowToObject\((tableData.headers, r
1 0 0 1 40 433 Tm (w\); })\);) Tj
1 0 0 1 40 422 Tm (        var found = null;) Tj
1 0 0 1 40 411 Tm (    }) Tj
1 0 0 1 40 400 Tm (    for \((var i = 0; i < records.length; i++\) {) Tj

```



```

1 0 0 1 40 510 Tm ( if \(!lock.tryLock\(\this._lockTimeoutMs\)\) {} Tj
1 0 0 1 40 499 Tm ( throw new TransactionError\('Failed to acquire script lock within timeout.\');) Tj
1 0 0 1 40 488 Tm ( }) Tj
1 0 0 1 40 477 Tm ( try {} Tj
1 0 0 1 40 466 Tm ( return callback\(\);) Tj
1 0 0 1 40 455 Tm ( } finally {} Tj
1 0 0 1 40 444 Tm ( lock.releaseLock\(\);) Tj
1 0 0 1 40 433 Tm ( }) Tj
1 0 0 1 40 422 Tm ( }) Tj
1 0 0 1 40 411 Tm (}) Tj
ET
endstream
endobj
196 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 195 0
endobj
197 0 obj
<< /Length 1449 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: sheet_db_lib/09_AuditLogger.gs) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (/**) Tj
1 0 0 1 40 708 Tm ( * Minimal pluggable audit logger.) Tj
1 0 0 1 40 697 Tm ( */) Tj
1 0 0 1 40 686 Tm (class AuditLogger {} Tj
1 0 0 1 40 675 Tm ( /**) Tj
1 0 0 1 40 664 Tm ( * @param {Object=} options) Tj
1 0 0 1 40 653 Tm ( * @param {Function=} options.sink Optional custom sink function.) Tj
1 0 0 1 40 642 Tm ( */) Tj
1 0 0 1 40 631 Tm ( constructor\(\options\) {} Tj
1 0 0 1 40 620 Tm ( var opts = options || {};) Tj
1 0 0 1 40 609 Tm ( this._sink = opts.sink || function\(\entry\) {} Tj
1 0 0 1 40 598 Tm ( Logger.log\(\JSON.stringify\(\entry\)\)\);) Tj
1 0 0 1 40 587 Tm ( );) Tj
1 0 0 1 40 576 Tm ( }) Tj
1 0 0 1 40 565 Tm ( ) Tj
1 0 0 1 40 554 Tm ( /**) Tj
1 0 0 1 40 543 Tm ( * Emits an audit entry.) Tj
1 0 0 1 40 532 Tm ( * @param {string} action) Tj
1 0 0 1 40 521 Tm ( * @param {string} tableName) Tj
1 0 0 1 40 510 Tm ( * @param {Object=} metadata) Tj
1 0 0 1 40 499 Tm ( */) Tj
1 0 0 1 40 488 Tm ( log\(\action, tableName, metadata\) {} Tj
1 0 0 1 40 477 Tm ( this._sink\(\{} Tj
1 0 0 1 40 466 Tm ( at: Util.nowIso\(\),) Tj
1 0 0 1 40 455 Tm ( action: action,) Tj
1 0 0 1 40 444 Tm ( table: tableName,) Tj
1 0 0 1 40 433 Tm ( metadata: Util.clone\(\metadata || {}\)\)) Tj
1 0 0 1 40 422 Tm ( );\);) Tj
1 0 0 1 40 411 Tm ( }) Tj
1 0 0 1 40 400 Tm (}) Tj
ET
endstream
endobj
198 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 197 0
endobj
199 0 obj
<< /Length 3271 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: sheet_db_lib/10_ScriptProperties.gs) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (/**) Tj
1 0 0 1 40 708 Tm ( * Script Properties helper for secure and environment-specific configuration.) Tj
1 0 0 1 40 697 Tm ( */) Tj
1 0 0 1 40 686 Tm (class ScriptPropertiesHelper {} Tj
1 0 0 1 40 675 Tm ( /**) Tj
1 0 0 1 40 664 Tm ( * Creates a helper bound to Script Properties store.) Tj
1 0 0 1 40 653 Tm ( */) Tj
1 0 0 1 40 642 Tm ( constructor\(\) {} Tj
1 0 0 1 40 631 Tm ( this._props = PropertiesService.getScriptProperties\(\);) Tj
1 0 0 1 40 620 Tm ( }) Tj
1 0 0 1 40 609 Tm ( ) Tj
1 0 0 1 40 598 Tm ( /**) Tj
1 0 0 1 40 587 Tm ( * Returns a property value, or a default when missing.) Tj
1 0 0 1 40 576 Tm ( * @param {string} key) Tj

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```

1 0 0 1 40 565 Tm ( * @param {*} defaultValue) Tj
1 0 0 1 40 554 Tm ( * @return {*}) Tj
1 0 0 1 40 543 Tm ( */) Tj
1 0 0 1 40 532 Tm ( get\(\key, defaultValue\) {}) Tj
1 0 0 1 40 521 Tm ( var value = this._props.getProperty\(\key\);) Tj
1 0 0 1 40 510 Tm ( return value === null ? defaultValue : value;) Tj
1 0 0 1 40 499 Tm ( }) Tj
1 0 0 1 40 488 Tm ( ) Tj
1 0 0 1 40 477 Tm ( /**) Tj
1 0 0 1 40 466 Tm ( * Returns a required property value or throws when missing/blank.) Tj
1 0 0 1 40 455 Tm ( * @param {string} key) Tj
1 0 0 1 40 444 Tm ( * @return {string}) Tj
1 0 0 1 40 433 Tm ( */) Tj
1 0 0 1 40 422 Tm ( getRequired\(\key\) {}) Tj
1 0 0 1 40 411 Tm ( var value = this.get\(\key, '\');) Tj
1 0 0 1 40 400 Tm ( if \(\value === '\') {} Tj
1 0 0 1 40 389 Tm ( throw new ConfigError\('Missing required script property: ' + key\);) Tj
1 0 0 1 40 378 Tm ( }) Tj
1 0 0 1 40 367 Tm ( return String\(\value\);) Tj
1 0 0 1 40 356 Tm ( }) Tj
1 0 0 1 40 345 Tm ( ) Tj
1 0 0 1 40 334 Tm ( /**) Tj
1 0 0 1 40 323 Tm ( * Returns a boolean property value.) Tj
1 0 0 1 40 312 Tm ( * Accepted true values: true, 1, yes, y, on.) Tj
1 0 0 1 40 301 Tm ( * Accepted false values: false, 0, no, n, off.) Tj
1 0 0 1 40 290 Tm ( * @param {string} key) Tj
1 0 0 1 40 279 Tm ( * @param {boolean=} defaultValue) Tj
1 0 0 1 40 268 Tm ( * @return {boolean}) Tj
1 0 0 1 40 257 Tm ( */) Tj
1 0 0 1 40 246 Tm ( getBoolean\(\key, defaultValue\) {}) Tj
1 0 0 1 40 235 Tm ( var raw = this.get\(\key, null\);) Tj
1 0 0 1 40 224 Tm ( if \(\raw === null || raw === '\') {} Tj
1 0 0 1 40 213 Tm ( return defaultValue === undefined ? false : !!defaultValue;) Tj
1 0 0 1 40 202 Tm ( }) Tj
1 0 0 1 40 191 Tm ( var normalized = String\(\raw\).trim\(\).toLowerCase\(\);) Tj
1 0 0 1 40 180 Tm ( if \(['true', '1', 'yes', 'y', 'on'].indexOf(normalized\) !== -1\) {} Tj
1 0 0 1 40 169 Tm ( return true;) Tj
1 0 0 1 40 158 Tm ( }) Tj
1 0 0 1 40 147 Tm ( if \(['false', '0', 'no', 'n', 'off'].indexOf(normalized\) !== -1\) {} Tj
1 0 0 1 40 136 Tm ( return false;) Tj
1 0 0 1 40 125 Tm ( }) Tj
1 0 0 1 40 114 Tm ( throw new ConfigError\('Script property is not a valid boolean: ' + key\);) Tj
1 0 0 1 40 103 Tm ( }) Tj
1 0 0 1 40 92 Tm ( ) Tj
1 0 0 1 40 81 Tm ( /**) Tj
1 0 0 1 40 70 Tm ( * Returns a numeric property value.) Tj
1 0 0 1 40 59 Tm ( * @param {string} key) Tj
1 0 0 1 40 48 Tm ( * @param {number=} defaultValue) Tj
ET
endstream
endobj
200 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 199 0
endobj
201 0 obj
<< /Length 2984 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm ( * @return {number}) Tj
1 0 0 1 40 730 Tm ( */) Tj
1 0 0 1 40 719 Tm ( getNumber\(\key, defaultValue\) {}) Tj
1 0 0 1 40 708 Tm ( var raw = this.get\(\key, null\);) Tj
1 0 0 1 40 697 Tm ( if \(\raw === null || raw === '\') {} Tj
1 0 0 1 40 686 Tm ( return defaultValue === undefined ? 0 : Number\(\defaultValue\);) Tj
1 0 0 1 40 675 Tm ( }) Tj
1 0 0 1 40 664 Tm ( var n = Number\(\raw\);) Tj
1 0 0 1 40 653 Tm ( if \(\isNaN\(\n\)\) {} Tj
1 0 0 1 40 642 Tm ( throw new ConfigError\('Script property is not a valid number: ' + key\);) Tj
1 0 0 1 40 631 Tm ( }) Tj
1 0 0 1 40 620 Tm ( return n;) Tj
1 0 0 1 40 609 Tm ( }) Tj
1 0 0 1 40 598 Tm ( ) Tj
1 0 0 1 40 587 Tm ( /**) Tj
1 0 0 1 40 576 Tm ( * Returns all script properties as key-value object.) Tj
1 0 0 1 40 565 Tm ( * @return {Object<string, string>}) Tj
1 0 0 1 40 554 Tm ( */) Tj
1 0 0 1 40 543 Tm ( getAll\(\) {}) Tj
1 0 0 1 40 532 Tm ( return this._props.getProperties\(\);) Tj
1 0 0 1 40 521 Tm ( }) Tj

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```

1 0 0 1 40 510 Tm ( ) Tj
1 0 0 1 40 499 Tm ( /**) Tj
1 0 0 1 40 488 Tm ( * Returns all script properties with sensitive values redacted.) Tj
1 0 0 1 40 477 Tm ( * @return {Object<string, string>}) Tj
1 0 0 1 40 466 Tm ( */) Tj
1 0 0 1 40 455 Tm ( getAllRedacted\(\) {}) Tj
1 0 0 1 40 444 Tm ( var all = this.getAll\(\);) Tj
1 0 0 1 40 433 Tm ( var out = {};) Tj
1 0 0 1 40 422 Tm ( Object.keys\(\all\).forEach\((function\(\key\) {}) Tj
1 0 0 1 40 411 Tm ( out[key] = ScriptPropertiesHelper.shouldRedactKey\(\key\ ? '[REDACTED]' : all[key];) Tj
1 0 0 1 40 400 Tm ( \));) Tj
1 0 0 1 40 389 Tm ( return out;) Tj
1 0 0 1 40 378 Tm ( }) Tj
1 0 0 1 40 367 Tm ( ) Tj
1 0 0 1 40 356 Tm ( /**) Tj
1 0 0 1 40 345 Tm ( * Sets a script property value.) Tj
1 0 0 1 40 334 Tm ( * @param {string} key) Tj
1 0 0 1 40 323 Tm ( * @param {*} value) Tj
1 0 0 1 40 312 Tm ( */) Tj
1 0 0 1 40 301 Tm ( set\(\key, value\() Tj
1 0 0 1 40 290 Tm ( this._props.setProperty\(\key, String\(\value\));) Tj
1 0 0 1 40 279 Tm ( }) Tj
1 0 0 1 40 268 Tm ( ) Tj
1 0 0 1 40 257 Tm ( /**) Tj
1 0 0 1 40 246 Tm ( * Sets multiple script properties.) Tj
1 0 0 1 40 235 Tm ( * @param {Object<string, *>} obj) Tj
1 0 0 1 40 224 Tm ( */) Tj
1 0 0 1 40 213 Tm ( setMany\(\obj\() Tj
1 0 0 1 40 202 Tm ( var stringified = {};) Tj
1 0 0 1 40 191 Tm ( Object.keys\(\obj || {}\).forEach\((function\(\key\() Tj
1 0 0 1 40 180 Tm ( stringified[key] = String\(\obj[key]);) Tj
1 0 0 1 40 169 Tm ( \));) Tj
1 0 0 1 40 158 Tm ( this._props.setProperties\(\stringified, false);) Tj
1 0 0 1 40 147 Tm ( }) Tj
1 0 0 1 40 136 Tm ( ) Tj
1 0 0 1 40 125 Tm ( /**) Tj
1 0 0 1 40 114 Tm ( * Deletes a script property.) Tj
1 0 0 1 40 103 Tm ( * @param {string} key) Tj
1 0 0 1 40 92 Tm ( */) Tj
1 0 0 1 40 81 Tm ( delete\(\key\() Tj
1 0 0 1 40 70 Tm ( this._props.deleteProperty\(\key);) Tj
1 0 0 1 40 59 Tm ( }) Tj
1 0 0 1 40 48 Tm ( ) Tj
ET
endstream
endobj
202 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 201 0
endobj
203 0 obj
<< /Length 3407 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm ( /**) Tj
1 0 0 1 40 730 Tm ( * Returns true when a script property exists and is non-empty.) Tj
1 0 0 1 40 719 Tm ( * @param {string} key) Tj
1 0 0 1 40 708 Tm ( * @return {boolean}) Tj
1 0 0 1 40 697 Tm ( */) Tj
1 0 0 1 40 686 Tm ( has\(\key\() Tj
1 0 0 1 40 675 Tm ( var value = this._props.getProperty\(\key);) Tj
1 0 0 1 40 664 Tm ( return value !== null && value !== '';) Tj
1 0 0 1 40 653 Tm ( }) Tj
1 0 0 1 40 642 Tm ( ) Tj
1 0 0 1 40 631 Tm ( /**) Tj
1 0 0 1 40 620 Tm ( * Returns true when key should be redacted in debug output.) Tj
1 0 0 1 40 609 Tm ( * @param {string} key) Tj
1 0 0 1 40 598 Tm ( * @return {boolean}) Tj
1 0 0 1 40 587 Tm ( */) Tj
1 0 0 1 40 576 Tm ( static shouldRedactKey\(\key\() Tj
1 0 0 1 40 565 Tm ( var upper = String\(\key || ''\).toUpperCase\(\);) Tj
1 0 0 1 40 554 Tm ( if \(\upper === 'APP_PASSCODE' || upper === 'SIGNING_SECRET' || upper === 'WEBHOOK_SECRET' || up)
1 0 0 1 40 543 Tm (per === 'API_KEY'\) {}) Tj
1 0 0 1 40 532 Tm ( return true;) Tj
1 0 0 1 40 521 Tm ( }) Tj
1 0 0 1 40 510 Tm ( return upper.indexOf\('TOKEN'\) !== -1 || upper.indexOf\('SECRET'\) !== -1 || upper.indexOf\('PAS
1 0 0 1 40 499 Tm (ORD'\) !== -1;) Tj
1 0 0 1 40 488 Tm ( }) Tj
1 0 0 1 40 477 Tm (}) Tj
1 0 0 1 40 466 Tm ( ) Tj

```



```

1 0 0 1 40 455 Tm (/**) Tj
1 0 0 1 40 444 Tm ( * Reads SheetDb client-related config from Script Properties.) Tj
1 0 0 1 40 433 Tm ( *) Tj
1 0 0 1 40 422 Tm ( * Supported keys:) Tj
1 0 0 1 40 411 Tm ( * - SHEET_DB_SPREADSHEET_ID) Tj
1 0 0 1 40 400 Tm ( * - SHEET_DB_ID_COLUMN) Tj
1 0 0 1 40 389 Tm ( * - SHEET_DB_CREATED_AT_COLUMN) Tj
1 0 0 1 40 378 Tm ( * - SHEET_DB_UPDATED_AT_COLUMN) Tj
1 0 0 1 40 367 Tm ( * - SHEET_DB_DELETED_AT_COLUMN) Tj
1 0 0 1 40 356 Tm ( * - SHEET_DB_LOCK_TIMEOUT_MS) Tj
1 0 0 1 40 345 Tm ( *) Tj
1 0 0 1 40 334 Tm ( * Also reserved for host-layer secure use:) Tj
1 0 0 1 40 323 Tm ( * - APP_PASSCODE) Tj
1 0 0 1 40 312 Tm ( * - SIGNING_SECRET) Tj
1 0 0 1 40 301 Tm ( * - WEBHOOK_SECRET) Tj
1 0 0 1 40 290 Tm ( * - API_KEY) Tj
1 0 0 1 40 279 Tm ( *) Tj
1 0 0 1 40 268 Tm ( * @return {Object}) Tj
1 0 0 1 40 257 Tm ( */) Tj
1 0 0 1 40 246 Tm (function readSheetDbConfigFromScriptProperties\(\) {}) Tj
1 0 0 1 40 235 Tm ( var helper = new ScriptPropertiesHelper\(\);) Tj
1 0 0 1 40 224 Tm ( var config = {};) Tj
1 0 0 1 40 213 Tm ( ) Tj
1 0 0 1 40 202 Tm ( if \(\helper.has\('SHEET_DB_SPREADSHEET_ID'\)\)\) {} Tj
1 0 0 1 40 191 Tm ( config.spreadsheetId = helper.get\('SHEET_DB_SPREADSHEET_ID', '\)\);) Tj
1 0 0 1 40 180 Tm ( }) Tj
1 0 0 1 40 169 Tm ( if \(\helper.has\('SHEET_DB_ID_COLUMN'\)\)\) {} Tj
1 0 0 1 40 158 Tm ( config.idColumn = helper.get\('SHEET_DB_ID_COLUMN', 'id'\);) Tj
1 0 0 1 40 147 Tm ( }) Tj
1 0 0 1 40 136 Tm ( if \(\helper.has\('SHEET_DB_CREATED_AT_COLUMN'\)\)\) {} Tj
1 0 0 1 40 125 Tm ( config.createdAtColumn = helper.get\('SHEET_DB_CREATED_AT_COLUMN', 'createdAt'\);) Tj
1 0 0 1 40 114 Tm ( }) Tj
1 0 0 1 40 103 Tm ( if \(\helper.has\('SHEET_DB_UPDATED_AT_COLUMN'\)\)\) {} Tj
1 0 0 1 40 92 Tm ( config.updatedAtColumn = helper.get\('SHEET_DB_UPDATED_AT_COLUMN', 'updatedAt'\);) Tj
1 0 0 1 40 81 Tm ( }) Tj
1 0 0 1 40 70 Tm ( if \(\helper.has\('SHEET_DB_DELETED_AT_COLUMN'\)\)\) {} Tj
1 0 0 1 40 59 Tm ( config.deletedAtColumn = helper.get\('SHEET_DB_DELETED_AT_COLUMN', 'deletedAt'\);) Tj
1 0 0 1 40 48 Tm ( }) Tj
ET
endstream
endobj
204 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 203 0
endobj
205 0 obj
<< /Length 1892 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm ( if \(\helper.has\('SHEET_DB_LOCK_TIMEOUT_MS'\)\)\) {} Tj
1 0 0 1 40 730 Tm ( config.lockTimeoutMs = helper.getNumber\('SHEET_DB_LOCK_TIMEOUT_MS', 30000\);) Tj
1 0 0 1 40 719 Tm ( }) Tj
1 0 0 1 40 708 Tm ( ) Tj
1 0 0 1 40 697 Tm ( return config;) Tj
1 0 0 1 40 686 Tm (}) Tj
1 0 0 1 40 675 Tm ( ) Tj
1 0 0 1 40 664 Tm (/**) Tj
1 0 0 1 40 653 Tm ( * Merges explicit client options with script properties-based config.) Tj
1 0 0 1 40 642 Tm ( * Precedence: explicit options > script properties > hardcoded defaults.) Tj
1 0 0 1 40 631 Tm ( *) Tj
1 0 0 1 40 620 Tm ( * @param {Object=} options) Tj
1 0 0 1 40 609 Tm ( * @return {Object}) Tj
1 0 0 1 40 598 Tm ( */) Tj
1 0 0 1 40 587 Tm (function mergeClientOptionsWithScriptProperties\(\options\) {}) Tj
1 0 0 1 40 576 Tm ( var fromProps = readSheetDbConfigFromScriptProperties\(\);) Tj
1 0 0 1 40 565 Tm ( var explicit = options || {};) Tj
1 0 0 1 40 554 Tm ( var merged = {};) Tj
1 0 0 1 40 543 Tm ( Object.keys\(\fromProps\).forEach\(\function\(\key\) {} Tj
1 0 0 1 40 532 Tm ( merged[key] = fromProps[key];) Tj
1 0 0 1 40 521 Tm ( }));) Tj
1 0 0 1 40 510 Tm ( Object.keys\(\explicit\).forEach\(\function\(\key\) {} Tj
1 0 0 1 40 499 Tm ( merged[key] = explicit[key];) Tj
1 0 0 1 40 488 Tm ( }));) Tj
1 0 0 1 40 477 Tm ( return merged;) Tj
1 0 0 1 40 466 Tm (}) Tj
1 0 0 1 40 455 Tm ( ) Tj
1 0 0 1 40 444 Tm (/**) Tj
1 0 0 1 40 433 Tm ( * Example setup helper for consumers \(\manual one-time setup\).) Tj
1 0 0 1 40 422 Tm ( *) Tj
1 0 0 1 40 411 Tm ( * Example:) Tj

```



```

ET
endstream
endobj
208 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 207 0
endobj
209 0 obj
<< /Length 1866 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm ( * @return {SlackRouter}) Tj
1 0 0 1 40 730 Tm ( */) Tj
1 0 0 1 40 719 Tm (function createHostRouter\(\db, hostConfig\) {} Tj
1 0 0 1 40 708 Tm ( var parser = new SlackPayloadParser\(\);) Tj
1 0 0 1 40 697 Tm ( var security = new SlackSecurity\(\hostConfig\);) Tj
1 0 0 1 40 686 Tm ( var contexts = new RequestContextFactory\(\hostConfig\);) Tj
1 0 0 1 40 675 Tm ( var formatter = new ResultFormatter\(\);) Tj
1 0 0 1 40 664 Tm ( ) Tj
1 0 0 1 40 653 Tm ( var enrollmentService = new LmsEnrollmentService\(\db\);) Tj
1 0 0 1 40 642 Tm ( var automationService = new LmsAutomationService\(\db\);) Tj
1 0 0 1 40 631 Tm ( var workflowHandlers = new WorkflowWebhookHandlers\(\enrollmentService, automationService\);) Tj
1 0 0 1 40 620 Tm ( var slackService = new SlackService\(\enrollmentService, workflowHandlers, formatter\);) Tj
1 0 0 1 40 609 Tm ( ) Tj
1 0 0 1 40 598 Tm ( return new SlackRouter\(\parser, security, contexts, slackService, formatter, db\);) Tj
1 0 0 1 40 587 Tm (}) Tj
1 0 0 1 40 576 Tm ( ) Tj
1 0 0 1 40 565 Tm (/**) Tj
1 0 0 1 40 554 Tm ( * Webhook endpoint.) Tj
1 0 0 1 40 543 Tm ( * @param {GoogleAppsScript.Events.DoPost} e) Tj
1 0 0 1 40 532 Tm ( * @return {GoogleAppsScript.Content.TextOutput}) Tj
1 0 0 1 40 521 Tm ( */) Tj
1 0 0 1 40 510 Tm (function doPost\(\e\) {} Tj
1 0 0 1 40 499 Tm ( var db = createHostDbClient\(\);) Tj
1 0 0 1 40 488 Tm ( var hostConfig = new HostConfig\(\new ScriptPropertiesHelper\(\)\);) Tj
1 0 0 1 40 477 Tm ( var router = createHostRouter\(\db, hostConfig\);) Tj
1 0 0 1 40 466 Tm ( var result = router.handle\(\e\);) Tj
1 0 0 1 40 455 Tm ( ) Tj
1 0 0 1 40 444 Tm ( return ContentService) Tj
1 0 0 1 40 433 Tm ( .createTextOutput\(\JSON.stringify\(\result.slack || result\)\)) Tj
1 0 0 1 40 422 Tm ( .setMimeType\(\ContentService.MimeType.JSON\);) Tj
1 0 0 1 40 411 Tm (}) Tj
ET
endstream
endobj
210 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 209 0
endobj
211 0 obj
<< /Length 3490 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: slack_host/01_SlackRouter.gs) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (/**) Tj
1 0 0 1 40 708 Tm ( * HTTP router boundary: parse -> verify -> context -> app service.) Tj
1 0 0 1 40 697 Tm ( */) Tj
1 0 0 1 40 686 Tm (class SlackRouter {} Tj
1 0 0 1 40 675 Tm ( /**) Tj
1 0 0 1 40 664 Tm ( * @param {SlackPayloadParser} parser) Tj
1 0 0 1 40 653 Tm ( * @param {SlackSecurity} security) Tj
1 0 0 1 40 642 Tm ( * @param {RequestContextFactory} contextFactory) Tj
1 0 0 1 40 631 Tm ( * @param {SlackService} slackService) Tj
1 0 0 1 40 620 Tm ( * @param {ResultFormatter} resultFormatter) Tj
1 0 0 1 40 609 Tm ( * @param {DbClient} db) Tj
1 0 0 1 40 598 Tm ( */) Tj
1 0 0 1 40 587 Tm ( constructor\(\parser, security, contextFactory, slackService, resultFormatter, db\) {} Tj
1 0 0 1 40 576 Tm ( this._parser = parser;) Tj
1 0 0 1 40 565 Tm ( this._security = security;) Tj
1 0 0 1 40 554 Tm ( this._contexts = contextFactory;) Tj
1 0 0 1 40 543 Tm ( this._service = slackService;) Tj
1 0 0 1 40 532 Tm ( this._results = resultFormatter;) Tj
1 0 0 1 40 521 Tm ( this._db = db;) Tj
1 0 0 1 40 510 Tm ( }) Tj
1 0 0 1 40 499 Tm ( ) Tj
1 0 0 1 40 488 Tm ( /**) Tj
1 0 0 1 40 477 Tm ( * @param {GoogleAppsScript.Events.DoPost} e) Tj
1 0 0 1 40 466 Tm ( * @return {Object}) Tj
1 0 0 1 40 455 Tm ( */) Tj

```

```

1 0 0 1 40 444 Tm ( handle\(\e\) {\} Tj
1 0 0 1 40 433 Tm ( var parsed = this._parser.parseEvent\(\e\);) Tj
1 0 0 1 40 422 Tm ( var context = this._contexts.create\((parsed.routeType, parsed.payload, parsed.envelope\);) Tj
1 0 0 1 40 411 Tm ( ) Tj
1 0 0 1 40 400 Tm ( var auth = this._security.verify\((parsed\);) Tj
1 0 0 1 40 389 Tm ( if \(!auth.ok\) {\} Tj
1 0 0 1 40 378 Tm ( var denied = this._results.failure\((auth.code, auth.message, context, {\}\);) Tj
1 0 0 1 40 367 Tm ( this._writeAudit\((context, denied\);) Tj
1 0 0 1 40 356 Tm ( return denied;) Tj
1 0 0 1 40 345 Tm ( }) Tj
1 0 0 1 40 334 Tm ( ) Tj
1 0 0 1 40 323 Tm ( var result;) Tj
1 0 0 1 40 312 Tm ( try {\} Tj
1 0 0 1 40 301 Tm ( if \((parsed.routeType === 'slash_command'\) {\} Tj
1 0 0 1 40 290 Tm ( result = this._service.handleSlashCommand\((context, parsed.payload\);) Tj
1 0 0 1 40 279 Tm ( } else if \((parsed.routeType === 'workflow_webhook'\) {\} Tj
1 0 0 1 40 268 Tm ( result = this._service.handleWorkflowWebhook\((context, parsed.payload\);) Tj
1 0 0 1 40 257 Tm ( } else {\} Tj
1 0 0 1 40 246 Tm ( result = this._results.failure\('UNSUPPORTED_PAYLOAD', 'Unsupported payload.', context, {\}\))
1 0 0 1 40 235 Tm ( ;) Tj
1 0 0 1 40 224 Tm ( }) Tj
1 0 0 1 40 213 Tm ( } catch \((err\) {\} Tj
1 0 0 1 40 202 Tm ( result = this._results.failure\((err.code || 'HOST_ERROR', err.message || 'Unexpected error.',)
1 0 0 1 40 191 Tm ( context, {\}\);) Tj
1 0 0 1 40 180 Tm ( }) Tj
1 0 0 1 40 169 Tm ( ) Tj
1 0 0 1 40 158 Tm ( this._writeAudit\((context, result\);) Tj
1 0 0 1 40 147 Tm ( return result;) Tj
1 0 0 1 40 136 Tm ( }) Tj
1 0 0 1 40 125 Tm ( ) Tj
1 0 0 1 40 114 Tm ( /** @private */) Tj
1 0 0 1 40 103 Tm ( _writeAudit\((context, result\) {\} Tj
1 0 0 1 40 92 Tm ( this._db.table\('audit_logs'\).insert\(({ Tj
1 0 0 1 40 81 Tm ( requestId: context.requestId,) Tj
1 0 0 1 40 70 Tm ( source: context.source,) Tj
1 0 0 1 40 59 Tm ( action: context.routeType,) Tj
1 0 0 1 40 48 Tm ( status: result.ok ? 'success' : 'failure',) Tj
ET
endstream
endobj
212 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 211 0
endobj
213 0 obj
<< /Length 260 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm ( message: result.message,) Tj
1 0 0 1 40 730 Tm ( metadata: JSON.stringify\(({ code: result.code, requestId: result.requestId }\)) Tj
1 0 0 1 40 719 Tm ( \);) Tj
1 0 0 1 40 708 Tm ( }) Tj
1 0 0 1 40 697 Tm (}) Tj
ET
endstream
endobj
214 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 213 0
endobj
215 0 obj
<< /Length 3669 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: slack_host/02_SlackPayloadParser.gs) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (/**) Tj
1 0 0 1 40 708 Tm ( * Parses and normalizes inbound Slack-related payloads.) Tj
1 0 0 1 40 697 Tm ( */) Tj
1 0 0 1 40 686 Tm (class SlackPayloadParser {\} Tj
1 0 0 1 40 675 Tm ( /**) Tj
1 0 0 1 40 664 Tm ( * @param {GoogleAppsScript.Events.DoPost} e) Tj
1 0 0 1 40 653 Tm ( * @return {{routeType: string, payload: Object, envelope: Object}}) Tj
1 0 0 1 40 642 Tm ( */) Tj
1 0 0 1 40 631 Tm ( parseEvent\(\e\) {\} Tj
1 0 0 1 40 620 Tm ( var rawBody = \((e && e.postData && e.postData.contents\) ? e.postData.contents : '');) Tj
1 0 0 1 40 609 Tm ( var params = this._parseFormEncoded\((rawBody\);) Tj
1 0 0 1 40 598 Tm ( ) Tj
1 0 0 1 40 587 Tm ( if \((params.command\) {\} Tj
1 0 0 1 40 576 Tm ( return {\} Tj

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1 0 0 1 40 565 Tm (      routeType: 'slash_command',) Tj
1 0 0 1 40 554 Tm (      payload: this._normalizeSlashCommand\(params\),) Tj
1 0 0 1 40 543 Tm (      envelope: { rawBody: rawBody, params: params }) Tj
1 0 0 1 40 532 Tm (    );) Tj
1 0 0 1 40 521 Tm (  }) Tj
1 0 0 1 40 510 Tm ( ) Tj
1 0 0 1 40 499 Tm (      var jsonPayload = this._parseJson\(rawBody\);) Tj
1 0 0 1 40 488 Tm (      if \(jsonPayload\) {) Tj
1 0 0 1 40 477 Tm (        return {) Tj
1 0 0 1 40 466 Tm (          routeType: 'workflow_webhook',) Tj
1 0 0 1 40 455 Tm (          payload: this._normalizeWorkflowWebhook\(jsonPayload\),) Tj
1 0 0 1 40 444 Tm (          envelope: { rawBody: rawBody, json: jsonPayload }) Tj
1 0 0 1 40 433 Tm (        );) Tj
1 0 0 1 40 422 Tm (      }) Tj
1 0 0 1 40 411 Tm ( ) Tj
1 0 0 1 40 400 Tm (      return { routeType: 'unknown', payload: {}, envelope: { rawBody: rawBody } };) Tj
1 0 0 1 40 389 Tm (    }) Tj
1 0 0 1 40 378 Tm ( ) Tj
1 0 0 1 40 367 Tm ( /** @private */) Tj
1 0 0 1 40 356 Tm ( _normalizeSlashCommand\(payload\) {) Tj
1 0 0 1 40 345 Tm (   return {) Tj
1 0 0 1 40 334 Tm (     command: String\(payload.command || ''\).trim\(\),) Tj
1 0 0 1 40 323 Tm (     text: String\(payload.text || ''\).trim\(\),) Tj
1 0 0 1 40 312 Tm (     teamId: String\(payload.team_id || ''\),) Tj
1 0 0 1 40 301 Tm (     channelId: String\(payload.channel_id || ''\),) Tj
1 0 0 1 40 290 Tm (     userId: String\(payload.user_id || ''\),) Tj
1 0 0 1 40 279 Tm (     triggerId: String\(payload.trigger_id || ''\),) Tj
1 0 0 1 40 268 Tm (     responseUrl: String\(payload.response_url || ''\),) Tj
1 0 0 1 40 257 Tm (     requestId: String\(payload.request_id || Util.generateId\( 'slash'\)\)\)) Tj
1 0 0 1 40 246 Tm (   );) Tj
1 0 0 1 40 235 Tm ( }) Tj
1 0 0 1 40 224 Tm ( ) Tj
1 0 0 1 40 213 Tm ( /** @private */) Tj
1 0 0 1 40 202 Tm ( _normalizeWorkflowWebhook\(payload\) {) Tj
1 0 0 1 40 191 Tm (   var data = payload.data || payload;) Tj
1 0 0 1 40 180 Tm (   return {) Tj
1 0 0 1 40 169 Tm (     workflow: String\(payload.workflow || payload.workflow_name || 'unknown_workflow'\),) Tj
1 0 0 1 40 158 Tm (     eventType: String\(payload.eventType || payload.event_type || 'workflow_event'\),) Tj
1 0 0 1 40 147 Tm (     requestId: String\(payload.requestId || payload.request_id || Util.generateId\( 'wf'\)\)\),) Tj
1 0 0 1 40 136 Tm (     studentId: String\(data.studentId || data.student_id || ''\),) Tj
1 0 0 1 40 125 Tm (     courseId: String\(data.courseId || data.course_id || ''\),) Tj
1 0 0 1 40 114 Tm (     actorId: String\(data.actorId || data.actor_id || ''\),) Tj
1 0 0 1 40 103 Tm (     status: String\(data.status || ''\)) Tj
1 0 0 1 40 92 Tm (   );) Tj
1 0 0 1 40 81 Tm ( }) Tj
1 0 0 1 40 70 Tm ( ) Tj
1 0 0 1 40 59 Tm ( /** @private */) Tj
1 0 0 1 40 48 Tm ( _parseJson\(raw\) {) Tj
ET
endstream
endobj
216 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 215 0
endobj
217 0 obj
<< /Length 1122 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (      try {) Tj
1 0 0 1 40 730 Tm (        if \(!raw\) return null;) Tj
1 0 0 1 40 719 Tm (        return JSON.parse\(raw\);) Tj
1 0 0 1 40 708 Tm (      } catch \(err\) {) Tj
1 0 0 1 40 697 Tm (        return null;) Tj
1 0 0 1 40 686 Tm (      }) Tj
1 0 0 1 40 675 Tm (    }) Tj
1 0 0 1 40 664 Tm ( ) Tj
1 0 0 1 40 653 Tm ( /** @private */) Tj
1 0 0 1 40 642 Tm ( _parseFormEncoded\(body\) {) Tj
1 0 0 1 40 631 Tm (   var out = {};) Tj
1 0 0 1 40 620 Tm (   if \(!body\) return out;) Tj
1 0 0 1 40 609 Tm (   body.split\('&'\).forEach\(function\(pair\) {) Tj
1 0 0 1 40 598 Tm (     var tokens = pair.split\('=') Tj
1 0 0 1 40 587 Tm (     var key = decodeURIComponent\(tokens[0] || ''\).replace\( /\+/g, ' '\);) Tj
1 0 0 1 40 576 Tm (     var rawValue = tokens.length > 1 ? tokens.slice(1).join\('=') : '';) Tj
1 0 0 1 40 565 Tm (     var value = decodeURIComponent\(rawValue\).replace\( /\+/g, ' '\);) Tj
1 0 0 1 40 554 Tm (     if \(key\) out[key] = value;) Tj
1 0 0 1 40 543 Tm (   });) Tj
1 0 0 1 40 532 Tm (   return out;) Tj
1 0 0 1 40 521 Tm ( }) Tj

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```

1 0 0 1 40 510 Tm (}) Tj
ET
endstream
endobj
218 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 217 0
endobj
219 0 obj
<< /Length 2464 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: slack_host/03_SlackService.gs) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (/**) Tj
1 0 0 1 40 708 Tm ( * Application service that orchestrates Slack routes into LMS services.) Tj
1 0 0 1 40 697 Tm ( */) Tj
1 0 0 1 40 686 Tm (class SlackService {) Tj
1 0 0 1 40 675 Tm ( /**) Tj
1 0 0 1 40 664 Tm ( * @param {LmsEnrollmentService} enrollmentService) Tj
1 0 0 1 40 653 Tm ( * @param {WorkflowWebhookHandlers} workflowHandlers) Tj
1 0 0 1 40 642 Tm ( * @param {ResultFormatter} resultFormatter) Tj
1 0 0 1 40 631 Tm ( */) Tj
1 0 0 1 40 620 Tm ( constructor\(enrollmentService, workflowHandlers, resultFormatter\) {) Tj
1 0 0 1 40 609 Tm ( this._enrollment = enrollmentService;) Tj
1 0 0 1 40 598 Tm ( this._workflow = workflowHandlers;) Tj
1 0 0 1 40 587 Tm ( this._results = resultFormatter;) Tj
1 0 0 1 40 576 Tm ( }) Tj
1 0 0 1 40 565 Tm ( ) Tj
1 0 0 1 40 554 Tm ( /**) Tj
1 0 0 1 40 543 Tm ( * @param {Object} context) Tj
1 0 0 1 40 532 Tm ( * @param {Object} payload) Tj
1 0 0 1 40 521 Tm ( * @return {Object}) Tj
1 0 0 1 40 510 Tm ( */) Tj
1 0 0 1 40 499 Tm ( handleSlashCommand\(context, payload\) {) Tj
1 0 0 1 40 488 Tm ( if \(payload.command === '/enroll-student\) {) Tj
1 0 0 1 40 477 Tm ( var result = this._enrollment.enrollFromSlash\(context, payload\);) Tj
1 0 0 1 40 466 Tm ( return result.ok ? this._results.success\(result.message, result.data, context\)) Tj
1 0 0 1 40 455 Tm ( : this._results.failure\(result.code, result.message, context, result.data\);) Tj
1 0 0 1 40 444 Tm ( }) Tj
1 0 0 1 40 433 Tm ( ) Tj
1 0 0 1 40 422 Tm ( return this._results.failure\('UNSUPPORTED_COMMAND', 'Unsupported command: ' + payload.command,)
1 0 0 1 40 411 Tm ( context, {}\);) Tj
1 0 0 1 40 400 Tm ( }) Tj
1 0 0 1 40 389 Tm ( ) Tj
1 0 0 1 40 378 Tm ( /**) Tj
1 0 0 1 40 367 Tm ( * @param {Object} context) Tj
1 0 0 1 40 356 Tm ( * @param {Object} payload) Tj
1 0 0 1 40 345 Tm ( * @return {Object}) Tj
1 0 0 1 40 334 Tm ( */) Tj
1 0 0 1 40 323 Tm ( handleWorkflowWebhook\(context, payload\) {) Tj
1 0 0 1 40 312 Tm ( var result = this._workflow.handle\(context, payload\);) Tj
1 0 0 1 40 301 Tm ( return result.ok ? this._results.success\(result.message, result.data, context\)) Tj
1 0 0 1 40 290 Tm ( : this._results.failure\(result.code, result.message, context, result.data\);) Tj
1 0 0 1 40 279 Tm ( }) Tj
1 0 0 1 40 268 Tm (}) Tj
ET
endstream
endobj
220 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 219 0
endobj
221 0 obj
<< /Length 1933 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: slack_host/04_SlackSecurity.gs) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (/**) Tj
1 0 0 1 40 708 Tm ( * Host-layer security checks for Slack/webhook calls.) Tj
1 0 0 1 40 697 Tm ( */) Tj
1 0 0 1 40 686 Tm (class SlackSecurity {) Tj
1 0 0 1 40 675 Tm ( /**) Tj
1 0 0 1 40 664 Tm ( * @param {HostConfig} hostConfig) Tj
1 0 0 1 40 653 Tm ( */) Tj
1 0 0 1 40 642 Tm ( constructor\(hostConfig\) {) Tj
1 0 0 1 40 631 Tm ( this._config = hostConfig;) Tj
1 0 0 1 40 620 Tm ( }) Tj
1 0 0 1 40 609 Tm ( ) Tj

```



```

1 0 0 1 40 224 Tm (    return this._upsertEnrollment(context, payload.studentId, payload.courseId, 'workflow_webhook')
1 0 0 1 40 213 Tm (, payload.status || 'enrolled');) Tj
1 0 0 1 40 202 Tm (    }) Tj
1 0 0 1 40 191 Tm (    ) Tj
1 0 0 1 40 180 Tm (    /** @private */) Tj
1 0 0 1 40 169 Tm (    _upsertEnrollment(context, studentId, courseId, source, status) {}) Tj
1 0 0 1 40 158 Tm (        var students = this._db.table('students');) Tj
1 0 0 1 40 147 Tm (        var enrollments = this._db.table('enrollments');) Tj
1 0 0 1 40 136 Tm (        var audit = this._db.table('audit_logs');) Tj
1 0 0 1 40 125 Tm (        var finalStatus = status || 'enrolled');) Tj
1 0 0 1 40 114 Tm (        var enrollmentRecord;) Tj
1 0 0 1 40 103 Tm (    ) Tj
1 0 0 1 40 92 Tm (        this._db.transaction(function() {) Tj
1 0 0 1 40 81 Tm (            var existingStudent = students.findById(studentId);) Tj
1 0 0 1 40 70 Tm (            if (!existingStudent) {) Tj
1 0 0 1 40 59 Tm (                students.insert({ id: studentId, name: studentId, source: source });) Tj
1 0 0 1 40 48 Tm (            }) Tj
ET
endstream
endobj
224 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 223 0
endobj
225 0 obj
<< /Length 2540 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (    ) Tj
1 0 0 1 40 730 Tm (        var existingEnrollment = enrollments.findAll().filter(function(r) {) Tj
1 0 0 1 40 719 Tm (            return r.studentId === studentId && r.courseId === courseId;) Tj
1 0 0 1 40 708 Tm (        })[0];) Tj
1 0 0 1 40 697 Tm (    ) Tj
1 0 0 1 40 686 Tm (        if !(existingEnrollment) {) Tj
1 0 0 1 40 675 Tm (            enrollmentRecord = enrollments.update(existingEnrollment.id, {) Tj
1 0 0 1 40 664 Tm (                status: finalStatus,) Tj
1 0 0 1 40 653 Tm (                source: source) Tj
1 0 0 1 40 642 Tm (            });) Tj
1 0 0 1 40 631 Tm (        } else {) Tj
1 0 0 1 40 620 Tm (            enrollmentRecord = enrollments.insert({) Tj
1 0 0 1 40 609 Tm (                studentId: studentId,) Tj
1 0 0 1 40 598 Tm (                courseId: courseId,) Tj
1 0 0 1 40 587 Tm (                status: finalStatus,) Tj
1 0 0 1 40 576 Tm (                source: source) Tj
1 0 0 1 40 565 Tm (            });) Tj
1 0 0 1 40 554 Tm (        }) Tj
1 0 0 1 40 543 Tm (    ) Tj
1 0 0 1 40 532 Tm (        audit.insert({) Tj
1 0 0 1 40 521 Tm (            requestId: context.requestId,) Tj
1 0 0 1 40 510 Tm (            source: context.source,) Tj
1 0 0 1 40 499 Tm (            action: 'enrollment_upsert',) Tj
1 0 0 1 40 488 Tm (            status: 'success',) Tj
1 0 0 1 40 477 Tm (            message: 'Enrollment upsert completed.',) Tj
1 0 0 1 40 466 Tm (            metadata: JSON.stringify({ studentId: studentId, courseId: courseId, enrollmentId: enrollm
1 0 0 1 40 455 Tm (entRecord.id }));) Tj
1 0 0 1 40 444 Tm (        });) Tj
1 0 0 1 40 433 Tm (    });) Tj
1 0 0 1 40 422 Tm (    ) Tj
1 0 0 1 40 411 Tm (        this._db.audit('enrollment_upsert', 'enrollments', {) Tj
1 0 0 1 40 400 Tm (            requestId: context.requestId,) Tj
1 0 0 1 40 389 Tm (            studentId: studentId,) Tj
1 0 0 1 40 378 Tm (            courseId: courseId,) Tj
1 0 0 1 40 367 Tm (            enrollmentId: enrollmentRecord.id) Tj
1 0 0 1 40 356 Tm (        });) Tj
1 0 0 1 40 345 Tm (    ) Tj
1 0 0 1 40 334 Tm (        return {};) Tj
1 0 0 1 40 323 Tm (        ok: true,) Tj
1 0 0 1 40 312 Tm (        code: 'ENROLLMENT_UPSERTED',) Tj
1 0 0 1 40 301 Tm (        message: 'Enrollment upserted successfully.',) Tj
1 0 0 1 40 290 Tm (        data: {}) Tj
1 0 0 1 40 279 Tm (        enrollmentId: enrollmentRecord.id,) Tj
1 0 0 1 40 268 Tm (        studentId: studentId,) Tj
1 0 0 1 40 257 Tm (        courseId: courseId,) Tj
1 0 0 1 40 246 Tm (        status: enrollmentRecord.status) Tj
1 0 0 1 40 235 Tm (    }) Tj
1 0 0 1 40 224 Tm (    });) Tj
1 0 0 1 40 213 Tm (    }) Tj
1 0 0 1 40 202 Tm (}) Tj
ET
endstream

```



```

endobj
226 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 225 0
endobj
227 0 obj
<< /Length 2928 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: slack_host/06_LmsAutomationService.gs) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (/**) Tj
1 0 0 1 40 708 Tm ( * LMS automation operations that persist workflow state via SheetDb.) Tj
1 0 0 1 40 697 Tm ( */) Tj
1 0 0 1 40 686 Tm (class LmsAutomationService {}) Tj
1 0 0 1 40 675 Tm ( /**) Tj
1 0 0 1 40 664 Tm ( * @param {DbClient} db) Tj
1 0 0 1 40 653 Tm ( */) Tj
1 0 0 1 40 642 Tm ( constructor\(\db\) {) Tj
1 0 0 1 40 631 Tm ( this._db = db;) Tj
1 0 0 1 40 620 Tm ( }) Tj
1 0 0 1 40 609 Tm ( ) Tj
1 0 0 1 40 598 Tm ( /**) Tj
1 0 0 1 40 587 Tm ( * Inserts/updates an automation row and writes an audit log row.) Tj
1 0 0 1 40 576 Tm ( * @param {Object} context) Tj
1 0 0 1 40 565 Tm ( * @param {Object} payload) Tj
1 0 0 1 40 554 Tm ( * @return {Object}) Tj
1 0 0 1 40 543 Tm ( */) Tj
1 0 0 1 40 532 Tm ( upsertAutomation\(\context, payload\) {) Tj
1 0 0 1 40 521 Tm ( var automations = this._db.table\('automations'\);) Tj
1 0 0 1 40 510 Tm ( var audit = this._db.table\('audit_logs'\);) Tj
1 0 0 1 40 499 Tm ( var workflow = payload.workflow || 'unknown_workflow';) Tj
1 0 0 1 40 488 Tm ( var eventType = payload.eventType || 'workflow_event';) Tj
1 0 0 1 40 477 Tm ( var state = payload.status || 'received';) Tj
1 0 0 1 40 466 Tm ( ) Tj
1 0 0 1 40 455 Tm ( var existing = automations.findAll\(\).filter\((function\(\r\) {) Tj
1 0 0 1 40 444 Tm ( return r.workflow === workflow && r.requestId === payload.requestId;) Tj
1 0 0 1 40 433 Tm ( })\)[0];) Tj
1 0 0 1 40 422 Tm ( ) Tj
1 0 0 1 40 411 Tm ( var row;) Tj
1 0 0 1 40 400 Tm ( if \(\existing\) {) Tj
1 0 0 1 40 389 Tm ( row = automations.update\(\existing.id, { eventType: eventType, state: state }\) );) Tj
1 0 0 1 40 378 Tm ( } else {) Tj
1 0 0 1 40 367 Tm ( row = automations.insert\(\{ requestId: payload.requestId, workflow: workflow, eventType: even)
1 0 0 1 40 356 Tm ( tType, state: state }\) );) Tj
1 0 0 1 40 345 Tm ( }) Tj
1 0 0 1 40 334 Tm ( ) Tj
1 0 0 1 40 323 Tm ( audit.insert\(\{) Tj
1 0 0 1 40 312 Tm ( requestId: context.requestId,) Tj
1 0 0 1 40 301 Tm ( source: context.source,) Tj
1 0 0 1 40 290 Tm ( action: 'automation_upsert',) Tj
1 0 0 1 40 279 Tm ( status: 'success',) Tj
1 0 0 1 40 268 Tm ( message: 'Automation row upserted.',) Tj
1 0 0 1 40 257 Tm ( metadata: JSON.stringify\(\{ automationId: row.id, workflow: workflow, eventType: eventType }\) )\);) Tj
1 0 0 1 40 246 Tm ( })\);) Tj
1 0 0 1 40 235 Tm ( ) Tj
1 0 0 1 40 224 Tm ( return {) Tj
1 0 0 1 40 213 Tm ( ok: true,) Tj
1 0 0 1 40 202 Tm ( code: 'AUTOMATION_UPSERTED',) Tj
1 0 0 1 40 191 Tm ( message: 'Automation row upserted.',) Tj
1 0 0 1 40 180 Tm ( data: { automationId: row.id, workflow: workflow, eventType: eventType, state: row.state } ) Tj
1 0 0 1 40 169 Tm ( );) Tj
1 0 0 1 40 158 Tm ( }) Tj
1 0 0 1 40 147 Tm ( }) Tj
ET
endstream
endobj
228 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 227 0
endobj
229 0 obj
<< /Length 1624 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: slack_host/07_RequestContextFactory.gs) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (/**) Tj
1 0 0 1 40 708 Tm ( * Creates normalized request-context objects for host handlers.) Tj
1 0 0 1 40 697 Tm ( */) Tj

```

```

1 0 0 1 40 686 Tm (class RequestContextFactory {}) Tj
1 0 0 1 40 675 Tm ( /**) Tj
1 0 0 1 40 664 Tm ( * @param {HostConfig} hostConfig) Tj
1 0 0 1 40 653 Tm ( */) Tj
1 0 0 1 40 642 Tm ( constructor\(\hostConfig\) {} Tj
1 0 0 1 40 631 Tm ( this._config = hostConfig;) Tj
1 0 0 1 40 620 Tm ( }) Tj
1 0 0 1 40 609 Tm ( ) Tj
1 0 0 1 40 598 Tm ( /**) Tj
1 0 0 1 40 587 Tm ( * @param {string} routeType) Tj
1 0 0 1 40 576 Tm ( * @param {Object} payload) Tj
1 0 0 1 40 565 Tm ( * @param {Object=} envelope) Tj
1 0 0 1 40 554 Tm ( * @return {Object}) Tj
1 0 0 1 40 543 Tm ( */) Tj
1 0 0 1 40 532 Tm ( create\(routeType, payload, envelope\) {} Tj
1 0 0 1 40 521 Tm ( var env = envelope || {};) Tj
1 0 0 1 40 510 Tm ( return {} Tj
1 0 0 1 40 499 Tm ( requestId: payload.requestId || Util.generateId\('req'\),) Tj
1 0 0 1 40 488 Tm ( receivedAt: Util.nowIso\(\),) Tj
1 0 0 1 40 477 Tm ( routeType: routeType,) Tj
1 0 0 1 40 466 Tm ( source: routeType,) Tj
1 0 0 1 40 455 Tm ( actorId: payload.userId || payload.actorId || 'unknown',) Tj
1 0 0 1 40 444 Tm ( teamId: payload.teamId || '',) Tj
1 0 0 1 40 433 Tm ( channelId: payload.channelId || this._config.getSlackDefaultChannel\(\),) Tj
1 0 0 1 40 422 Tm ( raw: env.rawBody || '') Tj
1 0 0 1 40 411 Tm ( );) Tj
1 0 0 1 40 400 Tm ( }) Tj
1 0 0 1 40 389 Tm (}) Tj
ET
endstream
endobj
230 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 229 0
endobj
231 0 obj
<< /Length 2156 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: slack_host/08_ResultFormatter.gs) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (/**) Tj
1 0 0 1 40 708 Tm ( * Formats host operation results into structured Slack-friendly objects.) Tj
1 0 0 1 40 697 Tm ( */) Tj
1 0 0 1 40 686 Tm (class ResultFormatter {}) Tj
1 0 0 1 40 675 Tm ( /**) Tj
1 0 0 1 40 664 Tm ( * @param {string} message) Tj
1 0 0 1 40 653 Tm ( * @param {Object=} data) Tj
1 0 0 1 40 642 Tm ( * @param {Object=} context) Tj
1 0 0 1 40 631 Tm ( * @return {Object}) Tj
1 0 0 1 40 620 Tm ( */) Tj
1 0 0 1 40 609 Tm ( success\(message, data, context\) {} Tj
1 0 0 1 40 598 Tm ( return {} Tj
1 0 0 1 40 587 Tm ( ok: true,) Tj
1 0 0 1 40 576 Tm ( code: 'OK',) Tj
1 0 0 1 40 565 Tm ( message: message,) Tj
1 0 0 1 40 554 Tm ( requestId: \(\context && context.requestId\) || '',) Tj
1 0 0 1 40 543 Tm ( data: data || {},) Tj
1 0 0 1 40 532 Tm ( slack: {} Tj
1 0 0 1 40 521 Tm ( response_type: 'ephemeral',) Tj
1 0 0 1 40 510 Tm ( text: message) Tj
1 0 0 1 40 499 Tm ( }) Tj
1 0 0 1 40 488 Tm ( );) Tj
1 0 0 1 40 477 Tm ( }) Tj
1 0 0 1 40 466 Tm ( ) Tj
1 0 0 1 40 455 Tm ( /**) Tj
1 0 0 1 40 444 Tm ( * @param {string} code) Tj
1 0 0 1 40 433 Tm ( * @param {string} message) Tj
1 0 0 1 40 422 Tm ( * @param {Object=} context) Tj
1 0 0 1 40 411 Tm ( * @param {Object=} data) Tj
1 0 0 1 40 400 Tm ( * @return {Object}) Tj
1 0 0 1 40 389 Tm ( */) Tj
1 0 0 1 40 378 Tm ( failure\(code, message, context, data\) {} Tj
1 0 0 1 40 367 Tm ( return {} Tj
1 0 0 1 40 356 Tm ( ok: false,) Tj
1 0 0 1 40 345 Tm ( code: code || 'ERROR',) Tj
1 0 0 1 40 334 Tm ( message: message || 'Unknown error.',) Tj
1 0 0 1 40 323 Tm ( requestId: \(\context && context.requestId\) || '',) Tj
1 0 0 1 40 312 Tm ( data: data || {},) Tj
1 0 0 1 40 301 Tm ( slack: {} Tj

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```
1 0 0 1 40 290 Tm (      response_type: 'ephemeral',) Tj
1 0 0 1 40 279 Tm (      text: ':warning: ' + \ (message || 'Unknown error.\')) Tj
1 0 0 1 40 268 Tm (      }) Tj
1 0 0 1 40 257 Tm (      );) Tj
1 0 0 1 40 246 Tm (    }) Tj
1 0 0 1 40 235 Tm (  }) Tj
ET
endstream
endobj
232 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 231 0
endobj
233 0 obj
<< /Length 1367 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: slack_host/09_WorkflowWebhookHandlers.gs) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (/**) Tj
1 0 0 1 40 708 Tm ( * Handles normalized workflow webhook events.) Tj
1 0 0 1 40 697 Tm ( */) Tj
1 0 0 1 40 686 Tm (class WorkflowWebhookHandlers {}) Tj
1 0 0 1 40 675 Tm ( /**) Tj
1 0 0 1 40 664 Tm ( * @param {LmsEnrollmentService} enrollmentService) Tj
1 0 0 1 40 653 Tm ( * @param {LmsAutomationService} automationService) Tj
1 0 0 1 40 642 Tm ( */) Tj
1 0 0 1 40 631 Tm ( constructor\ (enrollmentService, automationService\ ) {} ) Tj
1 0 0 1 40 620 Tm (   this._enrollment = enrollmentService;) Tj
1 0 0 1 40 609 Tm (   this._automation = automationService;) Tj
1 0 0 1 40 598 Tm ( }) Tj
1 0 0 1 40 587 Tm ( ) Tj
1 0 0 1 40 576 Tm ( /**) Tj
1 0 0 1 40 565 Tm ( * @param {Object} context) Tj
1 0 0 1 40 554 Tm ( * @param {Object} payload) Tj
1 0 0 1 40 543 Tm ( * @return {Object}) Tj
1 0 0 1 40 532 Tm ( */) Tj
1 0 0 1 40 521 Tm ( handle\ (context, payload\ ) {} ) Tj
1 0 0 1 40 510 Tm (   if \ (payload.eventType === 'enrollment_requested') {} ) Tj
1 0 0 1 40 499 Tm (   return this._enrollment.upsertFromWorkflow\ (context, payload\);) Tj
1 0 0 1 40 488 Tm ( }) Tj
1 0 0 1 40 477 Tm (   return this._automation.upsertAutomation\ (context, payload\);) Tj
1 0 0 1 40 466 Tm ( }) Tj
1 0 0 1 40 455 Tm ( {} ) Tj
ET
endstream
endobj
234 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 233 0
endobj
235 0 obj
<< /Length 1977 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (FILE PATH: slack_host/10_HostConfig.gs) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm (/**) Tj
1 0 0 1 40 708 Tm ( * Host-layer configuration backed by Script Properties.) Tj
1 0 0 1 40 697 Tm ( */) Tj
1 0 0 1 40 686 Tm (class HostConfig {}) Tj
1 0 0 1 40 675 Tm ( /**) Tj
1 0 0 1 40 664 Tm ( * @param {ScriptPropertiesHelper=} helper) Tj
1 0 0 1 40 653 Tm ( */) Tj
1 0 0 1 40 642 Tm ( constructor\ (helper\ ) {} ) Tj
1 0 0 1 40 631 Tm (   this._props = helper || new ScriptPropertiesHelper\ (\);) Tj
1 0 0 1 40 620 Tm ( }) Tj
1 0 0 1 40 609 Tm ( ) Tj
1 0 0 1 40 598 Tm ( /** @return {string} */) Tj
1 0 0 1 40 587 Tm ( getAppPasscode\ (\) { return this._props.get\ ('APP_PASSCODE', ''); }) Tj
1 0 0 1 40 576 Tm ( /** @return {string} */) Tj
1 0 0 1 40 565 Tm ( getSigningSecret\ (\) { return this._props.get\ ('SIGNING_SECRET', ''); }) Tj
1 0 0 1 40 554 Tm ( /** @return {string} */) Tj
1 0 0 1 40 543 Tm ( getWebhookSecret\ (\) { return this._props.get\ ('WEBHOOK_SECRET', ''); }) Tj
1 0 0 1 40 532 Tm ( /** @return {string} */) Tj
1 0 0 1 40 521 Tm ( getApiKey\ (\) { return this._props.get\ ('API_KEY', ''); }) Tj
1 0 0 1 40 510 Tm ( /** @return {string} */) Tj
1 0 0 1 40 499 Tm ( getSlackBotToken\ (\) { return this._props.get\ ('SLACK_BOT_TOKEN', ''); }) Tj
1 0 0 1 40 488 Tm ( /** @return {string} */) Tj
1 0 0 1 40 477 Tm ( getSlackSigningSecret\ (\) { return this._props.get\ ('SLACK_SIGNING_SECRET', this.getSigningSecret\ (
```



```

1 0 0 1 40 103 Tm (      metadata: { type: 'string' }) Tj
1 0 0 1 40 92 Tm (      }) Tj
1 0 0 1 40 81 Tm (    }});) Tj
1 0 0 1 40 70 Tm (    ) Tj
1 0 0 1 40 59 Tm (    var hostConfig = new HostConfig\(\props\);) Tj
1 0 0 1 40 48 Tm (    var router = createHostRouter\(\db, hostConfig\);) Tj
ET
endstream
endobj
238 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 237 0
endobj
239 0 obj
<< /Length 3124 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (    return { db: db, hostConfig: hostConfig, router: router };) Tj
1 0 0 1 40 730 Tm (    }) Tj
1 0 0 1 40 719 Tm (    ) Tj
1 0 0 1 40 708 Tm ( /**) Tj
1 0 0 1 40 697 Tm ( * Encodes form payload.) Tj
1 0 0 1 40 686 Tm ( * @param {Object} obj) Tj
1 0 0 1 40 675 Tm ( * @return {string}) Tj
1 0 0 1 40 664 Tm ( */) Tj
1 0 0 1 40 653 Tm (function _toFormBody\(\obj\) {) Tj
1 0 0 1 40 642 Tm (    return Object.keys\(\obj\).map\((function\(\key\) {) Tj
1 0 0 1 40 631 Tm (        return encodeURIComponent\(\key\) + '=' + encodeURIComponent\(\String\(\obj[key]\)\)\);) Tj
1 0 0 1 40 620 Tm (    }\).join\('&'\);) Tj
1 0 0 1 40 609 Tm (    }) Tj
1 0 0 1 40 598 Tm (    ) Tj
1 0 0 1 40 587 Tm ( /**) Tj
1 0 0 1 40 576 Tm ( * Builds a mock DoPost event object.) Tj
1 0 0 1 40 565 Tm ( * @param {string} body) Tj
1 0 0 1 40 554 Tm ( * @return {GoogleAppsScript.Events.DoPost}) Tj
1 0 0 1 40 543 Tm ( */) Tj
1 0 0 1 40 532 Tm (function _mockDoPostEvent\(\body\) {) Tj
1 0 0 1 40 521 Tm (    return {) Tj
1 0 0 1 40 510 Tm (        postData: {) Tj
1 0 0 1 40 499 Tm (            contents: body,) Tj
1 0 0 1 40 488 Tm (            type: 'application/x-www-form-urlencoded') Tj
1 0 0 1 40 477 Tm (        }) Tj
1 0 0 1 40 466 Tm (    };) Tj
1 0 0 1 40 455 Tm (    }) Tj
1 0 0 1 40 444 Tm (    ) Tj
1 0 0 1 40 433 Tm ( /**) Tj
1 0 0 1 40 422 Tm ( * testSlashEnrollMock) Tj
1 0 0 1 40 411 Tm ( * Simulates '/enroll-student' and runs full router path.) Tj
1 0 0 1 40 400 Tm ( */) Tj
1 0 0 1 40 389 Tm (function testSlashEnrollMock\(\) {) Tj
1 0 0 1 40 378 Tm (    var ctx = _buildHostTestContext\(\);) Tj
1 0 0 1 40 367 Tm (    var suffix = new Date\(\).getTime\(\);) Tj
1 0 0 1 40 356 Tm (    var passcode = ctx.hostConfig.getAppPasscode\(\);) Tj
1 0 0 1 40 345 Tm (    var payload = {) Tj
1 0 0 1 40 334 Tm (        command: '/enroll-student',) Tj
1 0 0 1 40 323 Tm (        text: 'student-' + suffix + ' course-' + suffix,) Tj
1 0 0 1 40 312 Tm (        user_id: 'U_TEST_' + suffix,) Tj
1 0 0 1 40 301 Tm (        team_id: 'T_TEST',) Tj
1 0 0 1 40 290 Tm (        channel_id: 'C_TEST') Tj
1 0 0 1 40 279 Tm (    };) Tj
1 0 0 1 40 268 Tm (    if \(\passcode\) {) Tj
1 0 0 1 40 257 Tm (        payload.passcode = passcode;) Tj
1 0 0 1 40 246 Tm (    }) Tj
1 0 0 1 40 235 Tm (    ) Tj
1 0 0 1 40 224 Tm (    var event = _mockDoPostEvent\(_toFormBody\(\payload\)\);) Tj
1 0 0 1 40 213 Tm (    var result = ctx.router.handle\(\event\);) Tj
1 0 0 1 40 202 Tm (    Logger.log\('[testSlashEnrollMock] ' + JSON.stringify\(\result\)\);) Tj
1 0 0 1 40 191 Tm (    }) Tj
1 0 0 1 40 180 Tm (    ) Tj
1 0 0 1 40 169 Tm ( /**) Tj
1 0 0 1 40 158 Tm ( * testWorkflowEnrollmentMock) Tj
1 0 0 1 40 147 Tm ( * Simulates 'enrollment_requested' workflow webhook and runs full router path.) Tj
1 0 0 1 40 136 Tm ( */) Tj
1 0 0 1 40 125 Tm (function testWorkflowEnrollmentMock\(\) {) Tj
1 0 0 1 40 114 Tm (    var ctx = _buildHostTestContext\(\);) Tj
1 0 0 1 40 103 Tm (    var suffix = new Date\(\).getTime\(\);) Tj
1 0 0 1 40 92 Tm (    var payload = {) Tj
1 0 0 1 40 81 Tm (        workflow: 'lms_enrollment',) Tj
1 0 0 1 40 70 Tm (        eventType: 'enrollment_requested',) Tj
1 0 0 1 40 59 Tm (        requestId: 'wf_req_' + suffix,) Tj

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```

1 0 0 1 40 48 Tm (    data: {}) Tj
ET
endstream
endobj
240 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 239 0
endobj
241 0 obj
<< /Length 3409 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (    studentId: 'wf-student-' + suffix,) Tj
1 0 0 1 40 730 Tm (    courseId: 'wf-course-' + suffix,) Tj
1 0 0 1 40 719 Tm (    actorId: 'WF_ACTOR') Tj
1 0 0 1 40 708 Tm (    }) Tj
1 0 0 1 40 697 Tm (    );) Tj
1 0 0 1 40 686 Tm (    var event = _mockDoPostEvent\(JSON.stringify\(payload\));) Tj
1 0 0 1 40 675 Tm (    var result = ctx.router.handle\(event\);) Tj
1 0 0 1 40 664 Tm (    Logger.log\('[testWorkflowEnrollmentMock] ' + JSON.stringify\(result\));) Tj
1 0 0 1 40 653 Tm (}) Tj
1 0 0 1 40 642 Tm ( ) Tj
1 0 0 1 40 631 Tm (/**) Tj
1 0 0 1 40 620 Tm ( * testAutomationMock) Tj
1 0 0 1 40 609 Tm ( * Simulates generic automation workflow webhook and runs full router path.) Tj
1 0 0 1 40 598 Tm ( */) Tj
1 0 0 1 40 587 Tm (function testAutomationMock\(\) {}) Tj
1 0 0 1 40 576 Tm (    var ctx = _buildHostTestContext\(\);) Tj
1 0 0 1 40 565 Tm (    var suffix = new Date\(\).getTime\(\);) Tj
1 0 0 1 40 554 Tm (    var payload = {}) Tj
1 0 0 1 40 543 Tm (        workflow: 'lms_automation',) Tj
1 0 0 1 40 532 Tm (        eventType: 'automation_ping',) Tj
1 0 0 1 40 521 Tm (        requestId: 'automation_req_' + suffix,) Tj
1 0 0 1 40 510 Tm (        data: {}) Tj
1 0 0 1 40 499 Tm (        status: 'queued',) Tj
1 0 0 1 40 488 Tm (        actorId: 'AUTO_ACTOR') Tj
1 0 0 1 40 477 Tm (    }) Tj
1 0 0 1 40 466 Tm (    );) Tj
1 0 0 1 40 455 Tm (    var event = _mockDoPostEvent\(JSON.stringify\(payload\));) Tj
1 0 0 1 40 444 Tm (    var result = ctx.router.handle\(event\);) Tj
1 0 0 1 40 433 Tm (    Logger.log\('[testAutomationMock] ' + JSON.stringify\(result\));) Tj
1 0 0 1 40 422 Tm (}) Tj
1 0 0 1 40 411 Tm ( ) Tj
1 0 0 1 40 400 Tm (/**) Tj
1 0 0 1 40 389 Tm ( * testHostConfig) Tj
1 0 0 1 40 378 Tm ( * Verifies host config reads script properties without logging raw secrets.) Tj
1 0 0 1 40 367 Tm ( */) Tj
1 0 0 1 40 356 Tm (function testHostConfig\(\) {}) Tj
1 0 0 1 40 345 Tm (    var props = new ScriptPropertiesHelper\(\);) Tj
1 0 0 1 40 334 Tm (    var hostConfig = new HostConfig\(props\);) Tj
1 0 0 1 40 323 Tm (    var keys = []) Tj
1 0 0 1 40 312 Tm (        'APP_PASSCODE',) Tj
1 0 0 1 40 301 Tm (        'SIGNING_SECRET',) Tj
1 0 0 1 40 290 Tm (        'WEBHOOK_SECRET',) Tj
1 0 0 1 40 279 Tm (        'API_KEY',) Tj
1 0 0 1 40 268 Tm (        'SLACK_BOT_TOKEN',) Tj
1 0 0 1 40 257 Tm (        'SLACK_SIGNING_SECRET',) Tj
1 0 0 1 40 246 Tm (        'SLACK_DEFAULT_CHANNEL') Tj
1 0 0 1 40 235 Tm (    ];) Tj
1 0 0 1 40 224 Tm ( ) Tj
1 0 0 1 40 213 Tm (    var values = {}) Tj
1 0 0 1 40 202 Tm (        APP_PASSCODE: hostConfig.getAppPasscode\(\),) Tj
1 0 0 1 40 191 Tm (        SIGNING_SECRET: hostConfig.getSigningSecret\(\),) Tj
1 0 0 1 40 180 Tm (        WEBHOOK_SECRET: hostConfig.getWebhookSecret\(\),) Tj
1 0 0 1 40 169 Tm (        API_KEY: hostConfig.getApiKey\(\),) Tj
1 0 0 1 40 158 Tm (        SLACK_BOT_TOKEN: hostConfig.getSlackBotToken\(\),) Tj
1 0 0 1 40 147 Tm (        SLACK_SIGNING_SECRET: hostConfig.getSlackSigningSecret\(\),) Tj
1 0 0 1 40 136 Tm (        SLACK_DEFAULT_CHANNEL: hostConfig.getSlackDefaultChannel\(\)) Tj
1 0 0 1 40 125 Tm (    );) Tj
1 0 0 1 40 114 Tm ( ) Tj
1 0 0 1 40 103 Tm (    var redacted = {};) Tj
1 0 0 1 40 92 Tm (    keys.forEach\(function\ (key\) {}) Tj
1 0 0 1 40 81 Tm (        var value = values[key] || '';) Tj
1 0 0 1 40 70 Tm (        redacted[key] = ScriptPropertiesHelper.shouldRedactKey\ (key\)) Tj
1 0 0 1 40 59 Tm (        ? \ (value ? '[REDACTED_SET]' : '[REDACTED_EMPTY]'\)) Tj
1 0 0 1 40 48 Tm (        : value;) Tj
ET
endstream
endobj
242 0 obj

```

```
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 241 0
endobj
243 0 obj
<< /Length 1905 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm ( }\\);) Tj
1 0 0 1 40 730 Tm ( ) Tj
1 0 0 1 40 719 Tm ( Logger.log\('[testHostConfig] ' + JSON.stringify\(\redacted\)\);) Tj
1 0 0 1 40 708 Tm (}) Tj
1 0 0 1 40 697 Tm ( ) Tj
1 0 0 1 40 686 Tm (/**) Tj
1 0 0 1 40 675 Tm ( * testAuditWrite) Tj
1 0 0 1 40 664 Tm ( * Executes one mutation and verifies an audit row is written.) Tj
1 0 0 1 40 653 Tm ( */) Tj
1 0 0 1 40 642 Tm (function testAuditWrite\(\) {}) Tj
1 0 0 1 40 631 Tm ( var ctx = _buildHostTestContext\(\);) Tj
1 0 0 1 40 620 Tm ( var auditTable = ctx.db.table\('audit_logs'\);) Tj
1 0 0 1 40 609 Tm ( var before = auditTable.findAll\(\).length;) Tj
1 0 0 1 40 598 Tm ( ) Tj
1 0 0 1 40 587 Tm ( var suffix = new Date\(\).getTime\(\);) Tj
1 0 0 1 40 576 Tm ( var passcode = ctx.hostConfig.getAppPasscode\(\);) Tj
1 0 0 1 40 565 Tm ( var payload = {} Tj
1 0 0 1 40 554 Tm ( command: '/enroll-student',) Tj
1 0 0 1 40 543 Tm ( text: 'audit-student-' + suffix + ' audit-course-' + suffix,) Tj
1 0 0 1 40 532 Tm ( user_id: 'U_AUDIT_' + suffix,) Tj
1 0 0 1 40 521 Tm ( team_id: 'T_AUDIT',) Tj
1 0 0 1 40 510 Tm ( channel_id: 'C_AUDIT') Tj
1 0 0 1 40 499 Tm ( );) Tj
1 0 0 1 40 488 Tm ( if \(\passcode\) {}) Tj
1 0 0 1 40 477 Tm ( payload.passcode = passcode;) Tj
1 0 0 1 40 466 Tm ( }) Tj
1 0 0 1 40 455 Tm ( ) Tj
1 0 0 1 40 444 Tm ( var event = _mockDoPostEvent\(_toFormBody\(\payload\)\);) Tj
1 0 0 1 40 433 Tm ( var result = ctx.router.handle\(\event\);) Tj
1 0 0 1 40 422 Tm ( var after = auditTable.findAll\(\).length;) Tj
1 0 0 1 40 411 Tm ( ) Tj
1 0 0 1 40 400 Tm ( Logger.log\('[testAuditWrite] ' + JSON.stringify\({} Tj
1 0 0 1 40 389 Tm ( mutationOk: !!result.ok,) Tj
1 0 0 1 40 378 Tm ( auditRowsBefore: before,) Tj
1 0 0 1 40 367 Tm ( auditRowsAfter: after,) Tj
1 0 0 1 40 356 Tm ( auditRowDelta: after - before) Tj
1 0 0 1 40 345 Tm ( }\\)\);) Tj
1 0 0 1 40 334 Tm (}) Tj
ET
endstream
endobj
244 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 243 0
endobj
245 0 obj
<< /Length 103 >>
stream
BT
/F1 9 Tf
1 0 0 1 40 741 Tm (Missing / Inaccessible Files Appendix) Tj
1 0 0 1 40 730 Tm (None) Tj
ET
endstream
endobj
246 0 obj
<< /Type /Page /Parent 247 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 1 0 R /F2 2 0 R >> >> /Contents 245 0
endobj
247 0 obj
<< /Type /Pages /Kids [4 0 R 6 0 R 8 0 R 10 0 R 12 0 R 14 0 R 16 0 R 18 0 R 20 0 R 22 0 R 24 0 R 26 0 R 28 0 R 30 0 R 32
endobj
248 0 obj
<< /Type /Catalog /Pages 247 0 R >>
endobj
xref
0 249
0000000000 65535 f
0000000015 00000 n
0000000083 00000 n
0000000158 00000 n
0000000429 00000 n
0000000567 00000 n
0000003729 00000 n
0000003867 00000 n
```

0000004450 00000 n  
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0000008881 00000 n  
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0000037402 00000 n  
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0000041052 00000 n  
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0000089840 00000 n





0000192649 00000 n  
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0000194041 00000 n  
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0000264289 00000 n  
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0000270291 00000 n  
0000271969 00000 n  
0000272111 00000 n  
0000274321 00000 n  
0000274463 00000 n  
0000275884 00000 n  
0000276026 00000 n  
0000278057 00000 n  
0000278199 00000 n  
0000281925 00000 n  
0000282067 00000 n  
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0000285387 00000 n  
0000288850 00000 n  
0000288992 00000 n  
0000290951 00000 n  
0000291093 00000 n  
0000291249 00000 n  
0000291391 00000 n

```
0000292371 00000 n
trailer
<< /Size 249 /Root 248 0 R >>
startxref
292424
%%EOF
```

```
architecture.md
# Slack LMS Architecture
```

## ## 1. System Overview

The Slack LMS system is a single-ingress integration where Slack sends all command, interactivity, and event traffic to

'Slack -> doPost(e) -> SlackPayloadParser -> SlackSecurity -> SlackRouter -> SlackService/domain services -> SheetDb run

The runtime data store is Google Sheets accessed through the shared 'SheetDb' / 'LMSLibrary' layer. Core operations are

## ## 2. Core Architecture Pattern

- \*\*Single ingress:\*\* 'code.gs#doPost(e)' is the only webhook ingress.
- \*\*Thin host shell:\*\* 'doPost(e)' only coordinates parse/verify/route and error response.
- \*\*Registry dispatch:\*\* '03\_SlackService.gs' maps slash commands and routes event/interactivity payload types.
- \*\*Domain services:\*\* enrollment, lesson, progress, completion, reminder, and report services own business behavior.
- \*\*Shared library reuse:\*\* 'SheetDb' and 'ScriptPropertiesHelper' are reused for DB and configuration helpers.
- \*\*Operational modules:\*\* scheduler, retry resolver, health monitor, and test harness are isolated modules.

## ## 3. Runtime Components

- \*\*Ingress / host shell\*\*
  - 'code.gs': 'doGet', 'doPost', host config/dependency accessors, safe response/log helpers.
- \*\*Dependency bootstrap\*\*
  - '00\_WebAppEntry.gs': DB schema registration and dependency graph wiring.
- \*\*Transport normalization and controls\*\*
  - '02\_SlackPayloadParser.gs': request normalization.
  - '04\_SlackSecurity.gs': signature/timestamp verification with fallback mode.
  - '01\_SlackRouter.gs': route-type dispatch.
- \*\*Application dispatch\*\*
  - '03\_SlackService.gs': slash/interactivity/event/workflow routing.
- \*\*Domain services\*\*
  - '09\_LmsEnrollmentService.gs'
  - '12\_LmsLessonService.gs'
  - '13\_LmsProgressService.gs'
  - '13a\_LmsCompletionService.gs'
  - '14\_LmsReminderService.gs'
  - 'ReportService.gs'
- \*\*Integrations and builders\*\*
  - '11\_SlackApiClient.gs'
  - '12\_SlackBlockKitBuilder.gs'
- \*\*State, jobs, resiliency\*\*
  - '13\_LearnerProgressStateMachine.gs'
  - '14\_Scheduler.gs'
  - '18\_RetryResolver.gs'
  - '17\_HealthMonitor.gs'
- \*\*Configuration and sync/testing helpers\*\*
  - '16\_ConfigBootstrap.gs'
  - '15\_SheetsDataSync.gs'
  - '19\_HostTestHarness.gs'

## ## 4. Workflow Map (detailed flows)

### ### 4.1 Onboarding

- \*\*Trigger:\*\* '/enroll' placeholder command flow (or service call path) and explicit enrollment service use.
- \*\*Flow steps:\*\*
  1. Payload parsed and security checked.
  2. Router dispatches to slash handler.
  3. 'LmsEnrollmentService.enrollLearner' ensures learner + enrollment, queues first lesson, sends welcome DM.
- \*\*Data reads/writes:\*\*
  - Read 'learners', 'enrollment'.
  - Write 'learners' (if new), 'enrollment' (if new), 'delivery\_queue', 'audit\_log'.
- \*\*Slack output:\*\* welcome DM via 'openDm' + 'postMessage'.
- \*\*Failure path:\*\* returns structured '{ ok:false, code, message }'; retry may apply for Slack API failures.

### ### 4.2 '/learn'

- \*\*Trigger:\*\* slash command '/learn' from Slack.
- \*\*Flow steps:\*\* parse -> verify -> route 'slash\_command' -> 'SlackService.handleSlashCommand' -> 'LmsLessonService.han
- \*\*Data reads/writes:\*\*
  - Read 'learners', 'learner\_progress', 'lessons'.
  - Optional write to 'audit\_log' through host audit helper.
- \*\*Slack output:\*\* ephemeral lesson payload with Block Kit lesson card.
- \*\*Failure path:\*\* learner/learn missing returns ephemeral error text.

### ### 4.3 '/submit'

- **Trigger:** slash command `/submit <lesson_id>`.
- **Flow steps:** parse -> verify -> route -> `LmsCompletionService.handleSubmit`.
- **Data reads/writes:**
  - Read `'learners'`, `'submission_log'`, `'learner_progress'`.
  - Write `'submission_log'` with idempotency key, update `'learner_progress'` state, enqueue `'delivery_queue'` next lesson.
  - Write audit via DB audit helper.
- **Slack output:** ephemeral submission confirmation.
- **Failure path:** missing args/learner/progress returns structured error response.

#### ### 4.4 `/progress`

- **Trigger:** slash command `/progress`.
- **Flow steps:** parse -> verify -> route -> `LmsProgressService.handleProgress`.
- **Data reads/writes:**
  - Read `'learners'`, `'learner_progress'`.
  - No required writes (scaffold).
- **Slack output:** ephemeral progress summary blocks.
- **Failure path:** learner not found -> ephemeral error.

#### ### 4.5 Interactivity

- **Trigger:** Slack interactive payload (`'payload'` form field).
- **Flow steps:** parse to `'interactivity'` -> verify -> route -> `SlackService.handleInteractivity`.
- **Data reads/writes:** Verify in code (current scaffold acknowledges payload type/action/callback).
- **Slack output:** ephemeral acknowledgement or view\_submission clear action.
- **Failure path:** unsupported interactive payload returns safe fallback response.

#### ### 4.6 `'message.im'`

- **Trigger:** Slack Events API `'message'` with `'channel_type=im'`.
- **Flow steps:** parse JSON -> verify -> route `'event_callback'` -> `SlackService.handleEventCallback` branch.
- **Data reads/writes:** none required in current scaffold (Verify in code for future enrichment).
- **Slack output:** acknowledgement text payload.
- **Failure path:** unsupported event type returns ignored response.

#### ### 4.7 `'app_mention'`

- **Trigger:** Slack event `'app_mention'`.
- **Flow steps:** same event flow as above, mention branch in event handler.
- **Data reads/writes:** none required in scaffold.
- **Slack output:** guidance message to use slash commands.
- **Failure path:** event falls back to ignored.

#### ### 4.8 Scheduled Delivery

- **Trigger:** `'runDailyLessonDelivery'` installable trigger.
- **Flow steps:** lock acquisition -> dependency bootstrap -> `LmsLessonService.deliverPendingLessons`.
- **Data reads/writes:**
  - Read `'delivery_queue'`, `'learners'`.
  - Write `'delivery_queue'` status updates, possibly `'learner_progress'` state transitions.
- **Slack output:** DMs for due lessons.
- **Failure path:** lock/contention or Slack failure -> structured error; retry queue can be populated by Slack client.

#### ### 4.9 Reminders

- **Trigger:** `'runHourlyReminderCheck'` trigger.
- **Flow steps:** lock -> `LmsReminderService.sendOverdueReminders`.
- **Data reads/writes:**
  - Read `'learner_progress'` overdue rows, `'learners'`.
  - Optional writes to audit/retry queues.
- **Slack output:** reminder DM blocks.
- **Failure path:** per-learner failure captured in batch result.

#### ### 4.10 Reporting

- **Trigger:** `'runWeeklyAdminReport'` trigger and optional `/report` command placeholder.
- **Flow steps:** dependency bootstrap -> `ReportService` summary methods.
- **Data reads/writes:**
  - Read `'learners'`, `'enrollment'`, `'learner_progress'`.
  - Optional write/audit in future implementation.
- **Slack output:** summary blocks for admin contexts.
- **Failure path:** returns structured report error result.

### ## 5. Data Architecture

#### ### 5.1 Runtime Tables (Google Sheets)

1. **learners**
  - Purpose: Slack user to LMS learner identity mapping.
  - Typical fields: `'id'`, `'slackUserId'`, `'email'`, `'name'`, `'status'`, timestamps.
2. **enrollment**
  - Purpose: learner-to-course/track enrollment state.
  - Typical fields: `'id'`, `'learnerId'`, `'courseId'`, `'track'`, `'status'`.
3. **lessons**
  - Purpose: runtime lesson catalog.
  - Typical fields: `'id'`, `'courseId'`, `'track'`, `'title'`, `'contentRef'`, `'active'`.
4. **learner\_progress**
  - Purpose: per-learner lesson state machine records.

- Typical fields: 'id', 'learnerId', 'lessonId', 'state', 'dueAt', 'completedAt'.
- 5. **\*\*submission\_log\*\***
  - Purpose: submission history and idempotency tracking.
  - Typical fields: 'id', 'learnerId', 'lessonId', 'submitKey', 'payload'.
- 6. **\*\*delivery\_queue\*\***
  - Purpose: async lesson delivery queue.
  - Typical fields: 'id', 'learnerId', 'lessonId', 'status', 'runAt'.
- 7. **\*\*retry\_queue\*\***
  - Purpose: retryable failures with attempt metadata.
  - Typical fields: 'id', 'jobType', 'payload', 'attempts', 'nextRunAt', 'status', 'lastError'.
- 8. **\*\*audit\_log\*\***
  - Purpose: operational trace and change logs.
  - Typical fields: 'id', 'actor', 'action', 'resourceType', 'resourceId', 'status', 'message', 'metadata'.

### 5.2 Script Properties Map

- **\*\*Required\*\***
  - 'SLACK\_BOT\_TOKEN'
  - 'SLACK\_SIGNING\_SECRET'
  - 'SPREADSHEET\_ID'
- **\*\*Optional\*\***
  - 'SLACK\_VERIFICATION\_TOKEN'
  - 'ADMIN\_USER\_IDS'
  - 'DEFAULT\_COURSE\_ID'
  - 'DEFAULT\_TRACK'
  - 'OPS\_ALERT\_CHANNEL'

### 5.3 Slack Payload Fields (normalized envelope)

'routeType', 'rawBody', 'body', 'params', 'command', 'payloadType', 'userId', 'channelId', 'teamId', 'triggerId', 'response'

### 5.4 Audit Fields

- Host and service audit entries include action/status metadata and redacted logs.
- 'db.audit(...)' emits structured entries through shared library audit sink.

### 5.5 Correlation IDs

- No dedicated global correlation ID key is hardcoded in current scaffold.
- Recommended operational correlation strategy: use queue row IDs / submission IDs / audit row IDs as trace anchors. **\*\*V\*\***

## 6. Tool / Module Map

### Ingress

- 'code.gs#doPost(e)', 'code.gs#doGet()'

### Parser

- '02\_SlackPayloadParser.gs' ('SlackPayloadParser.parse')

### Security

- '04\_SlackSecurity.gs' ('SlackSecurity.verify' + HMAC/replay logic)

### Router

- '01\_SlackRouter.gs' ('SlackRouter.route')

### Services

- '03\_SlackService.gs' command/event/interactivity registries
- '09\_LmsEnrollmentService.gs', '12\_LmsLessonService.gs', '13\_LmsProgressService.gs', '13a\_LmsCompletionService.gs', '14\_LmsCompletionService.gs'

### DB Layer

- '00\_WebAppEntry.gs#createHostDbClient'
- 'sheet\_db\_lib/\*' ('SheetDb', 'DbClient', 'TableRepository', transactions, audit)

### Slack API Client

- '11\_SlackApiClient.gs'

### Block Kit Builder

- '12\_SlackBlockKitBuilder.gs'

### Scheduler

- '14\_Scheduler.gs'

### Retry

- '18\_RetryResolver.gs'

### Health

- '17\_HealthMonitor.gs'

### Config

- '16\_ConfigBootstrap.gs' (authoritative host runtime/script property loading)
- '23\_Config.gs' (SheetDb client-only config contract)
- 'sheet\_db\_lib/10\_ScriptProperties.gs'

### Test Harness



```

    code: routed.code || ''
  });

  return jsonResponse_(routed.response || ErrorService.create('MISSING_RESPONSE', 'missing_response', false, correlationId));
} catch (err) {
  try {
    logAuditExact_({
      actor: CONFIG.ACTOR_SYSTEM,
      action: 'doPost_error',
      status: 'error',
      correlationId: correlationId,
      message: safeError_(err)
    });
  } catch (logErr) {}

  return jsonResponse_(ErrorService.create('HOST_ERROR', 'Something went wrong in the Slack handler.', true, correlationId));
}
}

function textResponse_(text) {
  return ContentService.createTextOutput(String(text || '')).setMimeType(ContentService.MimeType.TEXT);
}

function jsonResponse_(obj) {
  return ContentService.createTextOutput(JSON.stringify(obj || {})).setMimeType(ContentService.MimeType.JSON);
}

function logAuditExact_(entry) {
  Logger.log(JSON.stringify(sanitizeObjectForLog_(entry || {})));
}

function safeError_(err) {
  return (err && err.message) ? String(err.message) : String(err);
}

function sanitizeObjectForLog_(obj) {
  var clean = {};
  Object.keys(obj || {}).forEach(function(key) {
    var lower = String(key).toLowerCase();
    if (lower.indexOf('token') !== -1 || lower.indexOf('secret') !== -1 || lower.indexOf('authorization') !== -1) {
      clean[key] = '[REDACTED]';
      return;
    }
    clean[key] = String(obj[key] == null ? '' : obj[key]);
  });
  return clean;
}

```

## deployment.md

### # Deployment Guide

#### ## 1. Purpose

This guide describes how to deploy the Slack LMS Apps Script integration using the current single-endpoint Slack manifest.

#### ## 2. Deployment Architecture

- Slack routes `/learn`, `/submit`, `/progress`, `interactivity`, `message.im`, and `app_mention` to one Apps Script Web App.
- Apps Script host (`code.gs#doPost`) orchestrates parser -> security -> router -> services.
- Runtime persistence is Google Sheets through shared `'SheetDb'` (`'LMSLibrary'`) helpers.
- Triggered jobs run lesson delivery, reminders, and reporting.

#### ## 3. Prerequisites

- Google Workspace account with Apps Script deployment permissions.
- Slack app admin access for command/events/interactivity settings.
- Target Google Spreadsheet with edit access for script execution identity.
- Repository code synced into Apps Script project.

#### ## 4. Required Configuration

Set these Script Properties in Apps Script project settings.

| Key                      | Required | Description                                               |
|--------------------------|----------|-----------------------------------------------------------|
| SLACK_BOT_TOKEN          | Yes      | Bot token used by 'SlackApiClient'.                       |
| SLACK_SIGNING_SECRET     | Yes      | Secret for signature verification.                        |
| SPREADSHEET_ID           | Yes      | Runtime spreadsheet ID for SheetDb client.                |
| SLACK_VERIFICATION_TOKEN | Optional | Fallback verification token when headers are unavailable. |
| ADMIN_USER_IDS           | Optional | CSV of admin users for future gated operations.           |
| DEFAULT_COURSE_ID        | Optional | Default course assignment fallback.                       |
| DEFAULT_TRACK            | Optional | Default track assignment fallback.                        |
| OPS_ALERT_CHANNEL        | Optional | Escalation target for reminders/incidents.                |

#### ## 5. Google Sheets Setup

1. Create/select a Google Spreadsheet for runtime data.
2. Provide its ID via 'SPREADSHEET\_ID'.
3. Ensure Apps Script execution identity has edit access.
4. First execution of host bootstrap will register required runtime tabs:
  - 'learners'
  - 'enrollment'
  - 'lessons'
  - 'learner\_progress'
  - 'submission\_log'
  - 'delivery\_queue'
  - 'retry\_queue'
  - 'audit\_log'

#### ## 6. Apps Script Setup

1. Open/create Apps Script project.
2. Add repository '.gs' files and markdown docs.
3. Confirm 'code.gs' is present with 'doGet' and 'doPost'.
4. Set Script Properties from Section 4.
5. Save project and run a lightweight function (e.g., 'ConfigBootstrap.validate') to verify property access.

#### ## 7. Slack App Setup

Use existing 'manifest.json' values:

- Slash commands:
  - '/learn'
  - '/submit'
  - '/progress'
  - '/report'
  - '/onboard'
- Events:
  - 'message.im'
  - 'app\_mention'
- Interactivity: enabled.

If using manifest import/update in Slack:

1. Open Slack app settings -> App Manifest.
2. Paste current 'manifest.json' content.
3. Save/apply changes.

#### ## 8. Endpoint Wiring

1. Deploy Apps Script as **Web App**.
2. Copy deployment URL.
3. Ensure Slack manifest command/event/interactivity URLs match this deployment URL.
4. Reinstall/update app in workspace when required by Slack.

#### ## 9. Trigger Setup

After first deploy:

1. Run 'setupTriggers()' once from Apps Script editor.
2. Confirm triggers exist for:
  - 'runDailyLessonDelivery'
  - 'runHourlyReminderCheck'
  - 'runWeeklyAdminReport'
3. Verify project timezone matches reporting expectations.

#### ## 10. Smoke Test Procedure

Execute in order:

1. **Health check**: run 'runHealthCheck()' manually.
2. **Slash commands**:
  - '/learn'
  - '/submit <lesson\_id>'
  - '/progress'
  - '/report'
  - '/onboard'
3. **Interactivity**: submit a test interactive payload (or 'hostTest\_fakeInteractive').
4. **Events**:
  - 'hostTest\_fakeAppMention'
  - 'hostTest\_fakeMessageIm'
5. **Scheduler jobs**:
  - run 'runDailyLessonDelivery'
  - run 'runHourlyReminderCheck'
6. Confirm writes in 'audit\_log', 'submission\_log', queue tables.

#### ## 11. Troubleshooting

- **403/verification errors**: check 'SLACK\_SIGNING\_SECRET', timestamp skew, and fallback token behavior.
- **No sheet writes**: validate 'SPREADSHEET\_ID' and spreadsheet permissions.
- **Slack API failures**: inspect retry queue and 'SlackApiClient' error shape.
- **Triggers missing**: re-run 'setupTriggers()'.
- **Interactivity timeout**: keep router+service response path lightweight (<3s) and offload heavy work to queue/scheduler.

#### ## 12. Production Readiness Checklist

- [ ] Required Script Properties set.



- [ ] Slack endpoint URL matches current deployment.
- [ ] Commands and events enabled in Slack app.
- [ ] Trigger jobs installed.
- [ ] Audit table receiving host events.
- [ ] Retry queue monitored.
- [ ] Runbook shared with operations.
- [ ] Rollback path documented.

### ## 13. Safe Change Procedure

1. Create branch and apply minimal patch.
2. Run syntax checks and smoke harness functions.
3. Deploy to non-production/staging (if available).
4. Validate slash/event/interactivity flows.
5. Promote deployment and monitor 'audit\_log' + 'retry\_queue'.
6. Roll back to prior deployment version if error rate rises.

docs/csv-sheetdb-schema.md

# RWR LMS CSV and SheetDb Schema Inventory

### ## Section 1 â CSV File Inventory

#### ### CSV file scan result

- No physical '\*.csv' files were originally present in the repository; generated header-only CSV artifacts now live under
- The codebase is sheet-backed: tab names are registered through 'db.schema(...)' and materialized through 'SheetsGateway'
- No 'headers.indexOf(...)'-driven dynamic CSV parsing was found.
- No direct 'sheet.getDataRange().getValues()' consumers outside the shared 'SheetsGateway.readTable(...)' helper were found

### ## Section 2 â Sheet / Table Inventory

#### ### Discovery summary

Unique sheet-backed tables found across the repo:

- Main RWR LMS runtime: 'learners', 'enrollment', 'lessons', 'learner\_progress', 'submission\_log', 'delivery\_queue', 're'
- Split Slack host example app: 'students', 'enrollments', 'automations', 'audit\_logs'

#### ### Generated CSV artifacts

- Header-only CSV files were generated for every discovered table under 'docs/generated\_csv/' so the sheet/tab schemas could

#### ### Table: 'learners'

- \*\*Origin:\*\* 'db.schema()'
- \*\*Sheet/tab name:\*\* 'learners'
- \*\*Column order confidence:\*\* exact from schema registration
- \*\*Columns (ordered):\*\*
  1. 'id'
  2. 'slackUserId'
  3. 'email'
  4. 'name'
  5. 'status'
  6. 'createdAt'
  7. 'updatedAt'
  8. 'deletedAt'
- \*\*Column details:\*\*
  - 'id': String, required at persistence level, primary key. Auto-generated with 'Util.generateId('learners')' when omitted.
  - 'slackUserId': String, effectively required by application flow because lookups/enrollment depend on it.
  - 'email': String, optional, defaults to empty string in 'ensureLearnerRecord()'.
  - 'name': String, optional, defaults to empty string in 'ensureLearnerRecord()'.
  - 'status': String, required by business flow on insert, observed enum values: 'active'.
  - 'createdAt': DateTime, auto-populated on insert.
  - 'updatedAt': DateTime, auto-populated on insert/update.
  - 'deletedAt': DateTime, optional, soft-delete marker used by the repository filter.
- \*\*Foreign keys:\*\* none.
- \*\*Soft delete:\*\* '{ enabled: true, column: 'deletedAt', archivedValue: '<timestamp set on remove>' }'
- \*\*Observed active-record filter:\*\* implicit in 'TableRepository.findAll()' because rows with blank 'deletedAt' only are
- \*\*AppSheet consumer evidence:\*\* none found.

#### ### Table: 'enrollment'

- \*\*Origin:\*\* 'db.schema()'
- \*\*Sheet/tab name:\*\* 'enrollment'
- \*\*Column order confidence:\*\* exact from schema registration
- \*\*Columns (ordered):\*\*
  1. 'id'
  2. 'learnerId'
  3. 'courseId'
  4. 'track'
  5. 'status'
  6. 'createdAt'
  7. 'updatedAt'
  8. 'deletedAt'
- \*\*Column details:\*\*
  - 'id': String, required at persistence level, primary key. Auto-generated when omitted.
  - 'learnerId': String, required by business flow, foreign key to 'learners.id'.

- 'courseId': String, required by business flow.
- 'track': String, optional in code path, defaults to configured track or 'ONBOARDING'.
- 'status': String, required by business flow, observed enum values: 'active'.
- 'createdAt': DateTime, auto-populated.
- 'updatedAt': DateTime, auto-populated.
- 'deletedAt': DateTime, optional soft-delete marker.
- \*\*Foreign keys:\*\* 'learnerId' -> learners.id'.
- \*\*Soft delete:\*\* enabled via 'deletedAt'.
- \*\*Observed active-record filter:\*\* implicit repository 'deletedAt' filter; reports also filter 'status === 'active'' f
- \*\*AppSheet consumer evidence:\*\* none found.

### ### Table: 'lessons'

- \*\*Origin:\*\* 'db.schema()'
- \*\*Sheet/tab name:\*\* 'lessons'
- \*\*Column order confidence:\*\* exact from schema registration
- \*\*Columns (ordered):\*\*
  1. 'id'
  2. 'courseId'
  3. 'moduleId'
  4. 'sequenceNumber'
  5. 'track'
  6. 'title'
  7. 'topic'
  8. 'objective'
  9. 'difficulty'
  10. 'hook'
  11. 'coreContent'
  12. 'insight'
  13. 'takeaway'
  14. 'mission'
  15. 'missionType'
  16. 'missionDuration'
  17. 'verification'
  18. 'submitBlock'
  19. 'contentRef'
  20. 'slackPayload'
  21. 'active'
  22. 'createdAt'
  23. 'updatedAt'
  24. 'deletedAt'
- \*\*Column details:\*\*
  - 'id': String, primary key. Auto-generated when omitted, though runtime sync prefers caller-supplied 'row.id'.
  - 'courseId': String, optional in sync helper, defaults to empty string.
  - 'moduleId': String, optional module reference for Universal Lesson Canvas grouping.
  - 'sequenceNumber': String-encoded numeric sort index used for next-lesson sequencing.
  - 'track': String legacy routing field, mirrored from 'topic' when omitted.
  - 'title': String lesson title from metadata.
  - 'topic': String lesson topic from metadata.
  - 'objective': String lesson objective text.
  - 'difficulty': String level from metadata (default 'independent').
  - 'hook', 'coreContent', 'insight', 'takeaway', 'mission', 'verification': String blocks for sections 01â 06 of Unive
  - 'missionType': String mission response format (default 'text').
  - 'missionDuration': String mission duration (default '3 min').
  - 'submitBlock': String submit instructions block text.
  - 'contentRef': String optional external content pointer.
  - 'slackPayload': Stringified JSON Slack payload (e.g., '{ "text": "...", "blocks": [...] }') used for direct bot deli
  - 'active': String/Boolean-like flag. Sync helper persists 'String(...)'; observed values: 'true', inherited row val
  - 'createdAt': DateTime, auto-populated.
  - 'updatedAt': DateTime, auto-populated.
  - 'deletedAt': DateTime, optional soft-delete marker.
- \*\*Foreign keys:\*\* none enforced, though 'courseId' and 'moduleId' behave like curriculum references.
- \*\*Soft delete:\*\* enabled via 'deletedAt'.
- \*\*AppSheet consumer evidence:\*\* none found.

### ### Table: 'learner\_progress'

- \*\*Origin:\*\* 'db.schema()'
- \*\*Sheet/tab name:\*\* 'learner\_progress'
- \*\*Column order confidence:\*\* exact from schema registration
- \*\*Columns (ordered):\*\*
  1. 'id'
  2. 'learnerId'
  3. 'lessonId'
  4. 'state'
  5. 'dueAt'
  6. 'completedAt'
  7. 'createdAt'
  8. 'updatedAt'
  9. 'deletedAt'
- \*\*Column details:\*\*
  - 'id': String, primary key. Auto-generated when omitted.



```

10. 'deletedAt'
- **Column details:**
  - 'id': String, primary key. Auto-generated when omitted.
  - 'jobType': String, defaults to 'unknown'.
  - 'payload': String, JSON-serialized object, defaults to '{}' from 'JSON.stringify(payload.payload || {})'
  - 'attempts': Integer-like string. Stored as 'String(payload.attempts || 0)' and incremented as strings.
  - 'nextRunAt': DateTime, defaults to current timestamp if omitted.
  - 'status': String, observed enum values: 'queued', 'dead_letter'.
  - 'lastError': String, optional, defaults to empty string.
  - 'createdAt': DateTime, auto-populated.
  - 'updatedAt': DateTime, auto-populated.
  - 'deletedAt': DateTime, optional soft-delete marker.
- **Foreign keys:** none.
- **Soft delete:** enabled via 'deletedAt'.
- **Observed active-record filter:** implicit repository 'deletedAt' filter; retry scheduler additionally filters 'status'
- **AppSheet consumer evidence:** none found.

### Table: 'audit_log'
- **Origin:** 'db.schema()'
- **Sheet/tab name:** 'audit_log'
- **Column order confidence:** exact from schema registration
- **Columns (ordered):**
  1. 'id'
  2. 'actor'
  3. 'action'
  4. 'resourceType'
  5. 'resourceId'
  6. 'status'
  7. 'message'
  8. 'metadata'
  9. 'createdAt'
  10. 'updatedAt'
  11. 'deletedAt'
- **Column details:**
  - 'id': String, primary key. Auto-generated when omitted.
  - 'actor': String, observed values: 'system', 'scheduler'.
  - 'action': String, required by application writes.
  - 'resourceType': String, observed values: 'slack_host', 'job'.
  - 'resourceId': String, optional, commonly empty string.
  - 'status': String, observed enum values: 'info', 'error' plus caller-supplied job status.
  - 'message': String, optional human-readable audit message.
  - 'metadata': String, JSON-serialized object.
  - 'createdAt': DateTime, auto-populated.
  - 'updatedAt': DateTime, auto-populated.
  - 'deletedAt': DateTime, present in current implementation even though this is an audit/log table.
- **Foreign keys:** none.
- **Soft delete:** implementation supports 'deletedAt', but this table is logically append-only / audit-oriented.
- **Audit/log classification:** yes â should be treated as immutable in downstream design even though current generic
- **AppSheet consumer evidence:** none found.

### Table: 'students'
- **Origin:** 'db.schema()'
- **Sheet/tab name:** 'students'
- **Column order confidence:** exact from schema registration
- **Columns (ordered):**
  1. 'id'
  2. 'name'
  3. 'source'
  4. 'createdAt'
  5. 'updatedAt'
  6. 'deletedAt'
- **Column details:**
  - 'id': String, primary key, explicitly required by schema and caller-supplied in host flows.
  - 'name': String, required by schema.
  - 'source': String, optional with schema default 'slack'; observed values include 'slash_command', 'workflow_webhook'
  - 'createdAt': DateTime, auto-populated.
  - 'updatedAt': DateTime, auto-populated.
  - 'deletedAt': DateTime, optional soft-delete marker.
- **Foreign keys:** none.
- **Soft delete:** enabled via 'deletedAt'.
- **AppSheet consumer evidence:** none found.

### Table: 'enrollments'
- **Origin:** 'db.schema()'
- **Sheet/tab name:** 'enrollments'
- **Column order confidence:** exact from schema registration
- **Columns (ordered):**
  1. 'id'
  2. 'studentId'
  3. 'courseId'

```

```

4. 'status'
5. 'source'
6. 'createdAt'
7. 'updatedAt'
8. 'deletedAt'
- **Column details:**
- 'id': String, primary key. Auto-generated when omitted.
- 'studentId': String, required by schema, foreign key to 'students.id'.
- 'courseId': String, required by schema.
- 'status': String, optional with schema default 'enrolled'; observed enum values: 'enrolled' and workflow-provided
- 'source': String, optional with schema default 'slack'; observed values: 'slash_command', 'workflow_webhook'.
- 'createdAt': DateTime, auto-populated.
- 'updatedAt': DateTime, auto-populated.
- 'deletedAt': DateTime, optional soft-delete marker.
- **Foreign keys:** 'studentId -> students.id'.
- **Soft delete:** enabled via 'deletedAt'.
- **AppSheet consumer evidence:** none found.

```

### ### Table: 'automations'

```

- **Origin:** 'db.schema()'
- **Sheet/tab name:** 'automations'
- **Column order confidence:** exact from schema registration
- **Columns (ordered):**
1. 'id'
2. 'requestId'
3. 'workflow'
4. 'eventType'
5. 'state'
6. 'createdAt'
7. 'updatedAt'
8. 'deletedAt'
- **Column details:**
- 'id': String, primary key. Auto-generated when omitted.
- 'requestId': String, required by schema.
- 'workflow': String, required by schema, defaults in service to 'unknown_workflow' before insert path.
- 'eventType': String, required by schema, defaults in service to 'workflow_event' before insert path.
- 'state': String, optional with schema default 'received'; observed enum values: 'received', 'queued' and payload-d
- 'createdAt': DateTime, auto-populated.
- 'updatedAt': DateTime, auto-populated.
- 'deletedAt': DateTime, optional soft-delete marker.
- **Foreign keys:** none.
- **Soft delete:** enabled via 'deletedAt'.
- **AppSheet consumer evidence:** none found.

```

### ### Table: 'audit\_logs'

```

- **Origin:** 'db.schema()'
- **Sheet/tab name:** 'audit_logs'
- **Column order confidence:** exact from schema registration
- **Columns (ordered):**
1. 'id'
2. 'requestId'
3. 'source'
4. 'action'
5. 'status'
6. 'message'
7. 'metadata'
8. 'createdAt'
9. 'updatedAt'
10. 'deletedAt'
- **Column details:**
- 'id': String, primary key. Auto-generated when omitted.
- 'requestId': String, required by schema.
- 'source': String, required by schema; observed values include route types such as 'slash_command' / 'workflow_webhook
- 'action': String, required by schema; observed values include 'enrollment_upsert', 'automation_upsert', route type n
- 'status': String, required by schema; observed enum values: 'success', 'failure'.
- 'message': String, optional.
- 'metadata': String, optional JSON string.
- 'createdAt': DateTime, auto-populated.
- 'updatedAt': DateTime, auto-populated.
- 'deletedAt': DateTime, present in implementation even though this is an audit/log table.
- **Foreign keys:** none.
- **Soft delete:** implementation supports 'deletedAt', but the table is logically append-only / audit-oriented.
- **Audit/log classification:** yes.
- **AppSheet consumer evidence:** none found.

```

## ## Section 3 â Cross-cutting Patterns

### ### Header / column-order reconstruction notes

```

- All discovered tables had explicit 'db.schema(...)' registrations, so column order is known exactly from code.
- No table required fallback reconstruction from 'data[0]' or 'headers.indexOf(...)'.

```

### Generic SheetDb persistence behavior

- 'id' is the effective primary key for every registered table.
- 'createdAt', 'updatedAt', and 'deletedAt' are conventional system columns configured through 'Config' / script property.
- Insert behavior:
  - auto-generates 'id' when missing
  - auto-populates 'createdAt'
  - always refreshes 'updatedAt'
  - initializes 'deletedAt' to '' when omitted
- 'findAll()' only returns rows where 'deletedAt' is blank, confirming soft-delete behavior is active globally.

### Soft-delete and audit notes

- The current codebase uses 'deletedAt', not '\_rowStatus'.
- No table in this repo currently uses '\_rowStatus = 'active'' filtering.
- Audit/log tables ('audit\_log', 'audit\_logs') still include 'deletedAt' because the generic schema scaffold uses the same.

### AppSheet consumer scan

- No direct AppSheet integration, AppConfig, webhook handler, '\_Key' formula column, read-only AppSheet annotations, or
- Therefore no table can be positively flagged as AppSheet-consumed from current code.

## Section 4 â Ready-to-paste SheetDb schema block

```
'''javascript
function registerRwrLmsSchemas(db) {
  db.schema('learners', {
    columns: ['id', 'slackUserId', 'email', 'name', 'status', 'createdAt', 'updatedAt', 'deletedAt'],
    fields: {
      id: { type: 'string' },
      slackUserId: { type: 'string' },
      email: { type: 'string' },
      name: { type: 'string' },
      status: { type: 'string' }
    }
  });

  db.schema('enrollment', {
    columns: ['id', 'learnerId', 'courseId', 'track', 'status', 'createdAt', 'updatedAt', 'deletedAt'],
    fields: {
      id: { type: 'string' },
      learnerId: { type: 'string' },
      courseId: { type: 'string' },
      track: { type: 'string', default: 'ONBOARDING' },
      status: { type: 'string' }
    }
  });

  db.schema('lessons', {
    columns: [
      'id', 'courseId', 'moduleId', 'sequenceNumber', 'track',
      'title', 'topic', 'objective', 'difficulty',
      'hook', 'coreContent', 'insight', 'takeaway',
      'mission', 'missionType', 'missionDuration', 'verification', 'submitBlock',
      'contentRef', 'slackPayload', 'active', 'createdAt', 'updatedAt', 'deletedAt'
    ],
    fields: {
      id: { type: 'string' },
      courseId: { type: 'string' },
      moduleId: { type: 'string', default: '' },
      sequenceNumber: { type: 'string', default: '0' },
      track: { type: 'string' },
      title: { type: 'string' },
      topic: { type: 'string', default: '' },
      objective: { type: 'string', default: '' },
      difficulty: { type: 'string', default: 'independent' },
      hook: { type: 'string', default: '' },
      coreContent: { type: 'string', default: '' },
      insight: { type: 'string', default: '' },
      takeaway: { type: 'string', default: '' },
      mission: { type: 'string', default: '' },
      missionType: { type: 'string', default: 'text' },
      missionDuration: { type: 'string', default: '3 min' },
      verification: { type: 'string', default: '' },
      submitBlock: { type: 'string', default: '' },
      contentRef: { type: 'string' },
      slackPayload: { type: 'string', default: '' },
      active: { type: 'string', default: 'true' }
    }
  });

  db.schema('learner_progress', {
```

```

columns: ['id', 'learnerId', 'lessonId', 'state', 'dueAt', 'completedAt', 'createdAt', 'updatedAt', 'deletedAt'],
fields: {
  id: { type: 'string' },
  learnerId: { type: 'string' },
  lessonId: { type: 'string' },
  state: { type: 'string' },
  dueAt: { type: 'string' },
  completedAt: { type: 'string' }
}
});

db.schema('submission_log', {
  columns: ['id', 'learnerId', 'lessonId', 'submitKey', 'payload', 'createdAt', 'updatedAt', 'deletedAt'],
  fields: {
    id: { type: 'string' },
    learnerId: { type: 'string' },
    lessonId: { type: 'string' },
    submitKey: { type: 'string' },
    payload: { type: 'string', default: '' }
  }
});

db.schema('delivery_queue', {
  columns: ['id', 'learnerId', 'lessonId', 'status', 'runAt', 'createdAt', 'updatedAt', 'deletedAt'],
  fields: {
    id: { type: 'string' },
    learnerId: { type: 'string' },
    lessonId: { type: 'string' },
    status: { type: 'string' },
    runAt: { type: 'string' }
  }
});

db.schema('retry_queue', {
  columns: ['id', 'jobType', 'payload', 'attempts', 'nextRunAt', 'status', 'lastError', 'createdAt', 'updatedAt', 'deletedAt'],
  fields: {
    id: { type: 'string' },
    jobType: { type: 'string', default: 'unknown' },
    payload: { type: 'string', default: '{}' },
    attempts: { type: 'string', default: '0' },
    nextRunAt: { type: 'string' },
    status: { type: 'string', default: 'queued' },
    lastError: { type: 'string', default: '' }
  }
});

db.schema('audit_log', {
  columns: ['id', 'actor', 'action', 'resourceType', 'resourceId', 'status', 'message', 'metadata', 'createdAt', 'updatedAt', 'deletedAt'],
  fields: {
    id: { type: 'string' },
    actor: { type: 'string' },
    action: { type: 'string' },
    resourceType: { type: 'string' },
    resourceId: { type: 'string', default: '' },
    status: { type: 'string' },
    message: { type: 'string' },
    metadata: { type: 'string', default: '{}' }
  }
});

db.schema('students', {
  columns: ['id', 'name', 'source', 'createdAt', 'updatedAt', 'deletedAt'],
  fields: {
    id: { required: true, type: 'string' },
    name: { required: true, type: 'string' },
    source: { type: 'string', default: 'slack' }
  }
});

db.schema('enrollments', {
  columns: ['id', 'studentId', 'courseId', 'status', 'source', 'createdAt', 'updatedAt', 'deletedAt'],
  fields: {
    id: { type: 'string' },
    studentId: { required: true, type: 'string' },
    courseId: { required: true, type: 'string' },
    status: { type: 'string', default: 'enrolled' },
    source: { type: 'string', default: 'slack' }
  }
});

```





id,name,source,createdAt,updatedAt,deletedAt

docs/generated\_csv/submission\_log.csv

id, learnerId, lessonId, submitKey, payload, createdAt, updatedAt, deletedAt

included-files.txt

docs/csv-sheetdb-schema.md

docs/generated\_csv/audit\_log.csv

docs/generated\_csv/audit\_logs.csv

docs/generated\_csv/automations.csv

docs/generated\_csv/delivery\_queue.csv

docs/generated\_csv/enrollment.csv

docs/generated\_csv/enrollments.csv

docs/generated\_csv/learner\_progress.csv

docs/generated\_csv/learners.csv

docs/generated\_csv/lessons.csv

docs/generated\_csv/retry\_queue.csv

docs/generated\_csv/students.csv

docs/generated\_csv/submission\_log.csv

00\_WebAppEntry.gs

01\_SlackRouter.gs

02\_SlackPayloadParser.gs

03\_SlackService.gs

04\_SlackSecurity.gs

05\_LmsEnrollmentService.gs

06\_LmsLessonService.gs

07\_LmsProgressService.gs

08\_LmsCompletionService.gs

09\_LmsReminderService.gs

10\_LmsReportService.gs

11\_SlackApiClient.gs

12\_SlackBlockKitBuilder.gs

13\_LearnerProgressStateMachine.gs

14\_Scheduler.gs

15\_SheetsDataSync.gs

16\_ConfigBootstrap.gs

17\_HealthMonitor.gs

18\_RetryResolver.gs

19\_HostTestHarness.gs

20\_Errors.gs

21\_Util.gs

22\_ScriptProperties.gs

23\_Config.gs

24\_SheetsGateway.gs

25\_SchemaRegistry.gs

26\_TableRepository.gs

27\_TransactionManager.gs

28\_AuditLogger.gs

29\_DbClient.gs

30\_SheetDb.gs

code.gs

sheet\_db\_lib/00\_SheetDb.gs

sheet\_db\_lib/01\_DbClient.gs

sheet\_db\_lib/02\_Config.gs

sheet\_db\_lib/03\_Errors.gs

sheet\_db\_lib/04\_Util.gs

sheet\_db\_lib/05\_SchemaRegistry.gs

sheet\_db\_lib/06\_SheetsGateway.gs

sheet\_db\_lib/07\_TableRepository.gs

sheet\_db\_lib/08\_TransactionManager.gs

sheet\_db\_lib/09\_AuditLogger.gs

sheet\_db\_lib/10\_ScriptProperties.gs

slack\_host/00\_WebAppEntry.gs

slack\_host/01\_SlackRouter.gs

slack\_host/02\_SlackPayloadParser.gs

slack\_host/03\_SlackService.gs

slack\_host/04\_SlackSecurity.gs

slack\_host/05\_LmsEnrollmentService.gs

slack\_host/06\_LmsAutomationService.gs

slack\_host/07\_RequestContextFactory.gs

slack\_host/08\_ResultFormatter.gs

slack\_host/09\_WorkflowWebhookHandlers.gs

slack\_host/10\_HostConfig.gs

slack\_host/11\_HostTestHarness.gs

manifest.json

```
{
  "display_information": {
    "name": "RWR LMS Bot",
    "description": "Slack-based LMS for onboarding and daily lesson delivery",
```

```

"background_color": "#2E3A59"
},
"features": {
  "bot_user": {
    "display_name": "RWR LMS",
    "always_online": true
  },
  "slash_commands": [
    {
      "command": "/learn",
      "url": "https://script.google.com/macros/s/AKfycbxACBb8JvISTCL87-0LdKgxm2axtxRtCfzCW9XAT6m-j6oF7M-mFMBASjTDhG7N6",
      "description": "Get your current lesson",
      "should_escape": false
    },
    {
      "command": "/submit",
      "url": "https://script.google.com/macros/s/AKfycbxACBb8JvISTCL87-0LdKgxm2axtxRtCfzCW9XAT6m-j6oF7M-mFMBASjTDhG7N6",
      "description": "Submit a completed lesson",
      "should_escape": false
    },
    {
      "command": "/progress",
      "url": "https://script.google.com/macros/s/AKfycbxACBb8JvISTCL87-0LdKgxm2axtxRtCfzCW9XAT6m-j6oF7M-mFMBASjTDhG7N6",
      "description": "Check your progress",
      "should_escape": false
    },
    {
      "command": "/report",
      "url": "https://script.google.com/macros/s/AKfycbxACBb8JvISTCL87-0LdKgxm2axtxRtCfzCW9XAT6m-j6oF7M-mFMBASjTDhG7N6",
      "description": "Admin report dashboard",
      "should_escape": false
    },
    {
      "command": "/onboard",
      "url": "https://script.google.com/macros/s/AKfycbxACBb8JvISTCL87-0LdKgxm2axtxRtCfzCW9XAT6m-j6oF7M-mFMBASjTDhG7N6",
      "description": "Trigger onboarding pipeline",
      "should_escape": false
    },
    {
      "command": "/gaps",
      "url": "https://script.google.com/macros/s/AKfycbxACBb8JvISTCL87-0LdKgxm2axtxRtCfzCW9XAT6m-j6oF7M-mFMBASjTDhG7N6",
      "description": "Show overdue learning gaps",
      "should_escape": false
    },
    {
      "command": "/audit",
      "url": "https://script.google.com/macros/s/AKfycbxACBb8JvISTCL87-0LdKgxm2axtxRtCfzCW9XAT6m-j6oF7M-mFMBASjTDhG7N6",
      "description": "Show audit summary snapshot",
      "should_escape": false
    },
    {
      "command": "/mix",
      "url": "https://script.google.com/macros/s/AKfycbxACBb8JvISTCL87-0LdKgxm2axtxRtCfzCW9XAT6m-j6oF7M-mFMBASjTDhG7N6",
      "description": "Get mixed learning content",
      "should_escape": false
    },
    {
      "command": "/reinforce",
      "url": "https://script.google.com/macros/s/AKfycbxACBb8JvISTCL87-0LdKgxm2axtxRtCfzCW9XAT6m-j6oF7M-mFMBASjTDhG7N6",
      "description": "Get reinforcement guidance",
      "should_escape": false
    },
    {
      "command": "/offboard",
      "url": "https://script.google.com/macros/s/AKfycbxACBb8JvISTCL87-0LdKgxm2axtxRtCfzCW9XAT6m-j6oF7M-mFMBASjTDhG7N6",
      "description": "Record an offboarding request",
      "should_escape": false
    }
  ]
},
"oauth_config": {
  "scopes": {
    "bot": [
      "chat:write",
      "im:write",
      "im:history",
      "app_mentions:read",
      "commands"
    ]
  }
}

```

```

    },
    "settings": {
      "event_subscriptions": {
        "request_url": "https://script.google.com/macros/s/AKfycbxACBb8JvIstCL87-OLdKgxm2axtxRtCfzCW9XAT6m-j6oF7M-mFMBASjT",
        "bot_events": [
          "message.im",
          "app_mention"
        ]
      },
      "interactivity": {
        "is_enabled": true,
        "request_url": "https://script.google.com/macros/s/AKfycbxACBb8JvIstCL87-OLdKgxm2axtxRtCfzCW9XAT6m-j6oF7M-mFMBASjT"
      },
      "org_deploy_enabled": false,
      "socket_mode_enabled": false,
      "token_rotation_enabled": false
    }
  }
}

```

```

md_mirror/sheet_db_lib/00_SheetDb.md
# 00_SheetDb.gs

```

```

**Source path:** 'sheet_db_lib/00_SheetDb.gs'

```

```

'''javascript
/**
 * @fileoverview Public, library-visible exports for SheetDb.
 */

/**
 * Library semantic version.
 *
 * NOTE: Keep this in sync with release tags when publishing as an Apps Script library.
 * @type {string}
 */
var SHEET_DB_VERSION = '0.1.0';

```

```

/**
 * Creates a new SheetDb client instance.
 *
 * @param {Object=} options Client configuration options.
 * @return {DbClient}
 */

```

```

function _sheetDbCreateClient(options) {
  return new DbClient(new Config(options || {}));
}

```

```

/**
 * Bootstraps a client with one or more table schemas.
 *
 * @param {Object=} options
 * @param {Array<{name: string, schema: Object}>=} options.tables Tables to register.
 * @param {Object=} options.clientOptions Configuration passed to 'createClient'.
 * @return {DbClient}
 */

```

```

function _sheetDbBootstrap(options) {
  var opts = options || {};
  var client = _sheetDbCreateClient(opts.clientOptions || {});
  var tables = opts.tables || [];

  tables.forEach(function(table) {
    if (table && table.name && table.schema) {
      client.registerTable(table.name, table.schema);
    }
  });

  return client;
}

```

```

/**
 * Basic health check that can be called by consuming projects.
 *
 * @return {{ok: boolean, version: string, timestamp: string, details: Object}}
 */

```

```

function _sheetDbHealthCheck() {
  try {
    var client = _sheetDbCreateClient();
    return {
      ok: !!client,
      version: SHEET_DB_VERSION,

```

```

        timestamp: Util.nowIso(),
        details: {
            clientReady: true,
            spreadsheetMode: client._config && client._config.spreadsheetId ? 'by_id' : 'active'
        }
    };
} catch (err) {
    return {
        ok: false,
        version: SHEET_DB_VERSION,
        timestamp: new Date().toISOString(),
        details: {
            code: err.code || 'HEALTH_CHECK_FAILED',
            message: err.message || 'Unknown error.'
        }
    };
}
}
}

```

```

/**
 * Creates a client using Script Properties with optional explicit overrides.
 * Precedence: explicit options > script properties > defaults.
 * @param {Object=} options
 * @return {DbClient}
 */

```

```

function _sheetDbCreateClientFromScriptProperties(options) {
    var merged = mergeClientOptionsWithScriptProperties(options || {});
    return _sheetDbCreateClient(merged);
}

```

```

/**
 * Public Apps Script library namespace.
 *
 * Consumers should call library exports through this object:
 * 'LibraryIdentifier.SheetDb.createClient(...)'.
 *
 * @type {{
 *   createClient: function(Object=): DbClient,
 *   createClientFromScriptProperties: function(Object=): DbClient,
 *   bootstrap: function(Object=): DbClient,
 *   healthCheck: function(): Object,
 *   version: string
 * }}
 */

```

```

var SheetDb = {
    createClient: function(options) {
        return _sheetDbCreateClient(options);
    },

    createClientFromScriptProperties: function(options) {
        return _sheetDbCreateClientFromScriptProperties(options);
    },

    bootstrap: function(options) {
        return _sheetDbBootstrap(options);
    },

    healthCheck: function() {
        return _sheetDbHealthCheck();
    },

    version: SHEET_DB_VERSION
};

```

```

/**
 * Backward-compatible factory export.
 * @deprecated Use 'SheetDb.createClient' instead.
 * @param {Object=} options
 * @return {DbClient}
 */

```

```

function createSheetDbClient(options) {
    return SheetDb.createClient(options);
}

```

```

'''

```

```

md_mirror/sheet_db_lib/01_DbClient.md
# 01_DbClient.gs

```

```

**Source path:** 'sheet_db_lib/01_DbClient.gs'

```



```

        var self = this;
        return this._tx.run(function() {
            return callback(self);
        });
    }
}

```

'''

md\_mirror/sheet\_db\_lib/02\_Config.md  
# 02\_Config.gs

\*\*Source path:\*\* 'sheet\_db\_lib/02\_Config.gs'

```

'''javascript
/**
 * Immutable runtime configuration for the Sheet DB client.
 */
class Config {
    /**
     * @param {Object=} options
     * @param {string=} options.spreadsheetId Spreadsheet ID; defaults to active spreadsheet.
     * @param {string=} options.idColumn Record primary key column name.
     * @param {string=} options.createdAtColumn Created timestamp column name.
     * @param {string=} options.updatedAtColumn Updated timestamp column name.
     * @param {string=} options.deletedAtColumn Soft-delete timestamp column name.
     * @param {number=} options.lockTimeoutMs Lock acquisition timeout for transactions.
     */
    constructor(options) {
        var opts = options || {};
        this.spreadsheetId = opts.spreadsheetId || '';
        this.idColumn = opts.idColumn || 'id';
        this.createdAtColumn = opts.createdAtColumn || 'createdAt';
        this.updatedAtColumn = opts.updatedAtColumn || 'updatedAt';
        this.deletedAtColumn = opts.deletedAtColumn || 'deletedAt';
        this.lockTimeoutMs = Util.isNil(opts.lockTimeoutMs) ? 30000 : Number(opts.lockTimeoutMs);
        this.validate();
        Object.freeze(this);
    }

    /**
     * @return {void}
     */
    validate() {
        var columns = [
            this.idColumn,
            this.createdAtColumn,
            this.updatedAtColumn,
            this.deletedAtColumn
        ];
        var unique = {};
        columns.forEach(function(column) {
            if (!column || typeof column !== 'string') {
                throw new ConfigError('System column names must be non-empty strings.');
            }
            if (unique[column]) {
                throw new ConfigError('System column names must be unique. Duplicate: ' + column);
            }
            unique[column] = true;
        });

        if (isNaN(this.lockTimeoutMs) || this.lockTimeoutMs <= 0) {
            throw new ConfigError('lockTimeoutMs must be a positive number.');
        }
    }
}
'''

```

md\_mirror/sheet\_db\_lib/03\_Errors.md  
# 03\_Errors.gs

\*\*Source path:\*\* 'sheet\_db\_lib/03\_Errors.gs'

```

'''javascript
/**
 * Base error for the Sheet DB library.
 */
class SheetDbError extends Error {
    /**

```

```

    * @param {string} message Human-readable error message.
    * @param {string=} code Stable error code for programmatic handling.
    */
    constructor(message, code) {
        super(message);
        this.name = this.constructor.name;
        this.code = code || 'SHEET_DB_ERROR';
    }
}

/**
 * Raised when configuration is invalid or missing.
 */
class ConfigError extends SheetDbError {
    /**
     * @param {string} message
     */
    constructor(message) {
        super(message, 'CONFIG_ERROR');
    }
}

/**
 * Raised when schema definitions are invalid.
 */
class SchemaError extends SheetDbError {
    /**
     * @param {string} message
     */
    constructor(message) {
        super(message, 'SCHEMA_ERROR');
    }
}

/**
 * Raised when records fail validation.
 */
class ValidationError extends SheetDbError {
    /**
     * @param {string} message
     */
    constructor(message) {
        super(message, 'VALIDATION_ERROR');
    }
}

/**
 * Raised when underlying sheet operations fail.
 */
class StorageError extends SheetDbError {
    /**
     * @param {string} message
     */
    constructor(message) {
        super(message, 'STORAGE_ERROR');
    }
}

/**
 * Raised for optimistic transaction failures.
 */
class TransactionError extends SheetDbError {
    /**
     * @param {string} message
     */
    constructor(message) {
        super(message, 'TRANSACTION_ERROR');
    }
}

'''

md_mirror/sheet_db_lib/04_Util.md
# 04_Util.gs

**Source path:** 'sheet_db_lib/04_Util.gs'

'''javascript
/**
 * Utility helpers for Sheet DB.

```

```

*/
class Util {
  /**
   * @return {string} RFC3339 timestamp in UTC.
   */
  static nowIso() {
    return new Date().toISOString();
  }

  /**
   * Creates a shallow clone of an object.
   * @param {Object} source
   * @return {Object}
   */
  static clone(source) {
    return Object.assign({}, source || {});
  }

  /**
   * Creates a stable row ID with a prefix.
   * @param {string=} prefix
   * @return {string}
   */
  static generateId(prefix) {
    var p = prefix || 'row';
    var random = Math.random().toString(36).slice(2, 10);
    return p + '_' + new Date().getTime() + '_' + random;
  }

  /**
   * Returns true when value is null or undefined.
   * @param {*} value
   * @return {boolean}
   */
  static isNil(value) {
    return value === null || value === undefined;
  }

  /**
   * Returns true when value is empty string, null, or undefined.
   * @param {*} value
   * @return {boolean}
   */
  static isBlank(value) {
    return value === '' || Util.isNil(value);
  }

  /**
   * Throws when a required condition is not met.
   * @param {boolean} condition
   * @param {string} message
   */
  static assert(condition, message) {
    if (!condition) {
      throw new ValidationError(message);
    }
  }

  /**
   * Converts a sheet row to an object using the header order.
   * @param {string[]} headers
   * @param {Array<*>} row
   * @return {Object}
   */
  static rowToObject(headers, row) {
    var out = {};
    for (var i = 0; i < headers.length; i++) {
      out[headers[i]] = row[i];
    }
    return out;
  }

  /**
   * Converts an object to row array by header order.
   * @param {string[]} headers
   * @param {Object} obj
   * @return {Array<*>}
   */
  static objectToRow(headers, obj) {
    return headers.map(function(h) {

```



```

        return obj[h];
    });
}
}

'''
md_mirror/sheet_db_lib/05_SchemaRegistry.md
# 05_SchemaRegistry.gs

**Source path:** 'sheet_db_lib/05_SchemaRegistry.gs'

'''javascript
/**
 * Holds table schemas and applies field defaults/validation.
 */
class SchemaRegistry {
    /**
     * Creates a new in-memory schema registry.
     */
    constructor() {
        this._schemas = {};
    }

    /**
     * Registers a table schema.
     * @param {string} tableName
     * @param {Object} schema
     * @param {Array<string>} schema.columns Ordered columns for persistence.
     * @param {Object<string, Object>=} schema.fields Optional field constraints.
     * @return {SchemaRegistry}
     */
    register(tableName, schema) {
        if (!tableName) {
            throw new SchemaError('Table name is required.');
```



```

    if (!sheet) {
        throw new StorageError('Sheet not found for table: ' + tableName);
    }
    return sheet;
}

/**
 * @param {string} tableName
 * @return {{headers: string[], rows: Array<Array<*>>}}
 */
readTable(tableName) {
    var sheet = this.getSheet(tableName, false);
    var range = sheet.getDataRange();
    var values = range.getValues();
    if (values.length === 0) {
        return { headers: [], rows: [] };
    }
    return {
        headers: values[0],
        rows: values.slice(1)
    };
}

/**
 * Ensures table exists with given headers.
 * @param {string} tableName
 * @param {string[]} headers
 */
ensureTable(tableName, headers) {
    var sheet = this.getSheet(tableName, true);
    var lastRow = sheet.getLastRow();
    if (lastRow === 0) {
        sheet.getRange(1, 1, 1, headers.length).setValues([headers]);
        return;
    }

    var existingHeaders = sheet.getRange(1, 1, 1, sheet.getLastColumn()).getValues()[0]
        .map(function(value) { return String(value); });
    var expectedHeaders = headers.map(function(value) { return String(value); });
    if (existingHeaders.join('|') !== expectedHeaders.join('|')) {
        throw new StorageError('Header mismatch for table: ' + tableName);
    }
}

/**
 * Writes all rows for a table, replacing existing body rows.
 * @param {string} tableName
 * @param {string[]} headers
 * @param {Array<Array<*>>} rows
 */
writeAllRows(tableName, headers, rows) {
    var sheet = this.getSheet(tableName, true);
    this.ensureTable(tableName, headers);

    var maxRows = Math.max(sheet.getMaxRows(), rows.length + 1);
    if (sheet.getMaxRows() < maxRows) {
        sheet.insertRowsAfter(sheet.getMaxRows(), maxRows - sheet.getMaxRows());
    }

    if (sheet.getLastRow() > 1) {
        sheet.getRange(2, 1, sheet.getLastRow() - 1, headers.length).clearContent();
    }

    if (rows.length > 0) {
        sheet.getRange(2, 1, rows.length, headers.length).setValues(rows);
    }
}

'''
md_mirror/sheet_db_lib/07_TableRepository.md
# 07_TableRepository.gs

**Source path:** 'sheet_db_lib/07_TableRepository.gs'

'''javascript
/**
 * CRUD repository for a specific table.
 */

```

```

class TableRepository {
    /**
     * @param {string} tableName
     * @param {Config} config
     * @param {SchemaRegistry} schemaRegistry
     * @param {SheetsGateway} sheetsGateway
     * @param {AuditLogger} auditLogger
     */
    constructor(tableName, config, schemaRegistry, sheetsGateway, auditLogger) {
        this._tableName = tableName;
        this._config = config;
        this._schemas = schemaRegistry;
        this._sheets = sheetsGateway;
        this._audit = auditLogger;
    }

    /**
     * @return {Array<Object>}
     */
    findAll() {
        var schema = this._schemas.get(this._tableName);
        this._sheets.ensureTable(this._tableName, schema.columns);
        var data = this._sheets.readTable(this._tableName);
        return data.rows
            .map(function(row) { return Util.rowToObject(data.headers, row); })
            .filter(function(record) { return Util.isBlank(record[this._config.deletedAtColumn]); }, this);
    }

    /**
     * @param {string} id
     * @return {Object|null}
     */
    findById(id) {
        var rows = this.findAll();
        for (var i = 0; i < rows.length; i++) {
            if (rows[i][this._config.idColumn] === id) {
                return rows[i];
            }
        }
        return null;
    }

    /**
     * @param {Object} payload
     * @return {Object}
     */
    insert(payload) {
        var schema = this._schemas.get(this._tableName);
        this._sheets.ensureTable(this._tableName, schema.columns);
        var record = this._schemas.normalizeRecord(this._tableName, payload, false);
        var now = Util.nowIso();
        record[this._config.idColumn] = record[this._config.idColumn] || Util.generateId(this._tableName);
        record[this._config.createdAtColumn] = record[this._config.createdAtColumn] || now;
        record[this._config.updatedAtColumn] = now;
        if (!record.hasOwnProperty(this._config.deletedAtColumn)) {
            record[this._config.deletedAtColumn] = '';
        }

        var tableData = this._sheets.readTable(this._tableName);
        var existingRows = tableData.rows.map(function(row) { return Util.rowToObject(tableData.headers, row); });
        var idColumn = this._config.idColumn;
        var duplicate = existingRows.some(function(row) {
            return row[idColumn] === record[idColumn];
        });
        if (duplicate) {
            throw new ValidationError('Record ID already exists: ' + record[idColumn]);
        }
        existingRows.push(record);
        var rowValues = existingRows.map(function(r) { return Util.objectToRow(schema.columns, r); });
        this._sheets.writeAllRows(this._tableName, schema.columns, rowValues);
        this._audit.log('insert', this._tableName, { id: record[this._config.idColumn] });
        return record;
    }

    /**
     * @param {string} id
     * @param {Object} patch
     * @return {Object}
     */
    update(id, patch) {

```

```

var schema = this._schemas.get(this._tableName);
var normalizedPatch = this._schemas.normalizeRecord(this._tableName, patch, true);
var tableData = this._sheets.readTable(this._tableName);
var records = tableData.rows.map(function(row) { return Util.rowToObject(tableData.headers, row); });
var found = null;

for (var i = 0; i < records.length; i++) {
  if (records[i][this._config.idColumn] === id) {
    found = records[i];
    Object.keys(normalizedPatch).forEach(function(key) {
      if (key !== this._config.idColumn && key !== this._config.createdAtColumn) {
        found[key] = normalizedPatch[key];
      }
    }, this);
    found[this._config.updatedAtColumn] = Util.nowIso();
    break;
  }
}

if (!found) {
  throw new StorageError('Record not found for id: ' + id);
}

var rowValues = records.map(function(r) { return Util.objectToRow(schema.columns, r); });
this._sheets.writeAllRows(this._tableName, schema.columns, rowValues);
this._audit.log('update', this._tableName, { id: id, fields: Object.keys(normalizedPatch) });
return found;
}

/**
 * Soft-deletes a record.
 * @param {string} id
 * @return {Object}
 */
remove(id) {
  var result = this.update(id, (function(col, now) {
    var patch = {};
    patch[col] = now;
    return patch;
  })(this._config.deletedAtColumn, Util.nowIso()));
  this._audit.log('remove', this._tableName, { id: id });
  return result;
}
}

'''

md_mirror/sheet_db_lib/08_TransactionManager.md
# 08_TransactionManager.gs

**Source path:** 'sheet_db_lib/08_TransactionManager.gs'

'''javascript
/**
 * Runs operations under ScriptLock for coarse transaction semantics.
 */
class TransactionManager {
  /**
   * @param {number=} lockTimeoutMs
   */
  constructor(lockTimeoutMs) {
    this._lockTimeoutMs = lockTimeoutMs || 30000;
  }

  /**
   * Executes callback while holding a script lock.
   * @template T
   * @param {function(): T} callback
   * @return {T}
   */
  run(callback) {
    var lock = LockService.getScriptLock();
    if (!lock.tryLock(this._lockTimeoutMs)) {
      throw new TransactionError('Failed to acquire script lock within timeout.');
```

```
}  
}
```

```
'''
```

```
md_mirror/sheet_db_lib/09_AuditLogger.md  
# 09_AuditLogger.gs
```

```
**Source path:** 'sheet_db_lib/09_AuditLogger.gs'
```

```
'''javascript
```

```
/**
```

```
 * Minimal pluggable audit logger.
```

```
 */
```

```
class AuditLogger {
```

```
 /**
```

```
  * @param {Object=} options
```

```
  * @param {Function=} options.sink Optional custom sink function.
```

```
 */
```

```
  constructor(options) {
```

```
    var opts = options || {};
```

```
    this._sink = opts.sink || function(entry) {  
      Logger.log(JSON.stringify(entry));
```

```
    };
```

```
  }
```

```
 /**
```

```
  * Emits an audit entry.
```

```
  * @param {string} action
```

```
  * @param {string} tableName
```

```
  * @param {Object=} metadata
```

```
 */
```

```
  log(action, tableName, metadata) {
```

```
    this._sink({  
      at: Util.nowIso(),  
      action: action,  
      table: tableName,  
      metadata: Util.clone(metadata || {})  
    });
```

```
  }
```

```
}
```

```
'''
```

```
md_mirror/sheet_db_lib/10_ScriptProperties.md  
# 10_ScriptProperties.gs
```

```
**Source path:** 'sheet_db_lib/10_ScriptProperties.gs'
```

```
'''javascript
```

```
/**
```

```
 * Script Properties helper for secure and environment-specific configuration.
```

```
 */
```

```
class ScriptPropertiesHelper {
```

```
 /**
```

```
  * Creates a helper bound to Script Properties store.
```

```
 */
```

```
  constructor() {
```

```
    this._props = PropertiesService.getScriptProperties();
```

```
  }
```

```
 /**
```

```
  * Returns a property value, or a default when missing.
```

```
  * @param {string} key
```

```
  * @param {*} defaultValue
```

```
  * @return {*}
```

```
 */
```

```
  get(key, defaultValue) {
```

```
    var value = this._props.getProperty(key);  
    return value === null ? defaultValue : value;
```

```
  }
```

```
 /**
```

```
  * Returns a required property value or throws when missing/blank.
```

```
  * @param {string} key
```

```
  * @return {string}
```

```
 */
```

```
  getRequired(key) {
```

```
    var value = this.get(key, '');  
    if (value === '') {
```



```

setMany(obj) {
    var stringified = {};
    Object.keys(obj || {}).forEach(function(key) {
        stringified[key] = String(obj[key]);
    });
    this._props.setProperties(stringified, false);
}

/**
 * Deletes a script property.
 * @param {string} key
 */
delete(key) {
    this._props.deleteProperty(key);
}

/**
 * Returns true when a script property exists and is non-empty.
 * @param {string} key
 * @return {boolean}
 */
has(key) {
    var value = this._props.getProperty(key);
    return value !== null && value !== '';
}

/**
 * Returns true when key should be redacted in debug output.
 * @param {string} key
 * @return {boolean}
 */
static shouldRedactKey(key) {
    var upper = String(key || '').toUpperCase();
    if (upper === 'APP_PASSCODE' || upper === 'SIGNING_SECRET' || upper === 'WEBHOOK_SECRET' || upper === 'API_KEY') {
        return true;
    }
    return upper.indexOf('TOKEN') !== -1 || upper.indexOf('SECRET') !== -1 || upper.indexOf('PASSWORD') !== -1;
}

/**
 * Reads SheetDb client-related config from Script Properties.
 *
 * Supported keys:
 * - SHEET_DB_SPREADSHEET_ID
 * - SHEET_DB_ID_COLUMN
 * - SHEET_DB_CREATED_AT_COLUMN
 * - SHEET_DB_UPDATED_AT_COLUMN
 * - SHEET_DB_DELETED_AT_COLUMN
 * - SHEET_DB_LOCK_TIMEOUT_MS
 *
 * Also reserved for host-layer secure use:
 * - APP_PASSCODE
 * - SIGNING_SECRET
 * - WEBHOOK_SECRET
 * - API_KEY
 *
 * @return {Object}
 */
function readSheetDbConfigFromScriptProperties() {
    var helper = new ScriptPropertiesHelper();
    var config = {};

    if (helper.has('SHEET_DB_SPREADSHEET_ID')) {
        config.spreadsheetId = helper.get('SHEET_DB_SPREADSHEET_ID', '');
    }
    if (helper.has('SHEET_DB_ID_COLUMN')) {
        config.idColumn = helper.get('SHEET_DB_ID_COLUMN', 'id');
    }
    if (helper.has('SHEET_DB_CREATED_AT_COLUMN')) {
        config.createdAtColumn = helper.get('SHEET_DB_CREATED_AT_COLUMN', 'createdAt');
    }
    if (helper.has('SHEET_DB_UPDATED_AT_COLUMN')) {
        config.updatedAtColumn = helper.get('SHEET_DB_UPDATED_AT_COLUMN', 'updatedAt');
    }
    if (helper.has('SHEET_DB_DELETED_AT_COLUMN')) {
        config.deletedAtColumn = helper.get('SHEET_DB_DELETED_AT_COLUMN', 'deletedAt');
    }
    if (helper.has('SHEET_DB_LOCK_TIMEOUT_MS')) {
        config.lockTimeoutMs = helper.getNumber('SHEET_DB_LOCK_TIMEOUT_MS', 30000);
    }
}

```



```

    }

    return config;
}

/**
 * Merges explicit client options with script properties-based config.
 * Precedence: explicit options > script properties > hardcoded defaults.
 *
 * @param {Object=} options
 * @return {Object}
 */
function mergeClientOptionsWithScriptProperties(options) {
    var fromProps = readSheetDbConfigFromScriptProperties();
    var explicit = options || {};
    var merged = {};
    Object.keys(fromProps).forEach(function(key) {
        merged[key] = fromProps[key];
    });
    Object.keys(explicit).forEach(function(key) {
        merged[key] = explicit[key];
    });
    return merged;
}

/**
 * Example setup helper for consumers (manual one-time setup).
 *
 * Example:
 * PropertiesService.getScriptProperties().setProperty('SHEET_DB_SPREADSHEET_ID', '...');
 */
function exampleSetSheetDbScriptProperties() {
    // Intentionally no-op; serves as in-library usage documentation only.
}

'''

md_mirror/slack_host/00_WebAppEntry.md
# 00_WebAppEntry.gs

**Source path:** 'slack_host/00_WebAppEntry.gs'

'''javascript
/**
 * Slack host entryptpoint (web app).
 */

/**
 * Creates a DB client from SheetDb library and registers host schemas.
 * @return {DbClient}
 */
function createHostDbClient() {
    var db = SheetDb.createClientFromScriptProperties();

    db.schema('students', {
        columns: ['id', 'name', 'source', 'createdAt', 'updatedAt', 'deletedAt'],
        fields: {
            id: { required: true, type: 'string' },
            name: { required: true, type: 'string' },
            source: { default: 'slack', type: 'string' }
        }
    });

    db.schema('enrollments', {
        columns: ['id', 'studentId', 'courseId', 'status', 'source', 'createdAt', 'updatedAt', 'deletedAt'],
        fields: {
            studentId: { required: true, type: 'string' },
            courseId: { required: true, type: 'string' },
            status: { default: 'enrolled', type: 'string' },
            source: { default: 'slack', type: 'string' }
        }
    });

    db.schema('automations', {
        columns: ['id', 'requestId', 'workflow', 'eventType', 'state', 'createdAt', 'updatedAt', 'deletedAt'],
        fields: {
            requestId: { required: true, type: 'string' },
            workflow: { required: true, type: 'string' },
            eventType: { required: true, type: 'string' },
            state: { default: 'received', type: 'string' }
        }
    });
}

```

```

    }
  });

  db.schema('audit_logs', {
    columns: ['id', 'requestId', 'source', 'action', 'status', 'message', 'metadata', 'createdAt', 'updatedAt', 'deletedAt'],
    fields: {
      requestId: { required: true, type: 'string' },
      source: { required: true, type: 'string' },
      action: { required: true, type: 'string' },
      status: { required: true, type: 'string' },
      message: { type: 'string' },
      metadata: { type: 'string' }
    }
  });

  return db;
}

/**
 * Creates the host-layer router with all dependencies wired.
 * @param {DbClient} db
 * @param {HostConfig} hostConfig
 * @return {SlackRouter}
 */
function createHostRouter(db, hostConfig) {
  var parser = new SlackPayloadParser();
  var security = new SlackSecurity(hostConfig);
  var contexts = new RequestContextFactory(hostConfig);
  var formatter = new ResultFormatter();

  var enrollmentService = new LmsEnrollmentService(db);
  var automationService = new LmsAutomationService(db);
  var workflowHandlers = new WorkflowWebhookHandlers(enrollmentService, automationService);
  var slackService = new SlackService(enrollmentService, workflowHandlers, formatter);

  return new SlackRouter(parser, security, contexts, slackService, formatter, db);
}

/**
 * Webhook endpoint.
 * @param {GoogleAppsScript.Events.DoPost} e
 * @return {GoogleAppsScript.Content.TextOutput}
 */
function doPost(e) {
  var db = createHostDbClient();
  var hostConfig = new HostConfig(new ScriptPropertiesHelper());
  var router = createHostRouter(db, hostConfig);
  var result = router.handle(e);

  return ContentService
    .createTextOutput(JSON.stringify(result.slack || result))
    .setMimeType(ContentService.MimeType.JSON);
}

'''

md_mirror/slack_host/01_SlackRouter.md
# 01_SlackRouter.gs

**Source path:** 'slack_host/01_SlackRouter.gs'

'''javascript
/**
 * HTTP router boundary: parse -> verify -> context -> app service.
 */
class SlackRouter {
  /**
   * @param {SlackPayloadParser} parser
   * @param {SlackSecurity} security
   * @param {RequestContextFactory} contextFactory
   * @param {SlackService} slackService
   * @param {ResultFormatter} resultFormatter
   * @param {DbClient} db
   */
  constructor(parser, security, contextFactory, slackService, resultFormatter, db) {
    this._parser = parser;
    this._security = security;
    this._contexts = contextFactory;
    this._service = slackService;
    this._results = resultFormatter;
  }
}

```

```

    this._db = db;
}

/**
 * @param {GoogleAppsScript.Events.DoPost} e
 * @return {Object}
 */
handle(e) {
    var parsed = this._parser.parseEvent(e);
    var context = this._contexts.create(parsed.routeType, parsed.payload, parsed.envelope);

    var auth = this._security.verify(parsed);
    if (!auth.ok) {
        var denied = this._results.failure(auth.code, auth.message, context, {});
        this._writeAudit(context, denied);
        return denied;
    }

    var result;
    try {
        if (parsed.routeType === 'slash_command') {
            result = this._service.handleSlashCommand(context, parsed.payload);
        } else if (parsed.routeType === 'workflow_webhook') {
            result = this._service.handleWorkflowWebhook(context, parsed.payload);
        } else {
            result = this._results.failure('UNSUPPORTED_PAYLOAD', 'Unsupported payload.', context, {});
        }
    } catch (err) {
        result = this._results.failure(err.code || 'HOST_ERROR', err.message || 'Unexpected error.', context, {});
    }

    this._writeAudit(context, result);
    return result;
}

/** @private */
_writeAudit(context, result) {
    this._db.table('audit_logs').insert({
        requestId: context.requestId,
        source: context.source,
        action: context.routeType,
        status: result.ok ? 'success' : 'failure',
        message: result.message,
        metadata: JSON.stringify({ code: result.code, requestId: result.requestId })
    });
}
}

'''

md_mirror/slack_host/02_SlackPayloadParser.md
# 02_SlackPayloadParser.gs

**Source path:** 'slack_host/02_SlackPayloadParser.gs'

'''javascript
/**
 * Parses and normalizes inbound Slack-related payloads.
 */
class SlackPayloadParser {
    /**
     * @param {GoogleAppsScript.Events.DoPost} e
     * @return {{routeType: string, payload: Object, envelope: Object}}
     */
    parseEvent(e) {
        var rawBody = (e && e.postData && e.postData.contents) ? e.postData.contents : '';
        var params = this._parseFormEncoded(rawBody);

        if (params.command) {
            return {
                routeType: 'slash_command',
                payload: this._normalizeSlashCommand(params),
                envelope: { rawBody: rawBody, params: params }
            };
        }

        var jsonPayload = this._parseJson(rawBody);
        if (jsonPayload) {
            return {
                routeType: 'workflow_webhook',

```



```

        this._workflow = workflowHandlers;
        this._results = resultFormatter;
    }

    /**
     * @param {Object} context
     * @param {Object} payload
     * @return {Object}
     */
    handleSlashCommand(context, payload) {
        if (payload.command === '/enroll-student') {
            var result = this._enrollment.enrollFromSlash(context, payload);
            return result.ok ? this._results.success(result.message, result.data, context)
                : this._results.failure(result.code, result.message, context, result.data);
        }

        return this._results.failure('UNSUPPORTED_COMMAND', 'Unsupported command: ' + payload.command, context, {});
    }

    /**
     * @param {Object} context
     * @param {Object} payload
     * @return {Object}
     */
    handleWorkflowWebhook(context, payload) {
        var result = this._workflow.handle(context, payload);
        return result.ok ? this._results.success(result.message, result.data, context)
            : this._results.failure(result.code, result.message, context, result.data);
    }
}

'''

md_mirror/slack_host/04_SlackSecurity.md
# 04_SlackSecurity.gs

**Source path:** 'slack_host/04_SlackSecurity.gs'

'''javascript
/**
 * Host-layer security checks for Slack/webhook calls.
 */
class SlackSecurity {
    /**
     * @param {HostConfig} hostConfig
     */
    constructor(hostConfig) {
        this._config = hostConfig;
    }

    /**
     * Best-effort request verification for Apps Script web-app requests.
     * Uses passcode from query/form when available.
     *
     * @param {{routeType: string, payload: Object, envelope: Object}} parsed
     * @return {{ok: boolean, code: string, message: string}}
     */
    verify(parsed) {
        var configured = this._config.getAppPasscode();
        if (!configured) {
            return { ok: true, code: 'NO_PASSCODE_CONFIGURED', message: 'Passcode check skipped.' };
        }

        var params = (parsed.envelope && parsed.envelope.params) ? parsed.envelope.params : {};
        var supplied = String(params.passcode || params.api_key || '');
        if (supplied && supplied === configured) {
            return { ok: true, code: 'AUTHORIZED', message: 'Passcode validated.' };
        }

        return { ok: false, code: 'UNAUTHORIZED', message: 'Invalid or missing passcode.' };
    }
}

'''

md_mirror/slack_host/05_LmsEnrollmentService.md
# 05_LmsEnrollmentService.gs

**Source path:** 'slack_host/05_LmsEnrollmentService.gs'

```

```

'''javascript
/**
 * LMS enrollment operations implemented using the SheetDb library only.
 */
class LmsEnrollmentService {
  /**
   * @param {DbClient} db
   */
  constructor(db) {
    this._db = db;
  }

  /**
   * Slash-command flow example:
   * /enroll-student <studentId> <courseId>
   *
   * Upserts into 'students' and 'enrollments', then writes audit log row.
   * @param {Object} context
   * @param {Object} payload
   * @return {Object}
   */
  enrollFromSlash(context, payload) {
    var args = String(payload.text || '').split(/\s+/).filter(function(v) { return !!v; });
    if (args.length < 2) {
      return { ok: false, code: 'INVALID_ARGUMENTS', message: 'Usage: /enroll-student <studentId> <courseId>', data: {} };
    }

    var studentId = args[0];
    var courseId = args[1];
    var out = this._upsertEnrollment(context, studentId, courseId, 'slash_command');
    return out;
  }

  /**
   * Workflow webhook flow example for enrollment updates/inserts.
   * @param {Object} context
   * @param {Object} payload
   * @return {Object}
   */
  upsertFromWorkflow(context, payload) {
    if (!payload.studentId || !payload.courseId) {
      return { ok: false, code: 'INVALID_WORKFLOW_PAYLOAD', message: 'studentId and courseId are required.', data: {} };
    }

    return this._upsertEnrollment(context, payload.studentId, payload.courseId, 'workflow_webhook', payload.status || 'e');
  }

  /** @private */
  _upsertEnrollment(context, studentId, courseId, source, status) {
    var students = this._db.table('students');
    var enrollments = this._db.table('enrollments');
    var audit = this._db.table('audit_logs');
    var finalStatus = status || 'enrolled';
    var enrollmentRecord;

    this._db.transaction(function() {
      var existingStudent = students.findById(studentId);
      if (!existingStudent) {
        students.insert({ id: studentId, name: studentId, source: source });
      }

      var existingEnrollment = enrollments.findAll().filter(function(r) {
        return r.studentId === studentId && r.courseId === courseId;
      })[0];

      if (existingEnrollment) {
        enrollmentRecord = enrollments.update(existingEnrollment.id, {
          status: finalStatus,
          source: source
        });
      } else {
        enrollmentRecord = enrollments.insert({
          studentId: studentId,
          courseId: courseId,
          status: finalStatus,
          source: source
        });
      }
    });

    audit.insert({

```

```

        requestId: context.requestId,
        source: context.source,
        action: 'enrollment_upsert',
        status: 'success',
        message: 'Enrollment upsert completed.',
        metadata: JSON.stringify({ studentId: studentId, courseId: courseId, enrollmentId: enrollmentRecord.id })
    });
});

this._db.audit('enrollment_upsert', 'enrollments', {
    requestId: context.requestId,
    studentId: studentId,
    courseId: courseId,
    enrollmentId: enrollmentRecord.id
});

return {
    ok: true,
    code: 'ENROLLMENT_UPSERTED',
    message: 'Enrollment upserted successfully.',
    data: {
        enrollmentId: enrollmentRecord.id,
        studentId: studentId,
        courseId: courseId,
        status: enrollmentRecord.status
    }
};
}
}
}

'''

md_mirror/slack_host/06_LmsAutomationService.md
# 06_LmsAutomationService.gs

**Source path:** 'slack_host/06_LmsAutomationService.gs'

'''javascript
/**
 * LMS automation operations that persist workflow state via SheetDb.
 */
class LmsAutomationService {
    /**
     * @param {DbClient} db
     */
    constructor(db) {
        this._db = db;
    }

    /**
     * Inserts/updates an automation row and writes an audit log row.
     * @param {Object} context
     * @param {Object} payload
     * @return {Object}
     */
    upsertAutomation(context, payload) {
        var automations = this._db.table('automations');
        var audit = this._db.table('audit_logs');
        var workflow = payload.workflow || 'unknown_workflow';
        var eventType = payload.eventType || 'workflow_event';
        var state = payload.status || 'received';

        var existing = automations.findAll().filter(function(r) {
            return r.workflow === workflow && r.requestId === payload.requestId;
        });

        var row;
        if (existing) {
            row = automations.update(existing.id, { eventType: eventType, state: state });
        } else {
            row = automations.insert({ requestId: payload.requestId, workflow: workflow, eventType: eventType, state: state });
        }

        audit.insert({
            requestId: context.requestId,
            source: context.source,
            action: 'automation_upsert',
            status: 'success',
            message: 'Automation row upserted.',
            metadata: JSON.stringify({ automationId: row.id, workflow: workflow, eventType: eventType })
        });
    }
}

```

```

    });

    return {
      ok: true,
      code: 'AUTOMATION_UPSERTED',
      message: 'Automation row upserted.',
      data: { automationId: row.id, workflow: workflow, eventType: eventType, state: row.state }
    };
  }
}
'''

```

md\_mirror/slack\_host/07\_RequestContextFactory.md  
 # 07\_RequestContextFactory.gs

```

**Source path:** 'slack_host/07_RequestContextFactory.gs'

'''javascript
/**
 * Creates normalized request-context objects for host handlers.
 */
class RequestContextFactory {
  /**
   * @param {HostConfig} hostConfig
   */
  constructor(hostConfig) {
    this._config = hostConfig;
  }

  /**
   * @param {string} routeType
   * @param {Object} payload
   * @param {Object=} envelope
   * @return {Object}
   */
  create(routeType, payload, envelope) {
    var env = envelope || {};
    return {
      requestId: payload.requestId || Util.generateId('req'),
      receivedAt: Util.nowIso(),
      routeType: routeType,
      source: routeType,
      actorId: payload.userId || payload.actorId || 'unknown',
      teamId: payload.teamId || '',
      channelId: payload.channelId || this._config.getSlackDefaultChannel(),
      raw: env.rawBody || ''
    };
  }
}
'''

```

md\_mirror/slack\_host/08\_ResultFormatter.md  
 # 08\_ResultFormatter.gs

```

**Source path:** 'slack_host/08_ResultFormatter.gs'

'''javascript
/**
 * Formats host operation results into structured Slack-friendly objects.
 */
class ResultFormatter {
  /**
   * @param {string} message
   * @param {Object=} data
   * @param {Object=} context
   * @return {Object}
   */
  success(message, data, context) {
    return {
      ok: true,
      code: 'OK',
      message: message,
      requestId: (context && context.requestId) || '',
      data: data || {},
      slack: {
        response_type: 'ephemeral',
        text: message
      }
    }
  }
}
'''

```





```

/** @return {string} */
getSigningSecret() { return this._props.get('SIGNING_SECRET', ''); }
/** @return {string} */
getWebhookSecret() { return this._props.get('WEBHOOK_SECRET', ''); }
/** @return {string} */
getApiKey() { return this._props.get('API_KEY', ''); }
/** @return {string} */
getSlackBotToken() { return this._props.get('SLACK_BOT_TOKEN', ''); }
/** @return {string} */
getSlackSigningSecret() { return this._props.get('SLACK_SIGNING_SECRET', this.getSigningSecret()); }
/** @return {string} */
getSlackDefaultChannel() { return this._props.get('SLACK_DEFAULT_CHANNEL', ''); }

/**
 * @return {Object}
 */
getClientOverrides() {
  return mergeClientOptionsWithScriptProperties({});
}
}

```

```
'''
```

```

md_mirror/slack_host/11_HostTestHarness.md
# 11_HostTestHarness.gs

```

```
**Source path:** 'slack_host/11_HostTestHarness.gs'
```

```
'''javascript
```

```

/**
 * Manual Apps Script test harness for the split Slack host architecture.
 */

```

```

/**
 * Builds host dependencies for manual tests.
 * Uses TEST_SPREADSHEET_ID script property when available.
 * @return {{db: DbClient, hostConfig: HostConfig, router: SlackRouter}}
 */

```

```

function _buildHostTestContext() {
  var props = new ScriptPropertiesHelper();
  var testSpreadsheetId = props.get('TEST_SPREADSHEET_ID', '');
  var options = testSpreadsheetId ? { spreadsheetId: testSpreadsheetId } : {};

```

```
  var db = SheetDb.createClientFromScriptProperties(options);
```

```

  db.schema('students', {
    columns: ['id', 'name', 'source', 'createdAt', 'updatedAt', 'deletedAt'],
    fields: {
      id: { required: true, type: 'string' },
      name: { required: true, type: 'string' },
      source: { default: 'slack', type: 'string' }
    }
  });

```

```

  db.schema('enrollments', {
    columns: ['id', 'studentId', 'courseId', 'status', 'source', 'createdAt', 'updatedAt', 'deletedAt'],
    fields: {
      studentId: { required: true, type: 'string' },
      courseId: { required: true, type: 'string' },
      status: { default: 'enrolled', type: 'string' },
      source: { default: 'slack', type: 'string' }
    }
  });

```

```

  db.schema('automations', {
    columns: ['id', 'requestId', 'workflow', 'eventType', 'state', 'createdAt', 'updatedAt', 'deletedAt'],
    fields: {
      requestId: { required: true, type: 'string' },
      workflow: { required: true, type: 'string' },
      eventType: { required: true, type: 'string' },
      state: { default: 'received', type: 'string' }
    }
  });

```

```

  db.schema('audit_logs', {
    columns: ['id', 'requestId', 'source', 'action', 'status', 'message', 'metadata', 'createdAt', 'updatedAt', 'deletedAt'],
    fields: {
      requestId: { required: true, type: 'string' },
      source: { required: true, type: 'string' },
      action: { required: true, type: 'string' },

```

```

        status: { required: true, type: 'string' },
        message: { type: 'string' },
        metadata: { type: 'string' }
    }
});

var hostConfig = new HostConfig(props);
var router = createHostRouter(db, hostConfig);
return { db: db, hostConfig: hostConfig, router: router };
}

/**
 * Encodes form payload.
 * @param {Object} obj
 * @return {string}
 */
function _toFormBody(obj) {
    return Object.keys(obj).map(function(key) {
        return encodeURIComponent(key) + '=' + encodeURIComponent(String(obj[key]));
    }).join('&');
}

/**
 * Builds a mock DoPost event object.
 * @param {string} body
 * @return {GoogleAppsScript.Events.DoPost}
 */
function _mockDoPostEvent(body) {
    return {
        postData: {
            contents: body,
            type: 'application/x-www-form-urlencoded'
        }
    };
}

/**
 * testSlashEnrollMock
 * Simulates '/enroll-student' and runs full router path.
 */
function testSlashEnrollMock() {
    var ctx = _buildHostTestContext();
    var suffix = new Date().getTime();
    var passcode = ctx.hostConfig.getAppPasscode();
    var payload = {
        command: '/enroll-student',
        text: 'student-' + suffix + ' course-' + suffix,
        user_id: 'U_TEST_' + suffix,
        team_id: 'T_TEST',
        channel_id: 'C_TEST'
    };
    if (passcode) {
        payload.passcode = passcode;
    }

    var event = _mockDoPostEvent(_toFormBody(payload));
    var result = ctx.router.handle(event);
    Logger.log('[testSlashEnrollMock] ' + JSON.stringify(result));
}

/**
 * testWorkflowEnrollmentMock
 * Simulates 'enrollment_requested' workflow webhook and runs full router path.
 */
function testWorkflowEnrollmentMock() {
    var ctx = _buildHostTestContext();
    var suffix = new Date().getTime();
    var payload = {
        workflow: 'lms_enrollment',
        eventType: 'enrollment_requested',
        requestId: 'wf_req_' + suffix,
        data: {
            studentId: 'wf-student-' + suffix,
            courseId: 'wf-course-' + suffix,
            actorId: 'WF_ACTOR'
        }
    };
    var event = _mockDoPostEvent(JSON.stringify(payload));
    var result = ctx.router.handle(event);
    Logger.log('[testWorkflowEnrollmentMock] ' + JSON.stringify(result));
}

```

```

}

/**
 * testAutomationMock
 * Simulates generic automation workflow webhook and runs full router path.
 */
function testAutomationMock() {
  var ctx = _buildHostTestContext();
  var suffix = new Date().getTime();
  var payload = {
    workflow: 'lms_automation',
    eventType: 'automation_ping',
    requestId: 'automation_req_' + suffix,
    data: {
      status: 'queued',
      actorId: 'AUTO_ACTOR'
    }
  };
  var event = _mockDoPostEvent(JSON.stringify(payload));
  var result = ctx.router.handle(event);
  Logger.log('[testAutomationMock] ' + JSON.stringify(result));
}

/**
 * testHostConfig
 * Verifies host config reads script properties without logging raw secrets.
 */
function testHostConfig() {
  var props = new ScriptPropertiesHelper();
  var hostConfig = new HostConfig(props);
  var keys = [
    'APP_PASSCODE',
    'SIGNING_SECRET',
    'WEBHOOK_SECRET',
    'API_KEY',
    'SLACK_BOT_TOKEN',
    'SLACK_SIGNING_SECRET',
    'SLACK_DEFAULT_CHANNEL'
  ];

  var values = {
    APP_PASSCODE: hostConfig.getAppPasscode(),
    SIGNING_SECRET: hostConfig.getSigningSecret(),
    WEBHOOK_SECRET: hostConfig.getWebhookSecret(),
    API_KEY: hostConfig.getApiKey(),
    SLACK_BOT_TOKEN: hostConfig.getSlackBotToken(),
    SLACK_SIGNING_SECRET: hostConfig.getSlackSigningSecret(),
    SLACK_DEFAULT_CHANNEL: hostConfig.getSlackDefaultChannel()
  };

  var redacted = {};
  keys.forEach(function(key) {
    var value = values[key] || '';
    redacted[key] = ScriptPropertiesHelper.shouldRedactKey(key)
      ? (value ? '[REDACTED_SET]' : '[REDACTED_EMPTY]')
      : value;
  });

  Logger.log('[testHostConfig] ' + JSON.stringify(redacted));
}

/**
 * testAuditWrite
 * Executes one mutation and verifies an audit row is written.
 */
function testAuditWrite() {
  var ctx = _buildHostTestContext();
  var auditTable = ctx.db.table('audit_logs');
  var before = auditTable.findAll().length;

  var suffix = new Date().getTime();
  var passcode = ctx.hostConfig.getAppPasscode();
  var payload = {
    command: '/enroll-student',
    text: 'audit-student-' + suffix + ' audit-course-' + suffix,
    user_id: 'U_AUDIT_' + suffix,
    team_id: 'T_AUDIT',
    channel_id: 'C_AUDIT'
  };
  if (passcode) {

```

```

    payload.passcode = passcode;
  }

  var event = _mockDoPostEvent(_toFormBody(payload));
  var result = ctx.router.handle(event);
  var after = auditTable.findAll().length;

  Logger.log('[testAuditWrite] ' + JSON.stringify({
    mutationOk: !!result.ok,
    auditRowsBefore: before,
    auditRowsAfter: after,
    auditRowDelta: after - before
  }));
}

```

```
'''
```

```
pdf_mirror/CODEBASE_FULL.pdf
```

```
%PDF-1.4
```

```
1 0 obj
```

```
<< /Type /Catalog /Pages 2 0 R >>
```

```
endobj
```

```
2 0 obj
```

```
<< /Type /Pages /Count 32 /Kids [4 0 R 6 0 R 8 0 R 10 0 R 12 0 R 14 0 R 16 0 R 18 0 R 20 0 R 22 0 R 24 0 R 26 0 R 28 0 R
```

```
endobj
```

```
3 0 obj
```

```
<< /Type /Font /Subtype /Type1 /BaseFont /Courier >>
```

```
endobj
```

```
4 0 obj
```

```
<< /Type /Page /Parent 2 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 3 0 R >> >> /Contents 5 0 R >>
```

```
endobj
```

```
5 0 obj
```

```
<< /Length 3209 >>
```

```
stream
```

```
BT
```

```
/F1 9 Tf
```

```
1 0 0 1 36 756 Tm (# 00_SheetDb.gs) Tj
```

```
1 0 0 1 36 745 Tm ( ) Tj
```

```
1 0 0 1 36 734 Tm (**Source path:** 'sheet_db_lib/00_SheetDb.gs') Tj
```

```
1 0 0 1 36 723 Tm ( ) Tj
```

```
1 0 0 1 36 712 Tm (``javascript) Tj
```

```
1 0 0 1 36 701 Tm (/**) Tj
```

```
1 0 0 1 36 690 Tm ( * @fileoverview Public, library-visible exports for SheetDb.) Tj
```

```
1 0 0 1 36 679 Tm ( */) Tj
```

```
1 0 0 1 36 668 Tm ( ) Tj
```

```
1 0 0 1 36 657 Tm (/**) Tj
```

```
1 0 0 1 36 646 Tm ( * Library semantic version.) Tj
```

```
1 0 0 1 36 635 Tm ( *) Tj
```

```
1 0 0 1 36 624 Tm ( * NOTE: Keep this in sync with release tags when publishing as an Apps Script library.) Tj
```

```
1 0 0 1 36 613 Tm ( * @type {string}) Tj
```

```
1 0 0 1 36 602 Tm ( */) Tj
```

```
1 0 0 1 36 591 Tm (var SHEET_DB_VERSION = '0.1.0'); Tj
```

```
1 0 0 1 36 580 Tm ( ) Tj
```

```
1 0 0 1 36 569 Tm (/**) Tj
```

```
1 0 0 1 36 558 Tm ( * Creates a new SheetDb client instance.) Tj
```

```
1 0 0 1 36 547 Tm ( *) Tj
```

```
1 0 0 1 36 536 Tm ( * @param {Object=} options Client configuration options.) Tj
```

```
1 0 0 1 36 525 Tm ( * @return {DbClient}) Tj
```

```
1 0 0 1 36 514 Tm ( */) Tj
```

```
1 0 0 1 36 503 Tm (function _sheetDbCreateClient(options) { Tj
```

```
1 0 0 1 36 492 Tm ( return new DbClient(new Config(options || {})); Tj
```

```
1 0 0 1 36 481 Tm (}) Tj
```

```
1 0 0 1 36 470 Tm ( ) Tj
```

```
1 0 0 1 36 459 Tm (/**) Tj
```

```
1 0 0 1 36 448 Tm ( * Bootstraps a client with one or more table schemas.) Tj
```

```
1 0 0 1 36 437 Tm ( *) Tj
```

```
1 0 0 1 36 426 Tm ( * @param {Object=} options) Tj
```

```
1 0 0 1 36 415 Tm ( * @param {Array<{name: string, schema: Object}>} options.tables Tables to register.) Tj
```

```
1 0 0 1 36 404 Tm ( * @param {Object=} options.clientOptions Configuration passed to 'createClient'.) Tj
```

```
1 0 0 1 36 393 Tm ( * @return {DbClient}) Tj
```

```
1 0 0 1 36 382 Tm ( */) Tj
```

```
1 0 0 1 36 371 Tm (function _sheetDbBootstrap(options) { Tj
```

```
1 0 0 1 36 360 Tm ( var opts = options || {};) Tj
```

```
1 0 0 1 36 349 Tm ( var client = _sheetDbCreateClient(opts.clientOptions || {}); Tj
```

```
1 0 0 1 36 338 Tm ( var tables = opts.tables || [];) Tj
```

```
1 0 0 1 36 327 Tm ( ) Tj
```

```
1 0 0 1 36 316 Tm ( tables.forEach(function(table) { Tj
```

```
1 0 0 1 36 305 Tm ( if (table && table.name && table.schema) { Tj
```

```
1 0 0 1 36 294 Tm ( client.registerTable(table.name, table.schema); Tj
```

```
1 0 0 1 36 283 Tm ( }) Tj
```



```

1 0 0 1 36 239 Tm ( bootstrap: function\ (options\ ) {} ) Tj
1 0 0 1 36 228 Tm ( return _sheetDbBootstrap\ (options\);) Tj
1 0 0 1 36 217 Tm ( },) Tj
1 0 0 1 36 206 Tm ( ) Tj
1 0 0 1 36 195 Tm ( healthCheck: function\ ( ) {} ) Tj
1 0 0 1 36 184 Tm ( return _sheetDbHealthCheck\ ( );) Tj
1 0 0 1 36 173 Tm ( },) Tj
1 0 0 1 36 162 Tm ( ) Tj
1 0 0 1 36 151 Tm ( version: SHEET_DB_VERSION) Tj
1 0 0 1 36 140 Tm ( );) Tj
1 0 0 1 36 129 Tm ( ) Tj
1 0 0 1 36 118 Tm ( /**) Tj
1 0 0 1 36 107 Tm ( * Backward-compatible factory export.) Tj
1 0 0 1 36 96 Tm ( * @deprecated Use 'SheetDb.createClient' instead.) Tj
1 0 0 1 36 85 Tm ( * @param {Object=} options) Tj
1 0 0 1 36 74 Tm ( * @return {DbClient}) Tj
1 0 0 1 36 63 Tm ( */) Tj
1 0 0 1 36 52 Tm (function createSheetDbClient\ (options\ ) {} ) Tj
1 0 0 1 36 41 Tm ( return SheetDb.createClient\ (options\);) Tj
ET
endstream
endobj
8 0 obj
<< /Type /Page /Parent 2 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 3 0 R >> >> /Contents 9 0 R >>
endobj
9 0 obj
<< /Length 3028 >>
stream
BT
/F1 9 Tf
1 0 0 1 36 756 Tm ( {} ) Tj
1 0 0 1 36 745 Tm ( ) Tj
1 0 0 1 36 734 Tm ( ''' ) Tj
1 0 0 1 36 723 Tm ( ) Tj
1 0 0 1 36 712 Tm ( ) Tj
1 0 0 1 36 701 Tm ( # 01_DbClient.gs) Tj
1 0 0 1 36 690 Tm ( ) Tj
1 0 0 1 36 679 Tm ( **Source path:** 'sheet_db_lib/01_DbClient.gs') Tj
1 0 0 1 36 668 Tm ( ) Tj
1 0 0 1 36 657 Tm ( '''javascript) Tj
1 0 0 1 36 646 Tm ( /**) Tj
1 0 0 1 36 635 Tm ( * Main entrypoint for interacting with sheet-backed tables.) Tj
1 0 0 1 36 624 Tm ( */) Tj
1 0 0 1 36 613 Tm (class DbClient {} ) Tj
1 0 0 1 36 602 Tm ( /**) Tj
1 0 0 1 36 591 Tm ( * @param {Config=} config) Tj
1 0 0 1 36 580 Tm ( */) Tj
1 0 0 1 36 569 Tm ( constructor\ (config\ ) {} ) Tj
1 0 0 1 36 558 Tm ( this._config = config || new Config\ ( );) Tj
1 0 0 1 36 547 Tm ( this._schemas = new SchemaRegistry\ ( );) Tj
1 0 0 1 36 536 Tm ( this._sheets = new SheetsGateway\ (this._config\ );) Tj
1 0 0 1 36 525 Tm ( this._audit = new AuditLogger\ ( );) Tj
1 0 0 1 36 514 Tm ( this._tx = new TransactionManager\ (this._config.lockTimeoutMs\ );) Tj
1 0 0 1 36 503 Tm ( }) Tj
1 0 0 1 36 492 Tm ( ) Tj
1 0 0 1 36 481 Tm ( /**) Tj
1 0 0 1 36 470 Tm ( * Registers a table schema.) Tj
1 0 0 1 36 459 Tm ( * @param {string} tableName) Tj
1 0 0 1 36 448 Tm ( * @param {Object} schema) Tj
1 0 0 1 36 437 Tm ( * @return {DbClient}) Tj
1 0 0 1 36 426 Tm ( */) Tj
1 0 0 1 36 415 Tm ( registerTable\ (tableName, schema\ ) {} ) Tj
1 0 0 1 36 404 Tm ( this._schemas.register\ (tableName, schema\ );) Tj
1 0 0 1 36 393 Tm ( this._sheets.ensureTable\ (tableName, schema.columns\ );) Tj
1 0 0 1 36 382 Tm ( return this;) Tj
1 0 0 1 36 371 Tm ( }) Tj
1 0 0 1 36 360 Tm ( ) Tj
1 0 0 1 36 349 Tm ( /**) Tj
1 0 0 1 36 338 Tm ( * Returns a table repository.) Tj
1 0 0 1 36 327 Tm ( * @param {string} tableName) Tj
1 0 0 1 36 316 Tm ( * @return {TableRepository}) Tj
1 0 0 1 36 305 Tm ( */) Tj
1 0 0 1 36 294 Tm ( table\ (tableName\ ) {} ) Tj
1 0 0 1 36 283 Tm ( this._schemas.get\ (tableName\ );) Tj
1 0 0 1 36 272 Tm ( return new TableRepository\ ( ) Tj
1 0 0 1 36 261 Tm ( tableName,) Tj
1 0 0 1 36 250 Tm ( this._config,) Tj
1 0 0 1 36 239 Tm ( this._schemas,) Tj
1 0 0 1 36 228 Tm ( this._sheets,) Tj
1 0 0 1 36 217 Tm ( this._audit) Tj

```

```

1 0 0 1 36 206 Tm ( \);) Tj
1 0 0 1 36 195 Tm ( }) Tj
1 0 0 1 36 184 Tm ( ) Tj
1 0 0 1 36 173 Tm ( /**) Tj
1 0 0 1 36 162 Tm ( * Gets or registers a table schema.) Tj
1 0 0 1 36 151 Tm ( *) Tj
1 0 0 1 36 140 Tm ( * - When 'schema' is provided, registers/overwrites the schema and ensures the table.) Tj
1 0 0 1 36 129 Tm ( * - When 'schema' is omitted, returns the current schema.) Tj
1 0 0 1 36 118 Tm ( *) Tj
1 0 0 1 36 107 Tm ( * @param {string} tableName) Tj
1 0 0 1 36 96 Tm ( * @param {Object=} schema) Tj
1 0 0 1 36 85 Tm ( * @return {Object|DbClient}) Tj
1 0 0 1 36 74 Tm ( */) Tj
1 0 0 1 36 63 Tm ( schema\(tableName, schema\) {}) Tj
1 0 0 1 36 52 Tm ( if \(schema\) {}) Tj
1 0 0 1 36 41 Tm ( this.registerTable\(tableName, schema\);) Tj
ET
endstream
endobj
10 0 obj
<< /Type /Page /Parent 2 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 3 0 R >> >> /Contents 11 0 R >>
endobj
11 0 obj
<< /Length 3377 >>
stream
BT
/F1 9 Tf
1 0 0 1 36 756 Tm ( return this;) Tj
1 0 0 1 36 745 Tm ( }) Tj
1 0 0 1 36 734 Tm ( return this._schemas.get\(tableName\);) Tj
1 0 0 1 36 723 Tm ( }) Tj
1 0 0 1 36 712 Tm ( ) Tj
1 0 0 1 36 701 Tm ( /**) Tj
1 0 0 1 36 690 Tm ( * Writes an audit entry through the configured audit logger.) Tj
1 0 0 1 36 679 Tm ( * @param {string} action) Tj
1 0 0 1 36 668 Tm ( * @param {string} tableName) Tj
1 0 0 1 36 657 Tm ( * @param {Object=} metadata) Tj
1 0 0 1 36 646 Tm ( */) Tj
1 0 0 1 36 635 Tm ( audit\(action, tableName, metadata\) {}) Tj
1 0 0 1 36 624 Tm ( this._audit.log\(action, tableName, metadata || {})\);) Tj
1 0 0 1 36 613 Tm ( }) Tj
1 0 0 1 36 602 Tm ( ) Tj
1 0 0 1 36 591 Tm ( /**) Tj
1 0 0 1 36 580 Tm ( * Runs a set of operations under lock.) Tj
1 0 0 1 36 569 Tm ( * @template T) Tj
1 0 0 1 36 558 Tm ( * @param {function(DbClient\): T} callback) Tj
1 0 0 1 36 547 Tm ( * @return {T}) Tj
1 0 0 1 36 536 Tm ( */) Tj
1 0 0 1 36 525 Tm ( transaction\(callback\) {}) Tj
1 0 0 1 36 514 Tm ( var self = this;) Tj
1 0 0 1 36 503 Tm ( return this._tx.run\(function\(\) {}) Tj
1 0 0 1 36 492 Tm ( return callback\(self\);) Tj
1 0 0 1 36 481 Tm ( \);) Tj
1 0 0 1 36 470 Tm ( }) Tj
1 0 0 1 36 459 Tm (}) Tj
1 0 0 1 36 448 Tm ( ) Tj
1 0 0 1 36 437 Tm (``) Tj
1 0 0 1 36 426 Tm ( ) Tj
1 0 0 1 36 415 Tm ( ) Tj
1 0 0 1 36 404 Tm ( # 02_Config.gs) Tj
1 0 0 1 36 393 Tm ( ) Tj
1 0 0 1 36 382 Tm (**Source path: ** 'sheet_db_lib/02_Config.gs') Tj
1 0 0 1 36 371 Tm ( ) Tj
1 0 0 1 36 360 Tm (``javascript) Tj
1 0 0 1 36 349 Tm (/**) Tj
1 0 0 1 36 338 Tm ( * Immutable runtime configuration for the Sheet DB client.) Tj
1 0 0 1 36 327 Tm ( */) Tj
1 0 0 1 36 316 Tm (class Config {}) Tj
1 0 0 1 36 305 Tm ( /**) Tj
1 0 0 1 36 294 Tm ( * @param {Object=} options) Tj
1 0 0 1 36 283 Tm ( * @param {string=} options.spreadsheetId Spreadsheet ID; defaults to active spreadsheet.) Tj
1 0 0 1 36 272 Tm ( * @param {string=} options.idColumn Record primary key column name.) Tj
1 0 0 1 36 261 Tm ( * @param {string=} options.createdAtColumn Created timestamp column name.) Tj
1 0 0 1 36 250 Tm ( * @param {string=} options.updatedAtColumn Updated timestamp column name.) Tj
1 0 0 1 36 239 Tm ( * @param {string=} options.deletedAtColumn Soft-delete timestamp column name.) Tj
1 0 0 1 36 228 Tm ( * @param {number=} options.lockTimeoutMs Lock acquisition timeout for transactions.) Tj
1 0 0 1 36 217 Tm ( */) Tj
1 0 0 1 36 206 Tm ( constructor\(options\) {}) Tj
1 0 0 1 36 195 Tm ( var opts = options || {};) Tj
1 0 0 1 36 184 Tm ( this.spreadsheetId = opts.spreadsheetId || '');) Tj

```



```

1 0 0 1 36 173 Tm ( this.idColumn = opts.idColumn || 'id';) Tj
1 0 0 1 36 162 Tm ( this.createdAtColumn = opts.createdAtColumn || 'createdAt';) Tj
1 0 0 1 36 151 Tm ( this.updatedAtColumn = opts.updatedAtColumn || 'updatedAt';) Tj
1 0 0 1 36 140 Tm ( this.deletedAtColumn = opts.deletedAtColumn || 'deletedAt';) Tj
1 0 0 1 36 129 Tm ( this.lockTimeoutMs = Util.isNil\(\opts.lockTimeoutMs\) ? 30000 : Number\(\opts.lockTimeoutMs\);) Tj
1 0 0 1 36 118 Tm ( this.validate\(\);) Tj
1 0 0 1 36 107 Tm ( Object.freeze\(\this\);) Tj
1 0 0 1 36 96 Tm ( }) Tj
1 0 0 1 36 85 Tm ( ) Tj
1 0 0 1 36 74 Tm ( /**) Tj
1 0 0 1 36 63 Tm ( * @return {void}) Tj
1 0 0 1 36 52 Tm ( */) Tj
1 0 0 1 36 41 Tm ( validate\(\) {}) Tj
ET
endstream
endobj
12 0 obj
<< /Type /Page /Parent 2 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 3 0 R >> >> /Contents 13 0 R >>
endobj
13 0 obj
<< /Length 3019 >>
stream
BT
/F1 9 Tf
1 0 0 1 36 756 Tm ( var columns = []) Tj
1 0 0 1 36 745 Tm ( this.idColumn,) Tj
1 0 0 1 36 734 Tm ( this.createdAtColumn,) Tj
1 0 0 1 36 723 Tm ( this.updatedAtColumn,) Tj
1 0 0 1 36 712 Tm ( this.deletedAtColumn) Tj
1 0 0 1 36 701 Tm ( );) Tj
1 0 0 1 36 690 Tm ( var unique = {};) Tj
1 0 0 1 36 679 Tm ( columns.forEach\((function\(\column\) {) Tj
1 0 0 1 36 668 Tm ( if \(!column || typeof column !== 'string'\) {) Tj
1 0 0 1 36 657 Tm ( throw new ConfigError\('System column names must be non-empty strings.'\);) Tj
1 0 0 1 36 646 Tm ( }) Tj
1 0 0 1 36 635 Tm ( if \(!unique[column]\) {) Tj
1 0 0 1 36 624 Tm ( throw new ConfigError\('System column names must be unique. Duplicate: ' + column\);) Tj
1 0 0 1 36 613 Tm ( }) Tj
1 0 0 1 36 602 Tm ( unique[column] = true;) Tj
1 0 0 1 36 591 Tm ( })\);) Tj
1 0 0 1 36 580 Tm ( ) Tj
1 0 0 1 36 569 Tm ( if \(!isNaN\(\this.lockTimeoutMs\) || this.lockTimeoutMs <= 0\) {) Tj
1 0 0 1 36 558 Tm ( throw new ConfigError\('lockTimeoutMs must be a positive number.'\);) Tj
1 0 0 1 36 547 Tm ( }) Tj
1 0 0 1 36 536 Tm ( }) Tj
1 0 0 1 36 525 Tm ( }) Tj
1 0 0 1 36 514 Tm ( ) Tj
1 0 0 1 36 503 Tm ( '') Tj
1 0 0 1 36 492 Tm ( ) Tj
1 0 0 1 36 481 Tm ( ) Tj
1 0 0 1 36 470 Tm ( # 03_Errors.gs) Tj
1 0 0 1 36 459 Tm ( ) Tj
1 0 0 1 36 448 Tm ( **Source path:** 'sheet_db_lib/03_Errors.gs') Tj
1 0 0 1 36 437 Tm ( ) Tj
1 0 0 1 36 426 Tm ( '''javascript) Tj
1 0 0 1 36 415 Tm ( /**) Tj
1 0 0 1 36 404 Tm ( * Base error for the Sheet DB library.) Tj
1 0 0 1 36 393 Tm ( */) Tj
1 0 0 1 36 382 Tm ( class SheetDbError extends Error {) Tj
1 0 0 1 36 371 Tm ( /**) Tj
1 0 0 1 36 360 Tm ( * @param {string} message Human-readable error message.) Tj
1 0 0 1 36 349 Tm ( * @param {string=} code Stable error code for programmatic handling.) Tj
1 0 0 1 36 338 Tm ( */) Tj
1 0 0 1 36 327 Tm ( constructor\(\message, code\) {) Tj
1 0 0 1 36 316 Tm ( super\(\message\);) Tj
1 0 0 1 36 305 Tm ( this.name = this.constructor.name;) Tj
1 0 0 1 36 294 Tm ( this.code = code || 'SHEET_DB_ERROR';) Tj
1 0 0 1 36 283 Tm ( }) Tj
1 0 0 1 36 272 Tm ( }) Tj
1 0 0 1 36 261 Tm ( ) Tj
1 0 0 1 36 250 Tm ( /**) Tj
1 0 0 1 36 239 Tm ( * Raised when configuration is invalid or missing.) Tj
1 0 0 1 36 228 Tm ( */) Tj
1 0 0 1 36 217 Tm ( class ConfigError extends SheetDbError {) Tj
1 0 0 1 36 206 Tm ( /**) Tj
1 0 0 1 36 195 Tm ( * @param {string} message) Tj
1 0 0 1 36 184 Tm ( */) Tj
1 0 0 1 36 173 Tm ( constructor\(\message\) {) Tj
1 0 0 1 36 162 Tm ( super\(\message, 'CONFIG_ERROR'\);) Tj
1 0 0 1 36 151 Tm ( }) Tj

```

```

1 0 0 1 36 140 Tm (}) Tj
1 0 0 1 36 129 Tm ( ) Tj
1 0 0 1 36 118 Tm (/**) Tj
1 0 0 1 36 107 Tm ( * Raised when schema definitions are invalid.) Tj
1 0 0 1 36 96 Tm ( */) Tj
1 0 0 1 36 85 Tm (class SchemaError extends SheetDbError {}) Tj
1 0 0 1 36 74 Tm ( /**) Tj
1 0 0 1 36 63 Tm ( * @param {string} message) Tj
1 0 0 1 36 52 Tm ( */) Tj
1 0 0 1 36 41 Tm ( constructor\ (message\ ) {} Tj
ET
endstream
endobj
14 0 obj
<< /Type /Page /Parent 2 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 3 0 R >> >> /Contents 15 0 R >>
endobj
15 0 obj
<< /Length 2651 >>
stream
BT
/F1 9 Tf
1 0 0 1 36 756 Tm ( super\ (message, 'SCHEMA_ERROR'\);) Tj
1 0 0 1 36 745 Tm ( }) Tj
1 0 0 1 36 734 Tm (}) Tj
1 0 0 1 36 723 Tm ( ) Tj
1 0 0 1 36 712 Tm (/**) Tj
1 0 0 1 36 701 Tm ( * Raised when records fail validation.) Tj
1 0 0 1 36 690 Tm ( */) Tj
1 0 0 1 36 679 Tm (class ValidationError extends SheetDbError {}) Tj
1 0 0 1 36 668 Tm ( /**) Tj
1 0 0 1 36 657 Tm ( * @param {string} message) Tj
1 0 0 1 36 646 Tm ( */) Tj
1 0 0 1 36 635 Tm ( constructor\ (message\ ) {} Tj
1 0 0 1 36 624 Tm ( super\ (message, 'VALIDATION_ERROR'\);) Tj
1 0 0 1 36 613 Tm ( }) Tj
1 0 0 1 36 602 Tm (}) Tj
1 0 0 1 36 591 Tm ( ) Tj
1 0 0 1 36 580 Tm (/**) Tj
1 0 0 1 36 569 Tm ( * Raised when underlying sheet operations fail.) Tj
1 0 0 1 36 558 Tm ( */) Tj
1 0 0 1 36 547 Tm (class StorageError extends SheetDbError {}) Tj
1 0 0 1 36 536 Tm ( /**) Tj
1 0 0 1 36 525 Tm ( * @param {string} message) Tj
1 0 0 1 36 514 Tm ( */) Tj
1 0 0 1 36 503 Tm ( constructor\ (message\ ) {} Tj
1 0 0 1 36 492 Tm ( super\ (message, 'STORAGE_ERROR'\);) Tj
1 0 0 1 36 481 Tm ( }) Tj
1 0 0 1 36 470 Tm (}) Tj
1 0 0 1 36 459 Tm ( ) Tj
1 0 0 1 36 448 Tm (/**) Tj
1 0 0 1 36 437 Tm ( * Raised for optimistic transaction failures.) Tj
1 0 0 1 36 426 Tm ( */) Tj
1 0 0 1 36 415 Tm (class TransactionError extends SheetDbError {}) Tj
1 0 0 1 36 404 Tm ( /**) Tj
1 0 0 1 36 393 Tm ( * @param {string} message) Tj
1 0 0 1 36 382 Tm ( */) Tj
1 0 0 1 36 371 Tm ( constructor\ (message\ ) {} Tj
1 0 0 1 36 360 Tm ( super\ (message, 'TRANSACTION_ERROR'\);) Tj
1 0 0 1 36 349 Tm ( }) Tj
1 0 0 1 36 338 Tm (}) Tj
1 0 0 1 36 327 Tm ( ) Tj
1 0 0 1 36 316 Tm (''' ) Tj
1 0 0 1 36 305 Tm ( ) Tj
1 0 0 1 36 294 Tm ( ) Tj
1 0 0 1 36 283 Tm (# 04_Util.gs) Tj
1 0 0 1 36 272 Tm ( ) Tj
1 0 0 1 36 261 Tm (**Source path:** 'sheet_db_lib/04_Util.gs') Tj
1 0 0 1 36 250 Tm ( ) Tj
1 0 0 1 36 239 Tm ('''javascript) Tj
1 0 0 1 36 228 Tm (/**) Tj
1 0 0 1 36 217 Tm ( * Utility helpers for Sheet DB.) Tj
1 0 0 1 36 206 Tm ( */) Tj
1 0 0 1 36 195 Tm (class Util {}) Tj
1 0 0 1 36 184 Tm ( /**) Tj
1 0 0 1 36 173 Tm ( * @return {string} RFC3339 timestamp in UTC.) Tj
1 0 0 1 36 162 Tm ( */) Tj
1 0 0 1 36 151 Tm ( static nowIso\ (\ ) {} Tj
1 0 0 1 36 140 Tm ( return new Date\ (\ ).toISOString\ (\ );) Tj
1 0 0 1 36 129 Tm ( }) Tj
1 0 0 1 36 118 Tm ( ) Tj

```



```

1 0 0 1 36 74 Tm ( static objectToRow(headers, obj\ ) {} Tj
1 0 0 1 36 63 Tm ( return headers.map(function(h\ ) {} Tj
1 0 0 1 36 52 Tm ( return obj[h];) Tj
1 0 0 1 36 41 Tm ( }));) Tj
ET
endstream
endobj
18 0 obj
<< /Type /Page /Parent 2 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 3 0 R >> >> /Contents 19 0 R >>
endobj
19 0 obj
<< /Length 3291 >>
stream
BT
/F1 9 Tf
1 0 0 1 36 756 Tm ( ) Tj
1 0 0 1 36 745 Tm ( {} ) Tj
1 0 0 1 36 734 Tm ( ) Tj
1 0 0 1 36 723 Tm ( '' ) Tj
1 0 0 1 36 712 Tm ( ) Tj
1 0 0 1 36 701 Tm ( ) Tj
1 0 0 1 36 690 Tm ( # 05_SchemaRegistry.gs ) Tj
1 0 0 1 36 679 Tm ( ) Tj
1 0 0 1 36 668 Tm ( **Source path:** 'sheet_db_lib/05_SchemaRegistry.gs' ) Tj
1 0 0 1 36 657 Tm ( ) Tj
1 0 0 1 36 646 Tm ( '''javascript) Tj
1 0 0 1 36 635 Tm ( /**) Tj
1 0 0 1 36 624 Tm ( * Holds table schemas and applies field defaults/validation.) Tj
1 0 0 1 36 613 Tm ( */) Tj
1 0 0 1 36 602 Tm (class SchemaRegistry {} Tj
1 0 0 1 36 591 Tm ( /**) Tj
1 0 0 1 36 580 Tm ( * Creates a new in-memory schema registry.) Tj
1 0 0 1 36 569 Tm ( */) Tj
1 0 0 1 36 558 Tm ( constructor\(\ ) {} Tj
1 0 0 1 36 547 Tm ( this._schemas = {};) Tj
1 0 0 1 36 536 Tm ( }) Tj
1 0 0 1 36 525 Tm ( ) Tj
1 0 0 1 36 514 Tm ( /**) Tj
1 0 0 1 36 503 Tm ( * Registers a table schema.) Tj
1 0 0 1 36 492 Tm ( * @param {string} tableName) Tj
1 0 0 1 36 481 Tm ( * @param {Object} schema) Tj
1 0 0 1 36 470 Tm ( * @param {Array<string>} schema.columns Ordered columns for persistence.) Tj
1 0 0 1 36 459 Tm ( * @param {Object<string, Object>} schema.fields Optional field constraints.) Tj
1 0 0 1 36 448 Tm ( * @return {SchemaRegistry}) Tj
1 0 0 1 36 437 Tm ( */) Tj
1 0 0 1 36 426 Tm ( register\(tableName, schema\ ) {} Tj
1 0 0 1 36 415 Tm ( if \(!tableName\ ) {} Tj
1 0 0 1 36 404 Tm ( throw new SchemaError\('Table name is required.\');) Tj
1 0 0 1 36 393 Tm ( }) Tj
1 0 0 1 36 382 Tm ( if \(!schema || !Array.isArray\(schema.columns\ ) || schema.columns.length === 0\ ) {} Tj
1 0 0 1 36 371 Tm ( throw new SchemaError\('Schema for ' + tableName + ' must provide non-empty columns array.\');) Tj
1 0 0 1 36 360 Tm ( }) Tj
1 0 0 1 36 349 Tm ( var seen = {};) Tj
1 0 0 1 36 338 Tm ( schema.columns.forEach(function(column\ ) {} Tj
1 0 0 1 36 327 Tm ( if \(!column || typeof column !== 'string'\ ) {} Tj
1 0 0 1 36 316 Tm ( throw new SchemaError\('All columns must be non-empty strings for table: ' + tableName\);) Tj
1 0 0 1 36 305 Tm ( }) Tj
1 0 0 1 36 294 Tm ( if \(!seen[column]\ ) {} Tj
1 0 0 1 36 283 Tm ( throw new SchemaError\('Duplicate column "' + column + '" in table: ' + tableName\);) Tj
1 0 0 1 36 272 Tm ( }) Tj
1 0 0 1 36 261 Tm ( seen[column] = true;) Tj
1 0 0 1 36 250 Tm ( }));) Tj
1 0 0 1 36 239 Tm ( ) Tj
1 0 0 1 36 228 Tm ( var fields = Util.clone\(schema.fields || {});) Tj
1 0 0 1 36 217 Tm ( Object.keys\(fields\).forEach(function(fieldName\ ) {} Tj
1 0 0 1 36 206 Tm ( if \(!seen[fieldName]\ ) {} Tj
1 0 0 1 36 195 Tm ( throw new SchemaError\() Tj
1 0 0 1 36 184 Tm ( 'Field rules defined for unknown column "' + fieldName + '" in table: ' + tableName) Tj
1 0 0 1 36 173 Tm ( \);) Tj
1 0 0 1 36 162 Tm ( }) Tj
1 0 0 1 36 151 Tm ( }));) Tj
1 0 0 1 36 140 Tm ( ) Tj
1 0 0 1 36 129 Tm ( this._schemas[tableName] = {} Tj
1 0 0 1 36 118 Tm ( columns: schema.columns.slice\(\),) Tj
1 0 0 1 36 107 Tm ( fields: fields) Tj
1 0 0 1 36 96 Tm ( );) Tj
1 0 0 1 36 85 Tm ( return this;) Tj
1 0 0 1 36 74 Tm ( }) Tj
1 0 0 1 36 63 Tm ( ) Tj
1 0 0 1 36 52 Tm ( /**) Tj

```

```

1 0 0 1 36 41 Tm ( * Returns schema for a table.) Tj
ET
endstream
endobj
20 0 obj
<< /Type /Page /Parent 2 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 3 0 R >> >> /Contents 21 0 R >>
endobj
21 0 obj
<< /Length 3373 >>
stream
BT
/F1 9 Tf
1 0 0 1 36 756 Tm ( * @param {string} tableName) Tj
1 0 0 1 36 745 Tm ( * @return {Object}) Tj
1 0 0 1 36 734 Tm ( */) Tj
1 0 0 1 36 723 Tm ( get\(tableName\) {} Tj
1 0 0 1 36 712 Tm ( var schema = this._schemas[tableName];) Tj
1 0 0 1 36 701 Tm ( if \(!schema\) {} Tj
1 0 0 1 36 690 Tm ( throw new SchemaError\('No schema registered for table: ' + tableName\);) Tj
1 0 0 1 36 679 Tm ( }) Tj
1 0 0 1 36 668 Tm ( return {} Tj
1 0 0 1 36 657 Tm ( columns: schema.columns.slice\(\),) Tj
1 0 0 1 36 646 Tm ( fields: Util.clone\(schema.fields\)) Tj
1 0 0 1 36 635 Tm ( );) Tj
1 0 0 1 36 624 Tm ( }) Tj
1 0 0 1 36 613 Tm ( ) Tj
1 0 0 1 36 602 Tm ( /** Tj
1 0 0 1 36 591 Tm ( * Validates and normalizes a record.) Tj
1 0 0 1 36 580 Tm ( * @param {string} tableName) Tj
1 0 0 1 36 569 Tm ( * @param {Object} record) Tj
1 0 0 1 36 558 Tm ( * @param {boolean=} isPartial True for patch-like updates.) Tj
1 0 0 1 36 547 Tm ( * @return {Object}) Tj
1 0 0 1 36 536 Tm ( */) Tj
1 0 0 1 36 525 Tm ( normalizeRecord\(tableName, record, isPartial\) {} Tj
1 0 0 1 36 514 Tm ( var schema = this.get\(tableName\);) Tj
1 0 0 1 36 503 Tm ( var out = Util.clone\record || {});) Tj
1 0 0 1 36 492 Tm ( schema.columns.forEach\(function\column\) {} Tj
1 0 0 1 36 481 Tm ( var fieldRules = schema.fields[column] || {};) Tj
1 0 0 1 36 470 Tm ( var hasValue = !Util.isNil\out[column]\);) Tj
1 0 0 1 36 459 Tm ( ) Tj
1 0 0 1 36 448 Tm ( if \(!hasValue && fieldRules.hasOwnProperty\('default'\)\) {} Tj
1 0 0 1 36 437 Tm ( var defaultValue = fieldRules.default;) Tj
1 0 0 1 36 426 Tm ( out[column] = typeof defaultValue === 'function' ? defaultValue\(\) : defaultValue;) Tj
1 0 0 1 36 415 Tm ( }) Tj
1 0 0 1 36 404 Tm ( ) Tj
1 0 0 1 36 393 Tm ( if \(!isPartial && fieldRules.required && Util.isNil\out[column]\)\) {} Tj
1 0 0 1 36 382 Tm ( throw new ValidationError\('Field "' + column + '" is required.');
```

```

endobj
22 0 obj
<< /Type /Page /Parent 2 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 3 0 R >> >> /Contents 23 0 R >>
endobj
23 0 obj
<< /Length 3432 >>
stream
BT
/F1 9 Tf
1 0 0 1 36 756 Tm (    */) Tj
1 0 0 1 36 745 Tm (    constructor\(\config\) \) Tj
1 0 0 1 36 734 Tm (        this._config = config;) Tj
1 0 0 1 36 723 Tm (    }) Tj
1 0 0 1 36 712 Tm (    ) Tj
1 0 0 1 36 701 Tm (    /**) Tj
1 0 0 1 36 690 Tm (        * @return {GoogleAppsScript.Spreadsheet.Spreadsheet}) Tj
1 0 0 1 36 679 Tm (        */) Tj
1 0 0 1 36 668 Tm (    getSpreadsheet\(\) \) Tj
1 0 0 1 36 657 Tm (        if \(\this._config.spreadsheetId\) \) Tj
1 0 0 1 36 646 Tm (            return SpreadsheetApp.openById\(\this._config.spreadsheetId\);) Tj
1 0 0 1 36 635 Tm (        }) Tj
1 0 0 1 36 624 Tm (        return SpreadsheetApp.getActiveSpreadsheet\(\);) Tj
1 0 0 1 36 613 Tm (    }) Tj
1 0 0 1 36 602 Tm (    ) Tj
1 0 0 1 36 591 Tm (    /**) Tj
1 0 0 1 36 580 Tm (        * @param {string} tableName) Tj
1 0 0 1 36 569 Tm (        * @param {boolean=} createWhenMissing) Tj
1 0 0 1 36 558 Tm (        * @return {GoogleAppsScript.Spreadsheet.Sheet}) Tj
1 0 0 1 36 547 Tm (        */) Tj
1 0 0 1 36 536 Tm (    getSheet\(\tableName, createWhenMissing\) \) Tj
1 0 0 1 36 525 Tm (        var ss = this.getSpreadsheet\(\);) Tj
1 0 0 1 36 514 Tm (        var sheet = ss.getSheetByName\(\tableName\);) Tj
1 0 0 1 36 503 Tm (        if \(!sheet && createWhenMissing\) \) Tj
1 0 0 1 36 492 Tm (            sheet = ss.insertSheet\(\tableName\);) Tj
1 0 0 1 36 481 Tm (        }) Tj
1 0 0 1 36 470 Tm (        if \(!sheet\) \) Tj
1 0 0 1 36 459 Tm (            throw new StorageError\('Sheet not found for table: ' + tableName\);) Tj
1 0 0 1 36 448 Tm (        }) Tj
1 0 0 1 36 437 Tm (        return sheet;) Tj
1 0 0 1 36 426 Tm (    }) Tj
1 0 0 1 36 415 Tm (    ) Tj
1 0 0 1 36 404 Tm (    /**) Tj
1 0 0 1 36 393 Tm (        * @param {string} tableName) Tj
1 0 0 1 36 382 Tm (        * @return {{headers: string[], rows: Array<Array<*>>}}) Tj
1 0 0 1 36 371 Tm (        */) Tj
1 0 0 1 36 360 Tm (    readTable\(\tableName\) \) Tj
1 0 0 1 36 349 Tm (        var sheet = this.getSheet\(\tableName, false\);) Tj
1 0 0 1 36 338 Tm (        var range = sheet.getDataRange\(\);) Tj
1 0 0 1 36 327 Tm (        var values = range.getValues\(\);) Tj
1 0 0 1 36 316 Tm (        if \(\values.length === 0\) \) Tj
1 0 0 1 36 305 Tm (            return { headers: [], rows: [] };) Tj
1 0 0 1 36 294 Tm (        }) Tj
1 0 0 1 36 283 Tm (        return {} Tj
1 0 0 1 36 272 Tm (            headers: values[0],) Tj
1 0 0 1 36 261 Tm (            rows: values.slice\(\(1\)\) Tj
1 0 0 1 36 250 Tm (        );) Tj
1 0 0 1 36 239 Tm (    }) Tj
1 0 0 1 36 228 Tm (    ) Tj
1 0 0 1 36 217 Tm (    /**) Tj
1 0 0 1 36 206 Tm (        * Ensures table exists with given headers.) Tj
1 0 0 1 36 195 Tm (        * @param {string} tableName) Tj
1 0 0 1 36 184 Tm (        * @param {string[]} headers) Tj
1 0 0 1 36 173 Tm (        */) Tj
1 0 0 1 36 162 Tm (    ensureTable\(\tableName, headers\) \) Tj
1 0 0 1 36 151 Tm (        var sheet = this.getSheet\(\tableName, true\);) Tj
1 0 0 1 36 140 Tm (        var lastRow = sheet.getLastRow\(\);) Tj
1 0 0 1 36 129 Tm (        if \(\lastRow === 0\) \) Tj
1 0 0 1 36 118 Tm (            sheet.getRange\(\(1, 1, 1, headers.length\)\).setValues\(\[headers\]\);) Tj
1 0 0 1 36 107 Tm (        return;) Tj
1 0 0 1 36 96 Tm (    }) Tj
1 0 0 1 36 85 Tm (    ) Tj
1 0 0 1 36 74 Tm (        var existingHeaders = sheet.getRange\(\(1, 1, 1, sheet.getLastColumn\(\)\)\).getValues\(\)\[0\]) Tj
1 0 0 1 36 63 Tm (            .map\(\function\(\value\) { return String\(\value\); }\);) Tj
1 0 0 1 36 52 Tm (        var expectedHeaders = headers.map\(\function\(\value\) { return String\(\value\); }\);) Tj
1 0 0 1 36 41 Tm (        if \(\existingHeaders.join\('|\')\` !== expectedHeaders.join\('|\')\` \) Tj
ET
endstream
endobj
24 0 obj
<< /Type /Page /Parent 2 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 3 0 R >> >> /Contents 25 0 R >>

```

```

endobj
25 0 obj
<< /Length 3442 >>
stream
BT
/F1 9 Tf
1 0 0 1 36 756 Tm (      throw new StorageError\('Header mismatch for table: ' + tableName\);) Tj
1 0 0 1 36 745 Tm (    }) Tj
1 0 0 1 36 734 Tm (  }) Tj
1 0 0 1 36 723 Tm ( ) Tj
1 0 0 1 36 712 Tm ( /**) Tj
1 0 0 1 36 701 Tm (  * Writes all rows for a table, replacing existing body rows.) Tj
1 0 0 1 36 690 Tm (  * @param {string} tableName) Tj
1 0 0 1 36 679 Tm (  * @param {string[]} headers) Tj
1 0 0 1 36 668 Tm (  * @param {Array<Array<*>>} rows) Tj
1 0 0 1 36 657 Tm (  */) Tj
1 0 0 1 36 646 Tm ( writeAllRows\(tableName, headers, rows\) {) Tj
1 0 0 1 36 635 Tm (   var sheet = this.getSheet\(tableName, true\);) Tj
1 0 0 1 36 624 Tm (   this.ensureTable\(tableName, headers\);) Tj
1 0 0 1 36 613 Tm ( ) Tj
1 0 0 1 36 602 Tm (   var maxRows = Math.max\(sheet.getMaxRows\(), rows.length + 1\);) Tj
1 0 0 1 36 591 Tm (   if \sheet.getMaxRows\() < maxRows\) {) Tj
1 0 0 1 36 580 Tm (     sheet.insertRowsAfter\(sheet.getMaxRows\(), maxRows - sheet.getMaxRows\()\);) Tj
1 0 0 1 36 569 Tm (   }) Tj
1 0 0 1 36 558 Tm ( ) Tj
1 0 0 1 36 547 Tm (   if \sheet.getLastRow\() > 1\) {) Tj
1 0 0 1 36 536 Tm (     sheet.getRange\(2, 1, sheet.getLastRow\() - 1, headers.length\).clearContent\();) Tj
1 0 0 1 36 525 Tm (   }) Tj
1 0 0 1 36 514 Tm ( ) Tj
1 0 0 1 36 503 Tm (   if \rows.length > 0\) {) Tj
1 0 0 1 36 492 Tm (     sheet.getRange\(2, 1, rows.length, headers.length\).setValues\(rows\);) Tj
1 0 0 1 36 481 Tm (   }) Tj
1 0 0 1 36 470 Tm ( }) Tj
1 0 0 1 36 459 Tm (}) Tj
1 0 0 1 36 448 Tm ( ) Tj
1 0 0 1 36 437 Tm ( '') Tj
1 0 0 1 36 426 Tm ( ) Tj
1 0 0 1 36 415 Tm ( ) Tj
1 0 0 1 36 404 Tm ( # 07_TableRepository.gs) Tj
1 0 0 1 36 393 Tm ( ) Tj
1 0 0 1 36 382 Tm ( **Source path:** 'sheet_db_lib/07_TableRepository.gs') Tj
1 0 0 1 36 371 Tm ( ) Tj
1 0 0 1 36 360 Tm ( '''javascript) Tj
1 0 0 1 36 349 Tm ( /**) Tj
1 0 0 1 36 338 Tm (  * CRUD repository for a specific table.) Tj
1 0 0 1 36 327 Tm (  */) Tj
1 0 0 1 36 316 Tm (class TableRepository {) Tj
1 0 0 1 36 305 Tm ( /**) Tj
1 0 0 1 36 294 Tm (  * @param {string} tableName) Tj
1 0 0 1 36 283 Tm (  * @param {Config} config) Tj
1 0 0 1 36 272 Tm (  * @param {SchemaRegistry} schemaRegistry) Tj
1 0 0 1 36 261 Tm (  * @param {SheetsGateway} sheetsGateway) Tj
1 0 0 1 36 250 Tm (  * @param {AuditLogger} auditLogger) Tj
1 0 0 1 36 239 Tm (  */) Tj
1 0 0 1 36 228 Tm ( constructor\(tableName, config, schemaRegistry, sheetsGateway, auditLogger\) {) Tj
1 0 0 1 36 217 Tm (   this._tableName = tableName;) Tj
1 0 0 1 36 206 Tm (   this._config = config;) Tj
1 0 0 1 36 195 Tm (   this._schemas = schemaRegistry;) Tj
1 0 0 1 36 184 Tm (   this._sheets = sheetsGateway;) Tj
1 0 0 1 36 173 Tm (   this._audit = auditLogger;) Tj
1 0 0 1 36 162 Tm ( }) Tj
1 0 0 1 36 151 Tm ( ) Tj
1 0 0 1 36 140 Tm ( /**) Tj
1 0 0 1 36 129 Tm (  * @return {Array<Object>} ) Tj
1 0 0 1 36 118 Tm (  */) Tj
1 0 0 1 36 107 Tm ( findAll\()\) {) Tj
1 0 0 1 36 96 Tm (   var schema = this._schemas.get\(this._tableName\);) Tj
1 0 0 1 36 85 Tm (   this._sheets.ensureTable\this._tableName, schema.columns\);) Tj
1 0 0 1 36 74 Tm (   var data = this._sheets.readTable\this._tableName\);) Tj
1 0 0 1 36 63 Tm (   return data.rows) Tj
1 0 0 1 36 52 Tm (     .map\function\row\ { return Util.rowToObject\data.headers, row\; } \) ) Tj
1 0 0 1 36 41 Tm (     .filter\function\record\ { return Util.isBlank\record[this._config.deletedAtColumn]\; }, th
ET
endstream
endobj
26 0 obj
<< /Type /Page /Parent 2 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 3 0 R >> >> /Contents 27 0 R >>
endobj
27 0 obj
<< /Length 4053 >>

```





```

1 0 0 1 36 756 Tm (      }) Tj
1 0 0 1 36 745 Tm (      }, this\));) Tj
1 0 0 1 36 734 Tm (      found[this._config.updatedAtColumn] = Util.nowIso\(\));) Tj
1 0 0 1 36 723 Tm (      break;) Tj
1 0 0 1 36 712 Tm (      }) Tj
1 0 0 1 36 701 Tm (      }) Tj
1 0 0 1 36 690 Tm (    ) Tj
1 0 0 1 36 679 Tm (      if \(!found\) {) Tj
1 0 0 1 36 668 Tm (      throw new StorageError\('Record not found for id: ' + id\);) Tj
1 0 0 1 36 657 Tm (      }) Tj
1 0 0 1 36 646 Tm (    ) Tj
1 0 0 1 36 635 Tm (      var rowValues = records.map\(function\r\ { return Util.objectToRow\(schema.columns, r\); }\));) Tj
1 0 0 1 36 624 Tm (      this._sheets.writeAllRows\(this._tableName, schema.columns, rowValues\);) Tj
1 0 0 1 36 613 Tm (      this._audit.log\('update', this._tableName, { id: id, fields: Object.keys\normalizedPatch\ } \);) Tj
1 0 0 1 36 602 Tm (      return found;) Tj
1 0 0 1 36 591 Tm (    }) Tj
1 0 0 1 36 580 Tm (    ) Tj
1 0 0 1 36 569 Tm (    /**) Tj
1 0 0 1 36 558 Tm (      * Soft-deletes a record.) Tj
1 0 0 1 36 547 Tm (      * @param {string} id) Tj
1 0 0 1 36 536 Tm (      * @return {Object}) Tj
1 0 0 1 36 525 Tm (      */) Tj
1 0 0 1 36 514 Tm (      remove\id\ {) Tj
1 0 0 1 36 503 Tm (      var result = this.update\id, \function\col, now\ {) Tj
1 0 0 1 36 492 Tm (      var patch = {};) Tj
1 0 0 1 36 481 Tm (      patch[col] = now;) Tj
1 0 0 1 36 470 Tm (      return patch;) Tj
1 0 0 1 36 459 Tm (      })\this._config.deletedAtColumn, Util.nowIso\(\)\)\);) Tj
1 0 0 1 36 448 Tm (      this._audit.log\('remove', this._tableName, { id: id } \);) Tj
1 0 0 1 36 437 Tm (      return result;) Tj
1 0 0 1 36 426 Tm (    }) Tj
1 0 0 1 36 415 Tm (    }) Tj
1 0 0 1 36 404 Tm (    ) Tj
1 0 0 1 36 393 Tm (    '') Tj
1 0 0 1 36 382 Tm (    ) Tj
1 0 0 1 36 371 Tm (    ) Tj
1 0 0 1 36 360 Tm ( # 08_TransactionManager.gs) Tj
1 0 0 1 36 349 Tm (    ) Tj
1 0 0 1 36 338 Tm ( **Source path:** 'sheet_db_lib/08_TransactionManager.gs') Tj
1 0 0 1 36 327 Tm (    ) Tj
1 0 0 1 36 316 Tm (    ''javascript) Tj
1 0 0 1 36 305 Tm (    /**) Tj
1 0 0 1 36 294 Tm (      * Runs operations under ScriptLock for coarse transaction semantics.) Tj
1 0 0 1 36 283 Tm (      */) Tj
1 0 0 1 36 272 Tm (class TransactionManager {) Tj
1 0 0 1 36 261 Tm (    /**) Tj
1 0 0 1 36 250 Tm (      * @param {number=} lockTimeoutMs) Tj
1 0 0 1 36 239 Tm (      */) Tj
1 0 0 1 36 228 Tm (      constructor\lockTimeoutMs\ {) Tj
1 0 0 1 36 217 Tm (      this._lockTimeoutMs = lockTimeoutMs || 30000;) Tj
1 0 0 1 36 206 Tm (    }) Tj
1 0 0 1 36 195 Tm (    ) Tj
1 0 0 1 36 184 Tm (    /**) Tj
1 0 0 1 36 173 Tm (      * Executes callback while holding a script lock.) Tj
1 0 0 1 36 162 Tm (      * @template T) Tj
1 0 0 1 36 151 Tm (      * @param {function\(): T} callback) Tj
1 0 0 1 36 140 Tm (      * @return {T}) Tj
1 0 0 1 36 129 Tm (      */) Tj
1 0 0 1 36 118 Tm (      run\callback\ {) Tj
1 0 0 1 36 107 Tm (      var lock = LockService.getScriptLock\(\);) Tj
1 0 0 1 36 96 Tm (      if \(!lock.tryLock\this._lockTimeoutMs\)\ {) Tj
1 0 0 1 36 85 Tm (      throw new TransactionError\('Failed to acquire script lock within timeout.')\);) Tj
1 0 0 1 36 74 Tm (      }) Tj
1 0 0 1 36 63 Tm (      try {) Tj
1 0 0 1 36 52 Tm (      return callback\(\);) Tj
1 0 0 1 36 41 Tm (      } finally {) Tj
ET
endstream
endobj
30 0 obj
<< /Type /Page /Parent 2 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 3 0 R >> >> /Contents 31 0 R >>
endobj
31 0 obj
<< /Length 2779 >>
stream
BT
/F1 9 Tf
1 0 0 1 36 756 Tm (      lock.releaseLock\(\);) Tj
1 0 0 1 36 745 Tm (      }) Tj
1 0 0 1 36 734 Tm (    }) Tj

```

```

1 0 0 1 36 723 Tm (}) Tj
1 0 0 1 36 712 Tm ( ) Tj
1 0 0 1 36 701 Tm (') Tj
1 0 0 1 36 690 Tm ( ) Tj
1 0 0 1 36 679 Tm ( ) Tj
1 0 0 1 36 668 Tm (# 09_AuditLogger.gs) Tj
1 0 0 1 36 657 Tm ( ) Tj
1 0 0 1 36 646 Tm (**Source path:** 'sheet_db_lib/09_AuditLogger.gs') Tj
1 0 0 1 36 635 Tm ( ) Tj
1 0 0 1 36 624 Tm ('''javascript) Tj
1 0 0 1 36 613 Tm (/**) Tj
1 0 0 1 36 602 Tm ( * Minimal pluggable audit logger.) Tj
1 0 0 1 36 591 Tm ( */) Tj
1 0 0 1 36 580 Tm (class AuditLogger {}) Tj
1 0 0 1 36 569 Tm ( /**) Tj
1 0 0 1 36 558 Tm ( * @param {Object=} options) Tj
1 0 0 1 36 547 Tm ( * @param {Function=} options.sink Optional custom sink function.) Tj
1 0 0 1 36 536 Tm ( */) Tj
1 0 0 1 36 525 Tm ( constructor\(options\ ) {}) Tj
1 0 0 1 36 514 Tm ( var opts = options || {};) Tj
1 0 0 1 36 503 Tm ( this._sink = opts.sink || function\(entry\ ) {}) Tj
1 0 0 1 36 492 Tm ( Logger.log\(JSON.stringify\(entry\)\);) Tj
1 0 0 1 36 481 Tm ( );) Tj
1 0 0 1 36 470 Tm ( }) Tj
1 0 0 1 36 459 Tm ( ) Tj
1 0 0 1 36 448 Tm ( /**) Tj
1 0 0 1 36 437 Tm ( * Emits an audit entry.) Tj
1 0 0 1 36 426 Tm ( * @param {string} action) Tj
1 0 0 1 36 415 Tm ( * @param {string} tableName) Tj
1 0 0 1 36 404 Tm ( * @param {Object=} metadata) Tj
1 0 0 1 36 393 Tm ( */) Tj
1 0 0 1 36 382 Tm ( log\action, tableName, metadata\ ) {} Tj
1 0 0 1 36 371 Tm ( this._sink\({} Tj
1 0 0 1 36 360 Tm ( at: Util.nowIso\(\),) Tj
1 0 0 1 36 349 Tm ( action: action,) Tj
1 0 0 1 36 338 Tm ( table: tableName,) Tj
1 0 0 1 36 327 Tm ( metadata: Util.clone\(metadata || {})\)) Tj
1 0 0 1 36 316 Tm ( }\);) Tj
1 0 0 1 36 305 Tm ( }) Tj
1 0 0 1 36 294 Tm (}) Tj
1 0 0 1 36 283 Tm ( ) Tj
1 0 0 1 36 272 Tm (') Tj
1 0 0 1 36 261 Tm ( ) Tj
1 0 0 1 36 250 Tm ( ) Tj
1 0 0 1 36 239 Tm (# 10_ScriptProperties.gs) Tj
1 0 0 1 36 228 Tm ( ) Tj
1 0 0 1 36 217 Tm (**Source path:** 'sheet_db_lib/10_ScriptProperties.gs') Tj
1 0 0 1 36 206 Tm ( ) Tj
1 0 0 1 36 195 Tm ('''javascript) Tj
1 0 0 1 36 184 Tm (/**) Tj
1 0 0 1 36 173 Tm ( * Script Properties helper for secure and environment-specific configuration.) Tj
1 0 0 1 36 162 Tm ( */) Tj
1 0 0 1 36 151 Tm (class ScriptPropertiesHelper {}) Tj
1 0 0 1 36 140 Tm ( /**) Tj
1 0 0 1 36 129 Tm ( * Creates a helper bound to Script Properties store.) Tj
1 0 0 1 36 118 Tm ( */) Tj
1 0 0 1 36 107 Tm ( constructor\(\) {}) Tj
1 0 0 1 36 96 Tm ( this._props = PropertiesService.getScriptProperties\(\);) Tj
1 0 0 1 36 85 Tm ( }) Tj
1 0 0 1 36 74 Tm ( ) Tj
1 0 0 1 36 63 Tm ( /**) Tj
1 0 0 1 36 52 Tm ( * Returns a property value, or a default when missing.) Tj
1 0 0 1 36 41 Tm ( * @param {string} key) Tj
ET
endstream
endobj
32 0 obj
<< /Type /Page /Parent 2 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 3 0 R >> >> /Contents 33 0 R >>
endobj
33 0 obj
<< /Length 3385 >>
stream
BT
/F1 9 Tf
1 0 0 1 36 756 Tm ( * @param {*} default\value) Tj
1 0 0 1 36 745 Tm ( * @return {*}) Tj
1 0 0 1 36 734 Tm ( */) Tj
1 0 0 1 36 723 Tm ( get\key, default\value\ ) {} Tj
1 0 0 1 36 712 Tm ( var value = this._props.getProperty\key\);) Tj
1 0 0 1 36 701 Tm ( return value === null ? default\value : value;) Tj

```

```

1 0 0 1 36 690 Tm ( }) Tj
1 0 0 1 36 679 Tm ( ) Tj
1 0 0 1 36 668 Tm ( /**) Tj
1 0 0 1 36 657 Tm ( * Returns a required property value or throws when missing/blank.) Tj
1 0 0 1 36 646 Tm ( * @param {string} key) Tj
1 0 0 1 36 635 Tm ( * @return {string}) Tj
1 0 0 1 36 624 Tm ( */) Tj
1 0 0 1 36 613 Tm ( getRequired\(\key\) {) Tj
1 0 0 1 36 602 Tm ( var value = this.get\(\key, '\');) Tj
1 0 0 1 36 591 Tm ( if \(\value === '\') {) Tj
1 0 0 1 36 580 Tm ( throw new ConfigError\('Missing required script property: ' + key\);) Tj
1 0 0 1 36 569 Tm ( }) Tj
1 0 0 1 36 558 Tm ( return String\(\value\);) Tj
1 0 0 1 36 547 Tm ( }) Tj
1 0 0 1 36 536 Tm ( ) Tj
1 0 0 1 36 525 Tm ( /**) Tj
1 0 0 1 36 514 Tm ( * Returns a boolean property value.) Tj
1 0 0 1 36 503 Tm ( * Accepted true values: true, 1, yes, y, on.) Tj
1 0 0 1 36 492 Tm ( * Accepted false values: false, 0, no, n, off.) Tj
1 0 0 1 36 481 Tm ( * @param {string} key) Tj
1 0 0 1 36 470 Tm ( * @param {boolean=} defaultValue) Tj
1 0 0 1 36 459 Tm ( * @return {boolean}) Tj
1 0 0 1 36 448 Tm ( */) Tj
1 0 0 1 36 437 Tm ( getBoolean\(\key, defaultValue\) {) Tj
1 0 0 1 36 426 Tm ( var raw = this.get\(\key, null\);) Tj
1 0 0 1 36 415 Tm ( if \(\raw === null || raw === '\') {) Tj
1 0 0 1 36 404 Tm ( return defaultValue === undefined ? false : !!defaultValue;) Tj
1 0 0 1 36 393 Tm ( }) Tj
1 0 0 1 36 382 Tm ( var normalized = String\(\raw\).trim\(\).toLowerCase\(\);) Tj
1 0 0 1 36 371 Tm ( if \(['true', '1', 'yes', 'y', 'on'].indexOf\(\normalized\) !== -1\) {) Tj
1 0 0 1 36 360 Tm ( return true;) Tj
1 0 0 1 36 349 Tm ( }) Tj
1 0 0 1 36 338 Tm ( if \(['false', '0', 'no', 'n', 'off'].indexOf\(\normalized\) !== -1\) {) Tj
1 0 0 1 36 327 Tm ( return false;) Tj
1 0 0 1 36 316 Tm ( }) Tj
1 0 0 1 36 305 Tm ( throw new ConfigError\('Script property is not a valid boolean: ' + key\);) Tj
1 0 0 1 36 294 Tm ( }) Tj
1 0 0 1 36 283 Tm ( ) Tj
1 0 0 1 36 272 Tm ( /**) Tj
1 0 0 1 36 261 Tm ( * Returns a numeric property value.) Tj
1 0 0 1 36 250 Tm ( * @param {string} key) Tj
1 0 0 1 36 239 Tm ( * @param {number=} defaultValue) Tj
1 0 0 1 36 228 Tm ( * @return {number}) Tj
1 0 0 1 36 217 Tm ( */) Tj
1 0 0 1 36 206 Tm ( getNumber\(\key, defaultValue\) {) Tj
1 0 0 1 36 195 Tm ( var raw = this.get\(\key, null\);) Tj
1 0 0 1 36 184 Tm ( if \(\raw === null || raw === '\') {) Tj
1 0 0 1 36 173 Tm ( return defaultValue === undefined ? 0 : Number\(\defaultValue\);) Tj
1 0 0 1 36 162 Tm ( }) Tj
1 0 0 1 36 151 Tm ( var n = Number\(\raw\);) Tj
1 0 0 1 36 140 Tm ( if \(\isNaN\(\n\)\) {) Tj
1 0 0 1 36 129 Tm ( throw new ConfigError\('Script property is not a valid number: ' + key\);) Tj
1 0 0 1 36 118 Tm ( }) Tj
1 0 0 1 36 107 Tm ( return n;) Tj
1 0 0 1 36 96 Tm ( }) Tj
1 0 0 1 36 85 Tm ( ) Tj
1 0 0 1 36 74 Tm ( /**) Tj
1 0 0 1 36 63 Tm ( * Returns all script properties as key-value object.) Tj
1 0 0 1 36 52 Tm ( * @return {Object<string, string>}) Tj
1 0 0 1 36 41 Tm ( */) Tj
ET
endstream
endobj
34 0 obj
<< /Type /Page /Parent 2 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 3 0 R >> >> /Contents 35 0 R >>
endobj
35 0 obj
<< /Length 3144 >>
stream
BT
/F1 9 Tf
1 0 0 1 36 756 Tm ( getAll\(\) {) Tj
1 0 0 1 36 745 Tm ( return this._props.getProperties\(\);) Tj
1 0 0 1 36 734 Tm ( }) Tj
1 0 0 1 36 723 Tm ( ) Tj
1 0 0 1 36 712 Tm ( /**) Tj
1 0 0 1 36 701 Tm ( * Returns all script properties with sensitive values redacted.) Tj
1 0 0 1 36 690 Tm ( * @return {Object<string, string>}) Tj
1 0 0 1 36 679 Tm ( */) Tj
1 0 0 1 36 668 Tm ( getAllRedacted\(\) {) Tj

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1 0 0 1 36 657 Tm (    var all = this.getAll\(\);) Tj
1 0 0 1 36 646 Tm (    var out = {};) Tj
1 0 0 1 36 635 Tm (    Object.keys\(\all\).forEach\(\function\(\key\) {\} Tj
1 0 0 1 36 624 Tm (        out[key] = ScriptPropertiesHelper.shouldRedactKey\(\key\) ? '[REDACTED]' : all[key];) Tj
1 0 0 1 36 613 Tm (    }\);) Tj
1 0 0 1 36 602 Tm (    return out;) Tj
1 0 0 1 36 591 Tm (    }) Tj
1 0 0 1 36 580 Tm (    ) Tj
1 0 0 1 36 569 Tm (    /**) Tj
1 0 0 1 36 558 Tm (        * Sets a script property value.) Tj
1 0 0 1 36 547 Tm (        * @param {string} key) Tj
1 0 0 1 36 536 Tm (        * @param {*} value) Tj
1 0 0 1 36 525 Tm (        */) Tj
1 0 0 1 36 514 Tm (    set\(\key, value\) {\} Tj
1 0 0 1 36 503 Tm (        this._props.setProperty\(\key, String\(\value\)\);) Tj
1 0 0 1 36 492 Tm (    }) Tj
1 0 0 1 36 481 Tm (    ) Tj
1 0 0 1 36 470 Tm (    /**) Tj
1 0 0 1 36 459 Tm (        * Sets multiple script properties.) Tj
1 0 0 1 36 448 Tm (        * @param {Object<string, *>} obj) Tj
1 0 0 1 36 437 Tm (        */) Tj
1 0 0 1 36 426 Tm (    setMany\(\obj\) {\} Tj
1 0 0 1 36 415 Tm (        var stringified = {};) Tj
1 0 0 1 36 404 Tm (        Object.keys\(\obj || {\}\).forEach\(\function\(\key\) {\} Tj
1 0 0 1 36 393 Tm (            stringified[key] = String\(\obj[key]\);) Tj
1 0 0 1 36 382 Tm (        }\);) Tj
1 0 0 1 36 371 Tm (        this._props.setProperties\(\stringified, false\);) Tj
1 0 0 1 36 360 Tm (    }) Tj
1 0 0 1 36 349 Tm (    ) Tj
1 0 0 1 36 338 Tm (    /**) Tj
1 0 0 1 36 327 Tm (        * Deletes a script property.) Tj
1 0 0 1 36 316 Tm (        * @param {string} key) Tj
1 0 0 1 36 305 Tm (        */) Tj
1 0 0 1 36 294 Tm (    delete\(\key\) {\} Tj
1 0 0 1 36 283 Tm (        this._props.deleteProperty\(\key\);) Tj
1 0 0 1 36 272 Tm (    }) Tj
1 0 0 1 36 261 Tm (    ) Tj
1 0 0 1 36 250 Tm (    /**) Tj
1 0 0 1 36 239 Tm (        * Returns true when a script property exists and is non-empty.) Tj
1 0 0 1 36 228 Tm (        * @param {string} key) Tj
1 0 0 1 36 217 Tm (        * @return {boolean}) Tj
1 0 0 1 36 206 Tm (        */) Tj
1 0 0 1 36 195 Tm (    has\(\key\) {\} Tj
1 0 0 1 36 184 Tm (        var value = this._props.getProperty\(\key\);) Tj
1 0 0 1 36 173 Tm (        return value !== null && value !== '');) Tj
1 0 0 1 36 162 Tm (    }) Tj
1 0 0 1 36 151 Tm (    ) Tj
1 0 0 1 36 140 Tm (    /**) Tj
1 0 0 1 36 129 Tm (        * Returns true when key should be redacted in debug output.) Tj
1 0 0 1 36 118 Tm (        * @param {string} key) Tj
1 0 0 1 36 107 Tm (        * @return {boolean}) Tj
1 0 0 1 36 96 Tm (        */) Tj
1 0 0 1 36 85 Tm (    static shouldRedactKey\(\key\) {\} Tj
1 0 0 1 36 74 Tm (        var upper = String\(\key || ''\).toUpperCase\(\);) Tj
1 0 0 1 36 63 Tm (        if \(\upper === 'APP_PASSCODE' || upper === 'SIGNING_SECRET' || upper === 'WEBHOOK_SECRET' || upper === 'SECRET' || upper === 'TOKEN') Tj
1 0 0 1 36 52 Tm (            return true;) Tj
1 0 0 1 36 41 Tm (        }) Tj
ET
endstream
endobj
36 0 obj
<< /Type /Page /Parent 2 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 3 0 R >> >> /Contents 37 0 R >>
endobj
37 0 obj
<< /Length 3581 >>
stream
BT
/F1 9 Tf
1 0 0 1 36 756 Tm (    return upper.indexOf\('TOKEN'\) !== -1 || upper.indexOf\('SECRET'\) !== -1 || upper.indexOf\('PAS
1 0 0 1 36 745 Tm (    }) Tj
1 0 0 1 36 734 Tm (}) Tj
1 0 0 1 36 723 Tm (    ) Tj
1 0 0 1 36 712 Tm (    /**) Tj
1 0 0 1 36 701 Tm (        * Reads SheetDb client-related config from Script Properties.) Tj
1 0 0 1 36 690 Tm (        *) Tj
1 0 0 1 36 679 Tm (        * Supported keys:) Tj
1 0 0 1 36 668 Tm (        * - SHEET_DB_SPREADSHEET_ID) Tj
1 0 0 1 36 657 Tm (        * - SHEET_DB_ID_COLUMN) Tj
1 0 0 1 36 646 Tm (        * - SHEET_DB_CREATED_AT_COLUMN) Tj
1 0 0 1 36 635 Tm (        * - SHEET_DB_UPDATED_AT_COLUMN) Tj

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1 0 0 1 36 591 Tm ( ) Tj
1 0 0 1 36 580 Tm (# 00_WebAppEntry.gs) Tj
1 0 0 1 36 569 Tm ( ) Tj
1 0 0 1 36 558 Tm (**Source path: ** 'slack_host/00_WebAppEntry.gs') Tj
1 0 0 1 36 547 Tm ( ) Tj
1 0 0 1 36 536 Tm ('''javascript) Tj
1 0 0 1 36 525 Tm (/**) Tj
1 0 0 1 36 514 Tm ( * Slack host endpoint \ (web app\.) Tj
1 0 0 1 36 503 Tm ( */) Tj
1 0 0 1 36 492 Tm ( ) Tj
1 0 0 1 36 481 Tm (/**) Tj
1 0 0 1 36 470 Tm ( * Creates a DB client from SheetDb library and registers host schemas.) Tj
1 0 0 1 36 459 Tm ( * @return {DbClient}) Tj
1 0 0 1 36 448 Tm ( */) Tj
1 0 0 1 36 437 Tm (function createHostDbClient\(\) {) Tj
1 0 0 1 36 426 Tm ( var db = SheetDb.createClientFromScriptProperties\(\);) Tj
1 0 0 1 36 415 Tm ( ) Tj
1 0 0 1 36 404 Tm ( db.schema\('students', {) Tj
1 0 0 1 36 393 Tm ( columns: ['id', 'name', 'source', 'createdAt', 'updatedAt', 'deletedAt'],) Tj
1 0 0 1 36 382 Tm ( fields: {) Tj
1 0 0 1 36 371 Tm ( id: { required: true, type: 'string' },) Tj
1 0 0 1 36 360 Tm ( name: { required: true, type: 'string' },) Tj
1 0 0 1 36 349 Tm ( source: { default: 'slack', type: 'string' }) Tj
1 0 0 1 36 338 Tm ( }) Tj
1 0 0 1 36 327 Tm ( }\);) Tj
1 0 0 1 36 316 Tm ( ) Tj
1 0 0 1 36 305 Tm ( db.schema\('enrollments', {) Tj
1 0 0 1 36 294 Tm ( columns: ['id', 'studentId', 'courseId', 'status', 'source', 'createdAt', 'updatedAt', 'deletedAt'] Tj
1 0 0 1 36 283 Tm ( fields: {) Tj
1 0 0 1 36 272 Tm ( studentId: { required: true, type: 'string' },) Tj
1 0 0 1 36 261 Tm ( courseId: { required: true, type: 'string' },) Tj
1 0 0 1 36 250 Tm ( status: { default: 'enrolled', type: 'string' },) Tj
1 0 0 1 36 239 Tm ( source: { default: 'slack', type: 'string' }) Tj
1 0 0 1 36 228 Tm ( }) Tj
1 0 0 1 36 217 Tm ( }\);) Tj
1 0 0 1 36 206 Tm ( ) Tj
1 0 0 1 36 195 Tm ( db.schema\('automations', {) Tj
1 0 0 1 36 184 Tm ( columns: ['id', 'requestId', 'workflow', 'eventType', 'state', 'createdAt', 'updatedAt', 'deletedAt'] Tj
1 0 0 1 36 173 Tm ( fields: {) Tj
1 0 0 1 36 162 Tm ( requestId: { required: true, type: 'string' },) Tj
1 0 0 1 36 151 Tm ( workflow: { required: true, type: 'string' },) Tj
1 0 0 1 36 140 Tm ( eventType: { required: true, type: 'string' },) Tj
1 0 0 1 36 129 Tm ( state: { default: 'received', type: 'string' }) Tj
1 0 0 1 36 118 Tm ( }) Tj
1 0 0 1 36 107 Tm ( }\);) Tj
1 0 0 1 36 96 Tm ( ) Tj
1 0 0 1 36 85 Tm ( db.schema\('audit_logs', {) Tj
1 0 0 1 36 74 Tm ( columns: ['id', 'requestId', 'source', 'action', 'status', 'message', 'metadata', 'createdAt', 'updatedAt'] Tj
1 0 0 1 36 63 Tm ( fields: {) Tj
1 0 0 1 36 52 Tm ( requestId: { required: true, type: 'string' },) Tj
1 0 0 1 36 41 Tm ( source: { required: true, type: 'string' },) Tj
ET
endstream
endobj
40 0 obj
<< /Type /Page /Parent 2 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 3 0 R >> >> /Contents 41 0 R >>
endobj
41 0 obj
<< /Length 3428 >>
stream
BT
/F1 9 Tf
1 0 0 1 36 756 Tm ( action: { required: true, type: 'string' },) Tj
1 0 0 1 36 745 Tm ( status: { required: true, type: 'string' },) Tj
1 0 0 1 36 734 Tm ( message: { type: 'string' },) Tj
1 0 0 1 36 723 Tm ( metadata: { type: 'string' }) Tj
1 0 0 1 36 712 Tm ( }) Tj
1 0 0 1 36 701 Tm ( }\);) Tj
1 0 0 1 36 690 Tm ( ) Tj
1 0 0 1 36 679 Tm ( return db;) Tj
1 0 0 1 36 668 Tm (}) Tj
1 0 0 1 36 657 Tm ( ) Tj
1 0 0 1 36 646 Tm (/**) Tj
1 0 0 1 36 635 Tm ( * Creates the host-layer router with all dependencies wired.) Tj
1 0 0 1 36 624 Tm ( * @param {DbClient} db) Tj
1 0 0 1 36 613 Tm ( * @param {HostConfig} hostConfig) Tj
1 0 0 1 36 602 Tm ( * @return {SlackRouter}) Tj
1 0 0 1 36 591 Tm ( */) Tj
1 0 0 1 36 580 Tm (function createHostRouter\ (db, hostConfig\ ) {) Tj
1 0 0 1 36 569 Tm ( var parser = new SlackPayloadParser\(\);) Tj

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```

1 0 0 1 36 558 Tm ( var security = new SlackSecurity\(\hostConfig\);) Tj
1 0 0 1 36 547 Tm ( var contexts = new RequestContextFactory\(\hostConfig\);) Tj
1 0 0 1 36 536 Tm ( var formatter = new ResultFormatter\(\);) Tj
1 0 0 1 36 525 Tm ( ) Tj
1 0 0 1 36 514 Tm ( var enrollmentService = new LmsEnrollmentService\(\db\);) Tj
1 0 0 1 36 503 Tm ( var automationService = new LmsAutomationService\(\db\);) Tj
1 0 0 1 36 492 Tm ( var workflowHandlers = new WorkflowWebhookHandlers\(\enrollmentService, automationService\);) Tj
1 0 0 1 36 481 Tm ( var slackService = new SlackService\(\enrollmentService, workflowHandlers, formatter\);) Tj
1 0 0 1 36 470 Tm ( ) Tj
1 0 0 1 36 459 Tm ( return new SlackRouter\(\parser, security, contexts, slackService, formatter, db\);) Tj
1 0 0 1 36 448 Tm (}) Tj
1 0 0 1 36 437 Tm ( ) Tj
1 0 0 1 36 426 Tm (/**) Tj
1 0 0 1 36 415 Tm ( * Webhook endpoint.) Tj
1 0 0 1 36 404 Tm ( * @param {GoogleAppsScript.Events.DoPost} e) Tj
1 0 0 1 36 393 Tm ( * @return {GoogleAppsScript.Content.TextOutput}) Tj
1 0 0 1 36 382 Tm ( */) Tj
1 0 0 1 36 371 Tm (function doPost\(\e\) {}) Tj
1 0 0 1 36 360 Tm ( var db = createHostDbClient\(\);) Tj
1 0 0 1 36 349 Tm ( var hostConfig = new HostConfig\(\new ScriptPropertiesHelper\(\)\)\);) Tj
1 0 0 1 36 338 Tm ( var router = createHostRouter\(\db, hostConfig\);) Tj
1 0 0 1 36 327 Tm ( var result = router.handle\(\e\);) Tj
1 0 0 1 36 316 Tm ( ) Tj
1 0 0 1 36 305 Tm ( return ContentService) Tj
1 0 0 1 36 294 Tm ( .createTextOutput\(\JSON.stringify\(\result.slack || result\)\)\)) Tj
1 0 0 1 36 283 Tm ( .setMimeType\(\ContentService.MimeType.JSON\);) Tj
1 0 0 1 36 272 Tm (}) Tj
1 0 0 1 36 261 Tm ( ) Tj
1 0 0 1 36 250 Tm (```) Tj
1 0 0 1 36 239 Tm ( ) Tj
1 0 0 1 36 228 Tm ( ) Tj
1 0 0 1 36 217 Tm (# 01_SlackRouter.gs) Tj
1 0 0 1 36 206 Tm ( ) Tj
1 0 0 1 36 195 Tm (**Source path:** 'slack_host/01_SlackRouter.gs') Tj
1 0 0 1 36 184 Tm ( ) Tj
1 0 0 1 36 173 Tm (```) Tj
1 0 0 1 36 162 Tm (/**) Tj
1 0 0 1 36 151 Tm ( * HTTP router boundary: parse -> verify -> context -> app service.) Tj
1 0 0 1 36 140 Tm ( */) Tj
1 0 0 1 36 129 Tm (class SlackRouter {}) Tj
1 0 0 1 36 118 Tm ( /**) Tj
1 0 0 1 36 107 Tm ( * @param {SlackPayloadParser} parser) Tj
1 0 0 1 36 96 Tm ( * @param {SlackSecurity} security) Tj
1 0 0 1 36 85 Tm ( * @param {RequestContextFactory} contextFactory) Tj
1 0 0 1 36 74 Tm ( * @param {SlackService} slackService) Tj
1 0 0 1 36 63 Tm ( * @param {ResultFormatter} resultFormatter) Tj
1 0 0 1 36 52 Tm ( * @param {DbClient} db) Tj
1 0 0 1 36 41 Tm ( */) Tj
ET
endstream
endobj
42 0 obj
<< /Type /Page /Parent 2 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 3 0 R >> >> /Contents 43 0 R >>
endobj
43 0 obj
<< /Length 3472 >>
stream
BT
/F1 9 Tf
1 0 0 1 36 756 Tm ( constructor\(\parser, security, contextFactory, slackService, resultFormatter, db\) {}) Tj
1 0 0 1 36 745 Tm ( this._parser = parser;) Tj
1 0 0 1 36 734 Tm ( this._security = security;) Tj
1 0 0 1 36 723 Tm ( this._contexts = contextFactory;) Tj
1 0 0 1 36 712 Tm ( this._service = slackService;) Tj
1 0 0 1 36 701 Tm ( this._results = resultFormatter;) Tj
1 0 0 1 36 690 Tm ( this._db = db;) Tj
1 0 0 1 36 679 Tm ( }) Tj
1 0 0 1 36 668 Tm ( ) Tj
1 0 0 1 36 657 Tm ( /**) Tj
1 0 0 1 36 646 Tm ( * @param {GoogleAppsScript.Events.DoPost} e) Tj
1 0 0 1 36 635 Tm ( * @return {Object}) Tj
1 0 0 1 36 624 Tm ( */) Tj
1 0 0 1 36 613 Tm ( handle\(\e\) {}) Tj
1 0 0 1 36 602 Tm ( var parsed = this._parser.parseEvent\(\e\);) Tj
1 0 0 1 36 591 Tm ( var context = this._contexts.create\(\parsed.routeType, parsed.payload, parsed.envelope\);) Tj
1 0 0 1 36 580 Tm ( ) Tj
1 0 0 1 36 569 Tm ( var auth = this._security.verify\(\parsed\);) Tj
1 0 0 1 36 558 Tm ( if \(!auth.ok\) {}) Tj
1 0 0 1 36 547 Tm ( var denied = this._results.failure\(\auth.code, auth.message, context, {}\);) Tj
1 0 0 1 36 536 Tm ( this._writeAudit\(\context, denied\);) Tj

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```

1 0 0 1 36 525 Tm (      return denied;) Tj
1 0 0 1 36 514 Tm (    }) Tj
1 0 0 1 36 503 Tm (  ) Tj
1 0 0 1 36 492 Tm (    var result;) Tj
1 0 0 1 36 481 Tm (    try {} Tj
1 0 0 1 36 470 Tm (      if \((parsed.routeType === 'slash_command') {) Tj
1 0 0 1 36 459 Tm (        result = this._service.handleSlashCommand(context, parsed.payload\);) Tj
1 0 0 1 36 448 Tm (      } else if \((parsed.routeType === 'workflow_webhook') {) Tj
1 0 0 1 36 437 Tm (        result = this._service.handleWorkflowWebhook(context, parsed.payload\);) Tj
1 0 0 1 36 426 Tm (      } else {} Tj
1 0 0 1 36 415 Tm (        result = this._results.failure('UNSUPPORTED_PAYLOAD', 'Unsupported payload.', context, {})\);
1 0 0 1 36 404 Tm (      }) Tj
1 0 0 1 36 393 Tm (    } catch \((err\) {) Tj
1 0 0 1 36 382 Tm (      result = this._results.failure(err.code || 'HOST_ERROR', err.message || 'Unexpected error.', c
1 0 0 1 36 371 Tm (    }) Tj
1 0 0 1 36 360 Tm (  ) Tj
1 0 0 1 36 349 Tm (    this._writeAudit(context, result\);) Tj
1 0 0 1 36 338 Tm (    return result;) Tj
1 0 0 1 36 327 Tm (  }) Tj
1 0 0 1 36 316 Tm (  ) Tj
1 0 0 1 36 305 Tm (  /** @private */) Tj
1 0 0 1 36 294 Tm (    _writeAudit(context, result\) {) Tj
1 0 0 1 36 283 Tm (      this._db.table('audit_logs').insert({) Tj
1 0 0 1 36 272 Tm (        requestId: context.requestId,) Tj
1 0 0 1 36 261 Tm (        source: context.source,) Tj
1 0 0 1 36 250 Tm (        action: context.routeType,) Tj
1 0 0 1 36 239 Tm (        status: result.ok ? 'success' : 'failure',) Tj
1 0 0 1 36 228 Tm (        message: result.message,) Tj
1 0 0 1 36 217 Tm (        metadata: JSON.stringify({ code: result.code, requestId: result.requestId }))) Tj
1 0 0 1 36 206 Tm (      }\);) Tj
1 0 0 1 36 195 Tm (    }) Tj
1 0 0 1 36 184 Tm (  }) Tj
1 0 0 1 36 173 Tm (  ) Tj
1 0 0 1 36 162 Tm (  '') Tj
1 0 0 1 36 151 Tm (  ) Tj
1 0 0 1 36 140 Tm (  ) Tj
1 0 0 1 36 129 Tm (  # 02_SlackPayloadParser.gs) Tj
1 0 0 1 36 118 Tm (  ) Tj
1 0 0 1 36 107 Tm (  **Source path:** 'slack_host/02_SlackPayloadParser.gs') Tj
1 0 0 1 36 96 Tm (  ) Tj
1 0 0 1 36 85 Tm (  '''javascript) Tj
1 0 0 1 36 74 Tm (  /**) Tj
1 0 0 1 36 63 Tm (  * Parses and normalizes inbound Slack-related payloads.) Tj
1 0 0 1 36 52 Tm (  */) Tj
1 0 0 1 36 41 Tm (class SlackPayloadParser {}) Tj
ET
endstream
endobj
44 0 obj
<< /Type /Page /Parent 2 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 3 0 R >> >> /Contents 45 0 R >>
endobj
45 0 obj
<< /Length 3699 >>
stream
BT
/F1 9 Tf
1 0 0 1 36 756 Tm (  /**) Tj
1 0 0 1 36 745 Tm (    * @param {GoogleAppsScript.Events.DoPost} e) Tj
1 0 0 1 36 734 Tm (    * @return {{routeType: string, payload: Object, envelope: Object}}) Tj
1 0 0 1 36 723 Tm (    */) Tj
1 0 0 1 36 712 Tm (    parseEvent(e\) {) Tj
1 0 0 1 36 701 Tm (      var rawBody = \((e && e.postData && e.postData.contents) ? e.postData.contents : ''); Tj
1 0 0 1 36 690 Tm (      var params = this._parseFormEncoded(rawBody\);) Tj
1 0 0 1 36 679 Tm (    ) Tj
1 0 0 1 36 668 Tm (      if \((params.command\) {) Tj
1 0 0 1 36 657 Tm (        return {} Tj
1 0 0 1 36 646 Tm (        routeType: 'slash_command',) Tj
1 0 0 1 36 635 Tm (        payload: this._normalizeSlashCommand(params\),) Tj
1 0 0 1 36 624 Tm (        envelope: { rawBody: rawBody, params: params }) Tj
1 0 0 1 36 613 Tm (      );) Tj
1 0 0 1 36 602 Tm (    }) Tj
1 0 0 1 36 591 Tm (  ) Tj
1 0 0 1 36 580 Tm (    var jsonPayload = this._parseJson(rawBody\);) Tj
1 0 0 1 36 569 Tm (    if \((jsonPayload\) {) Tj
1 0 0 1 36 558 Tm (      return {} Tj
1 0 0 1 36 547 Tm (      routeType: 'workflow_webhook',) Tj
1 0 0 1 36 536 Tm (      payload: this._normalizeWorkflowWebhook(jsonPayload\),) Tj
1 0 0 1 36 525 Tm (      envelope: { rawBody: rawBody, json: jsonPayload }) Tj
1 0 0 1 36 514 Tm (    );) Tj
1 0 0 1 36 503 Tm (  }) Tj

```



```

1 0 0 1 36 492 Tm ( ) Tj
1 0 0 1 36 481 Tm ( return { routeType: 'unknown', payload: {}, envelope: { rawBody: rawBody } }); Tj
1 0 0 1 36 470 Tm ( }) Tj
1 0 0 1 36 459 Tm ( ) Tj
1 0 0 1 36 448 Tm ( /** @private */) Tj
1 0 0 1 36 437 Tm ( _normalizeSlashCommand\(payload\) { } Tj
1 0 0 1 36 426 Tm ( return { } Tj
1 0 0 1 36 415 Tm ( command: String\(payload.command || ''\).trim\(\),) Tj
1 0 0 1 36 404 Tm ( text: String\(payload.text || ''\).trim\(\),) Tj
1 0 0 1 36 393 Tm ( teamId: String\(payload.team_id || ''\),) Tj
1 0 0 1 36 382 Tm ( channelId: String\(payload.channel_id || ''\),) Tj
1 0 0 1 36 371 Tm ( userId: String\(payload.user_id || ''\),) Tj
1 0 0 1 36 360 Tm ( triggerId: String\(payload.trigger_id || ''\),) Tj
1 0 0 1 36 349 Tm ( responseUrl: String\(payload.response_url || ''\),) Tj
1 0 0 1 36 338 Tm ( requestId: String\(payload.request_id || Util.generateId\(\'slash\'\\)\) Tj
1 0 0 1 36 327 Tm ( ); Tj
1 0 0 1 36 316 Tm ( }) Tj
1 0 0 1 36 305 Tm ( ) Tj
1 0 0 1 36 294 Tm ( /** @private */) Tj
1 0 0 1 36 283 Tm ( _normalizeWorkflowWebhook\(payload\) { } Tj
1 0 0 1 36 272 Tm ( var data = payload.data || payload;) Tj
1 0 0 1 36 261 Tm ( return { } Tj
1 0 0 1 36 250 Tm ( workflow: String\(payload.workflow || payload.workflow_name || \'unknown_workflow\'\\),) Tj
1 0 0 1 36 239 Tm ( eventType: String\(payload.eventType || payload.event_type || \'workflow_event\'\\),) Tj
1 0 0 1 36 228 Tm ( requestId: String\(payload.requestId || payload.request_id || Util.generateId\(\'wf\'\\)\),) Tj
1 0 0 1 36 217 Tm ( studentId: String\(data.studentId || data.student_id || ''\),) Tj
1 0 0 1 36 206 Tm ( courseId: String\(data.courseId || data.course_id || ''\),) Tj
1 0 0 1 36 195 Tm ( actorId: String\(data.actorId || data.actor_id || ''\),) Tj
1 0 0 1 36 184 Tm ( status: String\(data.status || ''\)) Tj
1 0 0 1 36 173 Tm ( ); Tj
1 0 0 1 36 162 Tm ( }) Tj
1 0 0 1 36 151 Tm ( ) Tj
1 0 0 1 36 140 Tm ( /** @private */) Tj
1 0 0 1 36 129 Tm ( _parseJson\(raw\) { } Tj
1 0 0 1 36 118 Tm ( try { } Tj
1 0 0 1 36 107 Tm ( if \(!raw\) return null;) Tj
1 0 0 1 36 96 Tm ( return JSON.parse\(raw\);) Tj
1 0 0 1 36 85 Tm ( } catch \(err\) { } Tj
1 0 0 1 36 74 Tm ( return null;) Tj
1 0 0 1 36 63 Tm ( }) Tj
1 0 0 1 36 52 Tm ( }) Tj
1 0 0 1 36 41 Tm ( ) Tj
ET
endstream
endobj
46 0 obj
<< /Type /Page /Parent 2 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 3 0 R >> >> /Contents 47 0 R >>
endobj
47 0 obj
<< /Length 3496 >>
stream
BT
/F1 9 Tf
1 0 0 1 36 756 Tm ( /** @private */) Tj
1 0 0 1 36 745 Tm ( _parseFormEncoded\(body\) { } Tj
1 0 0 1 36 734 Tm ( var out = {};) Tj
1 0 0 1 36 723 Tm ( if \(!body\) return out;) Tj
1 0 0 1 36 712 Tm ( body.split\('&\'\\).forEach\(function\(pair\) { } Tj
1 0 0 1 36 701 Tm ( var tokens = pair.split\('=\'\\);) Tj
1 0 0 1 36 690 Tm ( var key = decodeURIComponent\(tokens[0] || ''\).replace\(/\\+/g, ' ');) Tj
1 0 0 1 36 679 Tm ( var rawValue = tokens.length > 1 ? tokens.slice(1).join\('=\'\\) : '';) Tj
1 0 0 1 36 668 Tm ( var value = decodeURIComponent\(rawValue\).replace\(/\\+/g, ' ');) Tj
1 0 0 1 36 657 Tm ( if \(key\) out[key] = value;) Tj
1 0 0 1 36 646 Tm ( );) Tj
1 0 0 1 36 635 Tm ( return out;) Tj
1 0 0 1 36 624 Tm ( }) Tj
1 0 0 1 36 613 Tm (}) Tj
1 0 0 1 36 602 Tm ( ) Tj
1 0 0 1 36 591 Tm (\'\'\'') Tj
1 0 0 1 36 580 Tm ( ) Tj
1 0 0 1 36 569 Tm ( ) Tj
1 0 0 1 36 558 Tm (# 03_SlackService.gs) Tj
1 0 0 1 36 547 Tm ( ) Tj
1 0 0 1 36 536 Tm (**Source path:** \'slack_host/03_SlackService.gs') Tj
1 0 0 1 36 525 Tm ( ) Tj
1 0 0 1 36 514 Tm (\'\'\'javascript) Tj
1 0 0 1 36 503 Tm (/**) Tj
1 0 0 1 36 492 Tm ( * Application service that orchestrates Slack routes into LMS services.) Tj
1 0 0 1 36 481 Tm ( */) Tj
1 0 0 1 36 470 Tm (class SlackService { } Tj

```



```

1 0 0 1 36 426 Tm ( ) Tj
1 0 0 1 36 415 Tm (   var params = \(\parsed.envelope && parsed.envelope.params\) ? parsed.envelope.params : {};) Tj
1 0 0 1 36 404 Tm (   var supplied = String\(\params.passcode || params.api_key || ''\);) Tj
1 0 0 1 36 393 Tm (   if \(\supplied && supplied === configured\) {}) Tj
1 0 0 1 36 382 Tm (       return { ok: true, code: 'AUTHORIZED', message: 'Passcode validated.' };) Tj
1 0 0 1 36 371 Tm (   }) Tj
1 0 0 1 36 360 Tm ( ) Tj
1 0 0 1 36 349 Tm (       return { ok: false, code: 'UNAUTHORIZED', message: 'Invalid or missing passcode.' };) Tj
1 0 0 1 36 338 Tm (   }) Tj
1 0 0 1 36 327 Tm (}) Tj
1 0 0 1 36 316 Tm ( ) Tj
1 0 0 1 36 305 Tm (``) Tj
1 0 0 1 36 294 Tm ( ) Tj
1 0 0 1 36 283 Tm ( ) Tj
1 0 0 1 36 272 Tm (# 05_LmsEnrollmentService.gs) Tj
1 0 0 1 36 261 Tm ( ) Tj
1 0 0 1 36 250 Tm (**Source path:** 'slack_host/05_LmsEnrollmentService.gs') Tj
1 0 0 1 36 239 Tm ( ) Tj
1 0 0 1 36 228 Tm (``javascript) Tj
1 0 0 1 36 217 Tm (/**) Tj
1 0 0 1 36 206 Tm ( * LMS enrollment operations implemented using the SheetDb library only.) Tj
1 0 0 1 36 195 Tm ( */) Tj
1 0 0 1 36 184 Tm (class LmsEnrollmentService {}) Tj
1 0 0 1 36 173 Tm ( /**) Tj
1 0 0 1 36 162 Tm (   * @param {DbClient} db) Tj
1 0 0 1 36 151 Tm (   */) Tj
1 0 0 1 36 140 Tm (   constructor\(\db\) {}) Tj
1 0 0 1 36 129 Tm (     this._db = db;) Tj
1 0 0 1 36 118 Tm (   }) Tj
1 0 0 1 36 107 Tm ( ) Tj
1 0 0 1 36 96 Tm ( /**) Tj
1 0 0 1 36 85 Tm (   * Slash-command flow example:) Tj
1 0 0 1 36 74 Tm (   * /enroll-student <studentId> <courseId>) Tj
1 0 0 1 36 63 Tm (   *) Tj
1 0 0 1 36 52 Tm (   * Upserts into 'students' and 'enrollments', then writes audit log row.) Tj
1 0 0 1 36 41 Tm (   * @param {Object} context) Tj
ET
endstream
endobj
50 0 obj
<< /Type /Page /Parent 2 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 3 0 R >> >> /Contents 51 0 R >>
endobj
51 0 obj
<< /Length 3799 >>
stream
BT
/F1 9 Tf
1 0 0 1 36 756 Tm (   * @param {Object} payload) Tj
1 0 0 1 36 745 Tm (   * @return {Object}) Tj
1 0 0 1 36 734 Tm (   */) Tj
1 0 0 1 36 723 Tm (   enrollFromSlash\(\context, payload\) {}) Tj
1 0 0 1 36 712 Tm (     var args = String\(\payload.text || ''\).split\(\(\\s+\\)\.filter\(\function\(\v\) { return !!v; })\);) Tj
1 0 0 1 36 701 Tm (     if \(\args.length < 2\) {}) Tj
1 0 0 1 36 690 Tm (       return { ok: false, code: 'INVALID_ARGUMENTS', message: 'Usage: /enroll-student <studentId> <courseId>' };) Tj
1 0 0 1 36 679 Tm (     }) Tj
1 0 0 1 36 668 Tm ( ) Tj
1 0 0 1 36 657 Tm (     var studentId = args[0];) Tj
1 0 0 1 36 646 Tm (     var courseId = args[1];) Tj
1 0 0 1 36 635 Tm (     var out = this._upsertEnrollment\(\context, studentId, courseId, 'slash_command'\);) Tj
1 0 0 1 36 624 Tm (     return out;) Tj
1 0 0 1 36 613 Tm (   }) Tj
1 0 0 1 36 602 Tm ( ) Tj
1 0 0 1 36 591 Tm ( /**) Tj
1 0 0 1 36 580 Tm (   * Workflow webhook flow example for enrollment updates/inserts.) Tj
1 0 0 1 36 569 Tm (   * @param {Object} context) Tj
1 0 0 1 36 558 Tm (   * @param {Object} payload) Tj
1 0 0 1 36 547 Tm (   * @return {Object}) Tj
1 0 0 1 36 536 Tm (   */) Tj
1 0 0 1 36 525 Tm (   upsertFromWorkflow\(\context, payload\) {}) Tj
1 0 0 1 36 514 Tm (     if \(!payload.studentId || !payload.courseId\) {}) Tj
1 0 0 1 36 503 Tm (       return { ok: false, code: 'INVALID_WORKFLOW_PAYLOAD', message: 'studentId and courseId are required' };) Tj
1 0 0 1 36 492 Tm (     }) Tj
1 0 0 1 36 481 Tm ( ) Tj
1 0 0 1 36 470 Tm (     return this._upsertEnrollment\(\context, payload.studentId, payload.courseId, 'workflow_webhook', 'worklog', payload\);) Tj
1 0 0 1 36 459 Tm (   }) Tj
1 0 0 1 36 448 Tm ( ) Tj
1 0 0 1 36 437 Tm ( /** @private */) Tj
1 0 0 1 36 426 Tm (   _upsertEnrollment\(\context, studentId, courseId, source, status\) {}) Tj
1 0 0 1 36 415 Tm (     var students = this._db.table\('students'\);) Tj
1 0 0 1 36 404 Tm (     var enrollments = this._db.table\('enrollments'\);) Tj

```

```

1 0 0 1 36 393 Tm (    var audit = this._db.table\('audit_logs'\);) Tj
1 0 0 1 36 382 Tm (    var finalStatus = status || 'enrolled';) Tj
1 0 0 1 36 371 Tm (    var enrollmentRecord;) Tj
1 0 0 1 36 360 Tm ( ) Tj
1 0 0 1 36 349 Tm (    this._db.transaction\((function\(\) {) Tj
1 0 0 1 36 338 Tm (        var existingStudent = students.findById\((studentId\);) Tj
1 0 0 1 36 327 Tm (        if \(!existingStudent\() {) Tj
1 0 0 1 36 316 Tm (            students.insert\({ id: studentId, name: studentId, source: source }\);) Tj
1 0 0 1 36 305 Tm (        }) Tj
1 0 0 1 36 294 Tm ( ) Tj
1 0 0 1 36 283 Tm (        var existingEnrollment = enrollments.findAll\(\).filter\((function\((r\() {) Tj
1 0 0 1 36 272 Tm (            return r.studentId === studentId && r.courseId === courseId;) Tj
1 0 0 1 36 261 Tm (        })[0];) Tj
1 0 0 1 36 250 Tm ( ) Tj
1 0 0 1 36 239 Tm (        if \((existingEnrollment\() {) Tj
1 0 0 1 36 228 Tm (            enrollmentRecord = enrollments.update\((existingEnrollment.id, {) Tj
1 0 0 1 36 217 Tm (                status: finalStatus,) Tj
1 0 0 1 36 206 Tm (                source: source) Tj
1 0 0 1 36 195 Tm (            })\);) Tj
1 0 0 1 36 184 Tm (        } else {) Tj
1 0 0 1 36 173 Tm (            enrollmentRecord = enrollments.insert\({) Tj
1 0 0 1 36 162 Tm (                studentId: studentId,) Tj
1 0 0 1 36 151 Tm (                courseId: courseId,) Tj
1 0 0 1 36 140 Tm (                status: finalStatus,) Tj
1 0 0 1 36 129 Tm (                source: source) Tj
1 0 0 1 36 118 Tm (            })\);) Tj
1 0 0 1 36 107 Tm (        }) Tj
1 0 0 1 36 96 Tm ( ) Tj
1 0 0 1 36 85 Tm (    audit.insert\({) Tj
1 0 0 1 36 74 Tm (        requestId: context.requestId,) Tj
1 0 0 1 36 63 Tm (        source: context.source,) Tj
1 0 0 1 36 52 Tm (        action: 'enrollment_upsert',) Tj
1 0 0 1 36 41 Tm (        status: 'success',) Tj
ET
endstream
endobj
52 0 obj
<< /Type /Page /Parent 2 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 3 0 R >> >> /Contents 53 0 R >>
endobj
53 0 obj
<< /Length 3261 >>
stream
BT
/F1 9 Tf
1 0 0 1 36 756 Tm (        message: 'Enrollment upsert completed.',) Tj
1 0 0 1 36 745 Tm (        metadata: JSON.stringify\({ studentId: studentId, courseId: courseId, enrollmentId: enrollmentId, } \);) Tj
1 0 0 1 36 734 Tm (        })\);) Tj
1 0 0 1 36 723 Tm (    })\);) Tj
1 0 0 1 36 712 Tm ( ) Tj
1 0 0 1 36 701 Tm (    this._db.audit\('enrollment_upsert', 'enrollments', {) Tj
1 0 0 1 36 690 Tm (        requestId: context.requestId,) Tj
1 0 0 1 36 679 Tm (        studentId: studentId,) Tj
1 0 0 1 36 668 Tm (        courseId: courseId,) Tj
1 0 0 1 36 657 Tm (        enrollmentId: enrollmentRecord.id) Tj
1 0 0 1 36 646 Tm (    })\);) Tj
1 0 0 1 36 635 Tm ( ) Tj
1 0 0 1 36 624 Tm (    return {) Tj
1 0 0 1 36 613 Tm (        ok: true,) Tj
1 0 0 1 36 602 Tm (        code: 'ENROLLMENT_UPSERTED',) Tj
1 0 0 1 36 591 Tm (        message: 'Enrollment upserted successfully.',) Tj
1 0 0 1 36 580 Tm (        data: {) Tj
1 0 0 1 36 569 Tm (            enrollmentId: enrollmentRecord.id,) Tj
1 0 0 1 36 558 Tm (            studentId: studentId,) Tj
1 0 0 1 36 547 Tm (            courseId: courseId,) Tj
1 0 0 1 36 536 Tm (            status: enrollmentRecord.status) Tj
1 0 0 1 36 525 Tm (        }) Tj
1 0 0 1 36 514 Tm (    })\);) Tj
1 0 0 1 36 503 Tm (    }) Tj
1 0 0 1 36 492 Tm (}) Tj
1 0 0 1 36 481 Tm ( ) Tj
1 0 0 1 36 470 Tm ('''') Tj
1 0 0 1 36 459 Tm ( ) Tj
1 0 0 1 36 448 Tm ( ) Tj
1 0 0 1 36 437 Tm (# 06_LmsAutomationService.gs) Tj
1 0 0 1 36 426 Tm ( ) Tj
1 0 0 1 36 415 Tm (**Source path:** 'slack_host/06_LmsAutomationService.gs') Tj
1 0 0 1 36 404 Tm ( ) Tj
1 0 0 1 36 393 Tm ('''javascript) Tj
1 0 0 1 36 382 Tm (/**) Tj
1 0 0 1 36 371 Tm ( * LMS automation operations that persist workflow state via SheetDb.) Tj

```

```

1 0 0 1 36 360 Tm ( */) Tj
1 0 0 1 36 349 Tm (class LmsAutomationService {}) Tj
1 0 0 1 36 338 Tm ( /**) Tj
1 0 0 1 36 327 Tm ( * @param {DbClient} db) Tj
1 0 0 1 36 316 Tm ( */) Tj
1 0 0 1 36 305 Tm ( constructor\ (db\ ) {} Tj
1 0 0 1 36 294 Tm ( this._db = db;) Tj
1 0 0 1 36 283 Tm ( }) Tj
1 0 0 1 36 272 Tm ( ) Tj
1 0 0 1 36 261 Tm ( /**) Tj
1 0 0 1 36 250 Tm ( * Inserts/updates an automation row and writes an audit log row.) Tj
1 0 0 1 36 239 Tm ( * @param {Object} context) Tj
1 0 0 1 36 228 Tm ( * @param {Object} payload) Tj
1 0 0 1 36 217 Tm ( * @return {Object}) Tj
1 0 0 1 36 206 Tm ( */) Tj
1 0 0 1 36 195 Tm ( upsertAutomation\ (context, payload\ ) {} Tj
1 0 0 1 36 184 Tm ( var automations = this._db.table\ ('automations'\);) Tj
1 0 0 1 36 173 Tm ( var audit = this._db.table\ ('audit_logs'\);) Tj
1 0 0 1 36 162 Tm ( var workflow = payload.workflow || 'unknown_workflow';) Tj
1 0 0 1 36 151 Tm ( var eventType = payload.eventType || 'workflow_event';) Tj
1 0 0 1 36 140 Tm ( var state = payload.status || 'received';) Tj
1 0 0 1 36 129 Tm ( ) Tj
1 0 0 1 36 118 Tm ( var existing = automations.findAll\ (\).filter\ (function\ (r\ ) {} Tj
1 0 0 1 36 107 Tm ( return r.workflow === workflow && r.requestId === payload.requestId;) Tj
1 0 0 1 36 96 Tm ( )\)[0];) Tj
1 0 0 1 36 85 Tm ( ) Tj
1 0 0 1 36 74 Tm ( var row;) Tj
1 0 0 1 36 63 Tm ( if \ (existing\ ) {} Tj
1 0 0 1 36 52 Tm ( row = automations.update\ (existing.id, { eventType: eventType, state: state } \);) Tj
1 0 0 1 36 41 Tm ( } else {} ) Tj
ET
endstream
endobj
54 0 obj
<< /Type /Page /Parent 2 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 3 0 R >> >> /Contents 55 0 R >>
endobj
55 0 obj
<< /Length 3117 >>
stream
BT
/F1 9 Tf
1 0 0 1 36 756 Tm ( row = automations.insert\ ({ requestId: payload.requestId, workflow: workflow, eventType: eventType) Tj
1 0 0 1 36 745 Tm ( }) Tj
1 0 0 1 36 734 Tm ( ) Tj
1 0 0 1 36 723 Tm ( audit.insert\ ({) Tj
1 0 0 1 36 712 Tm ( requestId: context.requestId,) Tj
1 0 0 1 36 701 Tm ( source: context.source,) Tj
1 0 0 1 36 690 Tm ( action: 'automation_upsert',) Tj
1 0 0 1 36 679 Tm ( status: 'success',) Tj
1 0 0 1 36 668 Tm ( message: 'Automation row upserted.',) Tj
1 0 0 1 36 657 Tm ( metadata: JSON.stringify\ ({ automationId: row.id, workflow: workflow, eventType: eventType } \);) Tj
1 0 0 1 36 646 Tm ( }) \);) Tj
1 0 0 1 36 635 Tm ( ) Tj
1 0 0 1 36 624 Tm ( return {} ) Tj
1 0 0 1 36 613 Tm ( ok: true,) Tj
1 0 0 1 36 602 Tm ( code: 'AUTOMATION_UPSERTED',) Tj
1 0 0 1 36 591 Tm ( message: 'Automation row upserted.',) Tj
1 0 0 1 36 580 Tm ( data: { automationId: row.id, workflow: workflow, eventType: eventType, state: row.state } ) Tj
1 0 0 1 36 569 Tm ( );) Tj
1 0 0 1 36 558 Tm ( }) Tj
1 0 0 1 36 547 Tm ( }) Tj
1 0 0 1 36 536 Tm ( ) Tj
1 0 0 1 36 525 Tm ( ''' ) Tj
1 0 0 1 36 514 Tm ( ) Tj
1 0 0 1 36 503 Tm ( ) Tj
1 0 0 1 36 492 Tm ( # 07_RequestContextFactory.gs) Tj
1 0 0 1 36 481 Tm ( ) Tj
1 0 0 1 36 470 Tm ( **Source path:** 'slack_host/07_RequestContextFactory.gs') Tj
1 0 0 1 36 459 Tm ( ) Tj
1 0 0 1 36 448 Tm ( '''javascript) Tj
1 0 0 1 36 437 Tm ( /**) Tj
1 0 0 1 36 426 Tm ( * Creates normalized request-context objects for host handlers.) Tj
1 0 0 1 36 415 Tm ( */) Tj
1 0 0 1 36 404 Tm (class RequestContextFactory {} ) Tj
1 0 0 1 36 393 Tm ( /**) Tj
1 0 0 1 36 382 Tm ( * @param {HostConfig} hostConfig) Tj
1 0 0 1 36 371 Tm ( */) Tj
1 0 0 1 36 360 Tm ( constructor\ (hostConfig\ ) {} Tj
1 0 0 1 36 349 Tm ( this._config = hostConfig;) Tj
1 0 0 1 36 338 Tm ( }) Tj

```

```

1 0 0 1 36 327 Tm ( ) Tj
1 0 0 1 36 316 Tm ( /**) Tj
1 0 0 1 36 305 Tm ( * @param {string} routeType) Tj
1 0 0 1 36 294 Tm ( * @param {Object} payload) Tj
1 0 0 1 36 283 Tm ( * @param {Object=} envelope) Tj
1 0 0 1 36 272 Tm ( * @return {Object}) Tj
1 0 0 1 36 261 Tm ( */) Tj
1 0 0 1 36 250 Tm ( create\(routeType, payload, envelope\) {}) Tj
1 0 0 1 36 239 Tm ( var env = envelope || {};) Tj
1 0 0 1 36 228 Tm ( return {}) Tj
1 0 0 1 36 217 Tm ( requestId: payload.requestId || Util.generateId('req'\),) Tj
1 0 0 1 36 206 Tm ( receivedAt: Util.nowIso\(\),) Tj
1 0 0 1 36 195 Tm ( routeType: routeType,) Tj
1 0 0 1 36 184 Tm ( source: routeType,) Tj
1 0 0 1 36 173 Tm ( actorId: payload.userId || payload.actorId || 'unknown',) Tj
1 0 0 1 36 162 Tm ( teamId: payload.teamId || '',) Tj
1 0 0 1 36 151 Tm ( channelId: payload.channelId || this._config.getSlackDefaultChannel\(\),) Tj
1 0 0 1 36 140 Tm ( raw: env.rawBody || '') Tj
1 0 0 1 36 129 Tm ( };) Tj
1 0 0 1 36 118 Tm ( }) Tj
1 0 0 1 36 107 Tm (}) Tj
1 0 0 1 36 96 Tm ( ) Tj
1 0 0 1 36 85 Tm ( '') Tj
1 0 0 1 36 74 Tm ( ) Tj
1 0 0 1 36 63 Tm ( ) Tj
1 0 0 1 36 52 Tm (# 08_ResultFormatter.gs) Tj
1 0 0 1 36 41 Tm ( ) Tj
ET
endstream
endobj
56 0 obj
<< /Type /Page /Parent 2 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 3 0 R >> >> /Contents 57 0 R >>
endobj
57 0 obj
<< /Length 2985 >>
stream
BT
/F1 9 Tf
1 0 0 1 36 756 Tm (**Source path:** 'slack_host/08_ResultFormatter.gs') Tj
1 0 0 1 36 745 Tm ( ) Tj
1 0 0 1 36 734 Tm ('''javascript) Tj
1 0 0 1 36 723 Tm (/**) Tj
1 0 0 1 36 712 Tm ( * Formats host operation results into structured Slack-friendly objects.) Tj
1 0 0 1 36 701 Tm ( */) Tj
1 0 0 1 36 690 Tm (class ResultFormatter {}) Tj
1 0 0 1 36 679 Tm ( /**) Tj
1 0 0 1 36 668 Tm ( * @param {string} message) Tj
1 0 0 1 36 657 Tm ( * @param {Object=} data) Tj
1 0 0 1 36 646 Tm ( * @param {Object=} context) Tj
1 0 0 1 36 635 Tm ( * @return {Object}) Tj
1 0 0 1 36 624 Tm ( */) Tj
1 0 0 1 36 613 Tm ( success\(message, data, context\) {}) Tj
1 0 0 1 36 602 Tm ( return {}) Tj
1 0 0 1 36 591 Tm ( ok: true,) Tj
1 0 0 1 36 580 Tm ( code: 'OK',) Tj
1 0 0 1 36 569 Tm ( message: message,) Tj
1 0 0 1 36 558 Tm ( requestId: \(context && context.requestId\) || '',) Tj
1 0 0 1 36 547 Tm ( data: data || {},) Tj
1 0 0 1 36 536 Tm ( slack: {}) Tj
1 0 0 1 36 525 Tm ( response_type: 'ephemeral',) Tj
1 0 0 1 36 514 Tm ( text: message) Tj
1 0 0 1 36 503 Tm ( }) Tj
1 0 0 1 36 492 Tm ( };) Tj
1 0 0 1 36 481 Tm ( }) Tj
1 0 0 1 36 470 Tm ( ) Tj
1 0 0 1 36 459 Tm ( /**) Tj
1 0 0 1 36 448 Tm ( * @param {string} code) Tj
1 0 0 1 36 437 Tm ( * @param {string} message) Tj
1 0 0 1 36 426 Tm ( * @param {Object=} context) Tj
1 0 0 1 36 415 Tm ( * @param {Object=} data) Tj
1 0 0 1 36 404 Tm ( * @return {Object}) Tj
1 0 0 1 36 393 Tm ( */) Tj
1 0 0 1 36 382 Tm ( failure\(code, message, context, data\) {}) Tj
1 0 0 1 36 371 Tm ( return {}) Tj
1 0 0 1 36 360 Tm ( ok: false,) Tj
1 0 0 1 36 349 Tm ( code: code || 'ERROR',) Tj
1 0 0 1 36 338 Tm ( message: message || 'Unknown error.',) Tj
1 0 0 1 36 327 Tm ( requestId: \(context && context.requestId\) || '',) Tj
1 0 0 1 36 316 Tm ( data: data || {},) Tj
1 0 0 1 36 305 Tm ( slack: {}) Tj

```



```

1 0 0 1 36 261 Tm ( getSlackBotToken\(\) { return this._props.get\('SLACK_BOT_TOKEN', '\'); }) Tj
1 0 0 1 36 250 Tm ( /** @return {string} */) Tj
1 0 0 1 36 239 Tm ( getSlackSigningSecret\(\) { return this._props.get\('SLACK_SIGNING_SECRET', this.getSigningSecret\(\)
1 0 0 1 36 228 Tm ( /** @return {string} */) Tj
1 0 0 1 36 217 Tm ( getSlackDefaultChannel\(\) { return this._props.get\('SLACK_DEFAULT_CHANNEL', '\'); }) Tj
1 0 0 1 36 206 Tm ( ) Tj
1 0 0 1 36 195 Tm ( /**) Tj
1 0 0 1 36 184 Tm ( * @return {Object}) Tj
1 0 0 1 36 173 Tm ( */) Tj
1 0 0 1 36 162 Tm ( getClientOverrides\(\) {}) Tj
1 0 0 1 36 151 Tm ( return mergeClientOptionsWithScriptProperties\({}\);) Tj
1 0 0 1 36 140 Tm ( }) Tj
1 0 0 1 36 129 Tm (}) Tj
1 0 0 1 36 118 Tm ( ) Tj
1 0 0 1 36 107 Tm (``) Tj
1 0 0 1 36 96 Tm ( ) Tj
1 0 0 1 36 85 Tm ( ) Tj
1 0 0 1 36 74 Tm (# 11_HostTestHarness.gs) Tj
1 0 0 1 36 63 Tm ( ) Tj
1 0 0 1 36 52 Tm (**Source path:** 'slack_host/11_HostTestHarness.gs') Tj
1 0 0 1 36 41 Tm ( ) Tj
ET
endstream
endobj
60 0 obj
<< /Type /Page /Parent 2 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 3 0 R >> >> /Contents 61 0 R >>
endobj
61 0 obj
<< /Length 3799 >>
stream
BT
/F1 9 Tf
1 0 0 1 36 756 Tm (``javascript) Tj
1 0 0 1 36 745 Tm (/**) Tj
1 0 0 1 36 734 Tm ( * Manual Apps Script test harness for the split Slack host architecture.) Tj
1 0 0 1 36 723 Tm ( */) Tj
1 0 0 1 36 712 Tm ( ) Tj
1 0 0 1 36 701 Tm (/**) Tj
1 0 0 1 36 690 Tm ( * Builds host dependencies for manual tests.) Tj
1 0 0 1 36 679 Tm ( * Uses TEST_SPREADSHEET_ID script property when available.) Tj
1 0 0 1 36 668 Tm ( * @return {{db: DbClient, hostConfig: HostConfig, router: SlackRouter}}) Tj
1 0 0 1 36 657 Tm ( */) Tj
1 0 0 1 36 646 Tm (function _buildHostTestContext\(\) {}) Tj
1 0 0 1 36 635 Tm ( var props = new ScriptPropertiesHelper\(\);) Tj
1 0 0 1 36 624 Tm ( var testSpreadsheetId = props.get\('TEST_SPREADSHEET_ID', '\');) Tj
1 0 0 1 36 613 Tm ( var options = testSpreadsheetId ? { spreadsheetId: testSpreadsheetId } : {};) Tj
1 0 0 1 36 602 Tm ( ) Tj
1 0 0 1 36 591 Tm ( var db = SheetDb.createClientFromScriptProperties\(options\);) Tj
1 0 0 1 36 580 Tm ( ) Tj
1 0 0 1 36 569 Tm ( db.schema\('students', {}) Tj
1 0 0 1 36 558 Tm ( columns: ['id', 'name', 'source', 'createdAt', 'updatedAt', 'deletedAt'],) Tj
1 0 0 1 36 547 Tm ( fields: {}) Tj
1 0 0 1 36 536 Tm ( id: { required: true, type: 'string' },) Tj
1 0 0 1 36 525 Tm ( name: { required: true, type: 'string' },) Tj
1 0 0 1 36 514 Tm ( source: { default: 'slack', type: 'string' }) Tj
1 0 0 1 36 503 Tm ( }) Tj
1 0 0 1 36 492 Tm ( }\);) Tj
1 0 0 1 36 481 Tm ( ) Tj
1 0 0 1 36 470 Tm ( db.schema\('enrollments', {}) Tj
1 0 0 1 36 459 Tm ( columns: ['id', 'studentId', 'courseId', 'status', 'source', 'createdAt', 'updatedAt', 'deletedAt']
1 0 0 1 36 448 Tm ( fields: {}) Tj
1 0 0 1 36 437 Tm ( studentId: { required: true, type: 'string' },) Tj
1 0 0 1 36 426 Tm ( courseId: { required: true, type: 'string' },) Tj
1 0 0 1 36 415 Tm ( status: { default: 'enrolled', type: 'string' },) Tj
1 0 0 1 36 404 Tm ( source: { default: 'slack', type: 'string' }) Tj
1 0 0 1 36 393 Tm ( }) Tj
1 0 0 1 36 382 Tm ( }\);) Tj
1 0 0 1 36 371 Tm ( ) Tj
1 0 0 1 36 360 Tm ( db.schema\('automations', {}) Tj
1 0 0 1 36 349 Tm ( columns: ['id', 'requestId', 'workflow', 'eventType', 'state', 'createdAt', 'updatedAt', 'deletedAt']
1 0 0 1 36 338 Tm ( fields: {}) Tj
1 0 0 1 36 327 Tm ( requestId: { required: true, type: 'string' },) Tj
1 0 0 1 36 316 Tm ( workflow: { required: true, type: 'string' },) Tj
1 0 0 1 36 305 Tm ( eventType: { required: true, type: 'string' },) Tj
1 0 0 1 36 294 Tm ( state: { default: 'received', type: 'string' }) Tj
1 0 0 1 36 283 Tm ( }) Tj
1 0 0 1 36 272 Tm ( }\);) Tj
1 0 0 1 36 261 Tm ( ) Tj
1 0 0 1 36 250 Tm ( db.schema\('audit_logs', {}) Tj
1 0 0 1 36 239 Tm ( columns: ['id', 'requestId', 'source', 'action', 'status', 'message', 'metadata', 'createdAt', 'u

```



```

1 0 0 1 36 228 Tm (    fields: {}) Tj
1 0 0 1 36 217 Tm (    requestId: { required: true, type: 'string' },) Tj
1 0 0 1 36 206 Tm (    source: { required: true, type: 'string' },) Tj
1 0 0 1 36 195 Tm (    action: { required: true, type: 'string' },) Tj
1 0 0 1 36 184 Tm (    status: { required: true, type: 'string' },) Tj
1 0 0 1 36 173 Tm (    message: { type: 'string' },) Tj
1 0 0 1 36 162 Tm (    metadata: { type: 'string' }) Tj
1 0 0 1 36 151 Tm (    }) Tj
1 0 0 1 36 140 Tm (  });) Tj
1 0 0 1 36 129 Tm ( ) Tj
1 0 0 1 36 118 Tm (  var hostConfig = new HostConfig\(\props\);) Tj
1 0 0 1 36 107 Tm (  var router = createHostRouter\(\db, hostConfig\);) Tj
1 0 0 1 36 96 Tm (  return { db: db, hostConfig: hostConfig, router: router };) Tj
1 0 0 1 36 85 Tm (}) Tj
1 0 0 1 36 74 Tm ( ) Tj
1 0 0 1 36 63 Tm (/**) Tj
1 0 0 1 36 52 Tm ( * Encodes form payload.) Tj
1 0 0 1 36 41 Tm ( * @param {Object} obj) Tj
ET
endstream
endobj
62 0 obj
<< /Type /Page /Parent 2 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 3 0 R >> >> /Contents 63 0 R >>
endobj
63 0 obj
<< /Length 3350 >>
stream
BT
/F1 9 Tf
1 0 0 1 36 756 Tm ( * @return {string}) Tj
1 0 0 1 36 745 Tm ( */) Tj
1 0 0 1 36 734 Tm (function _toFormBody\(\obj\) {}) Tj
1 0 0 1 36 723 Tm (  return Object.keys\(\obj\).map\((function\(\key\) {}) Tj
1 0 0 1 36 712 Tm (    return encodeURIComponent\(\key\) + '=' + encodeURIComponent\(\String\(\obj[key]\)\));) Tj
1 0 0 1 36 701 Tm (  }\).join\('&'\);) Tj
1 0 0 1 36 690 Tm (}) Tj
1 0 0 1 36 679 Tm ( ) Tj
1 0 0 1 36 668 Tm (/**) Tj
1 0 0 1 36 657 Tm ( * Builds a mock DoPost event object.) Tj
1 0 0 1 36 646 Tm ( * @param {string} body) Tj
1 0 0 1 36 635 Tm ( * @return {GoogleAppsScript.Events.DoPost}) Tj
1 0 0 1 36 624 Tm ( */) Tj
1 0 0 1 36 613 Tm (function _mockDoPostEvent\(\body\) {}) Tj
1 0 0 1 36 602 Tm (  return {} Tj
1 0 0 1 36 591 Tm (    postData: {} Tj
1 0 0 1 36 580 Tm (    contents: body,) Tj
1 0 0 1 36 569 Tm (    type: 'application/x-www-form-urlencoded') Tj
1 0 0 1 36 558 Tm (  }) Tj
1 0 0 1 36 547 Tm (  );) Tj
1 0 0 1 36 536 Tm (}) Tj
1 0 0 1 36 525 Tm ( ) Tj
1 0 0 1 36 514 Tm (/**) Tj
1 0 0 1 36 503 Tm ( * testSlashEnrollMock) Tj
1 0 0 1 36 492 Tm ( * Simulates '/enroll-student' and runs full router path.) Tj
1 0 0 1 36 481 Tm ( */) Tj
1 0 0 1 36 470 Tm (function testSlashEnrollMock\(\) {}) Tj
1 0 0 1 36 459 Tm (  var ctx = _buildHostTestContext\(\);) Tj
1 0 0 1 36 448 Tm (  var suffix = new Date\(\).getTime\(\);) Tj
1 0 0 1 36 437 Tm (  var passcode = ctx.hostConfig.getAppPasscode\(\);) Tj
1 0 0 1 36 426 Tm (  var payload = {} Tj
1 0 0 1 36 415 Tm (    command: '/enroll-student',) Tj
1 0 0 1 36 404 Tm (    text: 'student-' + suffix + ' course-' + suffix,) Tj
1 0 0 1 36 393 Tm (    user_id: 'U_TEST_' + suffix,) Tj
1 0 0 1 36 382 Tm (    team_id: 'T_TEST',) Tj
1 0 0 1 36 371 Tm (    channel_id: 'C_TEST') Tj
1 0 0 1 36 360 Tm (  );) Tj
1 0 0 1 36 349 Tm (  if \(\passcode\) {}) Tj
1 0 0 1 36 338 Tm (    payload.passcode = passcode;) Tj
1 0 0 1 36 327 Tm (  }) Tj
1 0 0 1 36 316 Tm ( ) Tj
1 0 0 1 36 305 Tm (  var event = _mockDoPostEvent\(_toFormBody\(\payload\)\);) Tj
1 0 0 1 36 294 Tm (  var result = ctx.router.handle\(\event\);) Tj
1 0 0 1 36 283 Tm (  Logger.log\('[testSlashEnrollMock] ' + JSON.stringify\(\result\)\);) Tj
1 0 0 1 36 272 Tm (}) Tj
1 0 0 1 36 261 Tm ( ) Tj
1 0 0 1 36 250 Tm (/**) Tj
1 0 0 1 36 239 Tm ( * testWorkflowEnrollmentMock) Tj
1 0 0 1 36 228 Tm ( * Simulates 'enrollment_requested' workflow webhook and runs full router path.) Tj
1 0 0 1 36 217 Tm ( */) Tj
1 0 0 1 36 206 Tm (function testWorkflowEnrollmentMock\(\) {}) Tj

```

```

1 0 0 1 36 195 Tm ( var ctx = _buildHostTestContext\(\);) Tj
1 0 0 1 36 184 Tm ( var suffix = new Date\(\).getTime\(\);) Tj
1 0 0 1 36 173 Tm ( var payload = {} Tj
1 0 0 1 36 162 Tm ( workflow: 'lms_enrollment',) Tj
1 0 0 1 36 151 Tm ( eventType: 'enrollment_requested',) Tj
1 0 0 1 36 140 Tm ( requestId: 'wf_req_' + suffix,) Tj
1 0 0 1 36 129 Tm ( data: {} Tj
1 0 0 1 36 118 Tm ( studentId: 'wf-student-' + suffix,) Tj
1 0 0 1 36 107 Tm ( courseId: 'wf-course-' + suffix,) Tj
1 0 0 1 36 96 Tm ( actorId: 'WF_ACTOR') Tj
1 0 0 1 36 85 Tm ( }) Tj
1 0 0 1 36 74 Tm ( );) Tj
1 0 0 1 36 63 Tm ( var event = _mockDoPostEvent\(JSON.stringify\(payload\)\);) Tj
1 0 0 1 36 52 Tm ( var result = ctx.router.handle\(event\);) Tj
1 0 0 1 36 41 Tm ( Logger.log\('[testWorkflowEnrollmentMock] ' + JSON.stringify\(result\)\);) Tj
ET
endstream
endobj
64 0 obj
<< /Type /Page /Parent 2 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 3 0 R >> >> /Contents 65 0 R >>
endobj
65 0 obj
<< /Length 3353 >>
stream
BT
/F1 9 Tf
1 0 0 1 36 756 Tm ({} Tj
1 0 0 1 36 745 Tm ( ) Tj
1 0 0 1 36 734 Tm (/**) Tj
1 0 0 1 36 723 Tm ( * testAutomationMock) Tj
1 0 0 1 36 712 Tm ( * Simulates generic automation workflow webhook and runs full router path.) Tj
1 0 0 1 36 701 Tm ( */) Tj
1 0 0 1 36 690 Tm (function testAutomationMock\(\) {} Tj
1 0 0 1 36 679 Tm ( var ctx = _buildHostTestContext\(\);) Tj
1 0 0 1 36 668 Tm ( var suffix = new Date\(\).getTime\(\);) Tj
1 0 0 1 36 657 Tm ( var payload = {} Tj
1 0 0 1 36 646 Tm ( workflow: 'lms_automation',) Tj
1 0 0 1 36 635 Tm ( eventType: 'automation_ping',) Tj
1 0 0 1 36 624 Tm ( requestId: 'automation_req_' + suffix,) Tj
1 0 0 1 36 613 Tm ( data: {} Tj
1 0 0 1 36 602 Tm ( status: 'queued',) Tj
1 0 0 1 36 591 Tm ( actorId: 'AUTO_ACTOR') Tj
1 0 0 1 36 580 Tm ( }) Tj
1 0 0 1 36 569 Tm ( );) Tj
1 0 0 1 36 558 Tm ( var event = _mockDoPostEvent\(JSON.stringify\(payload\)\);) Tj
1 0 0 1 36 547 Tm ( var result = ctx.router.handle\(event\);) Tj
1 0 0 1 36 536 Tm ( Logger.log\('[testAutomationMock] ' + JSON.stringify\(result\)\);) Tj
1 0 0 1 36 525 Tm ({} Tj
1 0 0 1 36 514 Tm ( ) Tj
1 0 0 1 36 503 Tm (/**) Tj
1 0 0 1 36 492 Tm ( * testHostConfig) Tj
1 0 0 1 36 481 Tm ( * Verifies host config reads script properties without logging raw secrets.) Tj
1 0 0 1 36 470 Tm ( */) Tj
1 0 0 1 36 459 Tm (function testHostConfig\(\) {} Tj
1 0 0 1 36 448 Tm ( var props = new ScriptPropertiesHelper\(\);) Tj
1 0 0 1 36 437 Tm ( var hostConfig = new HostConfig\(props\);) Tj
1 0 0 1 36 426 Tm ( var keys = [] Tj
1 0 0 1 36 415 Tm ( 'APP_PASSCODE',) Tj
1 0 0 1 36 404 Tm ( 'SIGNING_SECRET',) Tj
1 0 0 1 36 393 Tm ( 'WEBHOOK_SECRET',) Tj
1 0 0 1 36 382 Tm ( 'API_KEY',) Tj
1 0 0 1 36 371 Tm ( 'SLACK_BOT_TOKEN',) Tj
1 0 0 1 36 360 Tm ( 'SLACK_SIGNING_SECRET',) Tj
1 0 0 1 36 349 Tm ( 'SLACK_DEFAULT_CHANNEL') Tj
1 0 0 1 36 338 Tm ( ];) Tj
1 0 0 1 36 327 Tm ( ) Tj
1 0 0 1 36 316 Tm ( var values = {} Tj
1 0 0 1 36 305 Tm ( APP_PASSCODE: hostConfig.getAppPasscode\(\),) Tj
1 0 0 1 36 294 Tm ( SIGNING_SECRET: hostConfig.getSigningSecret\(\),) Tj
1 0 0 1 36 283 Tm ( WEBHOOK_SECRET: hostConfig.getWebhookSecret\(\),) Tj
1 0 0 1 36 272 Tm ( API_KEY: hostConfig.getApiKey\(\),) Tj
1 0 0 1 36 261 Tm ( SLACK_BOT_TOKEN: hostConfig.getSlackBotToken\(\),) Tj
1 0 0 1 36 250 Tm ( SLACK_SIGNING_SECRET: hostConfig.getSlackSigningSecret\(\),) Tj
1 0 0 1 36 239 Tm ( SLACK_DEFAULT_CHANNEL: hostConfig.getSlackDefaultChannel\(\)) Tj
1 0 0 1 36 228 Tm ( );) Tj
1 0 0 1 36 217 Tm ( ) Tj
1 0 0 1 36 206 Tm ( var redacted = {};) Tj
1 0 0 1 36 195 Tm ( keys.forEach\((function\(\key\)) {} Tj
1 0 0 1 36 184 Tm ( var value = values[key] || '';) Tj
1 0 0 1 36 173 Tm ( redacted[key] = ScriptPropertiesHelper.shouldRedactKey\(\key\)) Tj

```

```
1 0 0 1 36 162 Tm (      ? \ (value ? '[REDACTED_SET]' : '[REDACTED_EMPTY']\)) Tj
1 0 0 1 36 151 Tm (      : value;) Tj
1 0 0 1 36 140 Tm (      }\));) Tj
1 0 0 1 36 129 Tm (      ) Tj
1 0 0 1 36 118 Tm (      Logger.log\(['[testHostConfig]' + JSON.stringify\ (redacted)\));) Tj
1 0 0 1 36 107 Tm (      {} ) Tj
1 0 0 1 36 96 Tm (      ) Tj
1 0 0 1 36 85 Tm (      /**) Tj
1 0 0 1 36 74 Tm (      * testAuditWrite) Tj
1 0 0 1 36 63 Tm (      * Executes one mutation and verifies an audit row is written.) Tj
1 0 0 1 36 52 Tm (      */) Tj
1 0 0 1 36 41 Tm (function testAuditWrite\(\) {} ) Tj
ET
endstream
endobj
66 0 obj
<< /Type /Page /Parent 2 0 R /MediaBox [0 0 612 792] /Resources << /Font << /F1 3 0 R >> >> /Contents 67 0 R >>
endobj
67 0 obj
<< /Length 1530 >>
stream
BT
/F1 9 Tf
1 0 0 1 36 756 Tm (      var ctx = _buildHostTestContext\(\);) Tj
1 0 0 1 36 745 Tm (      var auditTable = ctx.db.table\('audit_logs'\);) Tj
1 0 0 1 36 734 Tm (      var before = auditTable.findAll\(\).length;) Tj
1 0 0 1 36 723 Tm (      ) Tj
1 0 0 1 36 712 Tm (      var suffix = new Date\(\).getTime\(\);) Tj
1 0 0 1 36 701 Tm (      var passcode = ctx.hostConfig.getAppPasscode\(\);) Tj
1 0 0 1 36 690 Tm (      var payload = {} ) Tj
1 0 0 1 36 679 Tm (      command: '/enroll-student',) Tj
1 0 0 1 36 668 Tm (      text: 'audit-student-' + suffix + ' audit-course-' + suffix,) Tj
1 0 0 1 36 657 Tm (      user_id: 'U_AUDIT_' + suffix,) Tj
1 0 0 1 36 646 Tm (      team_id: 'T_AUDIT',) Tj
1 0 0 1 36 635 Tm (      channel_id: 'C_AUDIT') Tj
1 0 0 1 36 624 Tm (      });) Tj
1 0 0 1 36 613 Tm (      if \ (passcode\) {} ) Tj
1 0 0 1 36 602 Tm (      payload.passcode = passcode;) Tj
1 0 0 1 36 591 Tm (      }) Tj
1 0 0 1 36 580 Tm (      ) Tj
1 0 0 1 36 569 Tm (      var event = _mockDoPostEvent\(_toFormBody\ (payload)\)\);) Tj
1 0 0 1 36 558 Tm (      var result = ctx.router.handle\ (event\);) Tj
1 0 0 1 36 547 Tm (      var after = auditTable.findAll\(\).length;) Tj
1 0 0 1 36 536 Tm (      ) Tj
1 0 0 1 36 525 Tm (      Logger.log\(['[testAuditWrite]' + JSON.stringify\ ({) Tj
1 0 0 1 36 514 Tm (      mutationOk: !!result.ok,) Tj
1 0 0 1 36 503 Tm (      auditRowsBefore: before,) Tj
1 0 0 1 36 492 Tm (      auditRowsAfter: after,) Tj
1 0 0 1 36 481 Tm (      auditRowDelta: after - before) Tj
1 0 0 1 36 470 Tm (      }\)\)\);) Tj
1 0 0 1 36 459 Tm (      }) Tj
1 0 0 1 36 448 Tm (      ) Tj
1 0 0 1 36 437 Tm (      ''') Tj
ET
endstream
endobj
xref
0 68
0000000000 65535 f
0000000009 00000 n
0000000058 00000 n
0000000331 00000 n
0000000399 00000 n
0000000525 00000 n
0000003786 00000 n
0000003912 00000 n
0000007201 00000 n
0000007327 00000 n
0000010407 00000 n
0000010535 00000 n
0000013965 00000 n
0000014093 00000 n
0000017165 00000 n
0000017293 00000 n
0000019997 00000 n
0000020125 00000 n
0000023218 00000 n
0000023346 00000 n
0000026690 00000 n
0000026818 00000 n
```



1. Run/validate 'runWeeklyAdminReport' output.
2. Check cohort progress consistency ('completed', 'submitted', 'overdue').
3. Data integrity checks:
  - learner records have valid 'slackUserId'
  - enrollment rows align to learner IDs
  - progress rows map to known lessons

## ## 5. Incident Response Playbooks

### ### Slack Not Responding

1. Verify Apps Script deployment URL is active.
2. Confirm Slack app is installed in workspace.
3. Check Apps Script executions/logs for incoming requests.
4. Re-deploy web app if deployment was revoked or outdated.

### ### Slash Command Fails

1. Verify slash command endpoint in Slack app settings.
2. Check Script Properties required keys.
3. Inspect parser output using host harness slash tests.
4. Validate security verification path (signature or fallback token).

### ### Interactivity Timeout

1. Ensure interactivity responses return quickly (<3s).
2. Check router dispatch for 'interactivity' route.
3. Verify payload parse and action/callback/type handling.
4. Offload heavy logic to queue/scheduler path where needed.

### ### Bot Cannot DM User

1. Ensure user has interacted with bot / app installed scope is valid.
2. Validate 'conversations.open' response in 'SlackApiClient.openDm'.
3. Confirm Slack scopes include required IM/chat permissions.

### ### Scheduler Not Running

1. Check trigger list in Apps Script project.
2. Re-run 'setupTriggers()'.
3. Verify Apps Script permissions/authorization for trigger owner.
4. Confirm lock contention is not blocking recurring jobs.

### ### Duplicate Lessons Sent

1. Inspect 'delivery\_queue' and duplicate queued rows.
2. Check idempotency strategy for queue insertion and submit keys.
3. Verify state transitions in 'LearnerProgressStateMachine'.

### ### Data Not Updating

1. Verify runtime sheet tab names exist and are writable.
2. Confirm 'SPREADSHEET\_ID' targets expected sheet.
3. Check 'SheetDb' client initialization and table schema registration.
4. Verify script execution identity has edit access.

## ## 6. Retry and Recovery

- Retry policy is centralized in 'RetryResolver'.
- Retryable Slack/API failures are queued through 'scheduleRetry'.
- Manual reprocess pattern:
  1. Inspect due rows ('resolveDueRetries').
  2. Re-run job payload through appropriate service.
  3. Update attempts/status via 'markAttempt'.
- Stuck jobs can be moved to dead-letter status after max attempts.

## ## 7. Manual Overrides

- Manually trigger lesson delivery: run 'runDailyLessonDelivery()'.
- Manually mark lesson complete: update 'learner\_progress' row carefully (Verify in code).
- Manually enroll learner: run enrollment service path through command/harness and verify 'learners' + 'enrollment' rows

## ## 8. Logs and Observability

- \*\*Apps Script execution logs:\*\* ingress and function execution traces.
- \*\*'audit\_log' table:\*\* structured operations and status metadata.
- \*\*Queue tables:\*\* 'delivery\_queue', 'retry\_queue' for asynchronous state visibility.
- Request tracing approach:
  - correlate by row IDs (audit/submission/queue IDs) and timestamps.
  - Verify in code if additional correlation key is introduced.

## ## 9. Health Monitoring

Check these metrics regularly:

- failed Slack API calls
- queue backlog ('delivery\_queue', 'retry\_queue')
- overdue learner count
- scheduler run success/failure
- config validation + SheetDb health snapshot

Use 'runHealthCheck()' / 'HealthMonitor.getSnapshot()' for consolidated status.

## ## 10. Maintenance Tasks

- Clean old audit/queue logs based on retention policy.
- Rotate secrets in Script Properties:
  - 'SLACK\_BOT\_TOKEN'
  - 'SLACK\_SIGNING\_SECRET'
  - optional fallback token
- Re-run trigger setup after ownership/permission changes.
- Review optional config defaults ('DEFAULT\_COURSE\_ID', 'DEFAULT\_TRACK').

## ## 11. Safe Update Procedure

1. Apply minimal scoped change in branch.
2. Run syntax checks and host harness smoke tests.
3. Deploy new web app version.
4. Validate '/learn', '/submit', '/progress', app mention, and interactivity.
5. Monitor logs/queues for 30-60 minutes.
6. Roll back to prior Apps Script deployment if error rate increases.

## ## 12. Escalation Path

When escalating, include:

- incident start time + timezone
- affected workflow(s): slash/event/interactivity/scheduler
- sample payload metadata (redacted)
- relevant audit row IDs and timestamps
- retry queue sample rows
- recent deployment or config changes

Notify:

- Slack app owner/admin
- Apps Script owner
- LMS operations owner

If ownership is unclear, \*\*Verify in code/repo ownership records\*\*.